Available on the CMS information server **CMS AN-19-090**

CMS Draft Analysis Note

The content of this note is intended for CMS internal use and distribution only

2020/09/17
Archive Hash: e1f3720
Archive Date: 2020/09/17

Search for Higgs boson decays into muons in proton-proton collisions at 13 TeV using full Run2 data

Stephane Cooperstein[1], Raffaele Angelo Gerosa[1], Vivek Sharma[1], and Adish Vartak[2]

[1] University of California San Diego (US)
[2] European Center for Nuclear Research (CERN)

DRAFT

Abstract

A search is performed for the decay of the 125 GeV Higgs boson to a pair of muons using $\sqrt{s} = 13\,\text{TeV}$ proton-proton collision data recorded by the CMS experiment at the CERN LHC. The data correspond to an integrated luminosity of $137.2\,\text{fb}^{-1}$. Assuming the branching ratio of the Higgs boson decay to a pair of muons to be the same as that predicted by the standard model, the signal is expected to produce an excess corresponding to 2.2 standard deviations with respect to the background only hypothesis for $m_{\text{H}} = 125\,\text{GeV}$. An excess of events over the background expectation is observed in data with a significance of 2.1 standard deviations. Conversely, the search puts an observed (expected) upper limit of 1.76 (0.93) on the branching ratio of the 125 GeV Higgs boson decay to a pair of muons at 95% confidence level. The measured signal strength, relative to the SM prediction, is $0.95^{+0.48}_{-0.46}$ for $m_{\text{H}} = 125\,\text{GeV}$.

This box is only visible in draft mode. Please make sure the values below make sense.

PDFAuthor: Raffaele Gerosa, Adish Vartak, Stephane Cooperstein
PDFTitle: Search for Higgs boson decays into muons in proton-proton collisions at 13 TeV using full Run2 data
PDFSubject: CMS
PDFKeywords: CMS, physics, your topics

Please also verify that the abstract does not use any user defined symbols

DRAFT

Contents

1 Introduction . . . 3
1.1 Main differences w.r.t. HIG-17-019 . . . 3
2 Data sets and simulation . . . 4
2.1 Simulation overview . . . 4
2.2 Signal simulation . . . 5
2.3 Background simulation . . . 6
2.4 Triggers . . . 8
2.5 Data sets . . . 10
3 Physics objects and event selection . . . 13
3.1 Primary vertex . . . 13
3.2 Muons . . . 13
3.3 Final state radiation recovery . . . 13
3.4 Jets . . . 14
3.5 Missing transverse momentum . . . 14
3.6 Electrons and b-tagged jets . . . 15
3.7 Baseline event selection . . . 15
4 Corrections to data and simulation . . . 15
4.1 Pileup reweighting . . . 15
4.2 Muon efficiency and trigger scale factors . . . 17
4.3 B-tagging scale factors . . . 17
4.4 Correction to Higgs and Z boson transverse momentum . . . 17
4.5 Muon momentum scale and resolution . . . 18
4.6 GeoFit correction to muon p_{T} . . . 20
4.7 Event-by-event mass resolution . . . 21
5 Multivariate discriminant . . . 24
5.1 Input variables to the ggH BDT . . . 26
5.2 Input variables for VBF BDT . . . 28
5.3 BDT discriminant trainings . . . 30
6 Event categorization . . . 37
6.1 Category optimization . . . 40
7 Signal extraction . . . 44
7.1 Signal lineshape and yields . . . 45
7.2 Uncertainties on the signal model . . . 50
7.2.1 Shape uncertainties . . . 52
7.2.2 Rate uncertainties from theory . . . 52
7.2.3 Rate uncertainties from experimental sources . . . 54
8 Background estimation . . . 58
8.1 Uncertainty on the background model . . . 63
8.1.1 Toy generation for bias studies . . . 64
8.1.2 Bias of the various background functions . . . 64
8.2 The core-PDF method . . . 67
9 Results . . . 71

DRAFT

9.1 Effect of systeamtic uncertainties on the background model 77
A Comparison between data and simulation in each era 80
A.1 Comparisons for events belonging to the ggH BDT 80
A.2 Comparisons for events belonging to the VBF BDT 89
B Jet energy scale uncertainties: full splitting 100
C Post-fit shape uncertainties vs background models 102
D Analysis performance in each era 107
E Bias studies for $m_{\mathrm{H}} = 120\,\mathrm{GeV}$ 108
F Bias studies for $m_{\mathrm{H}} = 130\,\mathrm{GeV}$ 110
G Likelihood scans in each category 112

DRAFT

1 Introduction

Since the discovery of the 125 GeV Higgs boson in 2012 [1, 2] various measurements of its interactions with standard model (SM) particles have been performed. The interactions of the Higgs boson with the electroweak gauge bosons and charged fermions belonging to the third generation of SM particles have now been observed, and have been found to be consistent with SM predictions [3–11]. The direct interactions of the Higgs boson with first and second generation fermions, however, have yet to be established experimentally. In fact, since the coupling of the Higgs boson to fermions, as predicted by the SM, is proportional to the fermion mass, the $\mathcal{B}(\mathrm{H} \to f\bar{f})$ is expected to be small for fermions belonging to the first and second generations. In particular, the $\mathrm{H} \to \mu\mu$ branching ratio for the 125 GeV Higgs boson is predicted by the SM to be 2.18×10^{-4} [12].

The most recent search from the CMS Collaboration for $\mathrm{H} \to \mu\mu$ decays was performed using a combination of proton-proton collision data collected at 7, 8 and 13 TeV, corresponding to integrated luminosities of 5.0, 19.8 and 35.9 fb^{-1}, respectively [13]. The expected significance for this search was 1.0 standard deviation w.r.t. the background, whereas the expected upper limit on the production cross section times $\mathcal{B}(\mathrm{H} \to \mu\mu)$ was 2.2 times the SM prediction. The observed significance and the observed upper limit were 0.9 standard deviations and 2.9 times the SM prediction, respectively. The most recent search for $\mathrm{H} \to \mu\mu$ decays from the ATLAS Collaboration has analyzed 13 TeV proton-proton collision data corresponding to a total integrated luminosity of 80 fb^{-1}. This search sets an expected (observed) upper limit on the production cross section times $\mathcal{B}(\mathrm{H} \to \mu\mu)$ of 2.0 (2.1) w.r.t. the SM prediction at 95% confidence level [14, 15].

In this analysis note, we describe the search for $\mathrm{H} \to \mu\mu$ decays performed using the complete data set collected by the CMS detector during Run-2, at a center of mass energy of 13 TeV. This corresponds to a total integrated luminosity of 137.2 fb^{-1}. The analysis described here specifically targets the gluon fusion and vector boson fusion production modes that offer the largest Higgs boson production cross sections at the LHC. The final states of interest comprise a pair of two prompt, isolated and opposite charge muons produced in association with 0, 1 or 2 jets. The largest background in this analysis consists of Drell-Yan events in which an off-shell Z boson decays to a pair of muons. The next most significant background contributions come from $\mathrm{t\bar{t}}$ events, in which both the top quarks decay to muons, and from diboson production (WW, WZ, ZZ). The vector boson fusion signal, although smaller in cross section compared to gluon fusion by an order of magnitude, provides a distinctive signature wherein the two muons from the Higgs boson decay are produced in association with two jets that have a large pseudorapidity gap ($\Delta\eta_{\mathrm{jj}}$), and form a large dijet invariant mass (m_{jj}). This feature helps to suppress the large Drell-Yan background, making the vector boson fusion channel a very sensitive probe of the $\mathrm{H} \to \mu\mu$ signal. In contrast, the gluon fusion signal is quite similar to the Drell-Yan background. The most important discriminating feature, apart from the dimuon resonance peak, is the transverse momentum (p_{T}) of the dimuon system, which has a harder spectrum in the case of the Higgs boson signal w.r.t. the Drell-Yan background. The analysis takes advantage of all of these features, and searches for a narrow dimuon resonance at the Higgs boson mass of 125 GeV on top of the background continuum.

1.1 Main differences w.r.t. HIG-17-019

Several changes and improvements in the analysis strategy have been introduced:

- **MVA classification:** In order to increase the signal-to-background ratio, two independent BDTs are trained targeting events with: $\mathrm{N_{jets}} = 0, 1$ and $\mathrm{N_{jets}} \geq 2$ but the

DRAFT

m_{jj} of the two highest p_{T} jets in the event is below 400 GeV, and $\mathrm{N_{jets}} \geq 2$ with $m_{\mathrm{jj}} > 400$ GeV. The first MVA aims to separate at best the gluon fusion signal from the Drell-Yan background in events with $\mathrm{N_{jets}} = 0, 1$, and 2 but, at the same time, to be sensitive to other Higgs production modes like VBF and VH-hadronic. In contrast, the second BDT exploits the characteristic features of the VBF process to increase the signal purity. The choice of input variables has been optimized in order to achieve the best signal to background separation without relying on observables known to be badly modelled by the simulation.

- **Per event mass uncertainties:** In HIG-17-019 [13], the MVA discriminator was blind to the information of the dimuon mass resolution. The η of the most forward muon in the event was used to gauge the $m_{\mu\mu}$ resolution. In contrast, in this analysis the dimuon mass resolution is embedded in the BDT training. In fact, signal events are weighted proportionally to the per-event mass uncertainty estimated from the covariance matrix of the muon tracks. This choice allows the BDT to learn how the mass resolution evolves with muon p_{T} and η.
- **Category boundaries:** A new strategy in order to maximize the sensitivity is used to define categories along the two MVA scores.
- **Event categorization:** due to the changes described above, the number of event categories is significantly reduced (from 16 in HIG-17-019) while improving the analysis sensitivity.
- **Background model:** The background is estimated directly from data (data-driven) by fitting the dimuon mass spectrum with analytical functions. A new strategy has been developed which allows to keep the bias on the signal strength under control and, at the same time, correlate some of the background parameters across categories, decreasing the number of freely floating parameters in the final fit.

In order to make an apple-to-apple comparison on 2016 data, this new analysis strategy allows improvement in the performance, in both expected significance and upper limit, by about 30% compared to HIG-17-019.

DRAFT

2 Data sets and simulation

Since this analysis targets a multitude of different final states, we take into account signal contributions from the main Higgs boson production modes at the LHC that pass our analysis selection requirements. The considered production modes are: gluon fusion (ggH), vector boson fusion (VBF), associated production with a W boson (WH), associated production with a Z boson (ZH), associated production with a top-quark pair (ttH), associated production with a b-quark pair (bbH), and associated production with single top (tHq and tHW). The WH and ZH processes are sometimes commonly referred to as VH modes. Table 1 provides the cross sections for each of these processes for the 125 GeV SM Higgs boson, along with the respective theoretical uncertainties, as recommended by the LHC Higgs Cross Section Working Group [12]. The SM $\mathrm{H} \rightarrow \mu\mu$ branching ratio for a 125 GeV Higgs boson is predicted to be 2.18×10^{-4} [12]. The uncertainty on this branching ratio is estimated to be $\pm 1.23\%$.

2.1 Simulation overview

Simulations are extremely useful and important to study the kinematic features of the signal and background processes, as well as to assess the impact of detector (mis)reconstruction on the physics objects used in the analysis. Simulated samples of signal and background events

Table 1: Higgs boson production cross sections for various modes at $\sqrt{s} = 13\,\text{TeV}$.

Process	Cross section (pb)	Perturbative Order	+QCD scale unc. (%)	-QCD scale unc. (%)	+(PDF+α_s) unc. (%)	-(PDF+α_s) unc. (%)
ggH	48.58	N3LO(QCD) NLO (EWK)	+4.6	-6.7	+3.2	-3.2
VBF	3.782	NNLO (QCD) NLO (EWK)	+0.4	-0.3	+2.1	-2.1
WH	1.373	NNLO (QCD) NLO (EWK)	+0.5	-0.7	+1.9	-1.9
qq → ZH	0.761	NNLO (QCD) NLO (EWK)	+0.5	-0.6	+1.9	-1.9
gg → ZH	0.123	NLO (QCD)	+25.1	-18.9	+2.4	-2.4
ttH	0.507	NLO (QCD) NLO (EWK)	+5.8	-9.2	+3.6	-3.6
bbH	0.488	NNLO (QCD)	+20.2	-23.9	—	—
tHq	0.074	NLO (QCD)	+6.5	-14.9	+3.7	-3.7
tHW	0.015	NLO (QCD)	+4.9	-6.7	+6.3	-6.3

are used to train MVA discriminants that helps to isolate the Higgs boson signal from the SM backgrounds.

The processes considered in this analysis have been simulated using either the MadGraph5_aMC@NLO [16] or the POWHEG (v2) [17] generators. The matrix element level Monte Carlo (MC) events from these generators are then interfaced with PYTHIA (v8.2 or greater) [18] in order to simulate the fragmentation and hadronization of partons in the initial and final states along with the underlying event. This is done using the CUETP8M1 tune [19] for simulations corresponding to the 2016 data taking era, and using the CP5 tune [20] for the 2017 and 2018 data taking periods.

In the case of the processes simulated with the MadGraph5_aMC@NLO generator at leading order (next-to-leading order), jets from the matrix element calculations are matched to the parton shower produced by PYTHIA following the MLM (FxFx) prescription [21, 22]. The 2016 (2017/18) era simulations use the NNPDF 3.0 (3.1) parton distribution functions [23, 24]. The interactions of all final state particles with the CMS detector are simulated using GEANT4 [25].

2.2 Signal simulation

The MC samples for the ggH production are simulated at next-to-leading order accuracy in QCD using both MadGraph5_aMC@NLO and POWHEG generators. Since the POWHEG simulation only contains events with positive weights, POWHEG samples have been used in the training of the BDT multivariate discriminants, which cannot correctly handle negative weights. On the other hand, the MadGraph5_aMC@NLO samples are used inthe final signal extraction. Simulated events for VBF, VH, ttH, and bbH are produced with NLO precision in QCD theory via the POWHEG generator. Additional signal samples, obtained by varying the tune parameters for the underlying event simulation, are also produced and used to estimate the corresponding systematic uncertainty.

The Higgs boson signal produced through gluon-fusion is simulated by Monte Carlo generators at NLO in QCD and then matched to the parton shower. The p_{T} of the Higgs boson is corrected to the prediction from the NNLOPS generator [26, 27], which is the highest order parton shower matched ggH simulation available that includes the finite top quark mass effects. The reweighting is performed in exclusive jet bins, $\mathrm{N}_{\text{jets}} = 0,\ 1,\ 2,\ 3$ or more jets, where jets are defined at the generator level and required to have $p_{\mathrm{T}} > 30\,\text{GeV}$. More details can be found in Section 4.1 of Ref. [6].

DRAFT

In the case of VBF processes, that have a distinct feature of two jets with a large $\Delta\eta_{jj}$ separation and no colour connection in the rapidity gap, it has been observed a significant difference in the parton showering done by PYTHIA and HERWIG [28] generators. The centrally produced VBF samples showered with PYTHIA have been used only for the training of the MVAs. In fact, as indicated by phenomenological studies shown in Ref. [29], the standard p_T-ordered shower implemented in PYTHIA, even when interfaced with a NLO generator like POWHEG, does not provide a good description of the additional hadronic activity in VBF events. In order to have a more correct description of the kinematics of VBF signal events when the final result is derived, VBF signal samples have been produced privately and showered by either PYTHIA in dipole shower mode[1] or HERWIG programs. The former configuration is used for the central prediction in each era, while the latter (events showered with HERWIG) is used to assign a conservative systematic uncertainty related to modelling of the parton shower. A detailed explanation of the VBF H signal modeling treatment including comparisons among many possible parton shower simulation configurations is available in Appendix A of Ref. [30]. Parton showering with HERWIG is done using HERWIG++ (HERWIG7) with the UE-EE-5C tune [19] (CH3 tune) in the 2016 (2017 and 2018) era. For all the other signal production modes (ggH, VH, and ttH), only PYTHIA has been considered for the parton showering. The signal simulated samples for 2016, 2017 and 2018 eras used in the analysis are listed in Tables 2, 3 and 4, respectively. In addition, simulated events have also been produced for Higgs boson masses of 120 and 130 GeV, allowing to interpolate signal acceptance and lineshape parameters over a 10 GeV mass range.

Simulated events for ggZH, tHq, and tHW processes have been produced privately because no official full-SIM samples were produced with exclusive decays of the Higgs boson into a pair of muons. The ggZ H process is produced at LO via POWHEG, while the remaining ones are also produced at LO accuracy with the MADGRAPH5_aMC@NLO generator.

2.3 Background simulation

The largest contribution to the background in this search comes from Drell-Yan events. We are particularly interested in the off-shell production of the Z boson in the mass range of 110 to 150 GeV. This is the typical range in which a fit is performed to the dimuon mass distribution in order to estimate the background from data and search for the Higgs boson signal peak. Therefore, a Drell-Yan MC sample has been generated via MADGRAPH5_aMC@NLO, with NLO precision in QCD and with up to 2 partons (jets) at the matrix element, applying a dimuon mass cut on LHE leptons in the range between 105 and 160 GeV. In addition, in order to gain statistics in a VBF-like phase space, a NLO QCD DY+2-jet sample has been produced via MADGRAPH5_aMC@NLO requiring two jets with $m_{jj} > 350$ GeV at the generator level. Dimuon events under the Z peak, however, are also of interest to study the kinematic properties of the Drell-Yan background, to compute certain data-based corrections, and, in general, to assess the reliability of the simulation. For such studies, a Drell-Yan MC sample with a dimuon mass cut of 50 GeV at the matrix element level, produced again using MADGRAPH5_aMC@NLO at NLO in QCD with up to 2 jets in the final state, is used.

When dimuon events with VBF-like jets are considered, the contribution from the electroweak production of the Z boson becomes significant (Zjj-EW). This process has been simulated at LO using MADGRAPH5_aMC@NLO and showered via HERWIG ++ (2016 simulated samples) or HERWIG7 (2017 and 2018 MC). The HERWIG parton shower is adopted instead of PYTHIA because it is known to better model purely electroweak process without color connection as reported in Ref. [31]. Two alternative set of samples for the Zjj-EW production are available:

---

[1] By default PYTHIA produces parton shower following a p_T-ordered logic. However, it can also produce "dipole-showers" by seeting the parameter "dipoleRecoil" to ON in the configuration.

DRAFT

Table 2: Signal simulated samples for the 2016 data taking era. The (*) stands for PUMoriond17_94X_mcRun2_asymptotic_v3.

Process	Sample	$\sigma \times$ BR (pb)
ggH	`/GluGlu_HToMuMu_M125_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v2/MINIAODSIM`	0.01057 pb
	`/GluGluHToMuMu_M125_TuneCP5_PSweights_13TeV_amcatnloFXFX_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.01057 pb
	`/GluGluHToMuMu_M125_TuneCP5up_PSweights_13TeV_amcatnloFXFX_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.01057 pb
	`/GluGluHToMuMu_M125_TuneCP5down_PSweights_13TeV_amcatnloFXFX_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.01057 pb
	`/GluGluHToMuMu_M120_TuneCP5_PSweights_13TeV_amcatnloFXFX_pythia8/⋆-v1/MINAODSIM`	0.012653 pb
	`/GluGluHToMuMu_M130_TuneCP5_PSweights_13TeV_amcatnloFXFX_pythia8/⋆-v1/MINAODSIM`	0.008505 pb
VBF	`/VBFHToMuMu_M-125_TuneCP5_PSweights_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.0008228 pb
	`/VBFHToMuMu_M-125⋆-powheg-pythia_dipole/rgerosa-RunIISummer16MiniAODv3⋆/USER`	0.0008228 pb
	`/VBFHToMuMu_M-125⋆-powheg-herwigpp/rgerosa-RunIISummer16MiniAODv3⋆/USER`	0.0008228 pb
	`/VBFHToMuMu_M-125_TuneCP5up_PSweights_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.0008228 pb
	`/VBFHToMuMu_M-125_TuneCP5down_PSweights_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.0008228 pb
	`/VBFHToMuMu_M-120_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM`	0.000953 pb
	`/VBFHToMuMu_M-130_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM`	0.000683 pb
WH	`/WPlusH_HToMuMu_M125_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v2/MINIAODSIM`	0.0001858 pb
	`/WPlusH_HToMuMu_WToAll_M125_TuneCP5up_PSweights_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.0001858 pb
	`/WPlusH_HToMuMu_WToAll_M125_TuneCP5down_PSweights_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.0001858 pb
	`/WplusH_HToMuMu_WToAll_M120_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM`	0.000232 pb
	`/WplusH_HToMuMu_WToAll_M130_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM`	0.000139 pb
	`/WMinusH_HToMuMu_M125_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v2/MINIAODSIM`	0.0001164 pb
	`/WMinusH_HToMuMu_WToAll_M125_TuneCP5up_PSweights_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.0001164 pb
	`/WMinusH_HToMuMu_WToAll_M125_TuneCP5down_PSweights_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.0001164 pb
	`/WminusH_HToMuMu_WToAll_M120_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM`	0.000148 pb
	`/WminusH_HToMuMu_WToAll_M130_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM`	0.000088 pb
qqZH	`/ZH_HToMuMu_M125_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v2/MINIAODSIM`	0.0001923 pb
	`/ZH_HToMuMu_ZToAll_M125_TuneCP5up_PSweights_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.0001923 pb
	`/ZH_HToMuMu_ZToAll_M125_TuneCP5down_PSweights_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.0001923 pb
	`/ZH_HToMuMu_ZToAll_M120_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆3-v1/MINIAODSIM`	0.000241 pb
	`/ZH_HToMuMu_ZToAll_M130_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆3-v1/MINIAODSIM`	0.000148 pb
ggZH	`/ggZH_HToMuMu_ZToLL_M125_13TeV_powheg_pythia8/rgerosa⋆RunIISummer16MiniAODv3⋆/USER`	2.684e-06 pb
ttH	`/ttHToMuMu_M125_TuneCP5_PSweights_13TeV-powheg-pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.0001103 pb
	`/ttHToMuMu_M125_TuneCP5up_PSweights_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.0001103 pb
	`/ttHToMuMu_M125_TuneCP5down_PSweights_13TeV_powheg_pythia8/RunIISummer16MiniAODv3⋆-v1/MINIAODSIM`	0.0001103 pb
	`/ttHToMuMu_M120_TuneCP5_PSweights_13TeV-powheg-pythia8/⋆-v1/MINIAODSIM`	0.000138 pb
	`/ttHToMuMu_M130_TuneCP5_PSweights_13TeV-powheg-pythia8/⋆-v1/MINIAODSIM`	0.000085 pb
bbH	`/bbHToMuMu_M-125_4FS_yb2_TuneCP5_13TeV-amcatnlo-pythia8/RunIISummer16MiniAODv3⋆/MINIAODSIM`	0.0001059 pb
tHq	`/THQ_HToMuMu_TuneCP5_13TeV-madgraph-pythia/rgerosa-RunIISummer16MiniAODv3⋆/USER`	1.612e-05 pb
tHW	`/THW_HToMuMu_TuneCP5_13TeV-madgraph-pythia/rgerosa-RunIISummer16MiniAODv3⋆/USER`	3.292e-06 pb

one in which the invariant mass of the two LHE muons is required to be larger than 50 GeV, the other one in which $m_{\mu\mu}$ is between 105 and 160 GeV. The former sample is used when data and simulation are compared under the Z-peak, while the latter is used for the signal region. A significant bias was observed for the prediction from the original centrally produced EW Zjj MC samples because of a generator-level selection applied that removed events that would have later passed the offline selections. This effect was studied in detail and a new set of EW Zjj samples was produced covering the full offline phase space, with the quark-level p_{T} requirement loosened from 25 GeV to 1 GeV. A detailed description of these studies is available in Appendix C of Ref. [30].

After Drell-Yan, one of the most significant background contribution comes from $\mathrm{t\bar{t}}$ events in which both the top quarks decay leptonically. The process has been generated at NLO using the POWHEG generator. Furthermore, there are minor contributions from other top quark processes such as single top (tW, t, and s-channel) production, ttZ, ttW, ttWW, tZq, and semi-leptonic $\mathrm{t\bar{t}}$ decays. These contributions are also taken into account via dedicated NLO simulation produced with MADGRAPH5_aMC@NLO or POWHEG. Remaining background can be attributed to the diboson processes (ZZ, WZ, WW) with some very small contributions from triboson production (WWW, WWZ, WZZ, ZZZ) that have also been taken into account. The diboson processes have been simulated using either MADGRAPH5_aMC@NLO or POWHEG at NLO in QCD, whereas the triboson processes have been simulated using MADGRAPH5_aMC@NLO with NLO precision in QCD corrections.

The centrally produced background simulated samples for 2016, 2017 and 2018 eras are listed in Tables 5, 6 and 7, respectively. The cross sections used for normalizing the background expected yields are obtained from the best available theoretical predictions. In particular, the

DRAFT

Table 3: Signal simulation samples for the 2017 data taking era. The (*) stands for PU2017_12Apr2018_94X_mc2017_realistic_v14.

Process	Sample	$\sigma \times$ BR (pb)
ggH	/GluGluHToMuMu_M-125_TuneCP5_PSweights_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.01057 pb
	/GluGluHToMuMu_M125_TuneCP5_PSweights_13TeV_amcatnloFXFX_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.01057 pb
	/GluGluHToMuMu_M125_TuneCP5up_PSweights_13TeV_amcatnloFXFX_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.01057 pb
	/GluGluHToMuMu_M125_TuneCP5down_PSweights_13TeV_amcatnloFXFX_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.01057 pb
	/GluGluHToMuMu_M120_TuneCP5_PSweights_13TeV_amcatnloFXFX_pythia8/⋆-v1/MINIAODSIM	0.012653 pb
	/GluGluHToMuMu_M130_TuneCP5_PSweights_13TeV_amcatnloFXFX_pythia8/⋆-v1/MINIAODSIM	0.008505 pb
VBF	/VBFHToMuMu_M-125_TuneCP5_PSweights_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0008228 pb
	/VBFHToMuMu_M-125⋆powheg-pythia_dipole/rgerosa-RunIIFall17MiniAODv2⋆MINIAODSIM⋆/USER	0.0008228 pb
	/VBFHToMuMu_M-125⋆powheg-herwig7_fixed/rgerosa-RunIIFall17MiniAODv2⋆MINIAODSIM⋆/USER	0.0008228 pb
	/VBFHToMuMu_M-125_TuneCP5up_PSweights_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0008228 pb
	/VBFHToMuMu_M-125_TuneCP5down_PSweights_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0008228 pb
	/VBFHToMuMu_M-120_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM	0.000953 pb
	/VBFHToMuMu_M-130_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM	0.000683 pb
WH	/WplusH_HToMuMu_WToAll_M125_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v3/MINIAODSIM	0.0001858 pb
	/WplusH_HToMuMu_WToAll_M125_TuneCP5up_PSweights_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0001858 pb
	/WplusH_HToMuMu_WToAll_M125_TuneCP5down_PSweights_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0001858 pb
	/WplusH_HToMuMu_WToAll_M120_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM	0.000232 pb
	/WplusH_HToMuMu_WToAll_M130_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM	0.000139 pb
	/WminusH_HToMuMu_WToAll_M125_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0001164 pb
	/WminusH_HToMuMu_WToAll_M125_TuneCP5up_PSweights_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0001164 pb
	/WminusH_HToMuMu_WToAll_M125_TuneCP5down_PSweights_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0001164 pb
	/WminusH_HToMuMu_WToAll_M120_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM	0.000148 pb
	/WminusH_HToMuMu_WToAll_M130_13TeV_powheg_pythia8/⋆-v1.MINIAODSIM	0.000088 pb
ZH	/ZH_HToMuMu_ZToAll_M125_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0001923 pb
	/ZH_HToMuMu_ZToAll_M125_TuneCP5up_PSweights_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0001923 pb
	/ZH_HToMuMu_ZToAll_M125_TuneCP5down_PSweights_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0001923 pb
	/ZH_HToMuMu_ZToAll_M120_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.000241 pb
	/ZH_HToMuMu_ZToAll_M130_13TeV_powheg_pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.000148 pb
ggZH	/ggZH_HToMuMu_ZToLL_M125_13TeV_powheg_pythia8/rgerosa⋆RunIIFall17MiniAODv2⋆/USER	2.684e-06 pb
ttH	/ttHToMuMu_M125_TuneCP5_PSweights_13TeV-powheg-pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0001103 pb
	/ttHToMuMu_M125_TuneCP5up_PSweights_13TeV-powheg-pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0001103 pb
	/ttHToMuMu_M125_TuneCP5down_PSweights_13TeV-powheg-pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.0001103 pb
	/ttHToMuMu_M120_TuneCP5_PSweights_13TeV-powheg-pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.000138 pb
	/ttHToMuMu_M130_TuneCP5_PSweights_13TeV-powheg-pythia8/RunIIFall17MiniAODv2⋆-v1/MINIAODSIM	0.000085 pb
bbH	/bbHToMuMu_M-125_4FS_yb2_TuneCP5_13TeV-amcatnlo-pythia8/RunIIFall17MiniAODv2⋆/MINIAODSIM	0.0001059 pb
tHq	/THQ_HToMuMu_TuneCP5_13TeV-madgraph-pythia/rgerosa-RunIIFall17MiniAODv2⋆/USER	1.612e-05 pb
tHW	/THW_HToMuMu_TuneCP5_13TeV-madgraph-pythia/rgerosa-RunIIFall17MiniAODv2⋆/USER	3.292e-06 pb

Drell-Yan cross section has been obtained from the FEWZ [32–34] generator at NNLO accuracy in QCD, and NLO accuracy in electroweak corrections. Similarly, the cross section for the $\mathrm{t\bar{t}}$ has been computed at NNLO+NNLL accuracy in QCD [35]. The cross section for WW production is taken from Ref. [36] calculated at NNLO in QCD, while the NLO cross section for the WZ process, as obtained from the simulated MC samples, is corrected with NNLO/NLO k-factors provided in Ref. [37]. The fully differential cross section for the qq → ZZ process has been computed at NNLO in QCD [38], and the NNLO/NLO K-factor as a function of m_{ZZ} has been applied to the simulated samples. This K-factor varies from 1.0 to 1.2 as a function of m_{ZZ}. Finally, the gg → ZZ cross section is corrected with NNLO/LO k-factors following the same strategy detailed in Ref. [39].

2.4 Triggers

Signal events in this analysis contain two prompt, isolated, and high p_{T} muons. When the Higgs boson is produced at rest in the transverse plane ($p_{\mathrm{T}}^{\mathrm{H}} \approx 0$), the two muons are emitted back-to-back with a transverse momentum of about $m_{\mathrm{H}}/2 \approx 60\,\mathrm{GeV}$. At the generator level, the p_{T} distribution for the leading muon produced in both ggH and VBF signal events turns on at about 40 GeV, peaking around 60 GeV as previously explained. Therefore, signal events can be selected with high efficiency in data using online single muon triggers. The high level trigger paths for the single muon triggers used in this search are listed in Table 8. These triggers impose a loose isolation requirement on each muon candidate, and a p_{T} threshold of 27 (24) GeV in 2017 (2016, 2018).

DRAFT

Table 4: Signal simulation samples for the 2018 data taking era. The (*) stands for 102X_upgrade2018_realistic_v15.

Process	Sample	$\sigma \times$ BR (pb)
ggH	`/GluGluHToMuMu_M125_TuneCP5_PSweights_13TeV_amcatnloFXFX_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.01057 pb
	`/GluGluHToMuMu_M-125_TuneCP5_PSweights_13TeV_powheg_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.01057 pb
	`/GluGluHToMuMu_M125_TuneCP5up_PSweights_13TeV_amcatnloFXFX_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.01057 pb
	`/GluGluHToMuMu_M125_TuneCP5down_PSweights_13TeV_amcatnloFXFX_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.01057 pb
	`/GluGluHToMuMu_M120_TuneCP5_PSweights_13TeV_amcatnloFXFX_pythia8/⋆-v1/MINIAODSIM`	0.012653 pb
	`/GluGluHToMuMu_M130_TuneCP5_PSweights_13TeV_amcatnloFXFX_pythia8/⋆-v1/MINIAODSIM`	0.008505 pb
VBF	`/VBFHToMuMu_M-125_TuneCP5_PSweights_13TeV_powheg_pythia8/RunIIAutumn18MiniAOD⋆-v1/MINIAODSIM`	0.0008228 pb
	`/VBFHToMuMu_M-125⋆-powheg-pythia_dipole_fixed/rgerosa-RunIIAutumn18MiniAOD⋆-MINIAODSIM⋆/USER`	0.0008228 pb
	`/VBFHToMuMu_M-125⋆-powheg-herwig7/rgerosa-RunIIAutumn18MiniAOD⋆-MINIAODSIM⋆/USER`	0.0008228 pb
	`/VBFHToMuMu_M125_TuneCP5up_PSweights_13TeV_amcatnlo_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0008228 pb
	`/VBFHToMuMu_M125_TuneCP5down_PSweights_13TeV_amcatnlo_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0008228 pb
	`/VBFHToMuMu_M120_TuneCP5_PSweights_13TeV_amcatnlo_pythia8/⋆-v1/MINIAODSIM`	0.000953 pb
	`/VBFHToMuMu_M130_TuneCP5_PSweights_13TeV_amcatnlo_pythia8/⋆-v1/MINIAODSIM`	0.000683 pb
WH	`/WplusH_HToMuMu_WToAll_M125_TuneCP5_PSweights_13TeV_powheg_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0001858 pb
	`/WplusH_HToMuMu_WToAll_M125_TuneCP5up_PSweights_13TeV_powheg_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0001858 pb
	`/WplusH_HToMuMu_WToAll_M125_TuneCP5down_PSweights_13TeV_powheg_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0001858 pb
	`/WplusH_HToMuMu_WToAll_M120_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM`	0.000232 pb
	`/WplusH_HToMuMu_WToAll_M130_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM`	0.000139 pb
	`/WminusH_HToMuMu_WToAll_M125_TuneCP5_PSweights_13TeV_powheg_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0001164 pb
	`/WminusH_HToMuMu_WToAll_M125_TuneCP5up_PSweights_13TeV_powheg_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0001164 pb
	`/WminusH_HToMuMu_WToAll_M125_TuneCP5down_PSweights_13TeV_powheg_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0001164 pb
	`/WminusH_HToMuMu_WToAll_M120_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM`	0.000148 pb
	`/WminusH_HToMuMu_WToAll_M130_TuneCP5_PSweights_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM`	0.000088 pb
ZH	`/ZH_HToMuMu_ZToAll_M125_TuneCP5_PSweights_13TeV_powheg_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0001923 pb
	`/ZH_HToMuMu_ZToAll_M125_TuneCP5up_PSweights_13TeV_powheg_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0001923 pb
	`/ZH_HToMuMu_ZToAll_M125_TuneCP5down_PSweights_13TeV_powheg_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0001923 pb
	`/ZH_HToMuMu_ZToAll_M120_TuneCP5_PSweights_13TeV_powheg_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.000241 pb
	`/ZH_HToMuMu_ZToAll_M130_TuneCP5_PSweights_13TeV_powheg_pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.000148 pb
ggZH	`/ggZH_HToMuMu_ZToLL_M125_13TeV_powheg_pythia8/rgerosa⋆RunIIAutumn18MiniAOD⋆/USER`	2.684e-06 pb
ttH	`/ttHToMuMu_M125_TuneCP5_PSweights_13TeV-powheg-pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0001103 pb
	`/ttHToMuMu_M125_TuneCP5up_PSweights_13TeV-powheg-pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0001103 pb
	`/ttHToMuMu_M125_TuneCP5down_PSweights_13TeV-powheg-pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.0001103 pb
	`/ttHToMuMu_M120_TuneCP5_PSweights_13TeV-powheg-pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.000138 pb
	`/ttHToMuMu_M130_TuneCP5_PSweights_13TeV-powheg-pythia8/RunIIAutumn18MiniAOD⋆-v2/MINIAODSIM`	0.000085 pb
bbH	`/bbHToMuMu_M-125_4FS_yb2_TuneCP5_13TeV-amcatnlo-pythia8/RunIIAutumn18MiniAOD⋆/MINIAODSIM`	0.0001059 pb
tHq	`/THQ_HToMuMu_TuneCP5_13TeV-madgraph-pythia/rgerosa-RunIIAutumn18MiniAOD⋆/USER`	1.612e-05 pb
tHW	`/THW_HToMuMu_TuneCP5_13TeV-madgraph-pythia/rgerosa-RunIIAutumn18MiniAOD⋆/USER`	3.292e-06 pb

Table 5: Background simulation samples for the 2016 data taking era. The (*) stands for RunIISummer16MiniAODv3-PUMoriond17_94X_mcRun2_asymptotic_v3.

Process	Sample	Cross section (pb)	Cross section Precision
Drell-Yan	`/DYJetsToLL_M-50_TuneCUETP8M1_13TeV-amcatnloFXFX-pythia8/⋆_ext2-v1/MINIAODSIM`	5954	NNLO (QCD) NLO (EWK)
	`/DYJetsToLL_M-105To160_TuneCUETP8M1_PSweights_13TeV-amcatnloFXFX-pythia8/⋆-v1/MINIAODSIM`	47.12	NNLO (QCD) NLO (EWK)
	`/DYJetsToLL_M-105To160_VBFFilter_TuneCUETP8M1_PSweights_13TeV-amcatnloFXFX-pythia8/⋆/MINIAODSIM`	1.774	NNLO (QCD) NLO (EWK)
	`/EWK_LLJJ_MLL_105-160_SM_5f_LO_TuneCH3_13TeV-madgraph-herwig7_corrected/⋆/MINIAODSIM`	0.075	LO
	`/EWK_LLJJ_MLL50_MJJ-120_TuneCH3_PSweights_13TeV-madgraph-herwig7_corrected/⋆/MINIAODSIM`	0.983	LO
Top	`/TTTo2L2Nu_TuneCUETP8M2_ttHtranche3_13TeV-powheg-pythia8/⋆-v2/MINIAODSIM`	86.61	NNLO (QCD)
	`/TTToSemilepton_TuneCUETP8M2_ttHtranche3_13TeV-powheg-pythia8/⋆-v2/MINIAODSIM`	358.57	NNLO (QCD)
	`/ST_tW_top_5f_inclusiveDecays_13TeV-powheg-pythia8_TuneCUETP8M1/⋆_ext1-v1/MINIAODSIM`	35.9	NNLO (QCD)
	`/ST_tW_antitop_5f_inclusiveDecays_13TeV-powheg-pythia8_TuneCUETP8M1/⋆_ext1-v1/MINIAODSIM`	35.9	NNLO (QCD)
	`/ST_t-channel_antitop_4f_inclusiveDecays_13TeV-powhegV2-madspin-pythia8_TuneCUETP8M1/⋆-v1/MINIAODSIM`	80.95	NLO (QCD)
	`/ST_t-channel_top_4f_inclusiveDecays_13TeV-powhegV2-madspin-pythia8_TuneCUETP8M1/⋆-v1/MINIAODSIM`	136.02	NLO (QCD)
	`/ST_s-channel_4f_leptonDecays_13TeV-amcatnlo-pythia8_TuneCUETP8M1/⋆-v1/MINIAODSIM`	3.40	NLO (QCD)
	`/tZq_ll_4f_13TeV-amcatnlo-pythia8/⋆_ext1-v1/MINIAODSIM`	0.0758	NLO (QCD)
	`/TTWJetsToLNu_TuneCUETP8M1_13TeV-amcatnloFXFX-madspin-pythia8/⋆_ext2-v1/MINIAODSIM`	0.2001	NLO (QCD)
	`/TTZToLLNuNu_M-10_TuneCUETP8M1_13TeV-amcatnlo-pythia8/⋆_ext2-v1/MINIAODSIM`	0.2529	NLO (QCD)
	`/TTTT_TuneCUETP8M2T4_13TeV-amcatnlo-pythia8/⋆-v1/MINIAODSIM`	0.0091	NLO (QCD)
	`/TTTW_TuneCUETP8M2T4_13TeV-amcatnlo-pythia8/⋆_ext1-v1/MINIAODSIM`	0.00073	NLO (QCD)
	`/TTWW_TuneCUETP8M2T4_13TeV-amcatnlo-pythia8/⋆_ext1-v1/MINIAODSIM`	0.0070	NLO (QCD)
	`/TTTJ_TuneCUETP8M2T4_13TeV-amcatnlo-pythia8/⋆_ext1-v1/MINIAODSIM`	0.000039	NLO (QCD)
Diboson	`/WWTo2L2Nu_13TeV-powheg/⋆-v2/MINIAODSIM`	12.178	NNLO (QCD)
	`/WZTo3LNu_TuneCUETP8M1_13TeV-powheg-pythia8/⋆_ext1-v1/MINIAODSIM`	4.102	NLO (QCD)
	`/WZTo2L2Q_13TeV_amcatnloFXFX_madspin_pythia8/⋆-v2/MINIAODSIM`	5.606	NLO (QCD)
	`/ZZTo2L2Nu_13TeV_powheg_pythia8_ext1/⋆-v2/MINIAODSIM`	0.5644	NLO (QCD)
	`/ZZTo2L2Q_13TeV_amcatnloFXFX_madspin_pythia8/⋆-v3/MINIAODSIM`	3.224	NLO (QCD)
	`/ZZTo4L_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM`	1.256	NLO (QCD)
	`/GluGluToContinToZZTo(2e2mu,2mu2tau,2e2tau)_13TeV_MCFM701_pythia8/⋆-v2/MINIAODSIM`	0.00318	LO
	`/GluGluToContinToZZTo(4mu,4tau)_13TeV_MCFM701_pythia8/⋆-v2/MINIAODSIM`	0.00157	LO
	`/GluGluToContinToZZTo(2mu2nu)_13TeV_MCFM701_pythia8/⋆-v2/MINIAODSIM`	0.00149	LO
Triboson	`/WWW_4F_TuneCUETP8M1_13TeV-amcatnlo-pythia8/⋆-v2/MINIAODSIM`	0.2086	NLO (QCD)
	`/WWZ_TuneCUETP8M1_13TeV-amcatnlo-pythia8/⋆-v2/MINIAODSIM`	0.1651	NLO (QCD)
	`/WZZ_TuneCUETP8M1_13TeV-amcatnlo-pythia8/⋆-v2/MINIAODSIM`	0.05565	NLO (QCD)
	`/ZZZ_TuneCUETP8M1_13TeV-amcatnlo-pythia8/⋆-v2/MINIAODSIM`	0.01398	NLO (QCD)

DRAFT

Table 6: Background simulation samples for the 2017 data taking era. The (*) stands for RunIIFall17MiniAODv2-PU2017_12Apr2018_94X_mc2017_realistic_v14.

Process	Sample	Cross section (pb)	Cross section Precision
Drell-Yan	`/DYJetsToLL_M-50_TuneCP5_13TeV-amcatnloFXFX-pythia8/⋆v1/MINIAODSIM`	6225.4	NNLO (QCD) NLO (EWK)
	`/DYJetsToLL_M-105To160_TuneCP5_PSweights_13TeV-amcatnloFXFX-pythia8/⋆-v1/MINIAODSIM`	47.12	NNLO (QCD) NLO (EWK)
	`/DYJetsToLL_M-105To160_VBFFilter_TuneCP5_PSweights_13TeV-amcatnloFXFX-pythia8/⋆/MINIAODSIM`	2.043	NNLO (QCD) NLO (EWK)
	`/EWK_LLJJ_MLL_105-160_SM_5f_LO_TuneCH3_13TeV-madgraph-herwig7_corrected/⋆/MINIAODSIM`	0.078	LO
	`/EWK_LLJJ_MLL50_MJJ-120_TuneCH3_PSweights_13TeV-madgraph-herwig7_corrected/⋆/MINIAODSIM`	1.029	LO
Top	`/TTTo2L2Nu_TuneCP5_PSweights_13TeV-powheg-pythia8/⋆-v1/MINIAODSIM`	86.61	NNLO (QCD)
	`/TTToSemiLeptonic_TuneCP5_PSweights_13TeV-powheg-pythia8/⋆-v2/MINIAODSIM`	358.57	NNLO (QCD)
	`/ST_tW_top_5f_inclusiveDecays_TuneCP5_PSweights_13TeV-powheg-pythia8/⋆-v1/MINIAODSIM`	35.9	NNLO (QCD)
	`/ST_tW_antitop_5f_inclusiveDecays_TuneCP5_PSweights_13TeV-powheg-pythia8/⋆-v2/MINIAODSIM`	35.9	NNLO (QCD)
	`/ST_t-channel_antitop_5f_TuneCP5_PSweights_13TeV-powheg-pythia8/⋆-v1/MINIAODSIM`	80.95	NLO (QCD)
	`/ST_t-channel_top_5f_TuneCP5_13TeV-powheg-pythia8/⋆-v1/MINIAODSIM`	136.02	NLO (QCD)
	`/ST_s-channel_antitop_leptonDecays_13TeV-PSweights_powheg-pythia/⋆-v1/MINIAODSIM`	1.293	NLO (QCD)
	`/ST_s-channel_top_leptonDecays_13TeV-PSweights_powheg-pythia/⋆-v1/MINIAODSIM`	2.069	NLO (QCD)
	`/tZq_ll_4f_ckm_NLO_TuneCP5_PSweights_13TeV-amcatnlo-pythia8/⋆-v1/MINIAODSIM`	0.0758	NLO (QCD)
	`/TTWJetsToLNu_TuneCP5_PSweights_13TeV-amcatnloFXFX-madspin-pythia8/⋆-v1/MINIAODSIM`	0.2001	NLO (QCD)
	`/TTZToLLNuNu_M-10_TuneCUETP8M1_13TeV-amcatnlo-pythia8/⋆/MINIAODSIM`	0.2529	NLO (QCD)
	`/TTTT_TuneCP5_13TeV-amcatnlo-pythia8/⋆-v1/MINIAODSIM`	0.0091	NLO (QCD)
	`/TTTW_TuneCP5_13TeV-amcatnlo-pythia8/⋆-v1/MINIAODSIM`	0.00073	NLO (QCD)
	`/TTTJ_TuneCP5_13TeV-amcatnlo-pythia8/⋆-v1/MINIAODSIM`	0.0070	NLO (QCD)
	`/TTWW_TuneCP5_13TeV-amcatnlo-pythia8/⋆-v1/MINIAODSIM`	0.000039	NLO (QCD)
Diboson	`/WWTo2L2Nu_NNPDF31_TuneCP5_PSweights_13TeV-powheg-pythia8/⋆_ext1-v1/MINIAODSIM`	12.178	NNLO (QCD)
	`/WZTo3LNu_13TeV-powheg-pythia8/⋆-v2/MINIAODSIM`	4.658	NLO (QCD)
	`/WZTo2L2Q_13TeV_amcatnloFXFX_madspin_pythia8/⋆-v1/MINIAODSIM`	6.331	NLO (QCD)
	`/ZZTo2L2Nu_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM`	0.601	NLO (QCD)
	`/ZZTo2L2Q_13TeV_amcatnloFXFX_madspin_pythia8/⋆-v1/MINIAODSIM`	3.688	NLO (QCD)
	`/ZZTo4L_13TeV_powheg_pythia8/⋆-v1/MINIAODSIM`	1.325	NLO (QCD)
	`/GluGluToContinToZZTo(2e2mu,2mu2tau,2e2tau)_13TeV_MCFM701_pythia8/⋆-v2/MINIAODSIM`	0.00329	LO
	`/GluGluToContinToZZTo(4mu,4tau)_13TeV_MCFM701_pythia8/⋆-v2/MINIAODSIM`	0.00140	LO
	`/GluGluToContinToZZTo(2mu2nu)_13TeV_MCFM701_pythia8/⋆-v2/MINIAODSIM`	0.00177	LO
Triboson	`/WWW_4F_TuneCP5_13TeV-amcatnlo-pythia8/⋆-v2/MINIAODSIM`	0.2086	NLO (QCD)
	`/WWZ_4F_TuneCP5_13TeV-amcatnlo-pythia8/⋆-v2/MINIAODSIM`	0.1651	NLO (QCD)
	`/WZZ_TuneCP5_13TeV-amcatnlo-pythia8/⋆-v1/MINIAODSIM`	0.05565	NLO (QCD)
	`/ZZZ_TuneCP5_13TeV-amcatnlo-pythia8/⋆-v2/MINIAODSIM`	0.01398	NLO (QCD)

2.5 Data sets

The analysis is performed using dimuon events in which at least one of the two muons fires the single muon triggers listed in Section 2.4. For this reason, offline events are selected from the SingleMuon data set in all the three years – 2016, 2017 and 2018.

The 2016 data used in this analysis correspond to an integrated luminosity of 35.9 fb^{-1}. The data are re-analyzed with the `MiniAODv3` and with the new analysis strategy, described in the following sections. Good run ranges and luminosity blocks are selected using the JSON file `Cert_271036-284044_13TeV_ReReco_07Aug2017_Collisions16_JSON.txt`.

The 2017 data used in this analysis correspond to an integrated luminosity of 41.5 fb^{-1}. Good run ranges and luminosity blocks are selected using the JSON file `Cert_294927-306462_13TeV_EOY2017ReReco_Collisions17_JSON_v1.txt`.

The 2018 data used in this analysis correspond to an integrated luminosity of 59.7 fb^{-1}. Good run ranges and luminosity blocks are selected using the JSON file `Cert_314472-325175_13TeV_17SeptEarlyReReco2018ABC_PromptEraD_Collisions18_JSON.txt`.

DRAFT

Table 7: Background simulation samples for the 2018 data taking era. The (*) stands for RunIIAutumn18MiniAOD-102X_upgrade2018_realistic_v15.

Process	Sample	Cross section (pb)	Cross section Precision
Drell-Yan	`/DYJetsToLL_M-50_TuneCP5_13TeV-amcatnloFXFX-pythia8/⋆v1/MINIAODSIM`	6225.4	NNLO (QCD) NLO (EWK)
	`/DYJetsToLL_M-105To160_TuneCP5_PSweights_13TeV-amcatnloFXFX-pythia8/⋆-v1/MINIAODSIM`	47.12	NNLO (QCD) NLO (EWK)
	`/DYJetsToLL_M-105To160_VBFFilter_TuneCP5_PSweights_13TeV-amcatnloFXFX-pythia8/⋆/MINIAODSIM`	2.029	NNLO (QCD) NLO (EWK)
	`/EWK_LLJJ_MLL_105-160_SM_5f_LO_TuneCH3_13TeV-madgraph-herwig7_corrected/⋆/MINIAODSIM`	0.078	LO
	`/EWK_LLJJ_MLL50_MJJ-120_TuneCH3_PSweights_13TeV-madgraph-herwig7_corrected/⋆/MINIAODSIM`	1.029	LO
Top	`/TT_DiLept_TuneCP5_13TeV-amcatnlo-pythia8/⋆_ext1-v2/MINIAODSIM`	86.61	NNLO (QCD)
	`/TTTo2L2Nu_TuneCP5_13TeV-powheg-pythia8/⋆-v1/MINIAODSIM`		NNLO (QCD)
	`/TTToSemiLeptonic_TuneCP5_13TeV-powheg-pythia8/⋆-v1/MINIAODSIM`	358.57	NNLO (QCD)
	`/ST_tW_top_5f_inclusiveDecays_TuneCP5_13TeV-powheg-pythia8/⋆_ext1-v1/MINIAODSIM`	35.9	NNLO (QCD)
	`/ST_tW_antitop_5f_inclusiveDecays_TuneCP5_13TeV-powheg-pythia8/⋆_ext1-v1/MINIAODSIM`	35.9	NNLO (QCD)
	`/ST_t-channel_antitop_5f_TuneCP5_13TeV-powheg-pythia8/⋆-v1/MINIAODSIM`	80.95	NLO (QCD)
	`/ST_t-channel_top_5f_TuneCP5_13TeV-powheg-pythia8/⋆-v1/MINIAODSIM`	136.02	NLO (QCD)
	`/ST_s-channel_4f_leptonDecays_TuneCP5_13TeV-madgraph-pythia8/⋆_ext1-v4/MINIAODSIM`	3.40	NLO (QCD)
	`/tZq_ll_4f_ckm_NLO_TuneCP5_13TeV-madgraph-pythia8/⋆_ext1-v2/MINIAODSIM`	0.0758	NLO (QCD)
	`/TTWJetsToLNu_TuneCP5_13TeV-amcatnloFXFX-madspin-pythia8/⋆_ext1-v2/MINIAODSIM`	0.2001	NLO (QCD)
	`/TTZToLLNuNu_M-10_TuneCP5_13TeV-amcatnlo-pythia8/⋆_ext1-v2/MINIAODSIM`	0.2529	NLO (QCD)
	`/TTTT_TuneCP5_13TeV-amcatnlo-pythia8/⋆_ext1-v2/MINIAODSIM`	0.0091	NLO (QCD)
	`/TTTW_TuneCP5_13TeV-amcatnlo-pythia8/⋆_ext1-v2/MINIAODSIM`	0.00073	NLO (QCD)
	`/TTWW_TuneCP5_13TeV-amcatnlo-pythia8/⋆_ext1-v2/MINIAODSIM`	0.00698	NLO (QCD)
	`/TTTJ_TuneCP5_13TeV-amcatnlo-pythia8/⋆_ext1-v2/MINIAODSIM`	0.00039	NLO (QCD)
Diboson	`/WWTo2L2Nu_NNPDF31_TuneCP5_13TeV-powheg-pythia8/⋆-v1/MINIAODSIM`	12.178	NNLO (QCD)
	`/WZTo3LNu_TuneCP5_13TeV-powheg-pythia8/⋆_ext1-v2/MINIAODSIM`	4.658	NLO (QCD)
	`/WZTo2L2Q_13TeV_amcatnloFXFX_madspin_pythia8/⋆-v1/MINIAODSIM`	6.321	NLO (QCD)
	`/ZZTo2L2Nu_TuneCP5_13TeV_powheg_pythia8/⋆_ext1-v2/MINIAODSIM`	0.601	NLO (QCD)
	`/ZZTo2L2Q_13TeV_amcatnloFXFX_madspin_pythia8/⋆-v1/MINIAODSIM`	3.696	NLO (QCD)
	`/ZZTo4L_TuneCP5_13TeV_powheg_pythia8/⋆_ext1-v2/MINIAODSIM`	1.325	NLO (QCD)
	`/GluGluToContinToZZTo(2e2mu,2mu2tau,2e2tau)_13TeV_MCFM701_pythia8/⋆-v2/MINIAODSIM`	0.00329	LO
	`/GluGluToContinToZZTo(4mu,4tau)_13TeV_MCFM701_pythia8/⋆-v2/MINIAODSIM`	0.001402	LO
	`/GluGluToContinToZZTo(2mu2nu)_13TeV_MCFM701_pythia8/⋆-v2/MINIAODSIM`	0.001772	LO
Triboson	`/WWW_4F_TuneCP5_13TeV-amcatnlo-pythia8/⋆_ext1-v2/MINIAODSIM`	0.2086	NLO (QCD)
	`/WWZ_TuneCP5_13TeV-amcatnlo-pythia8/⋆_ext1-v2/MINIAODSIM`	0.1651	NLO (QCD)
	`/WZZ_TuneCP5_13TeV-amcatnlo-pythia8/⋆_ext1-v2/MINIAODSIM`	0.05565	NLO (QCD)
	`/ZZZ_TuneCP5_13TeV-amcatnlo-pythia8/⋆_ext1-v2/MINIAODSIM`	0.01398	NLO (QCD)

Table 8: Single muon trigger paths used for selecting events for further analysis in 2016, 2017 and 2018 eras.

Year	Data set path
2016	HLT_IsoMu24_v* OR HLT_IsoTkMu24_v*
2017	HLT_IsoMu27_v*
2018	HLT_IsoMu24_v*

Table 9: Single muon data sets used in this analysis for the 2016 era.

Dataset	Run Range	Int. Lum.
`/SingleMuon/Run2016B-17Jul2018-v1/MINIAOD`	272007—275376	$5.8\,\mathrm{fb}^{-1}$
`/SingleMuon/Run2016C-17Jul2018-v1/MINIAOD`	275657—276283	$2.6\,\mathrm{fb}^{-1}$
`/SingleMuon/Run2016D-17Jul2018-v1/MINIAOD`	276315—276811	$4.2\,\mathrm{fb}^{-1}$
`/SingleMuon/Run2016E-17Jul2018-v1/MINIAOD`	276831—277420	$4.0\,\mathrm{fb}^{-1}$
`/SingleMuon/Run2016F-17Jul2018-v1/MINIAOD`	277772—278808	$3.1\,\mathrm{fb}^{-1}$
`/SingleMuon/Run2016G-17Jul2018-v1/MINIAOD`	278820—280385	$7.5\,\mathrm{fb}^{-1}$
`/SingleMuon/Run2016H-17Jul2018-v1/MINIAOD`	280919—284044	$8.6\,\mathrm{fb}^{-1}$

DRAFT

Table 10: Single muon data sets used in this analysis for the 2017 era.

Dataset	Run Range	Int. Lum.
`/SingleMuon/Run2017B-31Mar2018-v1/MINIAOD`	297047—299329	4.8 fb^{-1}
`/SingleMuon/Run2017C-31Mar2018-v1/MINIAOD`	299368—302029	9.6 fb^{-1}
`/SingleMuon/Run2017D-31Mar2018-v1/MINIAOD`	302031—302663	4.2 fb^{-1}
`/SingleMuon/Run2017E-31Mar2018-v1/MINIAOD`	303824—304797	9.2 fb^{-1}
`/SingleMuon/Run2017F-31Mar2018-v1/MINIAOD`	305040—306462	13.4 fb^{-1}

DRAFT

Table 11: Single muon data sets used in this analysis for the 2018 era.

Dataset	Run Range	Int. Lum.
`/SingleMuon/Run2018A-17Sep2018-v2/MINIAOD`	315257—316995	14.00 fb^{-1}
`/SingleMuon/Run2018B-17Sep2018-v1/MINIAOD`	317080—319310	7.10 fb^{-1}
`/SingleMuon/Run2018C-17Sep2018-v1/MINIAOD`	319337—320655	6.94 fb^{-1}
`/SingleMuon/Run2018D-31Mar2018-v1/MINIAOD`	320500—325175	31.93 fb^{-1}

3 Physics objects and event selection

The analysis primarily relies on muons and jets in order to target the final state signature of ggH and VBF signal events, but it also makes use of other physics objects such as electrons, and b-tagged jets in order to suppress certain backgrounds like $\mathrm{t\bar{t}}$ and dibosons, as well as to remain mutually exclusive from other searches that target VH and ttH production modes by requiring the presence of additional leptons or b-tagged jets to isolate signal events.

3.1 Primary vertex

Events are required to contain at least one well-identified primary vertex. Among the set of these vertices, the one with the largest value of summed physics-object p_{T}^2 is taken to be the *primary interaction vertex*. Moreover, the following requirements are used to identify the so-called "good vertices":

- Not identified as fake
- $|z| < 24$ mm
- $\rho < 2$ mm
- Number of degrees of freedom > 4

3.2 Muons

Events in this analysis are required to have *exactly* two opposite charge muons with $p_{\mathrm{T}} > 20\,\mathrm{GeV}$, $|\eta| < 2.4$, and passing the medium identification (ID) criteria defined by the MUO POG [40]. Each of the muons is required to pass the loose isolation requirement [41] in which the particle flow (PF) [42] based isolation sum of the p_{T} of charged and neutral PF hadrons, and PF photons, lying in a $\Delta R = \sqrt{\Delta\eta^2 + \Delta\phi^2}$ cone of 0.4 around the muon, is required to be less than 25% of the p_{T} of the muon. The charged hadrons included in this sum are required to originate from the primary interaction vertex. If a PF photon in the isolation cone is associated with the final state radiation (FSR) of the muon (see Section 3.3), then it is not included in the isolation sum. Furthermore, the $\Delta\beta$-correction is applied to the isolation sum which subtracts half of the p_{T} sum of charged hadron candidates, within the isolation cone, that belong to vertices other than the primary interaction one. This correction aims to deconvolve from the isolation calculation the contamination due to neutral particles produced by interactions other than the hard scattering (primary interaction vertex).

At least one of the two muons is required to be geometrically matched to the level-3 muon candidate corresponding to a single muon trigger that was fired in the event. This muon is required to have $p_{\mathrm{T}} > 29(26)$ GeV for 2017 (2016, 2018) eras to ensure that it lies on the single muon trigger plateau.

3.3 Final state radiation recovery

The sensitivity of this search depends critically on the resolution of the $\mathrm{H} \to \mu\mu$ mass peak. In a small fraction of signal events, a muon in the final state may radiate a photon, thereby losing some of its momentum. This causes a slight degradation in the resolution of the signal peak. A procedure has been employed that attempts to recover this FSR by looking for PF photons in the vicinity of the muon. This strategy is based on the one developed in the context of the $\mathrm{H} \to \mathrm{ZZ}$ searches [43] with some modifications tuned by Oliver Rieger in the context of the $\mathrm{H} \to \mu\mu$ search [44]. The procedure can be summarized as follows:

DRAFT

- Consider all reconstructed muons with $p_{\mathrm{T}} > 20\,\mathrm{GeV}$ and $|\eta| < 2.4$ as candidates for FSR recovery.c
- For a given muon, consider all PF photons with $0.0001 < \Delta R(\mu,\gamma) < 0.5$, $p_{\mathrm{T}} > 2\,\mathrm{GeV}$, $0.0 < |\eta| < 1.4442$ OR $1.566 < |\eta| < 2.5$ as possible FSR candidates.
- Ignore PF photons that are associated with a reconstructed electron, i.e. belong to the list of photons associated with the bremsstrahlung of the electron.
- In order to strongly suppress the contamination from $\mathrm{H} \to \mathrm{Z}\gamma \to \mu\mu\gamma$ decays, FSR photon candidates are required to have $p_{\mathrm{T}}^{\gamma}/p_{\mathrm{T}}^{\mu} < 0.4$.
- Only consider a photon with PF isolation sum, without pileup correction and computed in a cone of $\Delta R < 0.3$, less than 180% of its p_{T} as an FSR candidate ($I_{\gamma}/p_{\mathrm{T}}^{\mu}$).
- In case of multiple FSR photon candidates, only the one with the smallest value of $\Delta R(\mu,\gamma)/p_{\mathrm{T}}^{2}$ (p_{T} is that of the photon) is considered.
- Require $\Delta R(\mu,\gamma)/p_{\mathrm{T}}^{2} < 0.012$.

If an FSR photon is associated with a muon, its momentum is added to that of the original muon. The FSR-corrected momentum is promoted to be the new momentum of the muon. Furthermore, as already described in Section 3.2, this photon is ignored from the isolation sum of the muon when imposing the loose isolation requirement.

3.4 Jets

Jets used in the analysis are reconstructed by clustering the PF candidates in the event using the anti-k_{t} algorithm [45] with a distance parameters of 0.4. While clustering the jets, only the charged PF candidates that are associated with the primary interaction vertex are considered (charged hadron subtraction). Jet momentum is determined as the vector sum of all particle momenta inside the jet, and is found from simulation to vary, on average, between 5 and 10% of the true momentum over the whole p_{T} spectrum and detector acceptance. An offset correction (L1FastJet correction) is applied to jet energies to take into account the contribution from additional pp interactions within the same or adjacent bunch crossings (pileup) [46]. Additional energy corrections are derived from simulation and applied as a function of jet η and p_{T} (L2, L3 corrections). Finally, residual corrections are measured *in situ* from the energy balance in dijet, multijet, γ+jets, and leptonically decaying Z+jets events [47]. These are applied to jets in data in order to match the jet response with the simulation [48].

Offline, jets are required to have $p_{\mathrm{T}} > 25\,\mathrm{GeV}$ and $|\eta| < 4.7$. Furthermore, jets are required to pass the loose pileup jet ID and the loose (tight) jet ID requirements as recommended by the JME POG for the 2016 (2017, 2018) era. These requirements help to suppress the contamination from pileup jets and misreconstructed jets from detector noise in the fiducial region beyond the coverage of the tracker, respectively. Jets are also cleaned w.r.t. the selected muon candidates by requiring a geometrical separation of $\Delta R(j,\mu) > 0.4$. In order to mitigate the impact of jet misreconstruction due to ECAL endcap noise in 2017, jets that lie in the region $2.6 < |\eta| < 3.0$ are required to pass the tight pileup jet ID w.p. in both data and MC. Detailed studies explaining how this selection has been tuned in order to reject spurious jets in data and, at the same time, ensure the minimal loss in performance (signal acceptance) are reported in the following set of slides presented in one JME (JMAR) meeting [49].

3.5 Missing transverse momentum

The $p_{\mathrm{T}}^{\mathrm{miss}}$ is computed by taking the negative vector sum of the transverse momenta of all the reconstructed PF candidates in a given event. Corrections to the momenta of jets reconstructed

DRAFT

in the event are propagated to the p_T^{miss} (Type-1 correction). The analysis also employs all the p_T^{miss} filters recommended by the JME POG [50] to remove events with spurious p_T^{miss} due to misreconstruction, detector noise and non-collision background.

3.6 Electrons and b-tagged jets

A veto is applied on events with one or more electrons with $p_T > 20\,\text{GeV}$, $|\eta| < 2.5$ passing the WP90 MVA based ID developed by the EGamma POG [51]. A veto is also applied on events containing at least two jets in the final state, defined as in Section 3.4, in which either one of the jets lies in the tracker fiducial region ($|\eta| < 2.5$) and pass the medium w.p. of the DeepCSV algorithm developed by the BTV group [52–54] or at least two of them pass the loose w.p. of the same tagger . The purpose of these vetoes is to ensure that this analysis is mutually exclusive of other searches targeting other Higgs boson production modes such as leptonic VH and ttH.

3.7 Baseline event selection

We start by selecting events accepted by the single muon trigger in which there are exactly two opposite charged muons passing the requirements described in Section 3.2. Events containing electrons or b-tagged jets or additional muons[2] are vetoed. Three selection regions are defined based on the reconstructed dimuon invariant mass ($m_{\mu\mu}$). Events with $70 < m_{\mu\mu} < 110\,\text{GeV}$ belong to the *Z control region* as these events are largely dominated by on-shell Z+jets production. Events with $110 < m_{\mu\mu} < 150\,\text{GeV}$ belong to the *signal fit region* as this is the mass range that is used in the final fit for the signal and the background extraction. Lastly events with $115 < m_{\mu\mu} < 135\,\text{GeV}$ belong to the *signal region*, as the $\text{H} \to \mu\mu$ signal resides here. This mass range is blinded in the initial phase of the analysis.

4 Corrections to data and simulation

In this analysis we apply corrections to several physical observables in both data and simulation. This is essential to correct for certain biases in detector response, mismodeling of physical observables in simulation, and to ensure that the simulation correctly models the key features of the signal and background processes. The background in this search is determined in an entirely data-driven manner. However, we do categorize events as a function of a multivariate discriminant that is trained on simulated signal and background events. We also rely entirely on simulation to predict all features of our expected signal. Therefore, it is desirable to ensure that our signal and background processes are faithfully modelled by the simulation as much as possible.

4.1 Pileup reweighting

The simulation is produced with a pileup profile that is approximately similar to that measured in data. However, a reweigthing is performed to ensure that the profile of the pileup interactions in simulation matches exactly with the estimated pileup distribution integrated over the data taking period. The LUM POG provides the luminosity information for each certified lumisection [55] which is then multiplied by the estimated minimum bias cross section of 69.2 mb to obtain the pileup profile in data. In the case of the simulation, the true number of in-time pileup interactions are known, and these are used to obtain the pileup profile with which the simulated sample was produced. The ratio of the data and simulation profiles gives the weight

---

[2] Additional muons with $p_T > 20\,\text{GeV}$, passing the medium ID and the loose isolation requirement.

DRAFT

to be applied to a simulated event as a function of the true number of in-time interactions. These pileup weights are derived separately for the 2016, 2017 and 2018 eras.

The number of reconstructed primary vertices in a collision event is directly correlated with the pileup, and therefore, the distribution of the number of reconstructed vertices is indicative of the amount of pileup in a certain data taking period. Figure 1 shows the distribution of the number of vertices in the “signal fit region” (SR) in data and simulation, shown before (top row) after (bottom row) pileup reweighting for the 2016, 2017 and 2018 eras. These are dimuon events passing the trigger and muon selection requirements in which the dimuon mass falls between 110–150 GeV. Pileup reweighting improves visibly the agreement only in the 2016 era.

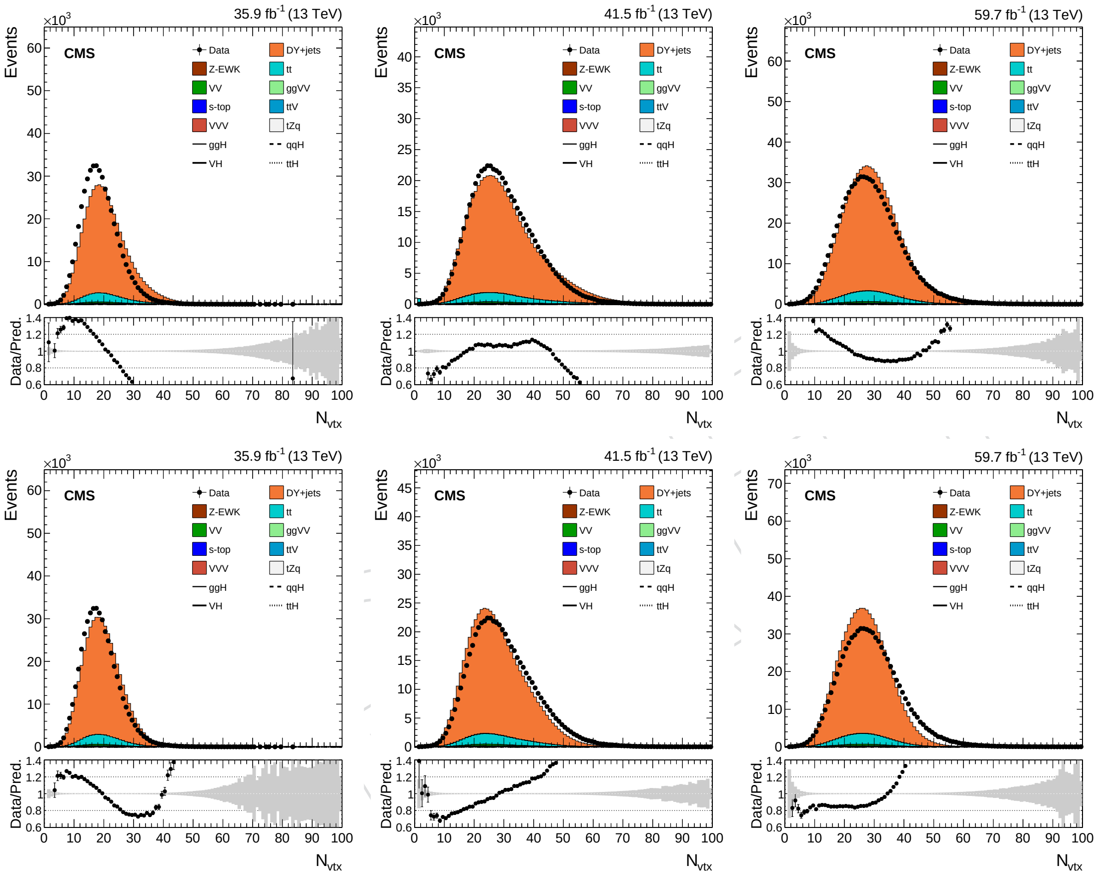

Figure 1: Distribution of the number of reconstructed primary vertexes in 2016 (left), 2017 (middle) and 2018 (right) eras reported for both data (black points) and simulation (solid histograms) before (first row) and after (second row) the pileup reweighting for events with $m_{\mu\mu}$ between 110 and 150 GeV. The pileup reweighting improves the agreement only in the 2016 era.

However, the remaining disagreements between data and simulation observed in the distribution of number of vertexes is expected to have a negligible impact in analysis. Therefore, no additional corrections, like re-weight the N_{vtx} in MC to match the one observed in data, are applied. In fact, all the experimental corrections for muons, electrons, jets, and b-jets have been derived applying only the official pileup reweight, hence they should already take into account the effect of residual differences in pileup between data and simulation. Moreover, observables more prone to the pileup model like p_T^{miss} are not used for the event classification.

4.2 Muon efficiency and trigger scale factors

The MUO POG has measured the efficiencies, in both data and simulation, for the identification and isolation requirements used in this analysis for selecting muons. These efficiencies have been measured using a tag-and-probe technique on Z+jets events in data and simulation, as a function of the muon p_{T} and η. These results are used to compute data-to-simulation scale factors that are then applied per muon, in simulated events, given its p_{T} and η.

The MUO POG also provides the efficiencies for a muon passing the baseline ID and isolation requirements to fire the single muon trigger. These are also computed as a function of the muon p_{T} and η. In a dimuon event, if the the per muon trigger efficiencies are ϵ_1 and ϵ_2, then the per-event efficiency to fire the trigger is given by $\epsilon = 1 - (1-\epsilon_1)(1-\epsilon_2)$. These efficiencies can be computed separately for data and for simulation, and their ratio gives the scale factor that is then applied as a weight to a simulated dimuon event.

4.3 B-tagging scale factors

In this analysis we veto events with b-tagged jets. We apply scale factors to correct for differences in the inefficiency of the b-jet veto in data and simulation. The scale factors, applied as event weights to simulated events, are derived using the method described as "event reweighting using scale factors and MC b-tagging efficiencies" in Ref. [56].

4.4 Correction to Higgs and Z boson transverse momentum

The Higgs boson signal produced through gluon fusion is simulated by Monte Carlo generators at NLO in QCD and then matched to the parton shower. The p_{T} of the Higgs boson is corrected to the prediction from the NNLOPS generator [26, 27], which is the highest order parton shower matched ggH simulation available that includes the finite top quark mass effects. The reweighting is performed in exclusive jet bins, $\mathrm{N_{jets}} = 0$, 1, 2, 3 or more jets, where jets are defined at the generator level and required to have $p_{\mathrm{T}} > 30\,\mathrm{GeV}$. More details can be found in Section 4.1 of Ref. [6].

We also perform a reweighting of the dimuon p_{T} distribution of the Drell-Yan background simulation. This reweighting is derived from data in the Z control region ($70 < m_{\mu\mu} < 110\,\mathrm{GeV}$, ZCR). All corrections described so far are applied to the simulated Drell-Yan events in the ZCR and their total yield is normalized to that observed in data. Figure 2 shows the dimuon p_{T} distribution in data and simulation for the 2016 (left), 2017 (middle) and 2018 (right) eras prior to any specific correction of the Z/γ^*+jets p_{T} spectrum and prior to $\mathrm{N_{jet}}$ binning. The distributions in data and simulation have visible differences, particularly at low p_{T}. These can be attributed to missing resummation effects that are not modelled in simulation. A correction for this effect is derived as follows:

- In each era, events are divided into three groups according to the number of jets, $\mathrm{N_{jets}} = 0, = 1, \geq 2$.
- The normalized data-to-simulation ratio in each jet bin is smoothed with a 12-th order polynomial in order to have a continuous correction as a function of the dimuon p_{T} up to 200 GeV.

Figure 3, 4, and 5 show the dimuon p_{T} distributions in the SR for 2016, 2017, and 2018 eras, respectively, obtained after the p_{T} reweighting for events with $\mathrm{N_{jets}} = 0$ (left), $\mathrm{N_{jets}} = 1$ (middle), and $\mathrm{N_{jets}} \geq 2$ (right). The re-weight allows to obtain a very good shape agreement between data and simulation in all eras and jet multiplicity.

DRAFT

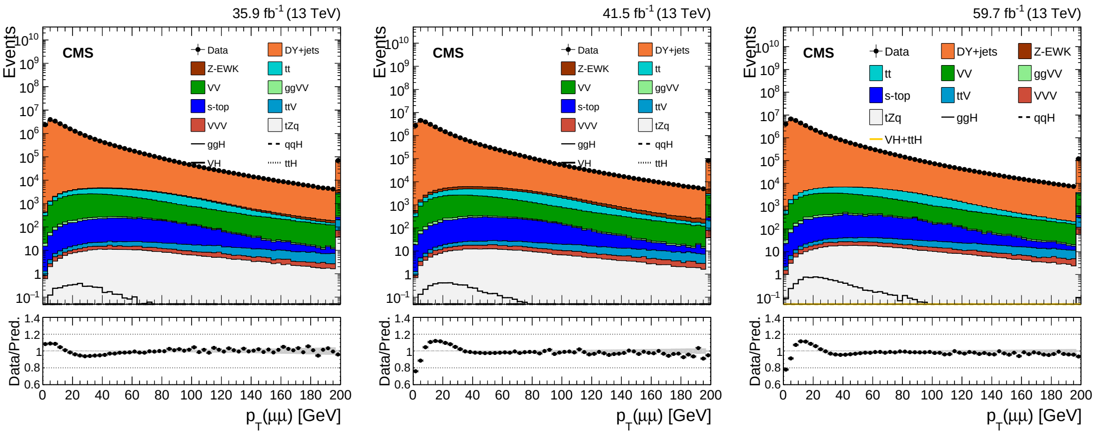

Figure 2: The dimuon p_T distribution in 2016 (left), 2017 (middle) and 2018 (right) eras reported for both data (black points) and simulation (solid histograms) for events with $m_{\mu\mu}$ between 70 and 110 GeV. A visible disagreement between data and MC is present for events with dimuon p_T smaller than 40 GeV. The different trend observed in the 2016 era compared to 2017 and 2018 is due to different parton shower and underlying event tunes used in the simulation as described in Section 2.1.

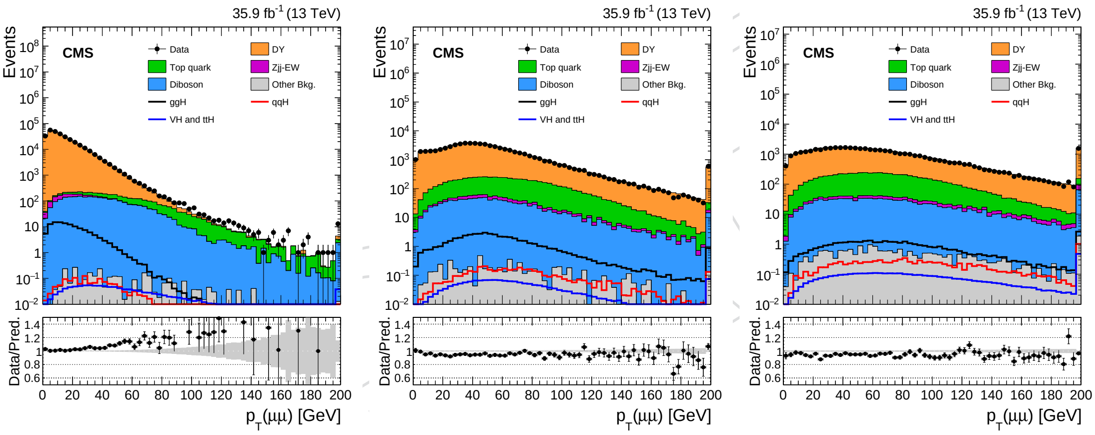

Figure 3: The dimuon p_T distribution in the 2016 era in events with $N_{\text{jets}} = 0$ (left), $N_{\text{jets}} = 1$ (middle), and $N_{\text{jets}} \geq 2$ reported for both data (black points) and simulation (solid histograms) for events in the SR after the Z/γ^*+jets p_T reweight. The agreement between data and simulation is significantly improved by the reweight defined in the Z+jets control region.

4.5 Muon momentum scale and resolution

There are differences in both the muon momentum scale and resolution between data and simulation. This causes a shift in the peak position of a dimuon resonance and a difference in its resolution between data and simulation. This is corrected by the *Rochester corrections*, which are applied to both data and simulation. The muon momentum scale is corrected in both data and simulation so that the Z boson peak matches the PDG value, while an additional smearing is applied to the muon momentum in the simulation so that the Z boson peak resolution in data and simulation agree. We use the Run2 v3 corrections as reported in Ref. [57]. Figure 6 shows the dimuon mass distributions for each era in the Z+jets control region after applying the Rochester corrections.

In order to check in more details whether the *Rochester corrections* are effectively resolving the disagreements in scale and resolution between data and simulation, the Z boson peak in data

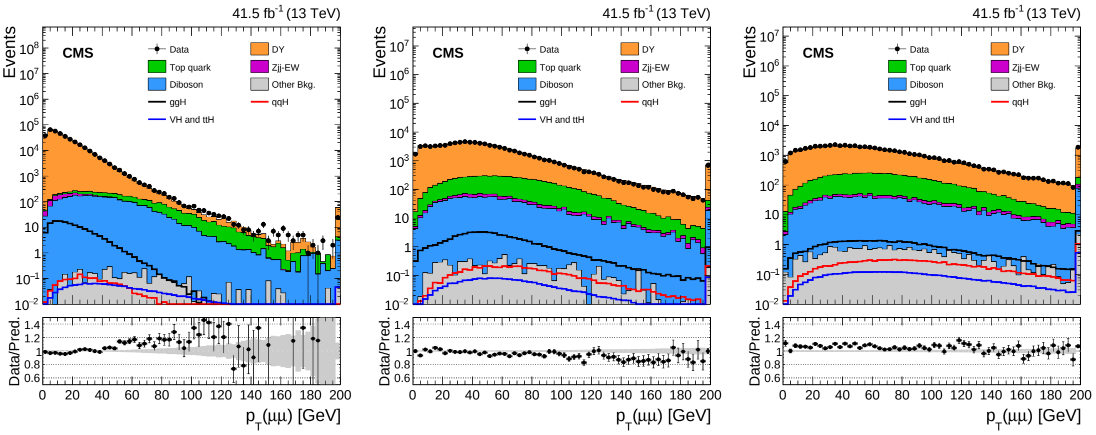

Figure 4: The dimuon p_T distribution in the 2017 era in events with $N_{jets} = 0$ (left), $N_{jets} = 1$ (middle), and $N_{jets} \geq 2$ reported for both data (black points) and simulation (solid histograms) for events in the SR after the Z/γ^*+jets p_T reweight. The agreement between data and simulation is significantly improved by the reweight defined in the Z+jets control region.

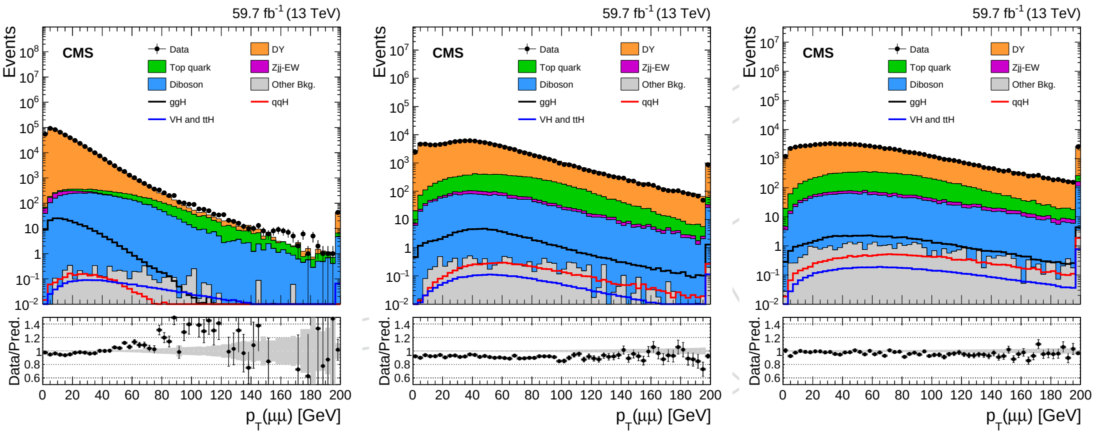

Figure 5: The dimuon p_T distribution in the 2018 era in events with $N_{jets} = 0$ (left), $N_{jets} = 1$ (middle), and $N_{jets} \geq 2$ reported for both data (black points) and simulation (solid histograms) for events in the SR after the Z/γ^*+jets p_T reweight. The agreement between data and simulation is significantly improved by the reweight defined in the Z+jets control region.

and MC is fitted with analytical functions after applying the *Rochester corrections* in the mass range $75 < m_{\mu\mu} < 105\,\mathrm{GeV}$. Dimuon events are categorized based on the η of each of the two muons. Three η categories are considered: barrel (B) with $|\eta| < 0.9$, overlap (O) with $0.9 < |\eta| < 1.8$, and endcap (E) $1.8 < |\eta| < 2.4$. This gives a total of nine possible η categories for a dimuon event. In each η-category, the dimuon mass distribution is fit with a functional form given by the convolution of a Breit-Wigner (Eq. 1) and a double-sided Crystal Ball (Eq. 2). The peak and width parameters of the Breit-Wigner function are fixed to the nominal mass and width of the Z boson, whereas the other parameters are allowed to float. Moreover, a background component is also included and is modelled with an exponential function. Figure 7 shows the fitted Z peak values in data and MC for each of the nine η-categories for 2016 (left), 2017 (middle), and 2018 (right) eras. Similarly, Fig. 8 reports the trends observed for the σ of the double-sided Crystal Ball, which is used as a proxy for the dimuon mass resolution. After the application of the *Rochester corrections*, the muon momentum scale, represented by the position of the Z boson peak, agrees within 0.1% between data and simulation. In contrast, the dimuon mass resolution in MC remains slightly optimistic, in particular for barrel muons in the 2016 era. Overall, the residual discrepancy between data and MC (not covered by the *Rochester energy smearing*) remains smaller than 5% across η-categories and eras. The maximum difference observed between data and simulation is covered by the mass resolution uncertainty assigned in the final fit, as described later in Section 7.

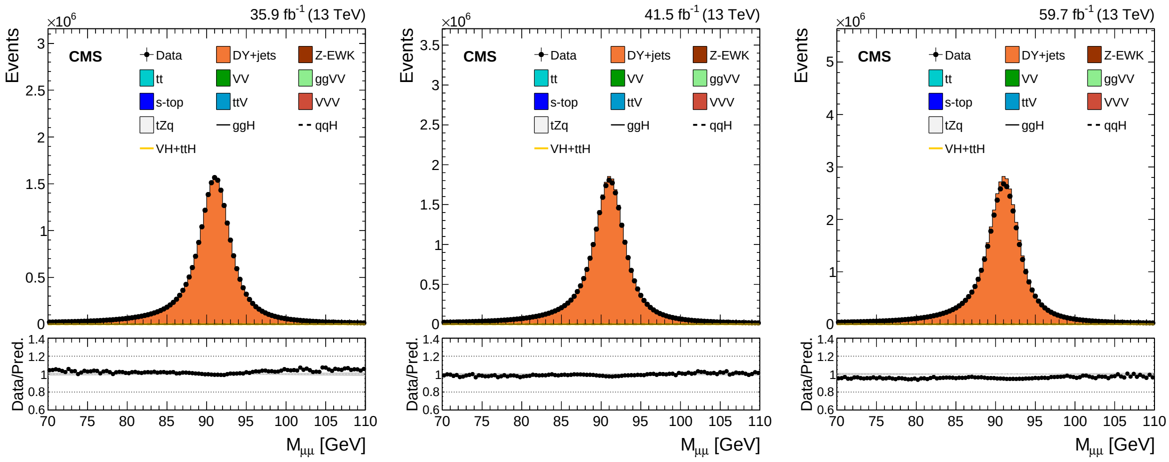

Figure 6: The dimuon mass distribution in 2016 (left), 2017 (middle) and 2018 (right) eras reported for both data (black points) and simulation (solid histograms) after the application of the Rochester corrections for events with $m_{\mu\mu}$ between 70 and 110 GeV. Besides a few percent shift in the overall normalization, the agreement in both peak position and resolution between data and simulation is good in all data taking periods.

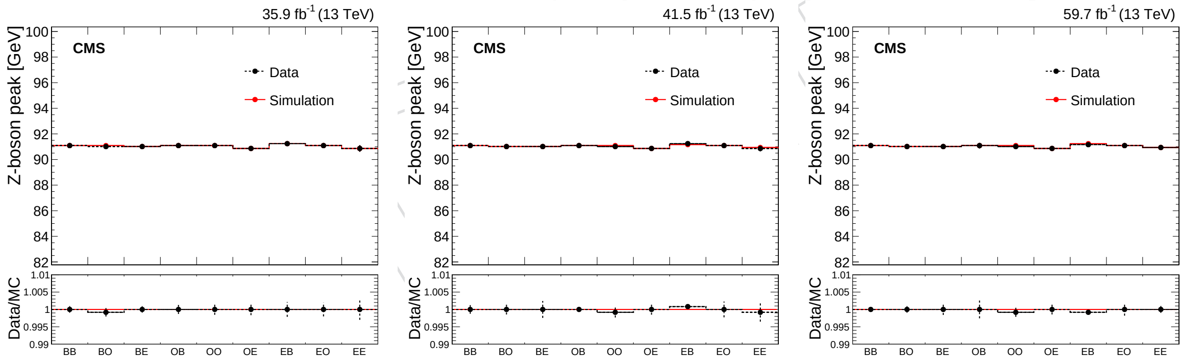

Figure 7: Z boson peak position extracted from an analytical fit to the dimuon mass distribution of events categoriezed in nine exclusive η-bins for 2016 (left), 2017 (middle), and 2018 (right) eras. Values are extracted independently for data and simulated events.

4.6 GeoFit correction to muon p_{T}

In CMS, the reconstructed muons can have non-zero displacements (d0) in the plane transverse to the beam direction with respect to the position of either the reconstructed primary vertex or the beam spot one. If this track-displacement is the result of a bias in the track reconstruction, this can cause a shift in the reconstructed muon p_{T} increasing the dimuon mass resolution due to an extra smearing for events with muons with non-zero d0. The effect of this residual d0 is observed as a trend in reconstructed p_{T} values of these muons, leading to an increase in the fitted dimuon mass peak in Z+jets events along the $d0_{\mu^+} - d0_{\mu^-}$ direction. A simple geometric inspection shows that in tracks that are bent by the magnetic field, d0 values are related to the radius of curvature of the tracks as $\Delta R/R^2$. Since in homogeneous magnetic fields $p_{\mathrm{T}} \approx R$,

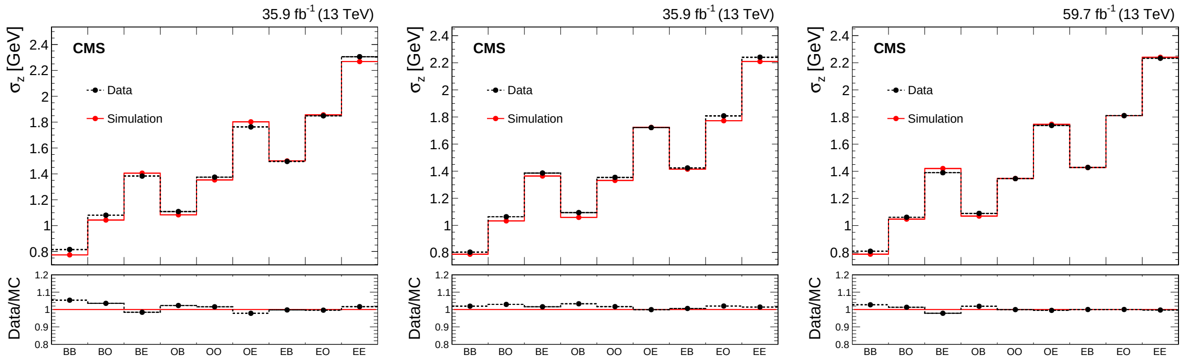

Figure 8: Z boson peak resolution extracted from an analytical fit to the dimuon mass distribution of events categoriezed in nine exclusive η-bins for 2016 (left), 2017 (middle), and 2018 (right) eras. Values are extracted independently for data and simulated events.

$d0 \approx \Delta p_{\mathrm{T}}/p_{\mathrm{T}}^2$. Therefore, this bias in the p_{T} of the muons can be corrected by finding the constant of proportionality of the aforementioned equation. The correction derived using this procedure is named GeoFit correction due to its geometric motivation.

The GeoFit correction is expected to be sensitive to the muon charge. In fact, since the charge of the muon affects the bending of the track in a magnetic field, the effect and the related correction must have a reversed sign for μ^- compared to μ^+. The constant of proportionality is found to evolve as a function of η. Therefore, corrections have been derived in three separate η-regions: $|\eta| < 0.9$, $0.9 < |\eta| < 1.7$, and $|\eta| > 1.7$. In deriving these GeoFit corrections, the position of the track with respect to the beam spot has been used as a reference ($d0_{BS}$). In addition, a strong dependence of the correction is observed as a function of the era of the MC samples that can be explained by the fact that detector conditions and reconstructed algorithms evolved during time. Due to improvements in inner tracker and BS calculation algorithms, we observe the performance of GeoFit correction to improve from 2016 to 2018. More details about these studies and results can be found in the Appendix A of Ref. [58]. Overall, the Geofit corrections allow to improve the dimuon mass resolution in signal events by about 3–10% depending on the era and the production mode.

4.7 Event-by-event mass resolution

The sensitivity of the search depends on the resolution of the Higgs boson mass peak, which in turn depends on the muon momentum resolution. The muon p_{T} resolution varies primarily depending on the η of the muon. Muons that pass through the central barrel region of the detector ($|\eta| < 0.9$) have the best p_{T} resolution, whereas muons passing through the forward region ($|\eta| > 1.8$) tend to have the worst resolution. In order to optimize the sensitivity of the search it is important to separate out events with good dimuon mass resolution from those in which the mass is measured less precisely.

In a given event, it is possible to estimate the uncertainty on the measured mass of the dimuon system from the covariance matrix of the fitted track of the individual muons. Figure 9 shows the shape of the distribution of the relative dimuon mass uncertainty in ggH signal events for different geometrical regions of the detector through which the muons traverse. We can see from this figure that the mass resolution tends to get worse, on average, as the maximum η of the muons increases. However, the mass resolution distribution for a given geometrical configuration of the two muons has a certain spread, therefore it is more suitable to make use of the per-event mass uncertainty information rather than categorizing events based on the muon

DRAFT

η, as was done in Ref. [13].

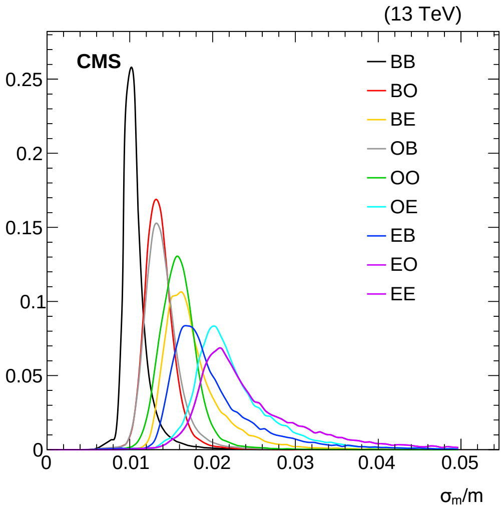

Figure 9: Distribution of the per-event dimuon mass resolution in ggH signal events for different combinations of the leading and subleading muon η: "B" corresponds to $|\eta| < 0.9$, "O" to $0.9 < |\eta| < 1.8$, and "E" to $|\eta| > 1.8$. Distributions are scaled in order to have unit area.

The Higgs boson mass peak is typically modelled via an analytical function that contains a Gaussian core and some non-Gaussian tails. We assume that the σ of this Gaussian core depends directly on the per-event mass resolution. The intrinsic width of the Higgs boson is expected to be extremely small, and its contribution can be neglected. However, the mass uncertainty obtained from the muon track fit covariance matrix is still an approximation and may have some bias w.r.t. the *actual* mass uncertainty. Therefore, we perform a calibration of the event-by-event mass uncertainty, separately in data and simulation, to ensure it correctly reflects the dimuon mass resolution in our events of interest.

This calibration is performed using events in the Z+jets control region, where the dimuon mass requirement used here is a bit tighter i.e. $75 < m_{\mu\mu} < 105\,\text{GeV}$. Furthermore, we categorize events based on the η of each of the two muons. Three η categories are considered: barrel (B) with $|\eta| < 0.9$, overlap (O) with $0.9 < |\eta| < 1.8$, and endcap (E) $1.8 < |\eta| < 2.4$. This gives a total of nine possible η categories for a dimuon event. Each η category is further subdivided into four p_T bins of the leading muon: [30, 45, 52, 62, 200] GeV.

For events with leading muon p_T smaller than 52 GeV, the dimuon mass distribution is caracterized by asymmetric tails. Therefore, the mass distribution of events in these low p_T categories is fit using a functional form given by the convolution of a Breit-Wigner function (Eq. 1) and the double-sided Crystal Ball (Eq. 2). In contrast, for events with leading muon $p_T > 52\,\text{GeV}$, the mass peak is more symmetric and therefore is modelled with a Voigtian function, which corresponds to the convolution of a Breit-Wigner and a Gaussian function.

$$\mathrm{BW}(x; m_Z, w) = \frac{1}{(x - m_Z)^2 + \frac{\Gamma_Z^2}{4}} \tag{1}$$

DRAFT

$$\mathrm{DCB}(x;s,\sigma,\alpha_{\mathrm{L}},n_{\mathrm{L}},\alpha_{\mathrm{R}},n_{\mathrm{R}}) = \begin{cases} e^{-\frac{(x-s)^2}{2\sigma^2}} & -\alpha_{\mathrm{L}} < \frac{x-s}{\sigma} < \alpha_{\mathrm{R}} \\ (\frac{n_{\mathrm{L}}}{|\alpha_{\mathrm{L}}|})^{n_{\mathrm{L}}} \times e^{-\frac{\alpha^2_{\mathrm{L}}}{2}} \times (\frac{n_{\mathrm{L}}}{|\alpha_{\mathrm{L}}|} - |\alpha_{\mathrm{L}}| - \frac{x-s}{\sigma})^{-n_{\mathrm{L}}} & \frac{x-s}{\sigma} \leq -\alpha_{\mathrm{L}} \\ (\frac{n_{\mathrm{R}}}{|\alpha_{\mathrm{R}}|})^{n_{\mathrm{R}}} \times e^{-\frac{\alpha^2_{\mathrm{R}}}{2}} \times (\frac{n_{\mathrm{R}}}{|\alpha_{\mathrm{R}}|} - |\alpha_{\mathrm{R}}| - \frac{x-s}{\sigma})^{-n_{\mathrm{R}}} & \frac{x-s}{\sigma} \geq \alpha_{\mathrm{R}} \end{cases} \tag{2}$$

The peak (m_{Z}) and width (Γ_{Z}) parameters of the Breit-Wigner function are fixed to the nominal mass and width of the Z boson, whereas the other parameters are allowed to float. Moreover, an additional component for the background is also included and is modelled with an exponential function in the two lowest p_{T} bins ($p_{\mathrm{T}} < 52\,\mathrm{GeV}$), while a Erf × Exp is adopted in events with an "high" p_{T} leading muon ($p_{\mathrm{T}} > 52\,\mathrm{GeV}$). The σ parameter of the Gaussian core can be compared to the median value of the event-by-event mass uncertainty for all the events in that category. The ratio of the two defines the calibration for that specific (η, p_{T}) category. Calibration factors are extracted separately in data and simulation and for each eras (2016, 2017, and 2018). Figure 10 shows three examples of fits to the dimuon invariant mass distribution in data in different η categories for events with leading muon p_{T} between 52 and 62 GeV recorded in 2016. The estimated calibration factors in data and MC for each era and for each (η, p_{T}) category are listed in Table 12.

Overall, in both data and MC, the calibration factors in each (η, p_{T}) category are positive, in the range between 1.05 and 1.30, indicating that the per-event mass resolution estimated from the covariance matrix of the track fit tends to overestimate the mass resolution of the Z peak by about 5–30%. In particular, the dimuon events that require the largest correction are those in which at least one of the two muons belong to the barrel region and the leading muon has a p_{T} above 50 GeV. The calibration factors extracted from simulated events are a few % larger than those from data. These corrections are also not identical across eras, in fact, in both data and MC, they progressively decrease moving from the 2016 to the 2018 era.

DRAFT

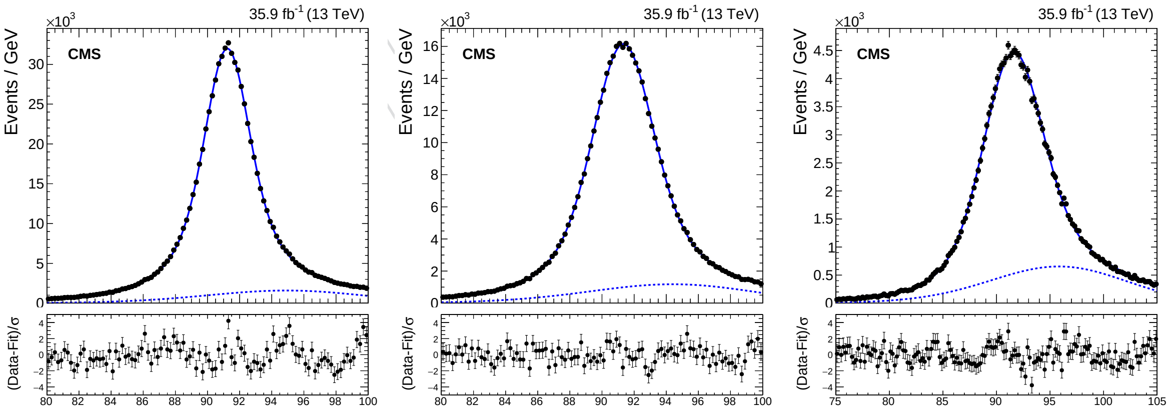

Figure 10: Example of fits to the the dimuon invariant mass distribution in data for events with leading muon p_{T} between 52 and 62 GeV recorded in 2016. From left to right, the distributions correspond to BB, OO, and EE categories.

In order to understand how well we are able to model the mass resolution, a closure test for the calibration procedure is performed. Using the same set of events in the Z+jets control region, we divide the distribution of the calibrated per-event mass resolution is several bins. We then fit the dimuon events in each resolution bin with the same analytical model previously described. Figure 11 shows the comparison between the fitted value of the σ parameter in each

Table 12: Estimated calibration factors, as extracted from the fit to the dimuon mass distribution in Z+jets events previously described, in data and MC for each era and for each (η, p_{T}) bin.

Leading muon p_{T}-bin (GeV)	Muon $\lvert\eta\rvert$-bin	Calibration factor in Data			Calibration factor in MC		
		2016	2017	2018	2016	2017	2018
30–45	BB+OB+EB	1.24	1.19	1.16	1.25	1.16	1.17
30–45	BO+OO+EO	1.18	1.13	1.11	1.19	1.11	1.12
30–45	BE+OE+EE	1.15	1.11	1.07	1.17	1.07	1.09
45–52	BB	1.15	1.19	1.16	1.14	1.16	1.10
45–52	BO	1.16	1.13	1.12	1.17	1.12	1.13
45–52	BE	1.16	1.10	1.08	1.18	1.08	1.12
45–52	OB	1.14	1.12	1.08	1.16	1.08	1.10
45–52	OO	1.12	1.09	1.07	1.11	1.11	1.09
45–52	OE	1.07	1.07	1.06	1.10	1.09	1.09
45–52	EB	1.11	1.05	1.03	1.15	1.09	1.05
45–52	EO	1.13	1.03	1.05	1.16	1.10	1.04
45–52	EE	1.00	0.98	0.96	1.04	1.02	1.00
52–62	BB	1.25	1.21	1.19	1.26	1.24	1.21
52–62	BO	1.26	1.24	1.24	1.28	1.28	1.26
52–62	BE	1.19	1.14	1.15	1.22	1.16	1.16
52–62	OB	1.24	1.24	1.22	1.26	1.27	1.24
52–62	OO	1.17	1.16	1.15	1.19	1.18	1.17
52–62	OE	1.17	1.17	1.15	1.20	1.21	1.17
52–62	EB	1.19	1.18	1.13	1.23	1.23	1.18
52–62	EO	1.19	1.15	1.13	1.23	1.20	1.17
52–62	EE	1.11	1.10	1.08	1.15	1.14	1.12
62–200	BB	1.24	1.21	1.18	1.26	1.25	1.19
62–200	BO	1.27	1.24	1.22	1.29	1.28	1.23
62–200	BE	1.15	1.12	1.14	1.17	1.17	1.16
62–200	OB	1.23	1.22	1.21	1.26	1.27	1.23
62–200	OO	1.16	1.15	1.14	1.18	1.18	1.16
62–200	OE	1.18	1.15	1.15	1.21	1.18	1.18
62–200	EB	1.15	1.10	1.12	1.17	1.16	1.16
62–200	EO	1.16	1.11	1.10	1.20	1.18	1.13
62–200	EE	1.09	1.08	1.06	1.14	1.15	1.10

DRAFT

resolution bin with the corresponding median value of the per-event mass uncertainty for both data (blue) and simulation (red). As can be seen, the calibrated per-event mass uncertainty and the measured resolution agree within 10% in both data and simulation.

Finally, Fig. 12 shows the distribution of the per-event mass resolution in data and simulation for events in the SR before (top row) and after (bottom row) the application of calibration factors. It is clear that the calibration procedure allows to significantly improve the agreement between data and MC. The residual discrepancy observed for low resolution events ($\sigma_m/m > 0.035$) has a negligible impact on the analysis.

5 Multivariate discriminant

The most important discriminating variable between signal and background processes in this search is the dimuon invariant mass, which forms a sharp peak around 125 GeV for the signal as opposed to the background in which it decreases monotonically within the same mass range. This feature also allows for a completely data-driven estimation of the background through an analytical fit to the dimuon mass spectrum. However, there are other kinematic variables that

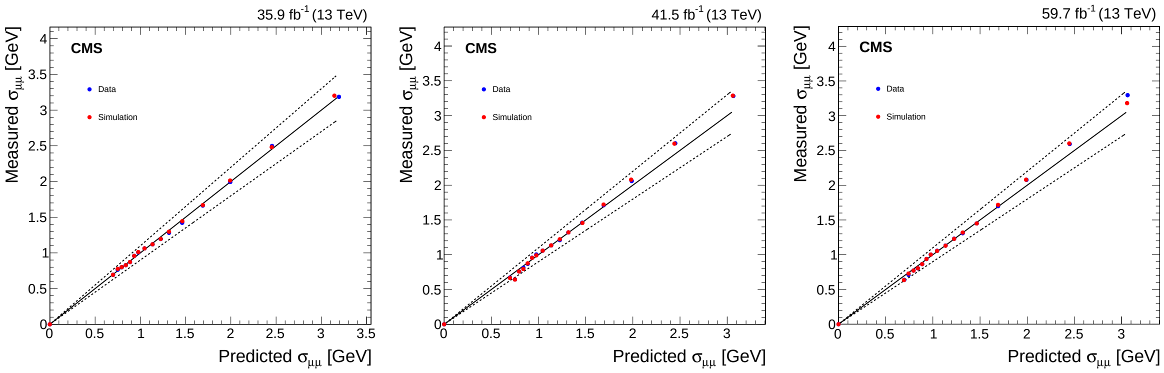

Figure 11: Results of the closure test for the per-event mass uncertainty calibration performed in the Z+jets control region for 2016 (left), 2017 (middle), and 2018 (right) eras. After categorizing the events in bins of resolution, the x-axis reports the median calibrated mass resolution in each bin, while the y-axis shows the resolution of the Z boson peak as extracted from an analytical fit to the $m_{\mu\mu}$ distribution.

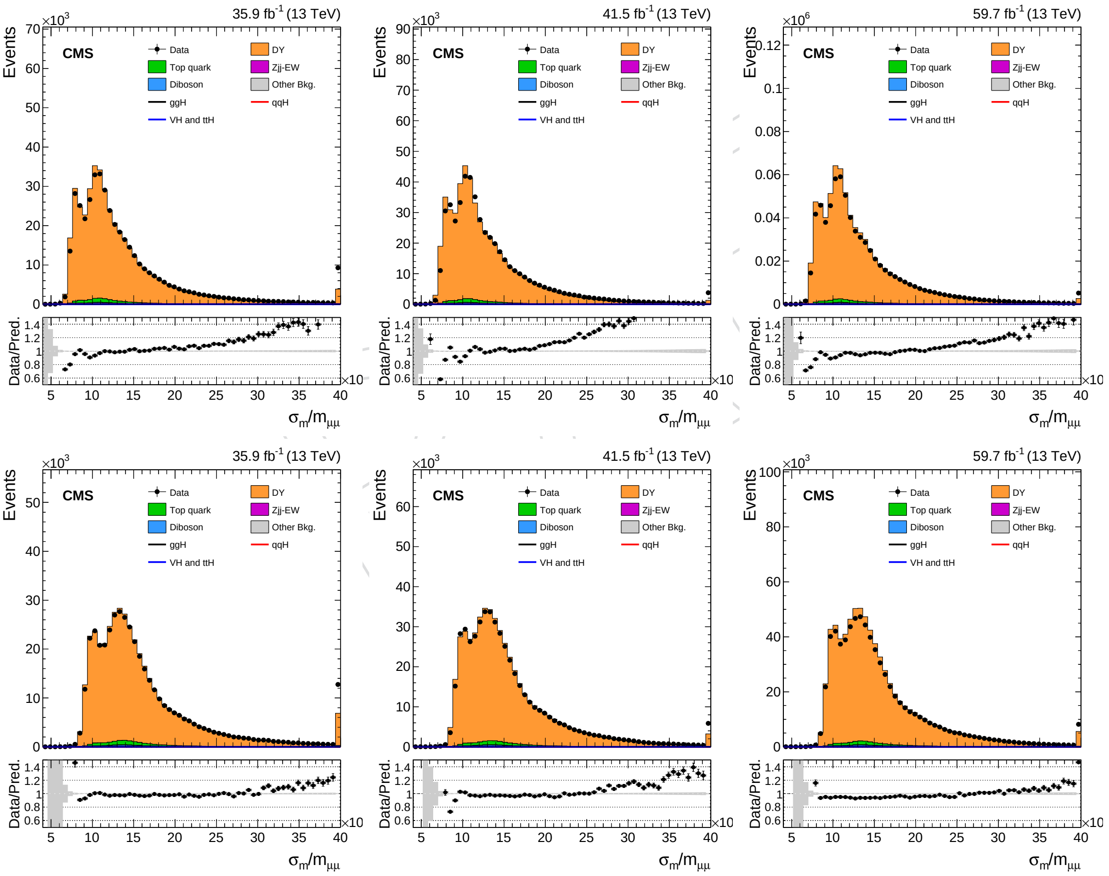

Figure 12: The per-event mass resolution distribution in 2016 (left), 2017 (middle) and 2018 (right) eras reported for both data (black points) and simulation (solid histograms) for events in the SR before (top) and after (bottom) the calibration performed in the Z+jets control region. The agreement between data and simulation is significantly improved by applying the calibration coefficients listed in Table 12.

can be used to enhance the signal-to-background separation. As has been previously mentioned, the VBF signal offers a distinctive signature involving two jets with a large η-gap and large invariant mass. In the case of the gluon fusion signal, the p_{T} of the dimuon system represents an important discriminating observable since it shows a harder spectrum in the signal compared to the Drell-Yan background.

Therefore, in order to enhance the signal-to-background separation, we construct two independent multivariate discriminants using boosted decision trees (these will be referred to as BDTs), which take as input several variables that capture the distinctive kinematic features of the signal and the background. One of the BDTs is trained exclusively on events containing 2 or more jets with with m_{jj} larger than 400 GeV (VBF BDT). In terms of signal, this category contains a significant contribution from VBF events (about 55% of the total expected signal). In the case of the background, these events are still largely dominated by the Drell-Yan process. However, the contribution from $\mathrm{t\bar{t}}$ and electroweak Z+jets production starts to become significant, particularly in events with a VBF-like topology. The other is trained on events belonging to the "residual" phase space i.e. events with either 0 or 1 jets, and events with two or more jets in which the m_{jj} of the two leading jets is smaller than 400 GeV. The dominant source of signal is ggH while the main background comes from Drell-Yan. This discriminator is referred to as the "ggH BDT" or "non-VBF BDT". The input variables have been specifically chosen so that the BDT output remains highly uncorrelated with the dimuon mass. We then categorize events based on BDT score in order to maximize the search sensitivity. Since the BDT output is highly uncorrelated with the dimuon mass, this categorization does not sculpt the background to peak around the Higgs boson mass. This in turn allows us to fit the dimuon mass distribution in each of the categories without a significant bias. Finally, we perform a statistical combination of all the categories to extract a final estimate of the analysis performance.

5.1 Input variables to the ggH BDT

We start with a suite of input variables constructed entirely from the two muons. A dimuon system essentially has six degrees of freedom corresponding to the momentum 3-vectors of the two muons. The ϕ-symmetry of the detector makes one of the degrees of freedom redundant. Since we want to explicitly avoid the use of the dimuon mass in the BDT discriminant, that leaves us with four degrees of freedom i.e. four possible input variables. Two of these variables are chosen to be the $p_{\mathrm{T}}^{\mu\mu}$ and $y_{\mu\mu}$, and they correspond to the p_{T} and rapidity of the dimuon system. The distributions of these variables are sensitive to the production mechanism in the case of the signal and the Drell-Yan background. The $p_{\mathrm{T}}^{\mu\mu}$ variable in particular is found to be the most discriminating between ggH and Drell-Yan events. The remaining two variables correspond to the decay of the boson in the rest frame of the dimuon system. We specifically choose the Collins-Soper rest frame [59] for this purpose, and take the azimuthal angle (ϕ_{CS}) and the cosine of the polar angle ($\cos\theta_{\mathrm{CS}}$) to be the two variables. These decay angles carry some discrimination owing to the difference in the spin properties of the signal (scalar) and the Drell-Yan background (spin-1). In addition to the four aforementioned variables, we also include single muon information represented by the transverse momenta of the two muons relative the dimuon mass, $p_{\mathrm{T}}^{\mu_1}/m_{\mu\mu}$ and $p_{\mathrm{T}}^{\mu_2}/m_{\mu\mu}$, as well as the individual muon η, η^{μ_1} and η^{μ_2}. Figure 13 shows the shapes of all the muon related variables used as inputs to the BDT training for signal and background events, where the signal is given by the sum of the expected contributions from the different Higgs boson production mechanisms (ggH, VBF, VH, and ttH) while the background is the total expected background (DY, top, ttV, diboson, triboson, Z+jets EW, etc..).

In addition to the observables describing to the dimuon system, we also include variables carrying information about jets in the event. In the case of events with only one jet in the final state with p_{T} larger than 30 GeV, the following variables are added: jet p_{T}, jet η, the η and the ϕ

DRAFT

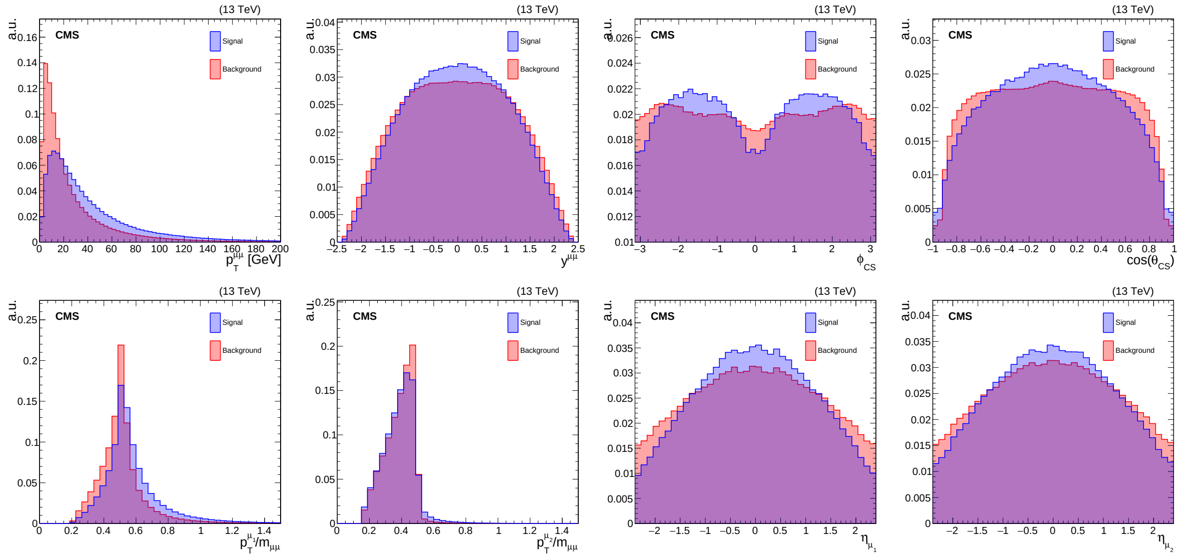

Figure 13: Distribution of the dimuon (top) and muon (bottom) variables used as input in the ggH BDT in signal (blue) and background (red) events. Distributions are scaled in order to have unit area. On top, from left to right: $p_T^{\mu\mu}$, $y_{\mu\mu}$, ϕ_{CS}, and $\cos\theta_{CS}$. Bottom, from left to right: $p_T^{\mu_1}/m_{\mu\mu}$, $p_T^{\mu_2}/m_{\mu\mu}$,η^{μ_1}, η^{μ_2}.

gaps between the dimuon system and the jet ($\Delta\eta(\mu\mu, j)$, $\Delta\phi(\mu\mu, j)$).

In contrast, in events with two or more jets, more variables are included to target the presence of VBF and V(qq)H signals, as well as to further exploit differences in the jet kinematics between ggH and Drell-Yan events. These are represented by the η-gap between the two leading jets in the event ($\Delta\eta_{jj}$), the dijet invariant mass (m_{jj}, required to be smaller than 400 GeV), their azimuthal angular separation ($\Delta\phi_{jj}$) which carries some spin-discrimination power [60], the Zeppenfeld variable [61] defined in Equation 3, where ys correspond to the rapidities of the dimuon system, the leading and the subleading jets, the p_T of the subleading jet, and the minimum difference in η and ϕ between the dimuon system and the two jets (min-$\Delta\eta(\mu\mu, j_1 j_2)$, min-$\Delta\phi(\mu\mu, j_1 j_2)$). Lastly, the distribution of the number of jets of the event is also used as input. In events with $N_{jet} = 0$, jet variables are set to zero apart from $\Delta\phi_{jj}$ which is set to be -1. Similarly, in one jet events dijet observables are defaulted to zero apart from $\Delta\phi_{jj}$ which is set again to be equal to -1.

$$\text{Zeppenfeld var.} = \frac{y_{\mu\mu} - (y_{j_1} + y_{j_2})/2}{|y_{j_1} - y_{j_2}|} \tag{3}$$

Fig 14 shows the shapes in signal and background events of jet related variables, as well as angular distances ratios between the dimuon system and the jets which are used as inputs to the ggH BDT trained on events with m_{jj} larger smaller than 400 GeV.

Figure 15 and 16 show the distributions of the ggH BDT inputs in data and simulation for events in the SR after requiring that no events with two or more jets have m_{jj} larger than 400 GeV, $\Delta\eta_{jj}$ above 2.5 units, and leading jet $p_T > 35$ GeV. The observation and the expectation in the three eras (2016, 2017, and 2018) have been summed together. Overall, the agreement between data and MC for both muon and jet variables used in input is good. In the $N_{jet} \geq 1$ phase-space, an overall normalization discrepancy of about 5% is observed between data and MC as reported by the N_{jets} distribution. Since the background in this search is directly esti-

DRAFT

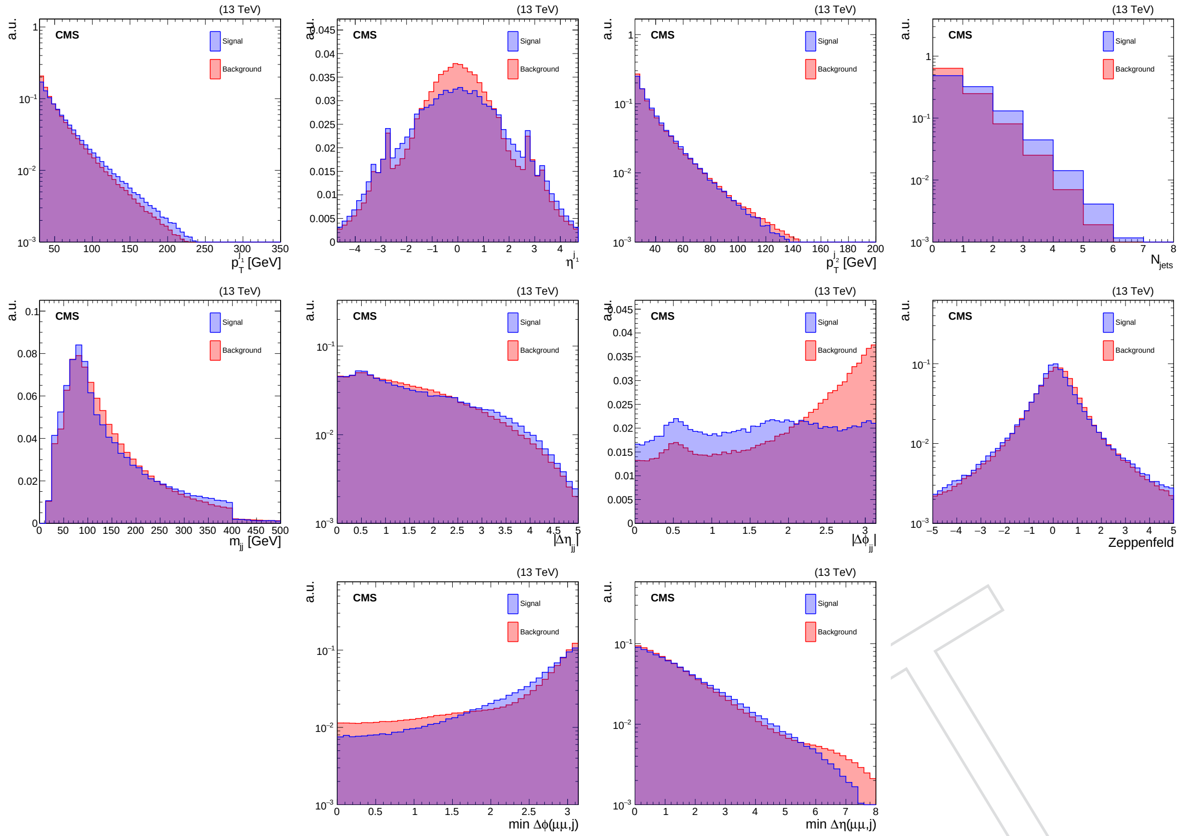

Figure 14: Distribution of the jet related variables used as input in the ggH BDT in signal (blue) and background (red) events. Distributions are scaled in order to have unit area. First row, from left to right: $p_T^{j_1}$, η^{j_1}, $p_T^{j_2}$, and N_{jets}. Second row, from left to right: m_{jj}, $\Delta\eta_{jj}$, $\Delta\phi_{jj}$, and Zeppenfeld. Last row, from left to right: min-$\Delta\phi(\mu\mu, js)$, and min-$\Delta\eta(\mu\mu, js)$.

mated from data, this small discrepancy in normalization between data and simulation does not have an impact in the analysis results. The largest differences between data and simulation are observed in the leading and subleading jet η distributions, when the jet falls either in the HF detector ($|\eta| > 3.0$) or is reconstructed within the most forward sectors of EE/HE ($2.7 < |\eta| < 3.0$). During Run2, this region registered a large increase in the noise vs time due to a strong radiation damage of both active materials (ECAL crystals) and light sensors. Moreover, discrepancies are also observed in the tails of m_{jj} and $\Delta\eta_{jj}$ distributions mainly due to the selection used to separate the ggH phase space from the VBF analysis one which will be described in the next Section 5.2. Comparisons are also reported independently for each era in Appendix A where a more detailed description of the observed trends in the main kinematic distributions is given.

5.2 Input variables for VBF BDT

The BDT discriminant used for events with 2 or more jets with m_{jj} larger than 400 GeV takes as input the same variables used in the ggH MVA described in Sec. 5.1. However, additional observables are included to further increase the separation between VBF and ggH signals. These are represented by the p_T-balance ratio (R_{p_T}), the p_T-centrality, the number of soft-jets with p_T larger than 2 GeV (N_{soft}), and the related H_T (H_T^{soft}). The R_{p_T} and p_T-centrality are defined in Equation 4. A soft-jet, instead, is defined as an anti-k_t jet, with a distance parameter R of 0.4 and p_T larger than 2 GeV, obtained by clustering only charged PF candidates assigned from the

DRAFT

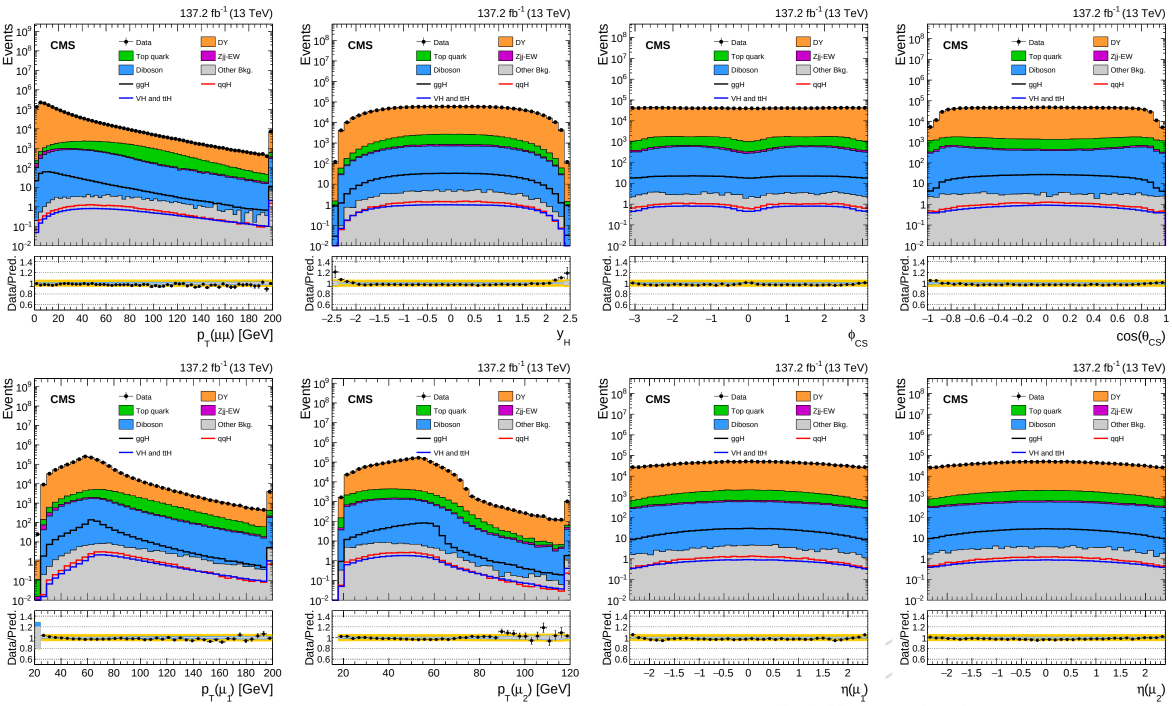

Figure 15: Distributions of muon variables used in input to the ggH BDT training for data (black points) and simulation (solid histograms) obtained from events in the SR. First line, from left to right, $p_T^{\mu\mu}$, $y_{\mu\mu}$, ϕ_{CS}, and $\cos\theta_{CS}$. Second line, from left to right: $p_T^{\mu_1}$, $p_T^{\mu_2}$, η^{μ_1}, and η^{μ_2}. In the ratio panel, the grey band indicates the uncertainty coming from the limited statistics of the simulated samples, the azure one corresponds to the sum in quadrature between statistical and systematic uncertainties from experimental sources, while the orange band includes the theoretical uncertainties in the background prediction. The asymmetry in the experimental uncertainty band is due to the one-sided effect of jet energy resolution.

primary vertex. The two identified muons and the charged PF candidates associated to the two leading jets in the event are not included in the soft-jet definition.

$$R_{p_T} = \frac{|\vec{p_T}^{\mu_1} + \vec{p_T}^{\mu_2} + \vec{p_T}^{j_1} + \vec{p_T}^{j_2}|}{p_T^{\mu_1} + p_T^{\mu_2} + p_T^{j_1} + p_T^{j_2}}, \; p_T\text{-centrality} = \frac{p_T^{\mu\mu} - |\vec{p_T}^{j_1} + \vec{p_T}^{j_2}|/2}{|\vec{p_T}^{j_1} - \vec{p_T}^{j_2}|} \quad (4)$$

Figure 17 shows the shapes in signal and background events of dijet and soft-jet related variables, as well as angular distances and p_T ratios between the dimuon system and jets which are used as inputs to the VBF BDT trained on events with m_{jj} larger than 400 GeV. Both the shapes and the separation power of muon observables are similar to what was shown for the ggH BDT in Fig. 13.

Figure 18 and 19 show the distributions of the VBF BDT inputs in data and simulation for events in the SR after requiring two or more jets in the event with m_{jj} larger than 400 GeV, $\Delta\eta_{jj} > 2.5$, and leading jet p_T greater than 35 GeV. The observation and the expectation in the three eras (2016, 2017, and 2018) have been summed together. Overall, the agreement between data and MC for muon and jet variables is good, but there are some visible discrepancies in the leading and subleading jet η distributions, in the $\Delta\phi_{jj}$, in the N$_{\text{jets}}$ and soft-jet variables. Comparisons are also reported independently for each era in Appendix A, where a more detailed description of the observed trends in the main kinematic distributions is reported.

DRAFT

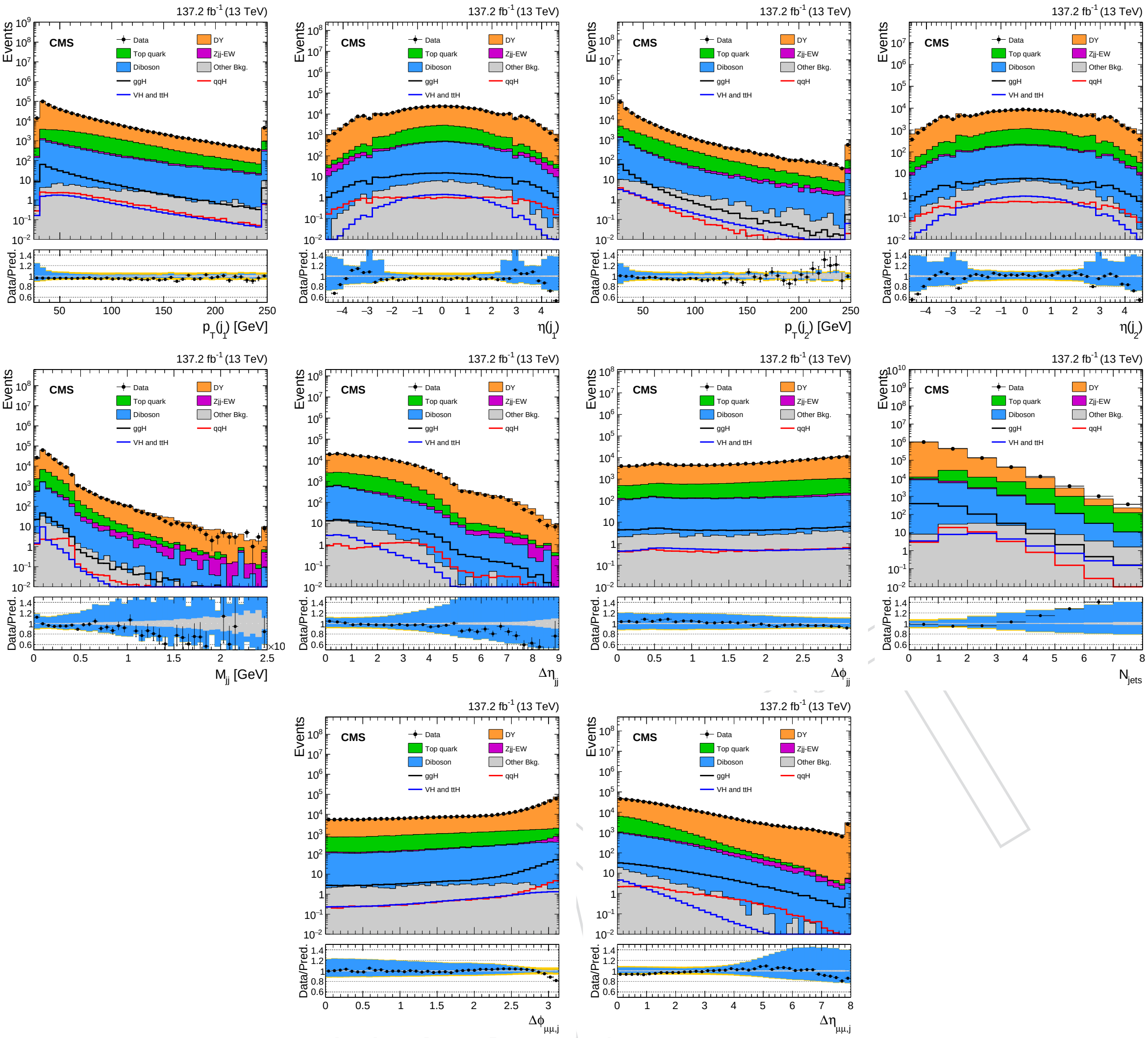

Figure 16: Distributions of jet variables used in input to ggH BDT training for data (black points) and simulation (solid histograms) obtained from events in the SR. First row, from left to right: $p_{\mathrm{T}}^{j_1}$, η^{j_1}, $p_{\mathrm{T}}^{j_2}$, and η^{j_2}. Second row, from left to right: m_{jj}, $\Delta\eta_{\mathrm{jj}}$, $\Delta\phi_{\mathrm{jj}}$, and $\mathrm{N_{jets}}$. Last row, from left to right: min-$\Delta\phi(\mu\mu, js)$, min-$\Delta\eta(\mu\mu, js)$. The description of the ratio panel is the same of Fig. 15.

5.3 BDT discriminant trainings

The training is performed separately on the MC samples for the three eras: 2016, 2017, and 2018. This helps to capture the impact of the differences in reconstruction and detector performance across the three data taking periods. For example, the occupancy and properties of forward jets $|\eta| > 2.6$ is pretty different between 2016 and the other years. In addition, in the 2017 era, a tight requirement on the pileup jet ID is adopted for jets in the range $2.6 < |\eta| < 3.0$ in order to mitigate the noise from the ECAL detector. The BDT discriminants are trained using the 'Gradient Boost' method. The training is done using simulated signal and background events, using approximatively half of the available statistics corresponding to events with odd event number. Events with even event number are used instead for testing purposes. In the case of the signal, POWHEG samples (centrally produced) for ggH and VBF are used in the training because they don't have negative weights. For background samples with negative weighted events (Drell-Yan for example), we observed similar performance on the testing sample follow-

DRAFT

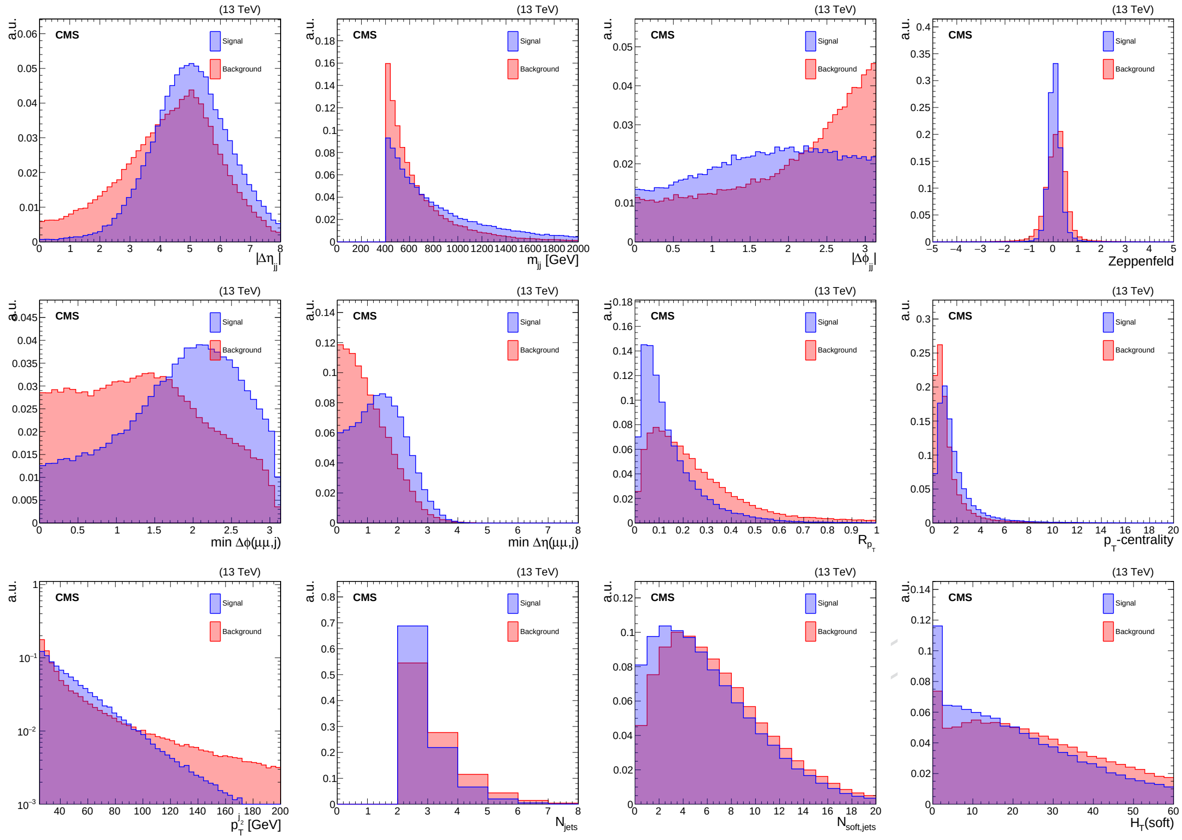

Figure 17: Distribution of the jet related variables used as inputs in the VBF BDT trained on events with m_{jj} larger than 400 GeV for signal (blue) and background (red) processes. Distributions are scaled in order to have unit area. From left to right on the top row: $\Delta\eta_{\mathrm{jj}}$, m_{jj}, $\Delta\phi_{\mathrm{jj}}$, and Zeppenfeld variable. From left to right on the middle row: min-$\Delta\phi(\mu\mu, j_1 j_2)$, min-$\Delta\eta(\mu\mu, j_1 j_2)$, $R_{p_{\mathrm{T}}}$, and p_{T}-centrality. From left to right on the bottom row: second jet p_{T}, number of jets, number of soft-jets with p_{T} larger than 2 GeV, and lastly their $\mathrm{H_T}$.

ing three alternative strategies: use both positive and negative weighted events in the training, discard events with a negative weight, and take the absolute value of the per-event weight[3]. In order to maximize the training statistics, the final BDTs have been trained by flipping the sign of negative weighted events. The training is performed targeting an Higgs boson signal with a mass of 125 GeV. The settings used for the BDT training are as follows:

- Number of trees = 1000.
- Minimum node size = 3%.
- Shrinkage = 0.10.
- Bagged sample fraction = 0.5.
- Number of cuts = 30.
- Maximum depth = 4.
- Transformation for inputs: (I, N).
- Separation type: CrossEntropy.

DRAFT

---

[3] In order not to bias the background composition, a correction factor is introduced to ensure that the same amount of background events are expected in the case negative weighted events are discarded or they are considered with a positive weight.

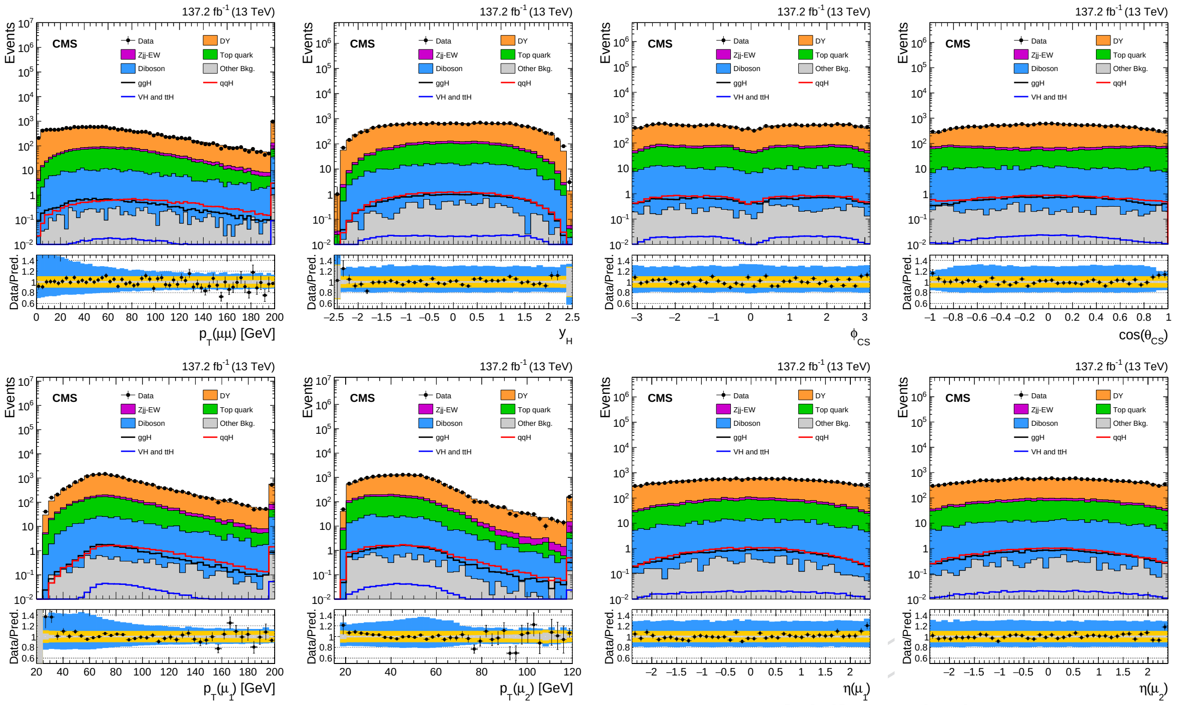

Figure 18: Distributions for some of the variables used in input to the VBF BDT training for data (black points) and simulation (solid histograms) obtained from events in the SR with $N_{\text{jets}} \geq 2$, $m_{\text{jj}} > 400\,\text{GeV}$ and $\Delta\eta_{\text{jj}} > 2.5$ and leading jet p_{T} greater than 35 GeV. First line, from left to right, $p_{\text{T}}^{\mu\mu}$, $y_{\mu\mu}$, ϕ_{CS}, and $\cos\theta_{\text{CS}}$. Second line, from left to right: $p_{\text{T}}^{\mu_1}$, $p_{\text{T}}^{\mu_2}$, η^{μ_1}, and η^{μ_2}. In the ratio panel, the grey band indicates the uncertainty coming from the limited statistics of the simulated samples, the orange one corresponds to the sum in quadrature between statistical and systematic uncertainties from theory sources, while the azure band includes the experimental uncertainties in the background prediction. The asymmetry in the experimental uncertainty band is due to the one-sided effect of jet energy resolution.

As explained in Section 5, the input variables have been specifically chosen so that the BDT output remains highly uncorrelated with the dimuon mass. The training is performed in a tighter mass range compared to the SR, i.e. $115 < m_{\mu\mu} < 135\,\text{GeV}$, which corresponds to a $\pm 10\,\text{GeV}$ window around the true Higgs boson mass.

The analysis sensitivity depends on the dimuon mass resolution, which in turn depends strongly on the muon kinematics as described in Section 4.7, particularly the η of the muon. Therefore, we make the BDT training aware of the dimuon mass resolution by weighting the signal events proportionally to $1/\sigma_{\mu\mu}$, where $\sigma_{\mu\mu}$ is the calibrated per-event dimuon mass resolution. It should be noted that the $\sigma_{\mu\mu}$ distribution is not used as an input to the BDT, but is only used as a weighting factor in the training. The weight is not applied in the evaluation of the BDT score. The performance of the BDT discriminant does not change significantly if we include the per-event mass resolution as one of the inputs. This is to be expected because the shape of the $\sigma_{\mu\mu}$ distribution is very similar between signal and background events. Therefore, the adopted use of the per-event mass uncertainties is a significant departure from the approach adopted in the earlier iteration of the analysis [13]. In fact, in the previous iteration, events were categorized both on the basis of their BDT score and the maximum of the $|\eta|$ of the two muons. The approach adopted here of using per-event mass uncertainties as a weight for signal events during the BDT training reduces the categorization to a single variable i.e. the BDT score. In addition, this also carries more precisely the information about the muon momen-

DRAFT

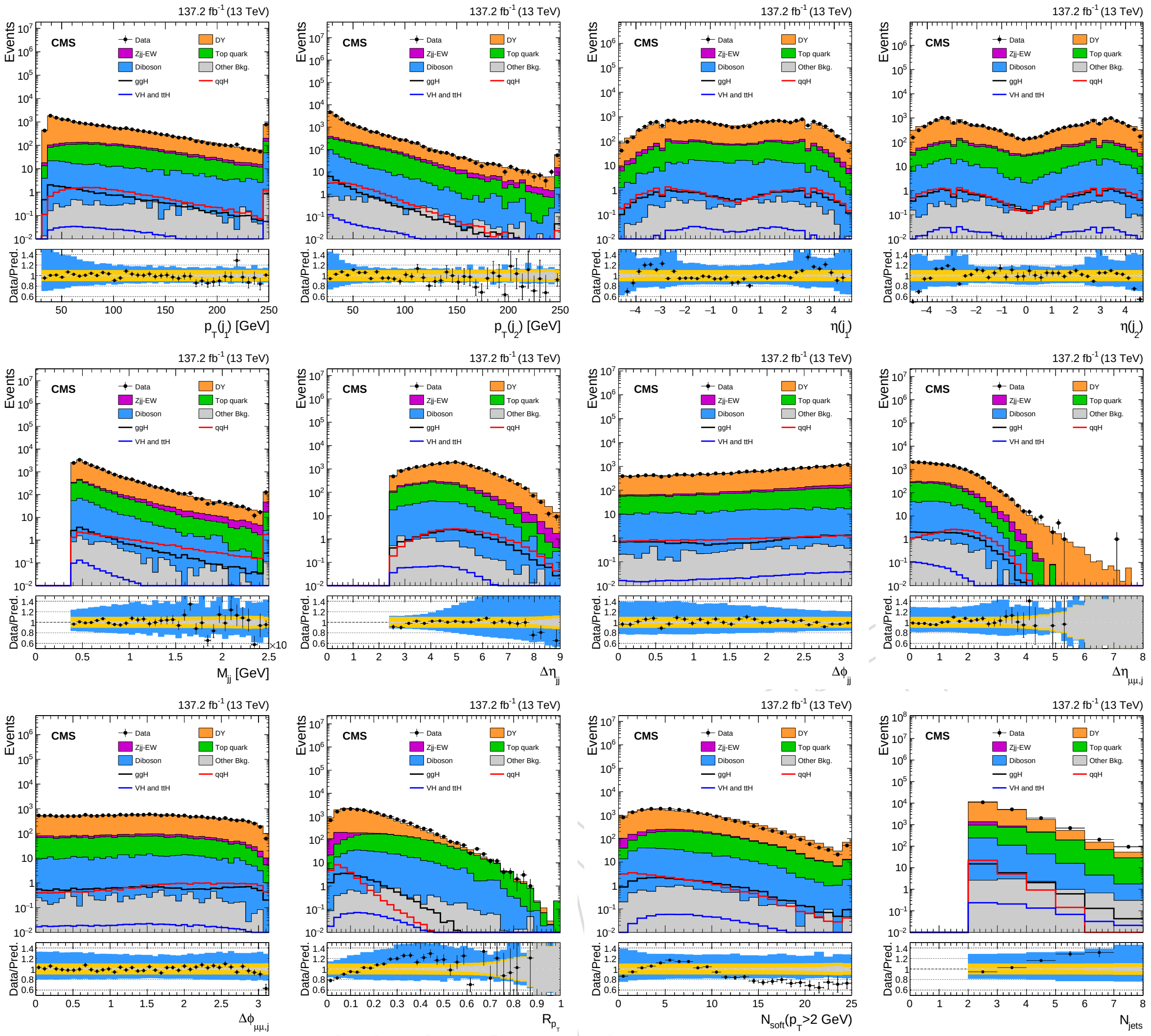

Figure 19: Distributions for some of the variables used in input to the 2-jet VBF BDT training for data (black points) and simulation (solid histograms) obtained from events in the SR with $N_{\text{jets}} \geq 2$, $m_{\text{jj}} > 400\,\text{GeV}$, $\Delta\eta_{\text{jj}} > 2.5$ and leading jet p_{T} larger than 35 GeV. First line, from left to right, $p_{\text{T}}^{j_1}$, $p_{\text{T}}^{j_2}$, η^{j_1}, and η^{j_2}. Second line, from left to right: m_{jj}, $\Delta\eta_{\text{jj}}$, $\Delta\phi_{\text{jj}}$, min-$\Delta\eta(\mu\mu, j_1 j_2)$. Third line, from left to right, min-$\Delta\phi(\mu\mu, j_1 j_2)$, $R_{p_{\text{T}}}$, N_{soft}, and N_{jets}. The description of the ratio panel is the same of Fig. 18.

tum resolution and its correlation with the event kinematics (muon p_{T} and η). Consequently, we will end up using significantly fewer categories but still having better performance (25% improved expected significance) on 2016 data w.r.t. Ref. [13].

Figure 20 shows a comparison between the output of the two BDT discriminants (left: ggH BDT, right: VBF BDT) in the case of the signal (top) and background (bottom) simulations for the three eras. The shapes of the ggH BDT outputs are found to be in good agreement for both signal and background processes apart from the high score region (BDT-score larger than 0.6), where the 2017 and 2018 BDTs show longer tails in both signal and background events. Similar differences in the high score region are also observed in the VBF BDTs (BDT-score greater than 0.85). Overall, the BDT discriminators related to 2017 and 2018 eras are quite similar in terms of shapes in both ggH and VBF trainings. In contrast, as already mentioned before, the 2016 shapes are more different particularly in the high score region. Since the properties of jets in the

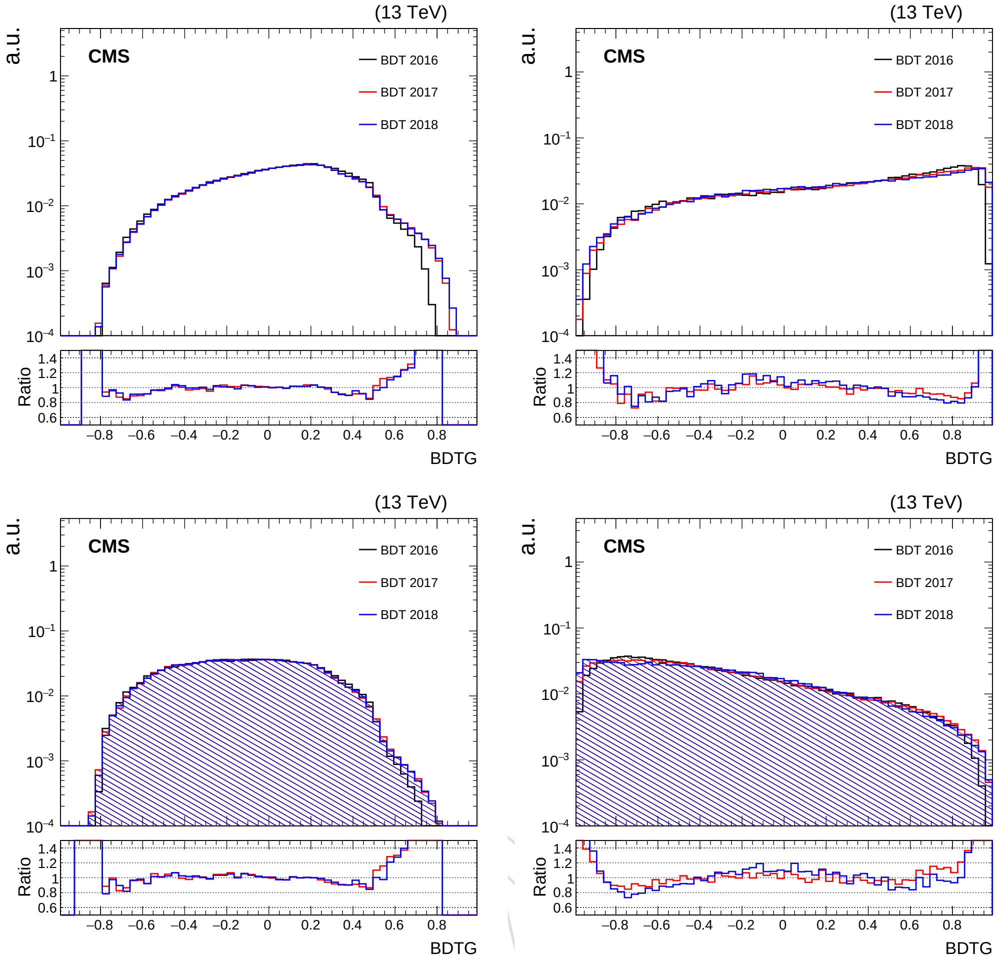

Figure 20: Distribution of the ggH (left), and VBF (right) BDT outputs in signal (top) and background (bottom) simulated events taken from the testing sample for three eras: 2016 (black), 2017 (red), and 2018 (blue). Distributions are scaled in order to have unit area. he red (blue) distribution in lower pad is defined as the ratio between the 2018 (2017) BDT shape and the 2016 one, which is used as reference. The shapes of the BDT outputs are found to be quite similar across the three eras. The largest differences are observed in the high score region of both the VBF and ggH phase spaces.

endcap and forward regions of the detector changed significantly after the 2016 data-taking, it is expected to see differences in regions more sensitive to the forward jet kinematics between the 2016 BDTs and those from the other eras.

Figure 21 shows the “ROC” (Receiver Operating Characteristic) curves for the ggH (left), and VBF (right) BDT discriminants. The performance of the ggH BDT discriminants are found to be very similar across the three eras, while the 2016 VBF BDT outperforms both the 2017 and the 2018 trainings. Figure 22 shows the dimuon mass resolution, expressed in terms of full width half maximum (FWHM), as a function of the BDT score. The dimuon mass resolution is extracted from a fit to the $m_{\mu\mu}$ distribution with a Double-sided Crystal Ball function. The signal resolution improves as a function of the BDT score, thereby demonstrating that the BDT is indeed aware of the importance of the dimuon mass resolution. This improvement in the resolution is more visible along the ggH BDT output than the VBF one. In fact, in the VBF phase space, there is a strong interplay between the per-event mass resolution and other kinematic variables used in input to the BDT (VBF sensitive variables), therefore the resolution more or less flattens out at high scores because a better separation between signal and background can be achieved by looking at kinematic properties rather than the mass resolution. In addition, the dimuon mass resolution is found to improve progressively moving from 2016 to 2018 eras, and it is also observed to be stable as a function of the BDT outputs. This gain in the muon p_T resolution is due to a combination of sources: installation of an additional layer in the Pixel detector in early 2017, improvements in muon and tracker detector alignments, as well as improvements in beam-spot and muon tracker reconstruction (particularly for high p_T muons) allowing the geo-fit corrections to be more effective in 2018 rather than in the 2016 data-taking period.

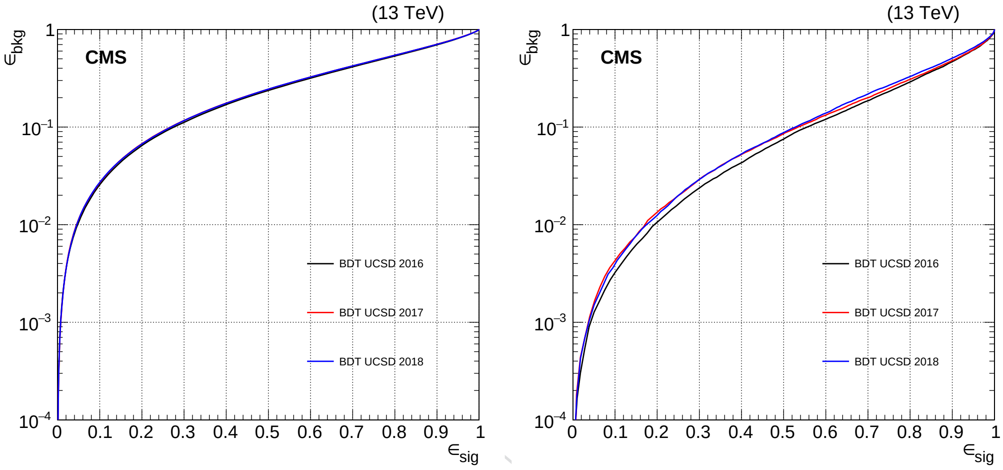

Figure 21: Signal efficiency (ϵ_{sig}) vs background efficiency (ϵ_{bkg}) ROC curves obtained from the ggH (left), and VBF (right) BDT output scores for events in the testing sample. The performance of the BDT discriminant is found to be stable across the three eras in the ggH phase space, while in VBF events the 2016 training outperforms the MVAs obtained from the other years.

DRAFT

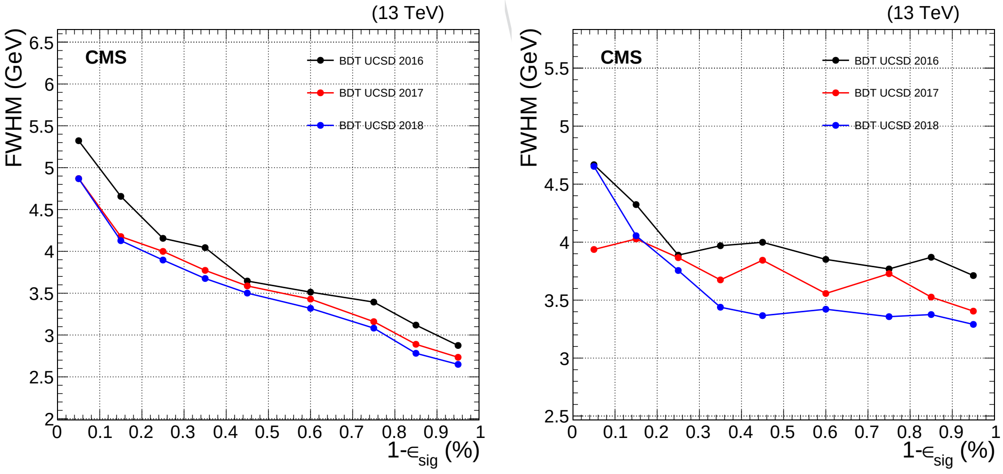

Figure 22: FWHM of the dimuon mass distribution in signal events computed as a function of the ggH (left), and VBF (right) BDT scores. The signal resolution improves similarly in the three eras as a function of the BDT score, proving that the BDT is able to assign high score to events with good mass resolution.

In order to understand how the BDT scores are correlated with the input observables, Fig. 23 and Fig. 24 show, for signal events, the evolution of the shapes of some of the input variables as a function of the ggH and VBF BDT scores, respectively. In events belonging to the ggH phase space, the high score region probes events with good dimuon mass resolution, as well as a large dimuon p_{T}. A good fraction of events at high are characterized by the presence of one jet or two jets in the final state with a large η-separation with respect to the dimuon system. In contrast, the high score region of the VBF BDT, which looks at events with at least two jets with m_{jj} larger than 400 GeV, probes events with a large dimuon p_{T}, large m_{jj}, large m_{jj}, and a small $R_{p_{\mathrm{T}}}$.

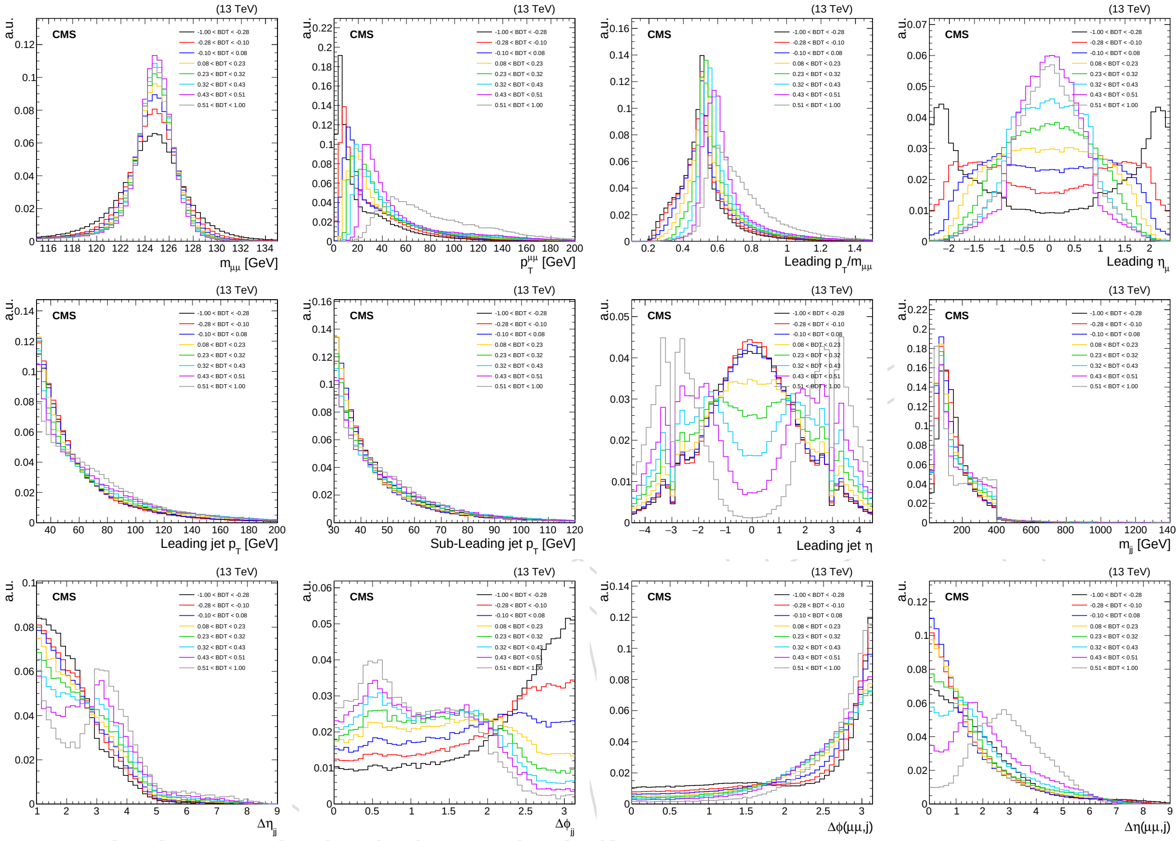

Figure 23: Evolution of the distributions of some of the key kinematic observables as a function of the ggH BDT output for signal events (ggH, VBF, VH, and ttH summed together). Distributions are scaled in order to have unit area. On top, from left to right: $m_{\mu\mu}$, $p_{\mathrm{T}}^{\mu\mu}$, $p_{\mathrm{T}}^{\mu_1}/m_{\mu\mu}$, and η_{μ_1}. Middle, from left to right: $p_{\mathrm{T}}^{j_1}$, $p_{\mathrm{T}}^{j_2}$, η^{j_1}, and m_{jj}. Bottom, from left to right: $\Delta\eta_{\mathrm{jj}}$, $\Delta\phi_{\mathrm{jj}}$, min-$\Delta\phi(\mu\mu, j)$, and min-$\Delta\eta(\mu\mu, j)$.

Moreover, it is also crucial to make sure that both the these BDTs haven't learnt the signal mass hypothesis used in the training, $m_{\mathrm{H}} = 125$ GeV, via hidden correlations between the invariant mass of the two muons and the other input variables. To do so, two independent cross checks are performed:

- Fig. 25 shows, for background events, the evolution of the dimuon mass shape as a function of the ggH (left), and VBF (right) BDT scores where no sculpting of the expected mass distribution is observed either at low or at high BDT score. This suggests that the BDT output is not sensitive to the dimuon mass distribution.
- Fig. 26 shows, for ggH and VBF signal events, the evolution of the ggH (left), and VBF (right) BDT shapes as a function of the signal mass hypothesis, i.e. for $m_{\mathrm{H}} =$ 120, 125, and 130 GeV. No differences as a function of the mass are observed for the two BDT outputs.

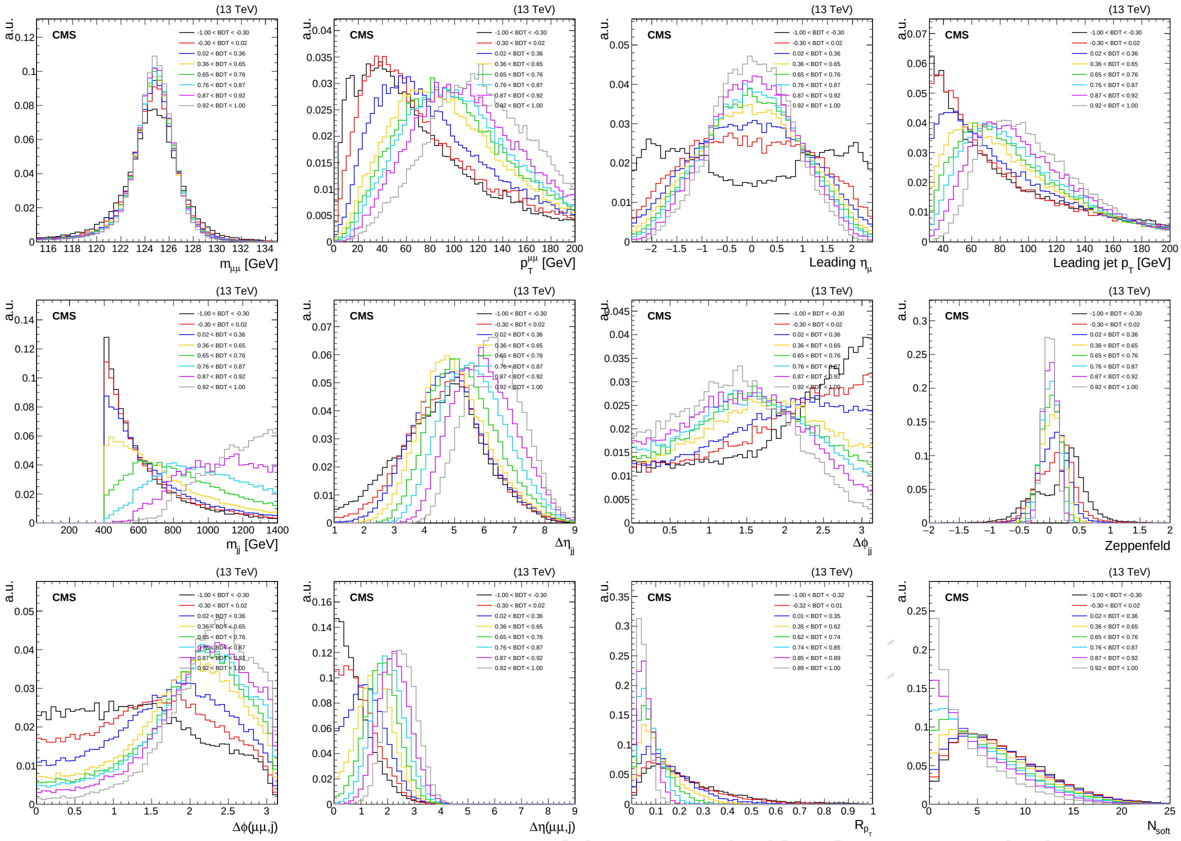

Figure 24: Evolution of the distributions of some of the key kinematic observables as a function of the VBF BDT output for signal events (ggH, VBF, VH, and ttH summed together). Distributions are scaled in order to have unit area. On top, from left to right: $m_{\mu\mu}$, $p_{\mathrm{T}}^{\mu\mu}$, η_{μ_1}, and $p_{\mathrm{T}}^{j_1}$. Middle, from left to right: m_{jj}, $\Delta\eta_{\mathrm{jj}}$, $\Delta\phi_{\mathrm{jj}}$, and Zeppenfeld. Bottom, from left to right: min-$\Delta\phi(\mu\mu, j)$, min-$\Delta\eta(\mu\mu, j)$, $R_{p_{\mathrm{T}}}$, and $\mathrm{N}_{\mathrm{soft}}$.

Therefore, we can conclude that both the ggH and VBF BDT are highly uncorrelated with the dimuon mass. Events can be further categorized based on the BDT scores in order to maximize the search sensitivity without sculpting or biasing the expected invariant mass distribution for signal and background events.

Finally, Fig. 27 shows a comparison between the ROC curves obtained in each era and in each training from training (black) and testing (red) events. In the ggH BDT (top) the ROC curves are essentially identical between training and testing events for all eras, indicating that no over-training is observed. On the other hand, in the case of the VBF BDT (bottom) a small overtraining is observed in 2017 and 2018 eras because the available Drell-Yan MC statistics is about 20% smaller than in 2016.

6 Event categorization

At this stage, we have pre-selected events into to two categories. VBF events are those with at least two jets in the final state with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, $p_{\mathrm{T}}^{j_1} > 35\,\mathrm{GeV}$, and $p_{\mathrm{T}}^{j_1} > 25\,\mathrm{GeV}$. In contrast, the ggH phase space collects all the remaining events which are not belonging to the VBF channel or to the other exclusive modes (VH-leptonic and ttH). Figure 28 shows the distributions of the ggH (left) and VBF (right) BDT scores in data and simulation for events belonging to the signal fit region (SR), $110 < m_{\mu\mu} < 150\,\mathrm{GeV}$. The observation and the expec-

DRAFT

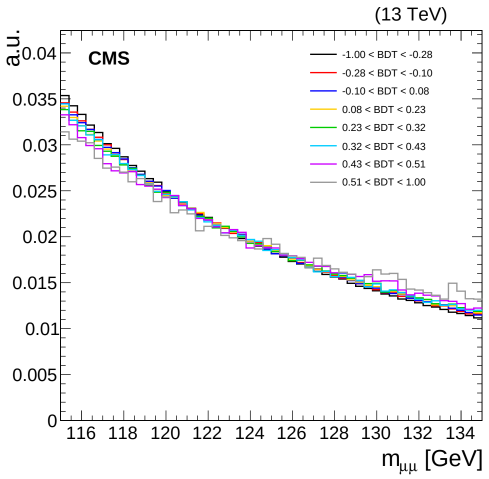

Figure 25: Dimuon mass distribution in simulated background events computed in a set of orthogonal ranges defined along the ggH (left), and VBF (right) BDT scores. Distributions are scaled in order to have unit area.

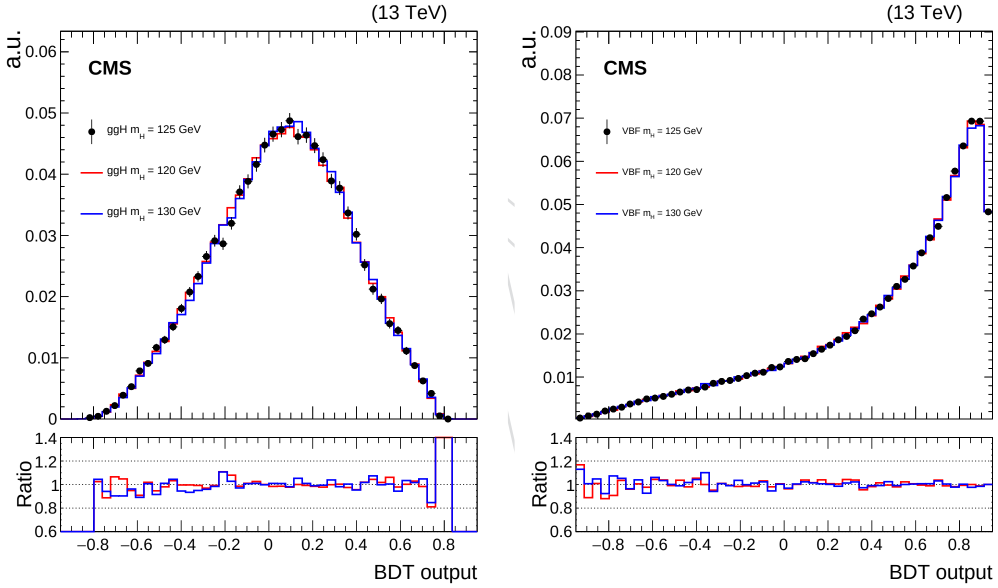

Figure 26: Distribution of the ggH (left), and VBF (right) BDT outputs in ggH or VBF simulated signal events for three different mass hypothesis: $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ (black), $m_{\mathrm{H}} = 120\,\mathrm{GeV}$ (red), and $m_{\mathrm{H}} = 130\,\mathrm{GeV}$ (blue). Distributions are scaled in order to have unit area.

tation in the three eras (2016, 2017, and 2018) have been summed together. In the case of the VBF BDT, the MVA-score the output score has been analytically transformed in order to better highlight events belonging to the high score region, where the VBF signal purity increases. The transformed BDT output is defined as follows:

$$\text{transformed-BDT} = \text{BDT}' = \tanh^{-1}\left(\frac{\text{BDT-score}+1}{2}\right)$$

In the ratio plot, the gray band corresponds to the uncertainty coming from the limited statistics of the simulated samples, while the azure one corresponds to the sum in quadrature between statistical and systematic uncertainties from experimental sources. The uncertainty due to jet

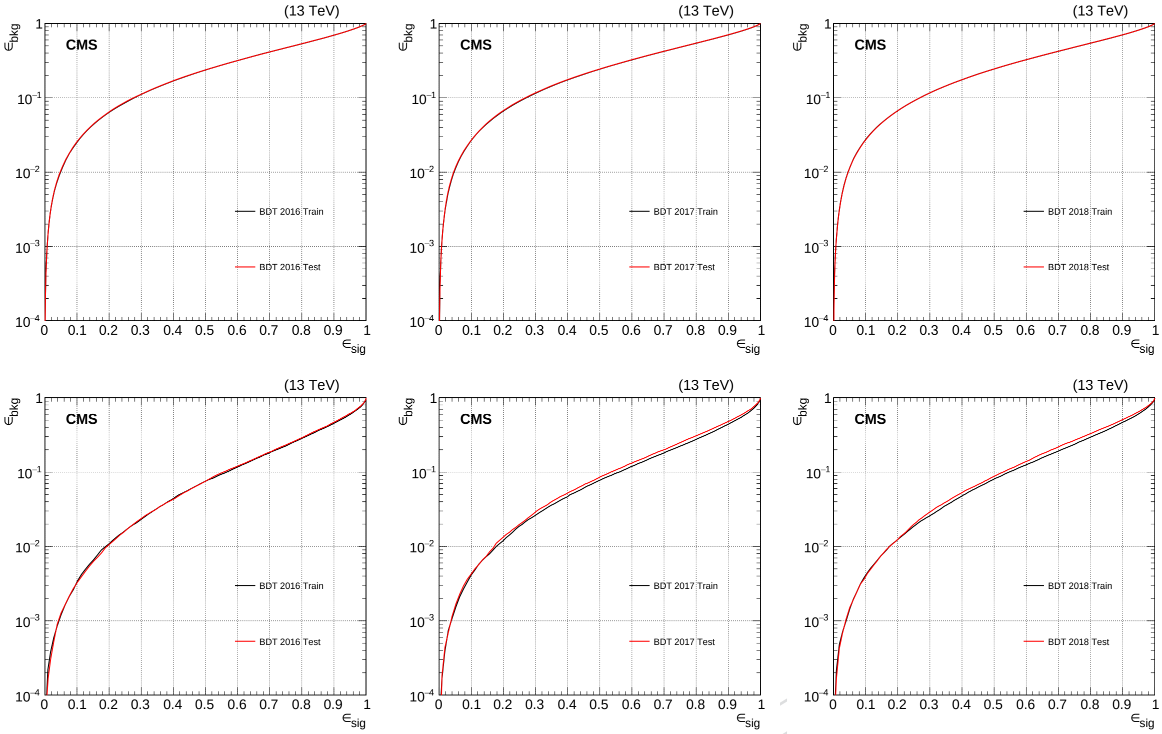

Figure 27: Signal efficiency (ϵ_{sig}) vs background efficiency (ϵ_{bkg}) ROC curves obtained in each eras from the ggH (top), and VBF (bottom) BDT output scores for events in the training (black) and testing (red) samples. The ROC curves shown in the first column refers to the 2016 era, while in the second and third ones the ROCs for 2017 and 2018 eras are reported. Besides the VBF BDT trained on 2017 and 2018 MC, no differences in performance are observed between training and testing events. Therefore, the trained BDTs are at most slightly affected by overfitting.

energy resolution is one-sided as described in more details in Sec. 7.2.3. Finally, the orange band includes an estimate of the theoretical uncertainties, which is added in quadrature to the statistical and the experimental systematic uncertainties. Overall, a good agreement between data and MC is observed in both cases and most of the discrepancies are covered by systematic uncertainties assigned to the background prediction. An excess in data of about 10-15% resides in the high score region (BDT-score larger than 0.65) of the ggH BDT. The high score part of the ggH BDT integrates mainly 1-jet events in which the jet belongs to the endcap region, or two-jet events with low m_{jj} but either a small $\Delta\phi_{\text{jj}}$ or a large min-$\Delta\eta(\mu\mu, j)$. The simulation does not provide a perfect description of data in these regions of the phase space as reported in Fig. 16. Therefore, the origin of the observed trend is understood and mainly related to jet occupancy in the forward part of the endcaps.

Comparisons are also reported independently for each year in Fig. 29, where the first column corresponds to the 2016, the second to the 2017, and the third to the 2018 era. The first row reports the ggH BDT outputs, while the second the transformed VBF BDT ones. A good agreement between data and MC is observed across all eras in both the ggH and VBF phase spaces. The excess in data observed in the high score region of the ggH BDT reported in Fig. 28 originates from 2017 and 2018 eras, where the description of endcap jets from simulation is known to be not as good as in 2016. In the VBF region, a 10% offset in the overall normalization is observed in the 2017 era which is well covered by both the theoretical uncertainties in the background prediction and the uncertainties attributed to the knowledge of the jet energy scale and

DRAFT

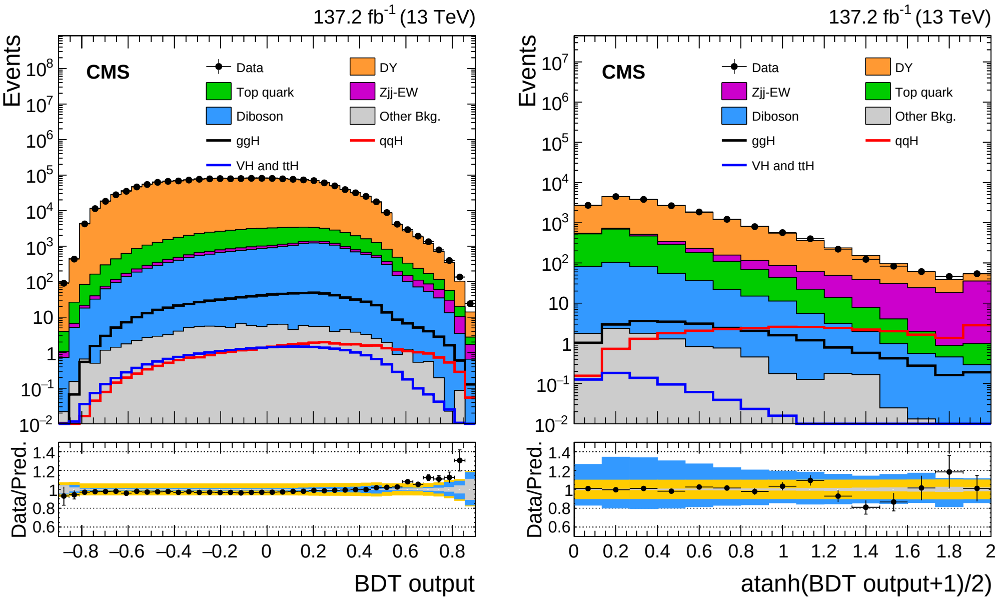

Figure 28: Distribution of ggH (left) and VBF (right) BDT scores reported for both data (black points) and simulation (solid histograms) for events with $m_{\mu\mu}$ between 110 and 150 GeV. The observation and the expectation in the three eras (2016, 2017, and 2018) have been summed together. In the ratio plot, the grey band indicates the uncertainty coming from the limited statistics of the simulated samples, the azure one corresponds to the sum in quadrature between statistical and systematic uncertainties from experimental sources, while the orange band includes the theoretical uncertainties in the background prediction. The asymmetry in the experimental uncertainty band is due to the one-sided effect of jet energy resolution.

resolution.

In order to test more precisely how the simulation models the data in the high score region of both the ggH and the transformed VBF BDTs, data and simulation are compared in the ZCR, i.e. in events passing the analysis selections with $m_{\mu\mu}$ between 70 and 110 GeV. In fact, the ZCR collects about 100 times more data across the full BDT range. Figure 30 reports these comparison where the first column corresponds to the 2016, the second to the 2017, and the third to the 2018 era. The first row reports the ggH BDT outputs, while the second the transformed VBF BDT ones. Overall, the trends observed in the ZCR are similar to those observed and already described for the signal region. Finally, Fig. 31 shows the the distributions of the ggH (left) and transformed VBF (right) BDT scores in data and simulation for events belonging to the ZCR region, $70 < m_{\mu\mu} < 110$ GeV. In this case, the observation and the expectation in the three eras (2016, 2017, and 2018) have been summed together.

6.1 Category optimization

Based on the output of the trained BDT discriminants reported in Fig. 28, we can further categorize events belonging to both the ggH and VBF phase spaces. In order to do that, we need to set up a procedure and choose a figure of merit to decide where to put a boundary along the BDT outputs. The designed procedure is iterative and the optimal boundaries are obtained by summing up the signal and background contributions expected from the three eras.

We start with a single category collecting all the events belonging to the ggH or the VBF phase space, and we compute the expected significance for a Higgs boson with $m_{\mathrm{H}} = 125$ GeV. This is obtained by modelling the $m_{\mu\mu}$ distribution of the total signal (all production modes are

DRAFT

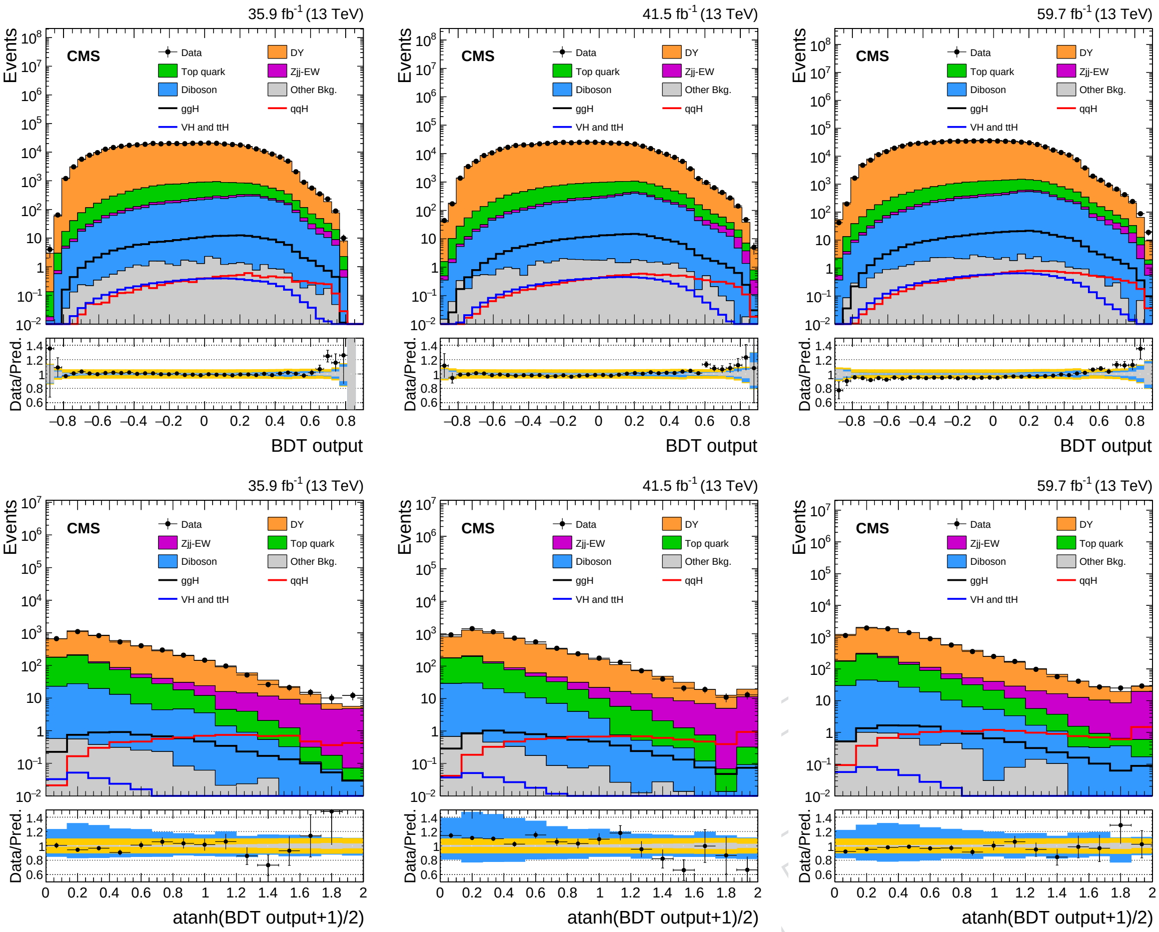

Figure 29: Distribution of the ggH (top) and VBF (bottom) BDT scores in 2016 (left), 2017 (middle) and 2018 (right) eras reported for both data (black points) and simulation (solid histograms) for events with $m_{\mu\mu}$ between 110 and 150 GeV. In the ratio plot, the grey band indicates the uncertainty coming from the limited statistics of the simulated samples, the azure one corresponds to the sum in quadrature between statistical and systematic uncertainties from experimental sources, while the orange band includes the theoretical uncertainties in the background prediction. The asymmetry in the experimental uncertainty band is due to the one-sided effect of jet energy resolution.

DRAFT

added together into a single distribution) with a double-sided Crystal Ball function. The mass range of the fit is chosen to be 110–150 GeV. In this optimization procedure, ggH events from the MADGRAPH5_aMC@NLO generator are considered because these simulated samples are the ones used for the final signal extraction. In order to avoid an over-optimization, MADGRAPH5_aMC@NLO events with odd event-number are considered, while even ones are used to build the signal model in each category for the final signal extraction. In contrast, VBF events are taken from the centrally produced POWHEG samples showered with PYTHIA and only even event-number are considered, which correspond to those belonging to the MVA testing sample. The total expected background yield is also obtained from simulation (background simulated samples listed in Sec. 2) and we model the shape of the dimuon mass distribution in background events via an analytical function called BWZRedux (three shapes degrees of freedom), defined in Equation 5, in the ggH phase space, while in the VBF one the BWZGamma function is chosen (two shape degrees of freedom) as defined in Equation 6. From earlier iterations of this analysis, both the BWZRedux and the BWZGamma functions are known to fit the back-

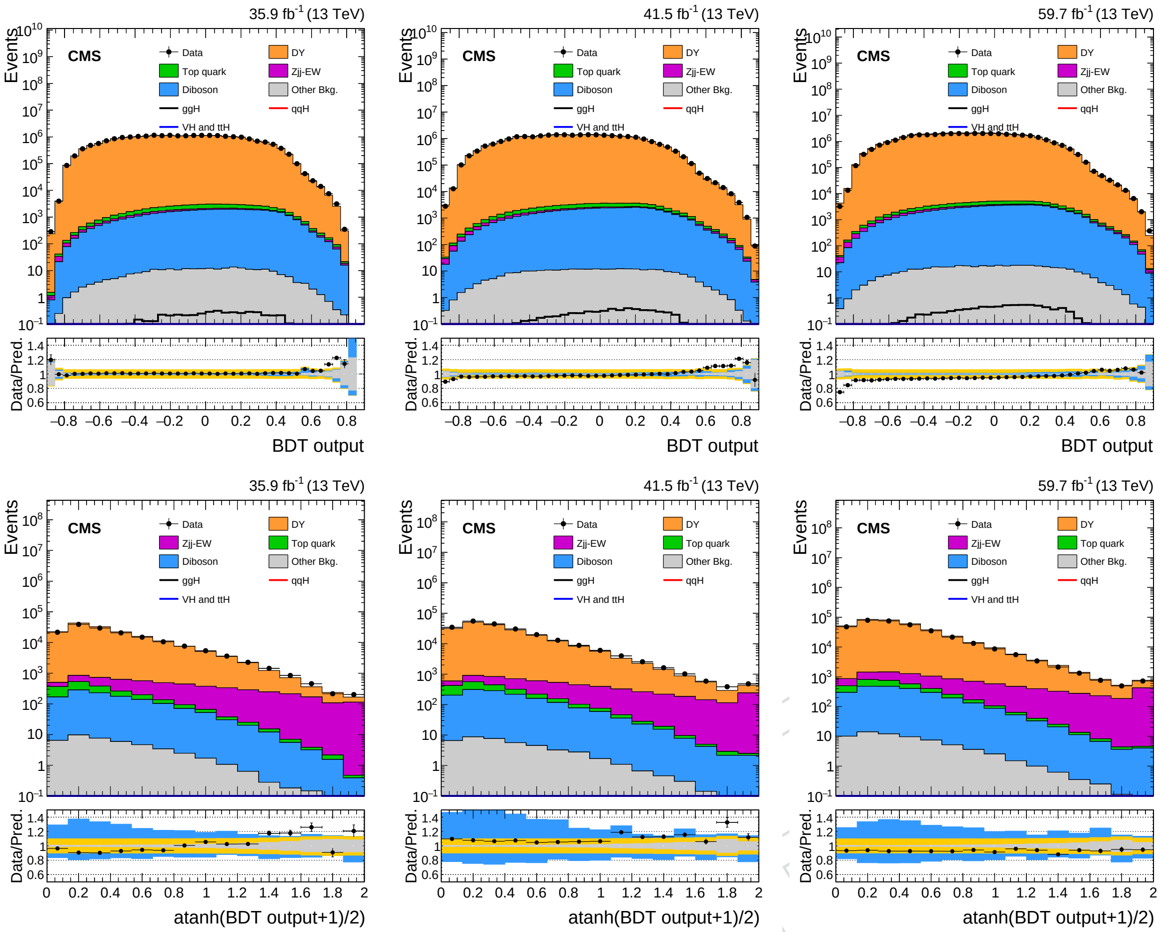

Figure 30: Distribution of the ggH (top) and VBF (bottom) BDT scores in 2016 (left), 2017 (middle) and 2018 (right) eras reported for both data (black points) and simulation (solid histograms) for events with $m_{\mu\mu}$ between 70 and 110 GeV. In the ratio plot, the grey band indicates the uncertainty coming from the limited statistics of the simulated samples, the azure one corresponds to the sum in quadrature between statistical and systematic uncertainties from experimental sources, while the orange band includes the theoretical uncertainties in the background prediction. The asymmetry in the experimental uncertainty band is due to the one-sided effect of jet energy resolution.

ground very well across the full phase-space.

$$\mathrm{BWZRedux}(m_{\mu\mu}; a_1, a_2, a_3) = \frac{e^{a_2 m_{\mu\mu} + a_3 m_{\mu\mu}^2}}{(m_{\mu\mu} - m_Z)^{a_1} + (\frac{\Gamma_Z}{2})^{a_1}} \tag{5}$$

$$\mathrm{BWZGamma}(m_{\mu\mu}; a_1, f) = f \times \frac{e^{a_1 m_{\mu\mu}}}{(m_{\mu\mu} - m_Z)^2 + (\frac{\Gamma_Z}{2})^2} + (1-f) \times \frac{e^{a_1 m_{\mu\mu}}}{m_{\mu\mu}^2} \tag{6}$$

After this, we compare the significance obtained with a single category with the one obtained by dividing the events into two categories. Since the BDT shapes for signal and background events are different across eras, as shown in Fig. 20, the BDT outputs in the ggH or in the VBF phase space are scanned in signal quantiles, i.e. in steps of 1% in signal efficiency. The optimal position for the category boundary corresponds to the one that gives the maximum

DRAFT

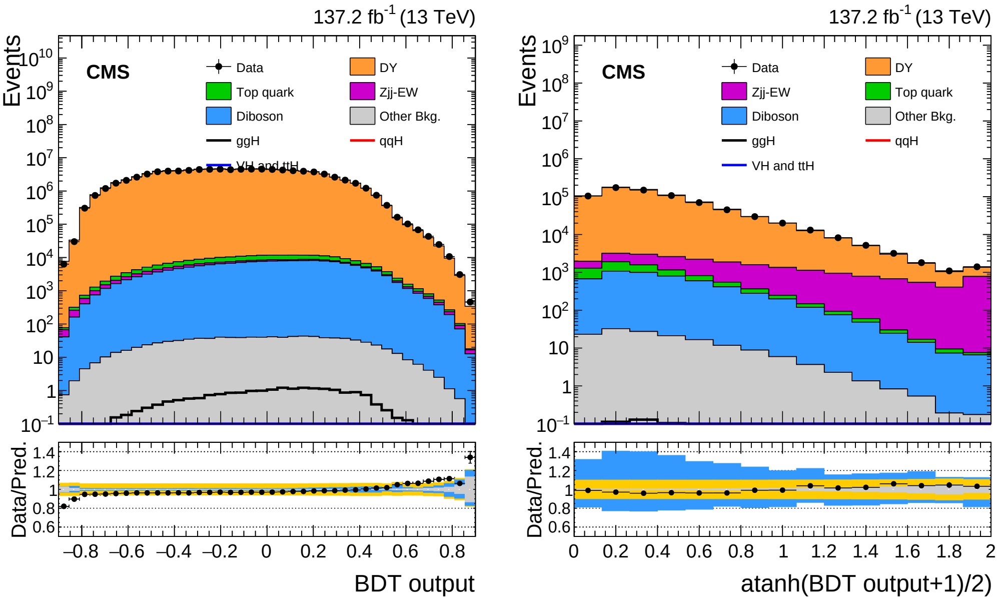

Figure 31: Distribution of ggH (left) and VBF (right) BDT scores reported for both data (black points) and simulation (solid histograms) for events with $m_{\mu\mu}$ between 70 and 110 GeV. The observation and the expectation in the three eras (2016, 2017, and 2018) have been summed together. In the ratio plot, the grey band indicates the uncertainty coming from the limited statistics of the simulated samples, the azure one corresponds to the sum in quadrature between statistical and systematic uncertainties from experimental sources, while the orange band includes the theoretical uncertainties in the background prediction. The asymmetry in the experimental uncertainty band is due to the one-sided effect of jet energy resolution.

significance obtained from a combined fit of the two categories, where the total signal and background contributions are modelled as described above. Once the optimal boundary has been found and two categories are defined, the iterative algorithm searches for a third optimal category, etc. This procedure stops when the n-th category does not provide an additional gain in significance larger than 1%.

By using this iterative strategy, nine categories are defined: five categories along the ggH BDT output and four along the transformed VBF BDT. The category boundaries defined in terms of signal efficiency quantiles and BDT ranges for the ggH and VBF BDT are reported in Table 13 and Table 14, respectively. However, since predictions from MADGRAPH5_aMC@NLO and POWHEG generators for the ggH process can be different by a few % depending on the phase space, we have repeated the same optimization procedure using POWHEG signal samples instead of MADGRAPH5_aMC@NLO ones. The same number of optimal categories is found, and the corresponding ranges in signal efficiency are in agreement within ±2%, suggesting that the optimal categories listed in Table 13 and Table 14 are robust and universal, i.e. not specific to a particular MC generator.

Figure 32 shows the evolution of the combined expected significance from categories defined along the ggH BDT output during the optimization procedure. The first plot on the left reports the trend of the significance as a function of the first boundary, defined in terms of signal efficiency quantiles. As described before, expectations in signal and background from the three eras are summed together. The significance is maximized by dividing the ggH phase space into two categories in a signal efficiency range between. Moreover, the other graphs reported in Fig. 32 show how the other category boundaries listed in Table 13 have been defined dur-

DRAFT

Table 13: Optimized category boundaries defined along the output of the ggH BDT discriminant.

Name	Signal eff. range	BDT range 2016	BDT range 2017	BDT range 2018
ggH-cat0	$[100, 70)\%$	$[-1.00, -0.125)$	$[-1.00, -0.113)$	$[-1.00, -0.115)$
ggH-cat1	$[70, 35)\%$	$[-0.125, 0.185)$	$[-0.113, 0.191)$	$[-0.115, 0.191)$
ggH-cat2	$[35, 20)\%$	$[0.185, 0.309)$	$[0.191, 0.315)$	$[0.191, 0.312)$
ggH-cat3	$[20, 5)\%$	$[0.309, 0.471)$	$[0.315, 0.498)$	$[0.312, 0.496)$
ggH-cat4	$\leq 5\%$	$[0.471, 1.00)$	$[0.498, 1.00)$	$[0.496, 1.00)$

Table 14: Optimized category boundaries defined along the output of the transformed VBF BDT discriminant.

Name	Signal eff. range	BDT range 2016	BDT range 2017	BDT range 2018
VBF-cat0	$[100, 45)\%$	$[0.000, 0.948)$	$[0.000, 0.932)$	$[0.000, 0.866)$
VBF-cat1	$[45, 22)\%$	$[0.948, 1.347)$	$[0.932, 1.372)$	$[0.866, 1.310)$
VBF-cat2	$[22, 10)\%$	$[1.347, 1.610)$	$[1.372, 1.728)$	$[1.310, 1.701)$
VBF-cat3	$\leq 10\%$	≥ 1.610	≥ 1.728	≥ 1.701

ing the iterative scan. The combined significance extracted during the optimization procedure from the five ggH categories is of about 1.63σ. Similarly, Fig. 33 shows the evolution of the combined expected significance from categories defined along the VBF transformed BDT output. The combined significance extracted during the optimization procedure from the four VBF categories is of about 1.45σ.

Figure 34 shows a comparison between the outputs of the two BDT discriminant (left: ggH BDT, right: VBF BDT) for signal and background simulated events. The black dashed vertical lines indicate, approximately, the category boundaries optimized with the iterative procedure previously described. In addition, inside the lower pad of each figure, the significance expressed as $\mathrm{S}/\sqrt{\mathrm{B}}$ of each category, calculated by integrating signal and background expected events inside the half width at half maximum (HWHM, where HWHM=FWHM/2) of the expected signal peak, is also reported.

Finally, Fig. 35 and 36 show the distributions of the dimuon invariant mass in the five ggH and in the four VBF categories, respectively, for data and simulated events in the range $110 < m_{\mu\mu} < 150\,\mathrm{GeV}$. The agreement between data and simulation in both shape and normalization, apart from statistical fluctuations in high score region of the VBF BDT, is good across all categories apart from a normalization offset of about 20% observed in the tightest VBF category. However, since the background in this search is directly extracted from data, as described later in Sec. 8, this discrepancy in normalization does not have any effect on the analyssi results. For the time being, the signal peak region remains blinded ($120 < m_{\mu\mu} < 130\,\mathrm{GeV}$).

7 Signal extraction

In this search the signal is extracted by fitting the dimuon invariant mass spectrum in data via analytical functions able to model the expected signal and background processes. In the case of the signal (ggH, VBF, VH, and ttH), these analytical functions contain several parameters whose values are determined from simulation and are not allowed to float in the final fit, i.e. they are treated as constants or constrained parameters when the fit to the data is performed. In the case of the background, we assume no prior knowledge of the parameters of the fit functions, therefore they are allowed to float freely in the signal extraction fit. The overall normalizations of the signal and the backgrounds are also floated freely.

DRAFT

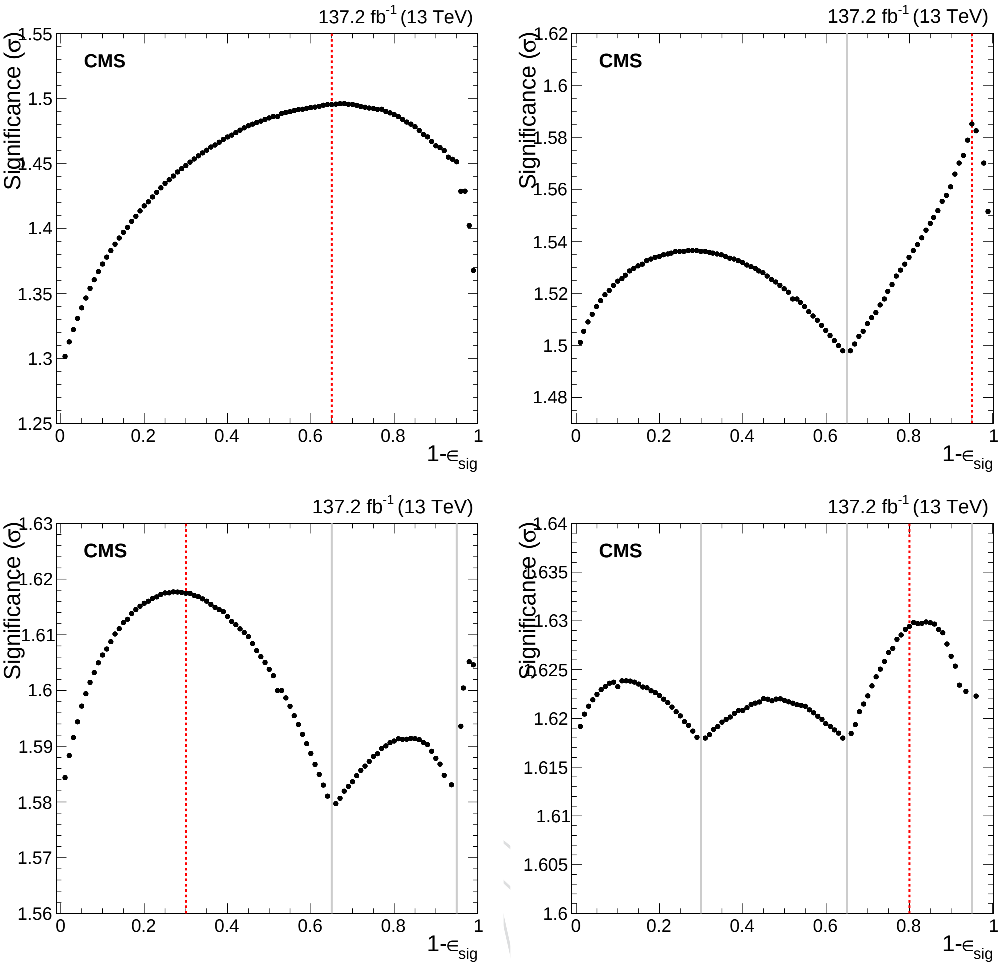

Figure 32: Evolution of the combined significance for a Higgs boson signal with $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ defined by scanning the ggH BDT output in signal efficiency quantiles. Top row, from left to right: improvements in the total expected significance by using two categories instead of one, and three categories instead of two. Second row, from left to right: improvements in the total expected significance with four categories instead of three, and five optimized categories instead of four. Dashed red lines indicate the category boundary giving optimal performance, while the solid gray ones indicate already defined boundaries from previous steps of the iterative procedure.

7.1 Signal lineshape and yields

The signal line shape is modelled using a double-sided Crystal Ball function defined in Equation 2. In each of the nine categories listed in Table 13 and 14, we fit the signal contribution from each the production modes (ggH, VBF, WH, ZH, and ttH) independently and we fix the corresponding shape parameters. Since POWHEG simulated events for ggH have been used in the ggH and VBF BDT trainings, both the lineshapes and the expected yields used in the ultimate signal extraction for this process are obtained from MADGRAPH5_aMC@NLO simulated samples. Predictions for the VBF production in both ggH and VBF BDT categories are taken instead from POWHEG interfaced with PYTHIA in dipole shower mode. Figure 37 shows the resulting signal shapes for the ggH process in ggH categories obtained by fitting the expected dimuon mass spectrum with a double-sided Crystal Ball, where the expectations in the three eras (2016, 2017, and 2018) have been summed together. Similarly, Fig. 38 reports the signal

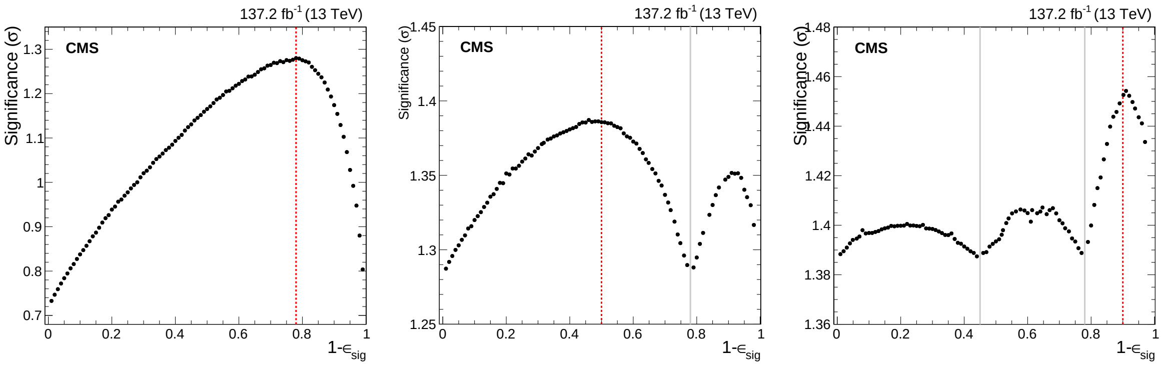

Figure 33: Evolution of the combined significance for a Higgs boson signal with $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ defined by scanning the transformed VBF BDT output in signal efficiency quantiles. From left to right: improvements in the total expected significance by using two categories instead of one, three categories instead of two, and four categories instead of three. Dashed red lines indicate the category boundary giving optimal performance, while the solid gray ones indicate already defined boundaries from previous steps of the iterative procedure.

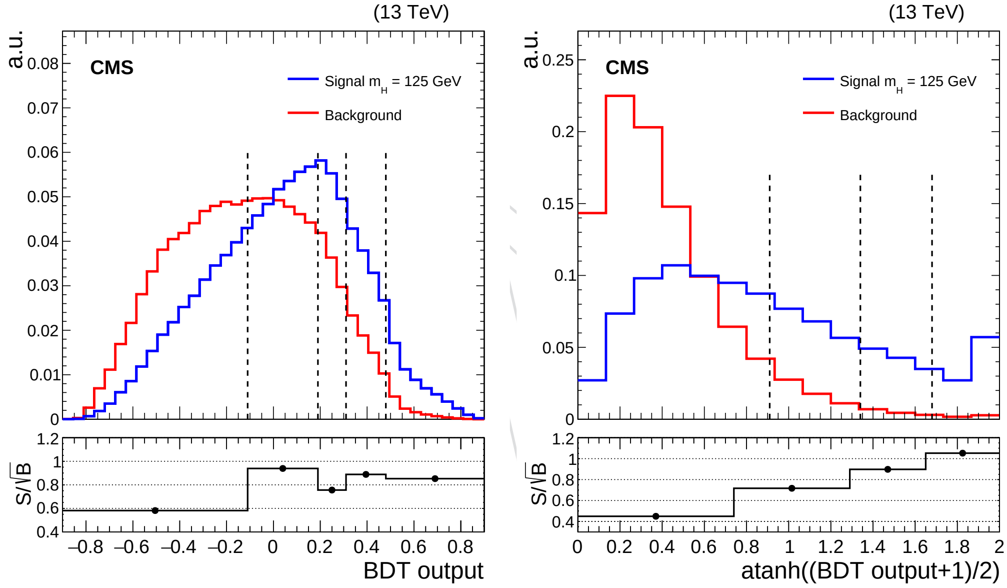

Figure 34: Distribution of the ggH (left) and transformed VBF (right) BDT output in signal (blue) and background (red) simulated events. The black dashed vertical lines indicate, approximately, the category boundaries optimized with the iterative procedure previously described. In the lower pad, the significance expressed as $\mathrm{S}/\sqrt{\mathrm{B}}$ of each category, calculated by integrating signal and background expected events inside the HWHM of the expected signal peak, is also reported.

shapes for VBF events in the VBF categories. Overall, the double-sided Crystal Ball provides a good description of both the bulk and the tails of the expected signal distributions in each category, $\chi^2/ndf \approx 1$.

The Higgs boson signal rates expected in $137.2\,\mathrm{fb}^{-1}$ of data are reported in Table 15 for each category defined along the ggH and VBF BDT outputs, as well as for each Higgs boson production mode. The ggH signal represents the dominant contribution in all the ggH categories, while the VBF mode overcomes ggH in the last three categories defined in the VBF phase space.

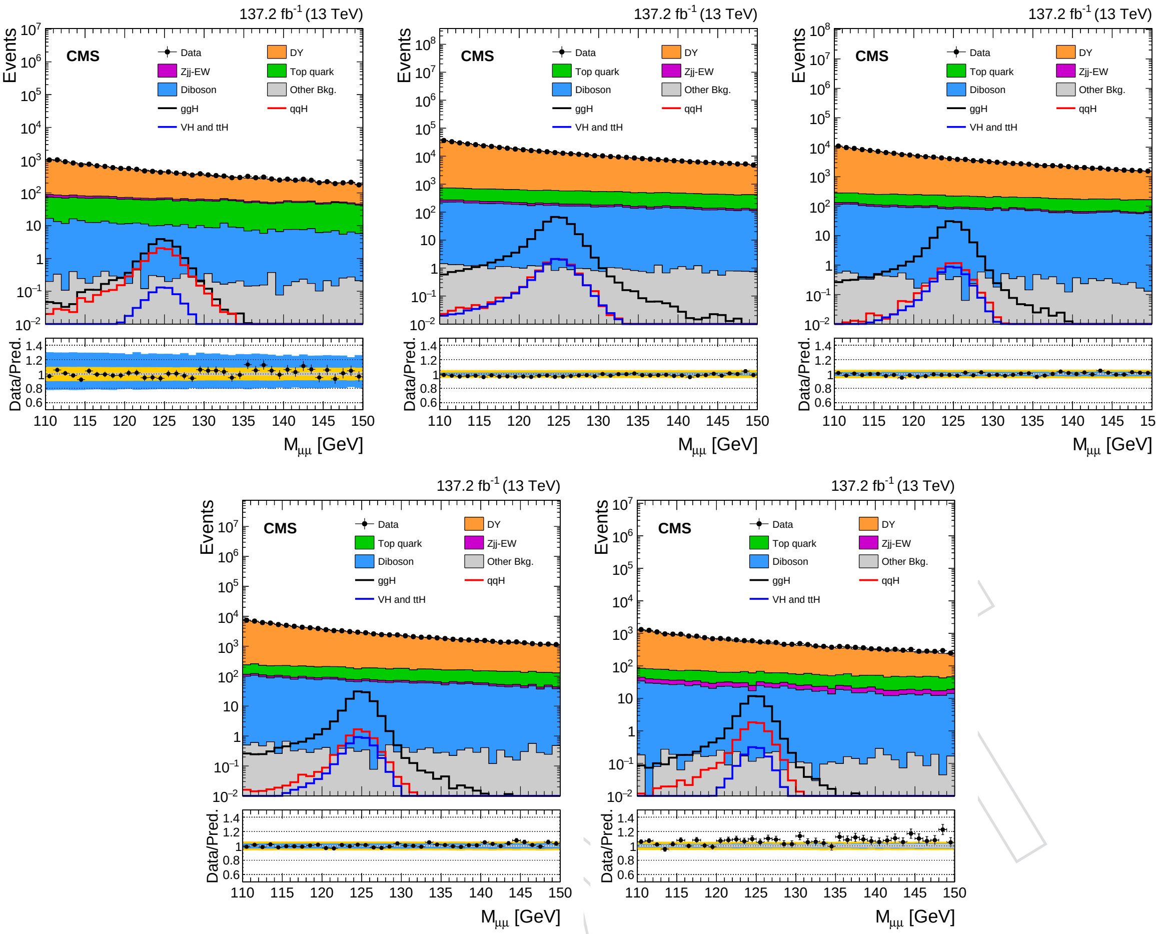

Figure 35: Distribution of the dimuon invariation mass in the ggH categories reported for both data (black points) and simulation (solid histograms) for events with $m_{\mu\mu}$ between 110 and 150 GeV. Top row: ggH-cat0 (left), ggH-cat1 (middle), and ggH-cat2 (right). Bottom row: ggH-cat3 (left), and ggH-cat4 (right). The expected signal distributions from the different Higgs boson production modes are also reported. In the ratio plot, the grey band indicates the uncertainty coming from the limited statistics of the simulated samples, the azure one corresponds to the sum in quadrature between statistical and systematic uncertainties from experimental sources, while the orange band includes the theoretical uncertainties in the background prediction.

Table 15: Higgs boson signal rates expected in 137.2 fb^{-1} of data in each category defined along the ggH and VBF BDT outputs for ggH, VBF, WH, ZH, and ttH production modes.

Category	ggH	VBF	WH	ZH	ttH	Total
ggH-cat0	254.52	7.81	5.48	3.42	0.27	271.53
ggH-cat1	295.21	10.84	5.83	3.67	0.19	315.74
ggH-cat2	124.04	5.35	2.19	1.50	0.07	133.14
ggH-cat3	116.49	7.03	2.07	1.60	0.07	127.26
ggH-cat4	45.42	7.70	0.60	0.61	0.02	54.36
VBF-cat0	18.38	9.74	0.39	0.23	0.04	28.79
VBF-cat1	3.99	7.38	0.02	0.01	< 0.01	11.42
VBF-cat2	1.16	4.75	< 0.01	< 0.01	< 0.01	5.94
VBF-cat3	0.48	5.17	< 0.01	< 0.01	< 0.01	5.67

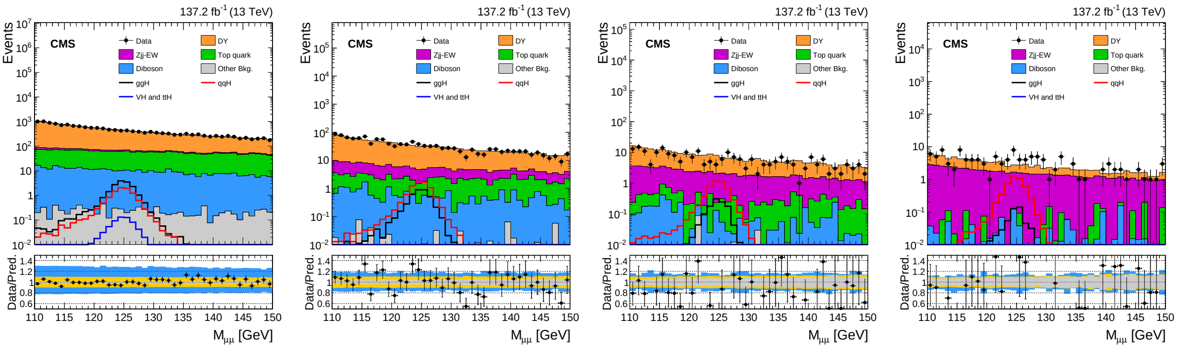

Figure 36: Distribution of the dimuon invariation mass in the VBF categories reported for both data (black points) and simulation (solid histograms) for events with $m_{\mu\mu}$ between 110 and 150 GeV. From left to right: VBF-cat0, VBF-cat1, VBF-cat2, and VBF-cat3. The expected signal distributions from the different Higgs boson production modes are also reported. In the ratio plot, the grey band indicates the uncertainty coming from the limited statistics of the simulated samples, the orange one corresponds to the sum in quadrature between statistical and systematic uncertainties from theoretical sources, while the azure band includes the experimental uncertainties in the background prediction.

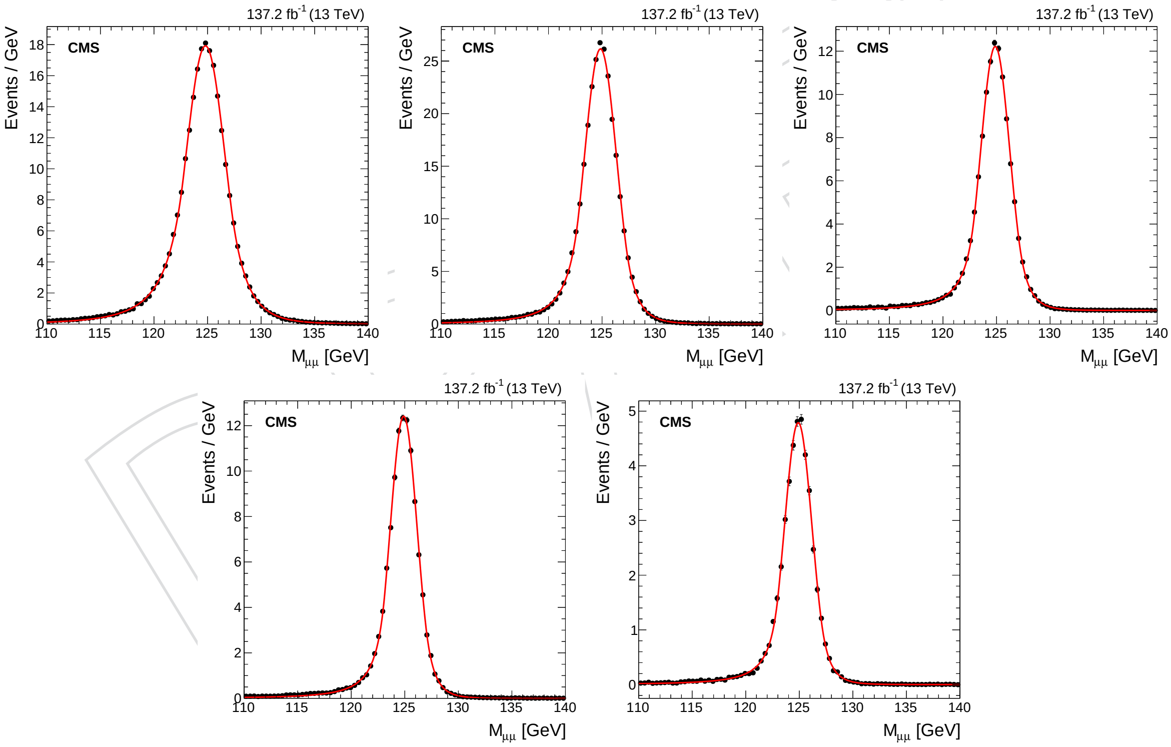

Figure 37: Distributions of the dimuon invariant mass for ggH signal events produced via the MADGRAPH5_aMC@NLO generator in the ggH BDT categories. Top row: ggH-cat0 (left), ggH-cat1 (middle), and ggH-cat2 (right). Bottom row: ggH-cat3 (left), and ggH-cat4 (right). The expectations in the three eras (2016, 2017, and 2018) have been summed together.

The same procedure is also applied to alternative signal samples produced with a different Higgs boson mass hypothesis, namely $m_H = 120$ and 130 GeV. This allows to interpolate both the expected signal rates and the double-sided Crystal Ball shape parameters as a function of the Higgs boson mass within the mass range of 120–130 GeV. The interpolation vs m_H is performed via a smooth spline function. The interpolation is crucial in order to compute local

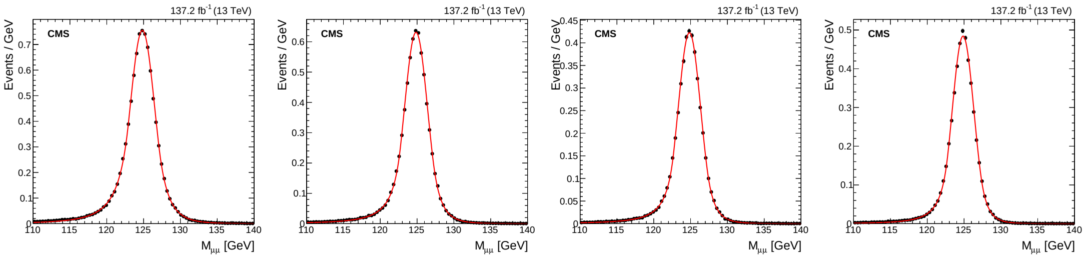

Figure 38: Distributions of the dimuon invariant mass for VBF signal events produced via the POWHEG generator (showered with PYTHIA in dipole mode) in the VBF BDT categories. From left to right: VBF-cat0, VBF-cat1, VBF-cat2, and VBF-cat3, where the expectations in the three eras (2016, 2017, and 2018) have been summed together.

950 p-values (upper limits) scanning the interesting mass range of a few GeV around the measured
951 Higgs boson mass of 125.09 GeV. Figure 39 and 40 show the interpolated signal shapes, nor-
952 malized to the expected rate in 137.2 fb^{-1} of data, for the ggH and VBF processes in the ggH
953 and VBF categories, respectively.

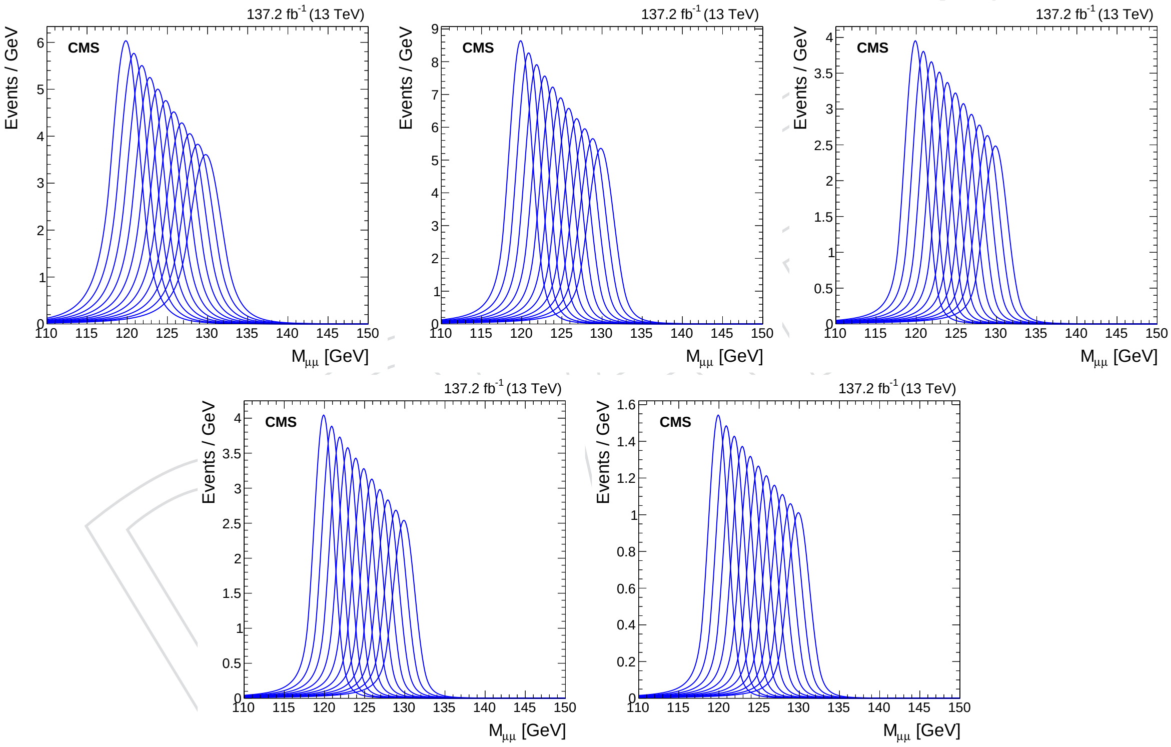

Figure 39: Distributions of the dimuon invariant mass for the ggH process in the ggH BDT categories as a function of m_H, resulting from an interpolation of the expected rates and shape parameters across the generated Higgs signal samples for $m_H = 120$, 125, and 130 GeV. Top row: ggH-cat0 (left), ggH-cat1 (middle), and ggH-cat2 (right). Bottom row: ggH-cat3 (left), and ggH-cat4 (right).

954 In addition, in order to better track the evolution of expected signal yield and shape as a func-
955 tion of m_H, the trends of the dimuon peak width (σ_{CB}) and of the acceptance times efficiency
956 are reported for the ggH signal in the ggH categories in Fig. 41 and 42, respectively. Similar
957 results are shown in Fig. 43 and 44 in the case of VBF signal in the VBF categories. As ex-
958 pected, the signal width increases linearly with the mass but the mass resolution, defined as
959 σ_{CB}/m_H, remains constant. The product of acceptance and efficiency also slightly increases

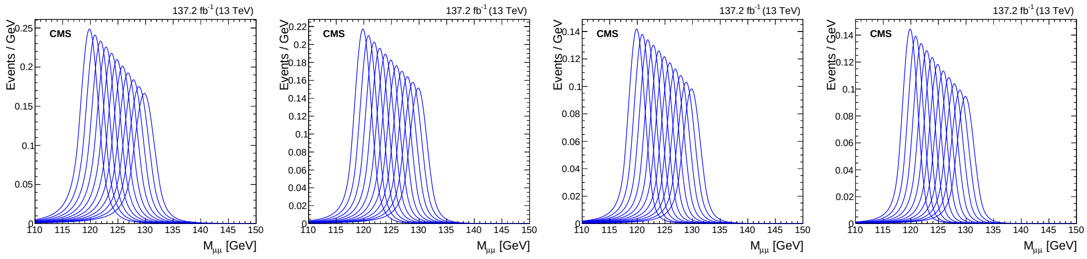

Figure 40: Distributions of the dimuon invariant mass for the VBF process in the VBF BDT categories as a function of m_H, resulting from an interpolation of the expected rates and shape parameters across the generated Higgs signal samples for $m_H = 120$, 125, and 130 GeV. From left to right: VBF-cat0, VBF-cat1, VBF-cat2, and VBF-cat3.

with the mass in each category. However, since $\sigma \times \mathcal{B}(H \to \mu\mu)$ decreases by about 30% moving from $m_H = 120$ to $m_H = 130$ GeV, the total expected signal rate given by $\sigma \times \mathcal{B}(H \to \mu\mu) \times$ acceptance $\times$ efficiency also decreases as already shown in Fig. 39 and 40.

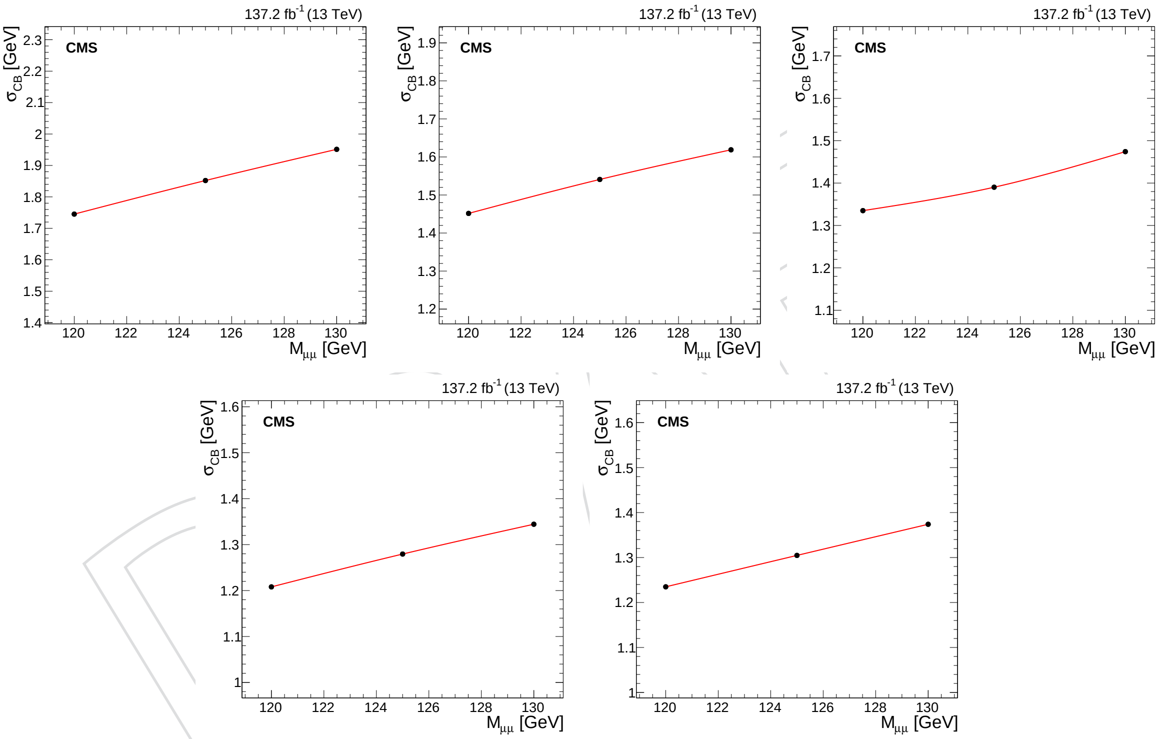

Figure 41: Evolution of the dimuon peak width (σ_{CB}) for the ggH process in the ggH BDT categories as a function of m_H, in the range $120 < m_{\mu\mu} < 130$ GeV. Top row: ggH-cat0 (left), ggH-cat1 (middle), and ggH-cat2 (right). Bottom row: ggH-cat3 (left), and ggH-cat4 (right).

7.2 Uncertainties on the signal model

Systematic uncertainties affecting of the expected signal prediction in each category are taken into account in the signal extraction and distinguished between *shape* and *rate* uncertainties. The *shape* uncertainties are those affecting the modelling of the signal peak, while the *rate* uncertainties are those affecting the predicted number of signal events in each category.

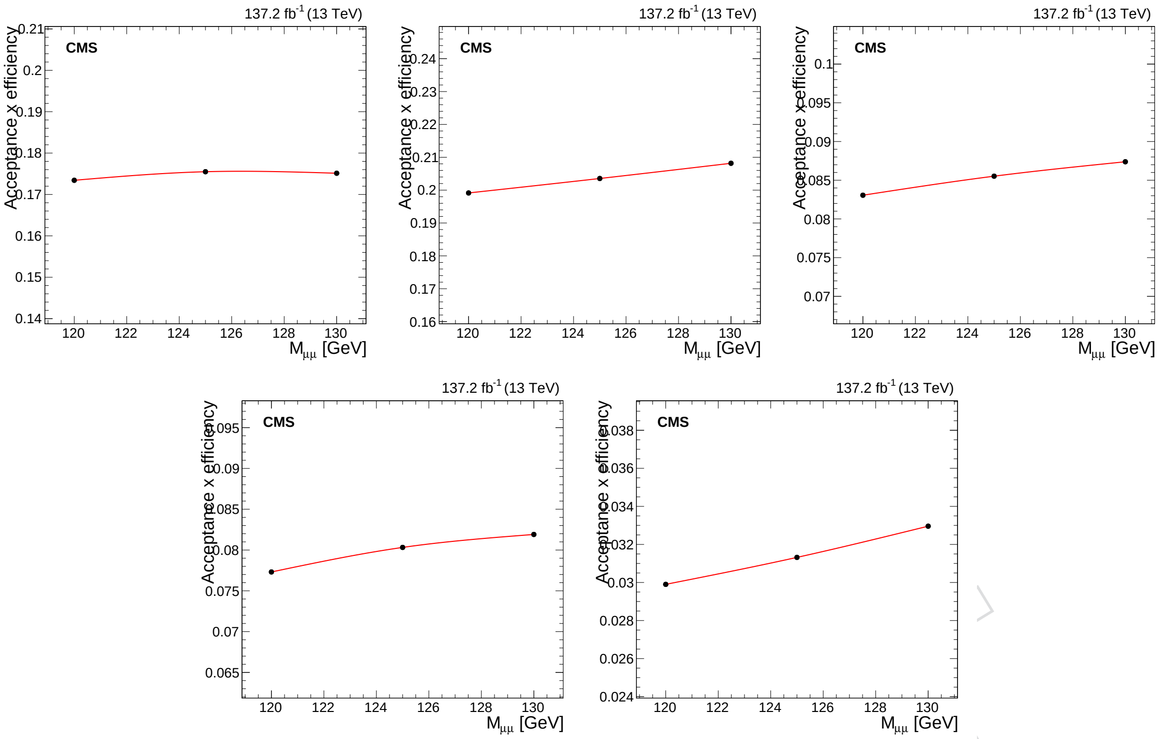

Figure 42: Evolution of the acceptance times efficiency for the ggH process in the ggH BDT categories as a function of m_H, in the range $120 < m_{\mu\mu} < 130\,\mathrm{GeV}$. Top row: ggH-cat0 (left), ggH-cat1 (middle), and ggH-cat2 (right). Bottom row: ggH-cat3 (left), and ggH-cat4 (right).

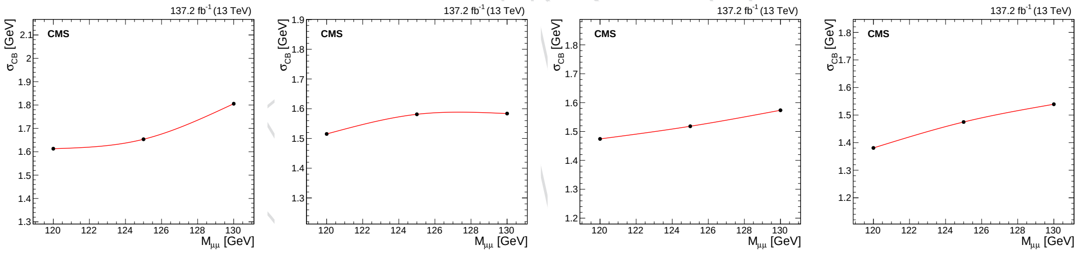

Figure 43: Evolution of the dimuon peak width (σ_CB) for the VBF process in the VBF BDT categories as a function of m_H, in the range $120 < m_{\mu\mu} < 130\,\mathrm{GeV}$. From left to right: VBF-cat0, VBF-cat1, VBF-cat2, and VBF-cat3.

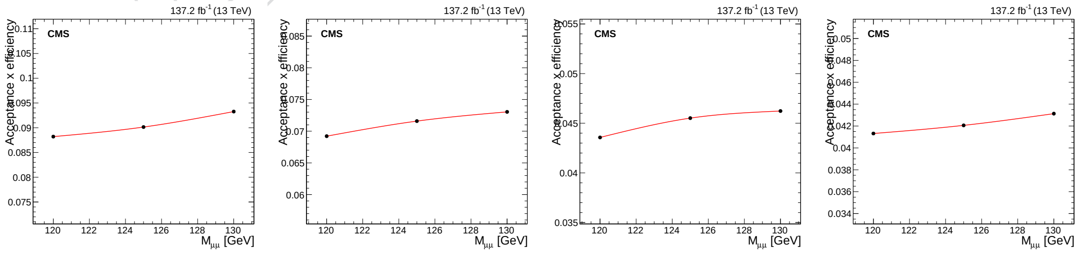

Figure 44: Evolution of the acceptance times efficiency for the VBF process in the VBF BDT categories as a function of m_H, in the range $120 < m_{\mu\mu} < 130\,\mathrm{GeV}$. From left to right: VBF-cat0, VBF-cat1, VBF-cat2, and VBF-cat3.

7.2.1 Shape uncertainties

For all Higgs boson production modes, the expected $m_{\mu\mu}$ signal shape is primarily affected by the uncertainties in the muon energy scale and resolution. The Rochester corrections, described in Section 4.5, are meant to correct for differences in both scale and resolution between data and simulated events. However, the uncertainties associated to the Rochester corrections, the uncertainties from the calibration of the per-event mass errors (closure test), as well as the assumption of a given analytical signal shape (double-sided Crystal Ball), have to be taken into account in the signal extraction.

The residual muon energy scale uncertainty has been estimated to be 0.1% of the muon momentum, and it affects the signal peak by changing its position w.r.t. the nominal expected value from simulation. This uncertainty is correlated across all signal production processes and eras, but decorrelated across categories. It is modelled via Gaussian constrained nuisance parameters. On the other hand, the uncertainty in the dimuon mass resolution has been estimated, conservatively, to be as large as 10% in each category. It is modelled via a Gaussian constrained nuisance parameter, correlated across processes and eras, but uncorrelated across categories whose effect is to shrink or enlarge the width of the signal peak (σ_{CB}).

7.2.2 Rate uncertainties from theory

Several sources from theory introduce an uncertainty in the predicted signal rate in each event category and for each production mode. Some of them affect the overall signal prediction, i.e. they affect either the production cross section or the value of $\mathcal{B}(\mathrm{H} \to \mu\mu)$, and are referred to as *normalization* uncertainties. In contrast, others only affect the prediction of the acceptance in single categories, causing migrations of events across ggH and VBF BDT categories. These are referred to as *acceptance* uncertainties.

As described in Section 2, the central values of the Higgs boson production cross section and $\mathcal{B}(\mathrm{H} \to \mu\mu)$ are taken from Ref. [12]. The associated uncertainties, listed in Table 1, are modelled via constrained log-Normal nuisance parameters and they are fully correlated across categories and eras. The uncertainty on $\mathcal{B}(\mathrm{H} \to \mu\mu)$ is also fully correlated across Higgs production modes. In addition, the uncertainty in the total cross section from PDF+α_s can be correlated only between VBF and VH modes because both processes are quark-initiated.

A set of acceptance uncertainties are specific for the ggH process and are estimated following the recommendations provided by the LHC Higgs cross-section working group (WG1) [62]. This recipe provides a set of independent sources of uncertainty defined in STXS bins, which are modelled via log-Normal nuisance parameters correlated across categories and eras, accounting mainly for variations in estimate of the ggH acceptance as a function of the Higgs boson p_{T} and $\mathrm{N}_{\mathrm{jets}}$. These uncertainties are estimated directly via the simulated sample used in the analysis, and are found to be in good agreement between MADGRAPH5_aMC@NLO and POWHEG generators. In addition, the uncertainty in the ggH acceptance due to the limited knowledge of the parton densities (PDF) has been also investigated using variations of the per-event weight stored inside in the simulated samples. Table 16 summarizes the aforementioned per-category uncertainties for the ggH production mode, as well as how they are correlated across categories. The total uncertainty in the acceptance from these sources is of about 6–8% in the ggH categories, and of about 12–15% in the VBF ones. As expected, specific $p_{\mathrm{T}}^{\mathrm{H}}$ and VBF phase-space uncertainties play either a limited or no role in the ggH BDT categories. In contrast, they represent the major sources of uncertainty in the VBF BDT categories, where VBF-2j and VBF-3j nuisances are strongly anti-correlated because the high-score VBF region is dominated by 2-jet events. Similarly, uncertainties related to $p_{\mathrm{T}}^{\mathrm{H}}$ are also strongly anti-correlated in

DRAFT

each category. Finally, uncertainties related to jet migrations (Mig-01, and Mig-12) are slightly anti-correlated between low-score ggH BDT categories (dominated by 0-jet events) and high-score ones (dominated by 1 and 2-jet events), while their effect is correlated in the acceptance VBF BDT categories.

Table 16: Acceptance uncertainties for ggH signal events in ggH and VBF BDT categories from WG1 uncertainties and PDF. Uncertainties smaller than 0.1% are not reported.

Category	ggH-Mu	ggH-Res	Mig-01	Mig-12	VBF-2j	VBF-3j	Pt60	Pt120	Mtop	PDF
ggH-cat0	0.954/1.046	0.980/1.020	1.004/0.996	—	—	—	1.008/0.992	0.998/1.002	0.999/1.001	1.003/0.997
ggH-cat1	0.953/1.047	0.978/1.022	0.998/1.002	1.004/0.996	0.999/1.001	—	—	1.001/0.999	0.999/1.001	1.002/0.998
ggH-cat2	0.953/1.047	0.979/1.021	1.001/0.999	1.001/0.999	0.999/1.001	—	0.995/1.005	1.005/0.995	0.998/1.002	1.002/0.998
ggH-cat3	0.952/1.048	0.975/1.025	0.995/1.005	0.999/1.001	0.999/1.001	—	0.991/1.009	1.008/0.992	0.997/1.003	1.002/0.998
ggH-cat4	0.945/1.055	0.957/1.043	0.955/1.045	1.004/0.996	0.997/1.003	—	0.971/1.029	1.017/0.983	0.996/1.004	0.997/1.003
VBF-cat0	0.963/1.037	0.968/1.032	0.967/1.033	0.991/1.009	0.919/1.081	0.999/1.001	0.996/1.004	1.019/0.981	0.990/1.010	0.993/1.007
VBF-cat1	0.969/1.031	0.972/1.028	0.968/1.032	0.997/1.003	0.901/1.099	1.047/0.953	0.974/1.026	1.033/0.967	0.989/1.011	0.990/1.010
VBF-cat2	0.975/1.025	0.978/1.022	0.974/1.026	0.999/1.001	0.880/1.120	1.108/0.892	0.969/1.031	1.039/0.961	0.990/1.010	0.983/1.021
VBF-cat3	0.977/1.023	0.979/1.021	0.977/1.023	0.993/1.007	0.872/1.128	1.133/0.867	0.959/1.041	1.055/0.945	0.986/1.014	0.976/1.043

Furthermore, acceptance uncertainties specifically designed for the VBF production mode are estimated following the recommendations provided by the LHC Higgs cross-section working group (WG2) [63]. This recipe provides a set of independent sources of uncertainty defined in different STXS bins, which are modelled via log-Normal nuisance parameters correlated across categories and eras, accounting mainly for variation of the VBF acceptance as a function of the Higgs boson p_{T}, m_{jj} and $\mathrm{N}_{\mathrm{jets}}$. These uncertainties are estimated directly via the simulated sample used in the analysis, and are found to be in good agreement between MADGRAPH5_aMC@NLO and POWHEG generators. In addition, the uncertainty in the VBF signal acceptance due to the limited knowledge of the parton densities (PDF) has been also investigated using variations of the per-event weight stored inside in the simulated samples. Table 17 summarizes the aforementioned per-category uncertainties for the VBF production mode, as well as how they are correlated across categories. The total uncertainty in the acceptance from these sources is of about 0.8–1.2% in the ggH categories, and of about 2.0–2.5% in the VBF ones. As expected from the coarse binning adopted in the STXS prescription, only the low-$p_{\mathrm{T}}^{\mathrm{H}}$ uncertainty (Pt25) plays a role in both ggH and VBF categories. For what concerns m_{jj} uncertainties, those related to low m_{jj} bins are driving the overall acceptance uncertainty in the VBF categories while Mjj-1000 and Mjj-1500 have no effect since the analysis is not very sensitive yet to the TeV tail of the m_{jj} distribution. The effect of Mjj-60 and Mjj-120 is correlated across ggH and VBF categories, while Mjj-350 is anti-correlated between VBF and ggH categories. In fact, given the offline selection of $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, this nuisance is mainly causing event migrations across the m_{jj} boundary. Similar behaviour is observed for Jet-01 which accounts for migrations across jet bins.

DRAFT

Table 17: Acceptance uncertainties for VBF signal events in ggH and VBF BDT categories from WG2 uncertainties and PDF. Uncertainties smaller than 0.1% are not reported.

Category	Pt25	Pt200	Mjj-60	Mjj-120	Mjj-350	Mjj-700	Mjj-1000	Mjj-1500	Jet-01	PDF
ggH-cat0	1.008/0.992	—	—	1.006/0.994	1.005/0.995	—	—	—	1.001/0.998	1.002/0.996
ggH-cat1	1.005/0.995	—	1.002/0.998	1.005/0.995	1.003/0.997	—	—	—	1.004/0.996	1.003/0.997
ggH-cat2	1.004/0.996	—	1.004/0.996	1.005/0.995	1.003/0.997	—	—	—	1.004/0.996	1.004/0.996
ggH-cat3	1.004/0.996	—	1.005/0.995	1.006/0.994	1.002/0.998	—	—	—	1.004/0.996	1.002/0.998
ggH-cat4	1.003/0.997	—	1.006/0.994	1.005/0.995	1.001/0.999	1.001/0.999	—	—	1.005/0.995	1.001/0.999
VBF-cat0	0.984/1.016	—	1.011/0.989	1.023/0.977	—	1.002/0.998	—	—	0.995/1.005	1.002/0.997
VBF-cat1	0.991/1.009	—	1.013/0.987	1.018/0.982	0.998/1.002	1.003/0.997	—	—	0.994/1.006	1.003/0.996
VBF-cat2	1.003/0.997	—	1.015/0.985	1.007/0.993	0.997/1.003	1.004/0.996	—	—	0.994/1.007	1.001/0.999
VBF-cat3	1.012/0.988	—	1.019/0.981	0.999/1.001	0.996/1.004	1.005/0.995	—	—	0.992/1.008	0.998/1.002

The acceptance uncertainties affecting the expected signal yields per-category for the remaining production processes (WH, ZH, and ttH), coming from QCD-scale and PDF variations, have been also estimated. In the case of QCD-scale, factorization and re-normalization scales are varied up and down by a factor two w.r.t. the nominal values used in the event generation. The most extreme variations, i.e. those for which $\mu_f/\mu_r = 0.25$ or 4, are excluded from the uncertainty computation. Table 18 summarizes the aforementioned per-category uncertainties for the VBF, VH, and ttH production modes. In the final signal extraction, QCD-scale and PDF acceptance uncertainties are modelled via log-Normal nuisance parameters, and their effect is correlated across categories and eras. For the ggH process, PDF uncertainties are mostly anti-correlated between ggH and VBF categories, while for the other modes the effect appears to be correlated.

Table 18: Acceptance uncertainties for VBF, WH, ZH, and ttH signal events in ggH and VBF BDT categories from QCD-scale and PDF variations. Uncertainties smaller than 0.1% are not reported. Uncertainties for WH, ZH, and ttH processes in the last two VBF categories are not estimated because the corresponding expected signal are smaller than 0.01 as reported in Table 15.

Category	QCD-scale uncertainty			PDF uncertainty		
	WH	ZH	ttH	WH	ZH	ttH
ggH-cat0	1.004/0.997	1.003/0.997	1.007/0.992	1.003/0.997	1.001/0.999	1.001/0.999
ggH-cat1	1.003/0.997	1.001/0.993	1.008/0.989	1.003/0.996	1.001/0.998	1.001/0.999
ggH-cat2	1.004/0.988	1.001/0.984	1.010/0.986	1.003/0.996	1.002/0.998	1.001/0.999
ggH-cat3	1.012/0.980	1.003/0.996	1.013/0.983	1.002/0.998	1.001/0.998	1.001/0.999
ggH-cat4	1.022/0.978	1.012/0.990	1.011/0.987	1.002/0.997	1.002/0.998	1.001/0.999
VBF-cat0	1.035/0.990	1.032/0.985	0.991/1.006	1.003/0.997	1.003/0.997	0.999/1.001
VBF-cat1	1.058/0.995	1.049/0.992	0.986/1.007	1.002/0.997	1.002/0.997	0.999/1.001
VBF-cat2	—	—	—	—	—	—
VBF-cat3	—	—	—	—	—	—

Uncertainties related to the parton shower model are derived by varying the per-event weights stored internally in each signal sample. Variations are those related to the CP5 tune [20] of PYTHIA (v8.2 or greater). Table 19 reports the parton shower related uncertainties in each category defined along the ggH and VBF BDTs for the different Higgs boson production modes. Uncertainties are fully correlated across processes and eras. The parton shower uncertainty is small, at the level of 2–3%, for the VBF production, while for ggH it progressively increases from 3–4% to 10–15% moving from low-score ggH categories to high-score VBF ones. The parton shower effect is anti-correlated between low-score ggH categories and VBF ones, which means that the associated nuisance parameter primarily induces migration of events across N_{jet} bins.

Finally, the uncertainty related to variations of the underlying event tune is estimated by using the alternative signal samples listed in Section 2.2. Values of this uncertainty in each category and for each production mode are reported in Table 20, and are fully correlated across processes and eras. In order to minimize the impact of the limited size of MC statistics available for the signal samples generated with the Up/Down variations of the underlying event tune, the observed variations in the signal yield have been symmetrized. For both ggH and VBF processes, the estimated uncertainty ranges between ±3% across ggH and VBF BDT categories.

7.2.3 Rate uncertainties from experimental sources

Several experimental sources of uncertainty affect the expected signal rate in each event category and for each production mode. A first set is represented by muon-related sources. Uncertainties in the knowledge of muon identification and isolation efficiencies, as well those related to the single muon trigger efficiency, are propagated in the analysis assuming full cor-

DRAFT

Table 19: Acceptance uncertainties for ggH, VBF, WH, ZH, and ttH signal events in ggH and VBF BDT categories from variations of the parton shower tune parameters. Uncertainties smaller than 0.1% are not reported. Uncertainties for WH, ZH, and ttH processes in the last two VBF categories are not estimated because the corresponding expected signal are smaller than 0.01 as reported in Table 15.

Category	ggH	VBF	WH	ZH	ttH
ggH-cat0	1.044/0.975	0.992/1.003	0.934/1.018	0.931/1.022	0.979/1.016
ggH-cat1	1.005/0.994	0.996/1.002	0.955/1.011	0.956/1.010	0.979/1.025
ggH-cat2	0.974/1.015	0.998/1.001	0.971/1.010	0.930/1.012	0.995/1.001
ggH-cat3	0.947/1.032	1.001/0.997	0.957/1.012	1.006/0.998	0.978/1.015
ggH-cat4	0.879/1.054	1.010/0.987	0.967/1.014	1.009/0.998	1.006/0.975
VBF-cat0	0.850/1.105	0.939/1.032	0.855/1.065	0.883/1.068	0.868/1.015
VBF-cat1	0.815/1.095	0.985/1.009	0.872/1.091	0.857/1.047	0.881/1.046
VBF-cat2	0.895/1.054	1.010/0.986	—	—	—
VBF-cat3	0.910/1.045	1.058/0.950	—	—	—

Table 20: Acceptance uncertainties for ggH, VBF, WH, ZH, and ttH signal events in ggH and VBF BDT categories from variations of the underlying event tune parameters. Uncertainties smaller than 0.1% are not reported. Uncertainties for WH, ZH, and ttH processes in the last two VBF categories are not estimated because the corresponding expected signal are smaller than 0.01 as reported in Table 15.

Category	ggH	VBF	WH	ZH	ttH
ggH-cat0	0.998/1.002	0.996/1.004	0.997/1.003	0.999/1.001	0.992/1.008
ggH-cat1	1.005/0.995	1.008/0.992	0.998/1.002	1.001/0.999	1.006/0.994
ggH-cat2	0.998/1.002	0.990/1.010	0.997/1.003	1.003/0.997	1.021/0.979
ggH-cat3	1.005/0.995	1.015/0.985	1.002/0.998	1.004/0.994	0.992/1.008
ggH-cat4	1.002/0.998	0.990/1.010	0.994/1.004	1.005/0.995	0.982/1.018
VBF-cat0	0.997/1.003	1.010/0.990	0.978/1.022	0.991/1.009	0.991/1.009
VBF-cat1	0.975/1.025	0.994/1.006	1.036/0.964	1.038/0.962	0.930/1.070
VBF-cat2	0.972/1.028	0.996/1.004	—	—	—
VBF-cat3	1.017/0.983	1.021/0.979	—	—	—

DRAFT

relation between the two muons in the event. Their effects are modelled via two independent log-Normal nuisance parameters correlated across processes, categories, and eras. Table 21 reports their values described for each production mode as well as for each BDT category. The uncertainty in the muon identification and isolation (single muon trigger) scale factors translates into an overall ±1% (±0.2%) uncertainty in the signal in each category. Moreover, the effect of the uncertainty in the knowledge of the muon p_{T} coming from the Rochester corrections is also taken into account and modelled via a log-Normal nuisance parameter correlated across processes, categories, and eras. Table 22 reports the estimate for this uncertainty for each production mode as well as for each BDT category. The muon p_{T} uncertainty induces a few permille effect mostly anti-correlated as a function of the dimuon p_{T} spectrum, i.e. anti-correlated moving from low to high score categories of both ggH and VBF BDTs.

The dominant experimental uncertainties are represented by the limited knowledge in the energy scale and resolution of the hadronic jets in the event, especially for those belonging to the forward region of the detector ($|\eta| > 2.5$). There are twenty-seven independent sources of uncertainty affecting the knowledge of the jet energy scale, whose effect is measured by the JME group and described in this twiki [64]. Depending on the uncertainty source, the effect

Table 21: Uncertainties for ggH, VBF, WH, ZH, and ttH processes in ggH and VBF BDT categories obtained from the uncertainties related to muon identification and isolation scale factors, and single muon trigger scale factors. Uncertainties smaller than 0.1% are not reported. Uncertainties for WH, ZH, and ttH processes in the last two VBF categories are not estimated because the corresponding expected signal are smaller than 0.01 as reported in Table 15.

Category	Muon ID and Iso uncertainty					Muon trigger uncertainty				
	ggH	VBF	WH	ZH	ttH	ggH	VBF	WH	ZH	ttH
ggH-cat0	0.994/1.006	0.993/1.008	0.992/1.008	0.992/1.008	0.992/1.008	0.998/1.002	0.998/1.002	0.998/1.002	0.998/1.002	0.998/1.002
ggH-cat1	0.993/1.007	0.992/1.008	0.992/1.008	0.992/1.008	0.991/1.009	0.998/1.002	0.998/1.002	0.998/1.002	0.998/1.002	0.998/1.002
ggH-cat2	0.993/1.007	0.992/1.008	0.991/1.009	0.992/1.008	0.990/1.010	0.999/1.001	0.999/1.001	0.998/1.002	0.998/1.002	0.998/1.002
ggH-cat3	0.992/1.008	0.992/1.008	0.991/1.009	0.991/1.009	0.990/1.011	0.999/1.001	0.999/1.001	0.998/1.002	0.998/1.002	0.998/1.002
ggH-cat4	0.991/1.009	0.991/1.009	0.990/1.010	0.990/1.010	0.990/1.010	0.999/1.001	0.998/1.002	0.999/1.001	0.999/1.001	0.998/1.002
VBF-cat0	0.992/1.008	0.992/1.008	0.992/1.008	0.992/1.008	0.990/1.010	0.998/1.002	0.998/1.002	0.998/1.002	0.998/1.002	0.998/1.002
VBF-cat1	0.991/1.009	0.991/1.009	0.990/1.010	0.990/1.010	0.990/1.010	0.999/1.001	0.998/1.002	0.998/1.002	0.998/1.002	0.998/1.002
VBF-cat2	0.990/1.010	0.990/1.010	—	—	—	0.998/1.002	0.998/1.002	—	—	—
VBF-cat3	0.990/1.010	0.991/1.009	—	—	—	0.998/1.002	0.998/1.002	—	—	—

Table 22: Uncertainties for ggH, VBF, WH, ZH, and ttH processes in ggH and VBF BDT categories obtained from the uncertainties related to muon energy scale. Uncertainties smaller than 0.1% are not reported. Uncertainties for WH, ZH, and ttH processes in the last two VBF categories are not estimated because the corresponding expected signal are smaller than 0.01 as reported in Table 15.

Category	Muon scale uncertainty				
	ggH	VBF	WH	ZH	ttH
ggH-cat0	1.002/0.998	1.001/0.998	1.001/0.999	1.001/0.999	1.003/0.998
ggH-cat1	1.001/0.999	1.001/0.999	1.001/0.999	1.001/0.999	1.002/0.999
ggH-cat2	0.999/1.002	0.998/1.002	0.999/1.001	0.998/1.002	0.999/1.001
ggH-cat3	0.998/1.002	0.998/1.002	0.998/1.002	0.999/1.001	0.999/1.001
ggH-cat4	0.998/1.003	0.999/1.001	0.998/1.003	0.998/1.002	0.999/1.003
VBF-cat0	1.001/0.999	1.002/0.999	1.001/0.999	1.001/0.999	1.002/0.999
VBF-cat1	1.001/0.998	1.001/0.999	0.997/1.003	0.998/1.002	0.996/1.002
VBF-cat2	0.995/1.003	0.998/1.002	—	—	—
VBF-cat3	0.998/1.008	0.998/1.003	—	—	—

DRAFT

can be either correlated or un-correlated across data taking periods. Each source of uncertainty is varied independently up and down in order to change the transverse momentum of all the jets found in the event. The analysis selections are re-applied after each variation, and both the ggH and VBF BDTs are re-evaluated to account for the corresponding changes in the jet kinematics. Variations in the expected signal rate in each category are modelled via Log-Normal nuisance parameters, whose effects are correlated across categories, signal production modes, and eras. Similarly, the uncertainty in the jet energy resolution is used to smear the p_{T} of the jets in simulated events. The overall uncertainty is then further divided into six independent components depending on the p_{T} and η of each jet: barrel ($|\eta| < 1.9$, "jerBarrel"), endcap within the tracker coverage ($1.9 < |\eta| < 2.5$, "jerEC1"), endcap outiside the tracker coverage with low p_{T} ($2.5 < |\eta| < 3.14$ and $p_{\mathrm{T}} < 50\,\mathrm{GeV}$, "jerEC2low"), endcap outside the tracker coverage with high p_{T} ($2.5 < |\eta| < 3.14$ and $p_{\mathrm{T}} > 50\,\mathrm{GeV}$, "jerEC2high"), forward and low p_{T} ($|\eta| > 3.14$ and $p_{\mathrm{T}} < 50\,\mathrm{GeV}$, "jerHFlow"), and forward with high p_{T} ($|\eta| > 3.14$ and $p_{\mathrm{T}} > 50\,\mathrm{GeV}$, "jerHFhigh"). The corresponding variations in the expected signal rate are modelled via independent log-Normal nuisance parameters, whose effect is correlated across categories, signal processes, but not correlated across eras. The uncertainties related to jet-energy-resolution are found to be one-sided because we don't apply the resolution scale-factors (smearing) to correct for the central value, but energy smearing is applied only to estimate an associated uncertainty (conservative).

Table 33 and Table 34 reported in Appendix B, in each BDT category (columns), the estimated values for each jet-energy-scale and jet-energy-resolution uncertainty source (rows) for the ggH and VBF processes, respectively. However, instead of using all the twenty-seven independent per-year in the final fit, we have followed a recent recommendation from the JetMET group for analysis which are not constraining jet-energy related uncertainties. This recipe is described in the following twiki page [65]. The reduced set of uncertainties roughly groups sources belonging to the same detector region, so that analyses using both central and forward jets get a reasonable modelling of the correlations between central and forward region, without taking into account the full set of sources. This uncertainty grouping results into twenty-one nuisance parameters which are either correlated or un-correlated across eras. Table 23 and Table 24 list for each BDT category the estimated values for each jet-energy-scale and jet-energy-resolution uncertainty belonging to the reduced set of sources for the ggH and VBF processes, respectively. Uncertainties in the jet energy scale cause significant migration of events between the ggH and the VBF categories, therefore the effect of these nuisances is found to be largely anti-correlated across N_{jet} bins. The sources showing the largest impact on the predicted signal yields are those correlated across eras and not specific to a particular detector region: flavorQCD, relativeBal, and absolute. According to Table 23, the total uncertainty due to jet-energy-scale affecting the ggH process ranges between 0.2–2.5% as a function of the ggH BDT-score, increasing up to 10–16% in the VBF categories. In contrast, the total uncertainty on the predict VBF signal yields due to jet-energy-scale is found to be larger than ggH in the ggH categories, ranging from 1.5 to 7%, and significantly smaller in the VBF ones ranging between 3–5%. On the other hand, the uncertainty due to jet energy resolution has a slightly smaller impact on the predicted signal yields. In the ggH categories, the total jet-energy-resolution uncertainty affecting the ggH (VBF) prediction ranges between 0.1–1% (0.1–2.5%) as a function of the BDT score. In contrast, the effect on the ggH (VBF) prediction in VBF categories moves from 8–5% (4–6%) as a function of the VBF BDT score.

The jet energy scale and resolution uncertainties affecting the other production modes (WH, ZH, and ttH) are not reported in this note because they have a small impact on the final result. However, they are included in the “datacards” used in the final fit for the signal extraction and, overall, their size is similar to what has been measured for the ggH process and reported in Table 23.

Other sources of uncertainty are represented by the uncertainty in the ECAL pre-firing correction introduced to emulate an inefficiency in the L1 trigger efficiency affecting 2016 and 2017 eras, and the uncertainty in the minimum bias cross section which affects the pileup re-weight procedure described in Section 4.1. In the former case, the pre-firing uncertainty is estimated by varying the per-event pre-firing correction by $\pm 20\%$, following the official recommendations from the JME group. In the latter one, the minimum bias cross section of 69.2 mb at 13 TeV is varied by its experimental uncertainty of 4.6%. The pre-firing uncertainty has a small impact ($\leq 0.3\%$) on the predicted signal rate in the ggH categories. In contrast, it increases as a function of the VBF BDT output from 0.4% to 0.8%. In fact, in the high score region of the VBF BDT a good fraction of signal events have one energetic jet in the endcap region. The pileup uncertainty instead is found to be quite flat as a function of both ggH and VBF categories, ranging between 0.1% and 0.5%. Table 25 summarizes the estimated value of pre-firing and pileup uncertainties for each production mode and for each category. Finally, the uncertainty on the luminosity measurement is of about 2.5% in each era, and is modelled via a log-Normal nuisance parameter whose effect is correlated across BDT categories and processes.

Table 23: Uncertainties for the ggH process in both the ggH and VBF BDT categories obtained by propagating the effect of each source of uncertainty related to jet energy scale and jet energy resolution to the expected signal yields. Uncertainties smaller than 0.1% are not reported.

Source of Uncertainty	ggH Categories					VBF Categories			
Jet energy scale	cat0	cat1	cat2	cat3	cat4	cat0	cat1	cat2	cat3
flavorQCD	0.994/1.006	—	1.009/0.991	1.012/0.989	1.013/0.988	0.957/1.044	0.974/1.035	0.929/1.023	0.992/1.042
relativeBal	—	0.999/1.001	0.998/1.003	0.996/1.004	0.997/1.002	1.047/0.957	1.027/0.969	1.047/0.951	1.042/0.951
absolute	0.998/1.002	1.001/0.999	1.008/0.992	1.008/0.992	1.003/0.996	0.934/1.074	0.948/1.051	0.935/1.032	0.947/1.071
BBEC1	0.998/1.002	0.999/1.000	1.002/0.998	1.004/0.996	1.005/0.996	0.997/1.006	0.993/1.002	0.993/1.001	0.998/1.006
EC2	—	—	1.001/0.999	1.001/0.999	—	0.994/1.007	0.993/1.005	1.001/0.995	0.978/1.002
HF	1.001/0.998	1.001/0.999	1.002/0.998	—	0.996/1.004	0.970/1.032	0.983/1.026	0.952/1.009	0.993/1.045
relativeSample_2016	—	—	1.001/0.998	1.002/0.998	1.001/0.999	0.982/1.015	0.996/1.010	0.983/1.009	0.998/1.016
absolute_2016	—	—	1.001/0.999	1.001/0.999	—	0.993/1.003	0.999/1.002	—	0.998/1.021
BBEC1_2016	—	—	—	1.001/0.999	1.001/0.999	0.998/1.001	0.996/1.001	0.994/1.001	0.999/1.005
EC2_2016	—	—	—	1.001/0.999	—	0.995/1.005	0.998/1.001	0.999/1.003	0.990/1.004
HF_2016	—	—	—	—	0.998/1.001	0.997/1.001	—	1.001/0.997	0.995/1.013
relativeSample_2017	—	1.001/0.999	1.002/0.999	1.001/0.999	1.001/0.999	0.981/1.020	0.988/1.014	0.968/1.013	0.995/1.022
absolute_2017	—	—	1.001/0.999	1.001/0.999	1.001/0.999	0.991/1.010	0.993/1.003	0.978/1.008	0.999/1.014
BBEC1_2017	—	—	—	—	1.001/0.999	—	0.997/1.001	0.996/1.001	0.999/1.001
EC2_2017	—	—	1.001/0.999	1.001/0.999	1.001/0.999	0.994/1.006	0.995/1.003	0.997/1.009	0.991/1.002
HF_2017	—	—	—	—	—	0.997/1.003	0.996/1.001	0.998/1.001	0.999/1.006
relativeSample_2018	0.999/1.001	1.001/0.999	1.004/0.995	1.006/0.994	1.005/0.994	0.956/1.049	0.968/1.042	0.934/1.028	0.983/1.050
absolute_2018	0.999/1.001	—	1.002/0.997	1.002/0.998	1.004/0.997	0.990/1.011	0.989/1.007	0.981/1.020	1.011/0.983
BBEC1_2018	—	—	—	—	1.001/0.999	0.996/1.005	0.999/1.003	0.988/1.002	1.003/0.993
EC2_2018	—	—	—	—	1.001/0.999	0.991/1.008	0.998/1.004	0.992/1.001	0.999/1.008
HF_2018	—	—	—	—	—	0.996/1.005	0.994/1.003	0.987/1.013	1.008/0.997
Jet energy resolution	cat0	cat1	cat2	cat3	cat4	cat0	cat1	cat2	cat3
jerBarrel	—/1.005	—	—/0.996	—/0.994	—/0.994	—/1.007	—/0.998	—/0.986	—/1.012
jerEC1	—/1.001	—	—/0.999	—/0.997	—/0.997	—/1.010	—/1.006	—/0.986	—/1.013
jerEC2low	—	—/0.999	—/0.997	—/0.997	—/0.995	—/1.068	—/1.038	—/1.009	—/1.007
jerEC2high	—	—	—/0.999	—	—/0.999	—/1.002	—/1.002	—/0.983	—/0.977
jerHFlow	—/0.998	—/0.999	—	—	—/1.004	—/1.026	—/1.025	—/1.014	—/0.998
jerHFhigh	—	—/0.999	—	—	—	—/1.001	—/0.998	—/0.997	—/0.996

Table 24: Uncertainties for the VBF process in both the ggH and VBF BDT categories obtained by propagating the effect of each source of uncertainty related to jet energy scale and jet energy resolution to the expected signal yields. Uncertainties smaller than 0.1% are not reported.

Source of Uncertainty	ggH Categories					VBF Categories			
Jet energy scale	cat0	cat1	cat2	cat3	cat4	cat0	cat1	cat2	cat3
flavorQCD	0.997/1.001	1.009/0.996	1.013/0.980	1.014/0.978	1.026/0.976	0.985/1.023	0.986/1.008	0.984/1.012	0.982/1.011
relativeBal	0.996/1.004	0.993/1.014	0.982/1.008	0.984/1.006	0.982/1.022	1.023/0.983	1.001/0.995	1.014/0.982	1.017/0.985
absolute	1.001/0.997	1.015/0.990	1.019/0.977	1.016/0.974	1.029/0.972	0.978/1.030	0.984/1.013	0.981/1.017	0.983/1.019
BBEC1	0.998/1.001	1.002/0.999	0.999/0.997	1.004/0.996	1.004/0.995	0.999/1.005	0.998/1.001	0.997/1.001	0.999/1.002
EC2	—	1.002/0.999	1.002/0.995	1.001/0.998	1.004/0.997	0.997/1.004	0.997/1.001	—	0.999/1.002
HF	1.004/0.996	1.006/0.994	1.010/0.990	1.006/0.990	1.009/0.990	0.990/1.015	0.994/1.004	0.987/1.010	0.996/1.007
relativeSample_2016	1.003/1.000	1.007/0.998	1.000/0.992	1.002/0.995	1.011/0.992	0.990/1.009	0.999/1.005	0.990/1.001	0.999/1.004
absolute_2016	—	1.002/1.000	1.000/0.994	—	1.003/0.999	0.999/1.003	—	0.999/1.004	0.998/1.021
BBEC1_2016	—	1.001/0.999	—	1.000/0.997	—	—	—	0.998/1.002	—
EC2_2016	—	1.002/0.999	0.999/0.997	—	1.002/0.999	1.000/1.003	0.997/0.999	0.998/0.998	0.999/1.003
HF_2016	—	—	1.001/0.998	0.999/1.000	—	—	—	0.999/1.003	1.001/0.999
relativeSample_2017	1.002/0.997	1.005/0.997	1.005/0.997	1.006/0.995	1.009/0.993	0.993/1.009	0.997/1.003	0.991/1.005	0.990/1.007
absolute_2017	—	1.003/0.999	1.002/0.996	1.002/0.996	1.005/0.997	0.997/1.005	0.998/1.000	0.994/1.005	0.997/1.002
BBEC1_2017	—	—	—	—	1.001/0.999	0.999/1.001	—	—	—
EC2_2017	—	—	1.002/0.997	1.002/0.998	1.003/0.998	0.998/1.002	0.997/1.001	0.999/1.001	0.998/1.002
HF_2017	—	—	—	—	—	0.997/1.003	0.996/1.001	0.998/1.001	—
relativeSample_2018	1.004/0.996	1.008/0.993	1.015/0.986	1.013/0.986	1.020/0.978	0.985/1.020	0.991/1.012	0.987/1.010	0.982/1.011
absolute_2018	0.999/1.001	1.003/0.999	1.003/0.997	1.004/0.995	1.005/0.997	0.997/1.005	0.996/1.002	0.998/1.004	0.998/1.001
BBEC1_2018	—	—	1.001/0.998	1.001/0.998	1.002/0.999	0.998/1.002	—	0.997/1.002	—
EC2_2018	1.001/0.999	—	1.001/0.998	1.003/0.998	1.003/0.996	0.997/1.004	0.998/1.001	0.999/1.002	0.997/1.001
HF_2018	—	—	1.002/0.999	1.001/0.998	1.001/0.998	0.999/1.002	0.999/1.001	0.999/1.003	0.999/1.002
Jet energy resolution	cat0	cat1	cat2	cat3	cat4	cat0	cat1	cat2	cat3
jerBarrel	—/1.005	—	—/0.996	—/0.994	—/0.994	—/1.007	—/0.998	—/0.986	—/1.012
jerEC1	—/1.001	—	—/0.999	—/0.997	—/0.997	—/1.010	—/1.006	—/0.986	—/1.013
jerEC2low	—	—/0.999	—/0.997	—/0.997	—/0.995	—/1.068	—/1.038	—/1.009	—/1.007
jerEC2high	—	—	—/0.999	—	—/0.999	—/1.002	—/1.002	—/0.983	—/0.977
jerHFlow	—/0.998	—/0.999	—	—	—/1.004	—/1.026	—/1.025	—/1.014	—/0.998
jerHFhigh	—	—/0.999	—	—	—	—/1.001	—/0.998	—/0.997	—/0.996

DRAFT

1154 8 Background estimation

1155 Given the low signal-to-background ratio expected in most of the event categories defined in 1156 Section 6, the key aspect of this search is represented by a robust estimation of the $m_{\mu\mu}$ shape

Table 25: Uncertainties for ggH, VBF, WH, ZH, and ttH processes in ggH and VBF BDT categories obtained from the uncertainties related to pre-firing correction and pileup re-weight. Uncertainties smaller than 0.1% are not reported. Uncertainties for WH, ZH, and ttH processes in the last two VBF categories are not estimated because the corresponding expected signal are smaller than 0.01 as reported in Table 15.

Category	Pileup re-weight					Pre-firing				
	ggH	VBF	WH	ZH	ttH	ggH	VBF	WH	ZH	ttH
ggH-cat0	1.001/0.998	1.001/0.999	1.002/0.998	1.002/0.997	1.004/0.996	1.002/0.998	1.004/0.996	1.003/0.997	1.004/0.996	1.005/0.995
ggH-cat1	1.001/0.999	1.001/0.998	1.002/0.997	1.003/0.997	1.005/0.995	1.001/0.999	1.002/0.998	1.001/0.999	1.001/0.999	1.004/0.996
ggH-cat2	1.001/0.999	1.001/0.999	1.002/0.998	1.002/0.998	1.006/0.994	1.001/0.999	1.002/0.998	1.001/0.999	1.001/0.999	1.003/0.997
ggH-cat3	1.001/0.999	1.001/0.999	1.002/0.998	1.002/0.998	1.005/0.995	1.001/0.999	1.003/0.997	1.001/0.999	1.001/0.999	1.003/0.997
ggH-cat4	1.002/0.997	1.002/0.997	1.001/0.999	1.001/0.998	1.006/0.994	1.002/0.998	1.005/0.995	1.002/0.998	1.001/0.999	1.005/0.995
VBF-cat0	0.997/1.003	0.999/1.001	0.993/1.007	0.994/1.006	1.003/0.995	1.004/0.996	1.006/0.994	1.006/0.994	1.006/0.994	1.011/0.989
VBF-cat1	0.996/1.004	1.003/0.997	0.991/1.009	0.996/1.005	1.006/0.993	1.004/0.996	1.005/0.995	1.006/0.994	1.005/0.995	1.012/0.987
VBF-cat2	0.990/1.010	1.003/0.997	—	—	—	1.006/0.993	1.007/0.993	—	—	—
VBF-cat3	0.998/1.004	1.003/0.996	—	—	—	1.006/0.994	1.008/0.992	—	—	—

of the total background. The expected background in each category is mainly composed of Z/γ^*+jets events, with smaller contaminations arising from $t\bar{t}$, single-top, and diboson processes. In the high score region of the VBF BDT, the contribution from the electroweak Z production becomes also relevant. Table 26 reports the expected background composition from simulation in each category. Processes involving dimuon decays of a Z boson represent more than 90% of the total expected background across both the ggH and VBF phase space.

Table 26: Background composition in (%) expected in the ggH and VBF BDT categories for events in the SR, $110 < m_{\mu\mu} < 150\,\text{GeV}$. Event yields are normalized to the measured luminosity of $137.2\,\text{fb}^{-1}$.

Category	Z/γ^*+jets	Z+jets EW	$t\bar{t}$	Diboson	Other (s-top, VVV, etc)
ggH-cat0	97.5%	—	1.5%	0.6%	0.4%
ggH-cat1	96.0%	0.1%	2.1%	1.2%	0.6%
ggH-cat2	95.0%	0.2%	2.3%	1.9%	0.6%
ggH-cat3	93.7%	0.3%	2.8%	2.2%	1.0%
ggH-cat4	88.8%	1.3%	4.6%	3.6%	1.7%
VBF-cat0	84.9%	1.6%	10.0%	2.2%	1.2%
VBF-cat1	83.2%	9.6%	4.8%	1.7%	0.6%
VBF-cat2	70.8%	24.6%	2.9%	1.2%	0.4%
VBF-cat3	43.6%	54.3%	1.2%	0.1%	—

DRAFT

As described in Section 2.1, simulated samples for the main background processes have been generated with NLO QCD precision but they cannot be directly used to model the background for several reasons:

- Simulated samples are missing higher order corrections in both QCD (NNLO) and EW (NLO) theory which are known to have a sizable impact on the predicted dimuon mass spectrum.
- Theory uncertainties due to resummation effects, QCD scale variations, and PDF are typically large enough to be over-constrained by data. They primarily provide an acceptance uncertainty in each category but they cannot be sufficiently trusted to provide a shape uncertainty for the background $m_{\mu\mu}$ spectrum.

The $m_{\mu\mu}$ distribution for the total background in each category is expected to be smoothly falling, driven by the Breit-Wigner nature of the Z boson lineshape. In this context, we choose to estimate the background via a data driven approach, smoothing the dimuon mass spectrum in the SR ($110 < m_{\mu\mu} < 150\,\text{GeV}$) via empirical functional forms. According to this strategy, the background contribution under the peak is therefore estimated from signal-free mass sidebands. We assume no prior knowledge of the parameters of the fit functions (in both shape and normalization), i.e. they are allowed to float freely in the fit to the data. Therefore, the uncertainty in the estimate of the background under the peak is mainly affected by the statistical power of the mass sidebands, as well as on the assumption of the background functional form.

In order to model the background $m_{\mu\mu}$ spectrum in each category, two classes of analytical functional forms are considered, based on their ability to fit the mass spectrum with a χ^2/dof of about 1:

- **Physics-inspired models:** given that the majority of the background comes from dimuon decays of a Z boson, this class of functions is characterized by a Breit-Wigner core modified by either polynomial, exponential, or power-law terms. These functions have been validated by fitting the Drell-Yan mass spectrum predicted by the FEWZ [32–34] generator at NNLO accuracy in QCD, and NLO accuracy in electroweak corrections, within $110 < m_{\mu\mu} < 150\,\text{GeV}$ as described in Ref. [13]. The following parametrizations are considered:

$$\text{BWZ} \times \text{Bernstein}(m_{\mu\mu}) = \frac{e^{a\cdot m_{\mu\mu}}\Gamma_{\text{Z}}}{(m_{\mu\mu}-m_{\text{Z}})^2+(\Gamma_{\text{Z}}/2)^2} \times \text{Bern}_n(m_{\mu\mu}) \tag{7}$$

$$\text{BWZGamma}(m_{\mu\mu}) = f\cdot\frac{e^{a\cdot m_{\mu\mu}}\Gamma_{\text{Z}}}{(m_{\mu\mu}-m_{\text{Z}})^2+(\Gamma_{\text{Z}}/2)^2} + (1-f)\cdot\frac{e^{a\cdot m_{\mu\mu}}}{m_{\mu\mu}^2} \tag{8}$$

$$\text{BWZRedux}(m_{\mu\mu}) = \frac{e^{a\cdot m_{\mu\mu}+b\cdot m_{\mu\mu}^2}\Gamma_{\text{Z}}}{(m_{\mu\mu}-m_{\text{Z}})^c+(\Gamma_{\text{Z}}/2)^c} \tag{9}$$

  where $\text{Bern}_n(m_{\mu\mu})$ corresponds to a Bernstein polynomial of order n, whose coefficients are chosen such that the integral is normalized to unity. In addition, the predictions from FEWZ, derived independently for ggH and VBF events, have been also considered as alternatives to the Breit-Wigner core. The FEWZ prediction in discrete bins of $m_{\mu\mu}$ is first interpolated by means of a spline in the range $110 < m_{\mu\mu} < 150\,\text{GeV}$. Then, in order to account for the presence of non Z/γ^*+jets backgrounds as well as for detector effects, the spline is multiplied by a Bernstein polynomial as follows:

$$\text{FEWZ} \times \text{Bernstein}(m_{\mu\mu}) = \text{FEWZ-spline}(m_{\mu\mu}) \times \text{Bern}_n(m_{\mu\mu}) \tag{10}$$

- **Agnostic models:** this class of functions consists in a set of general parametrizations able, in principle, to describe any functional form by choosing the right number of degrees of freedom (dof). We consider three main families: polynomials in the Bernstein basis, series of power laws, and series of exponentials, which are defined as follows:

$$\text{Bernstein}(m_{\mu\mu}) = \sum_i^n a_i \cdot \binom{n}{i} m_{\mu\mu}^i (1-m_{\mu\mu})^{n-i} \tag{11}$$

$$\text{S-PowerLaw}(m_{\mu\mu}) = \sum_i^n a_i \cdot m_{\mu\mu}^{b_i} \tag{12}$$

$$\text{S-Exponential}(m_{\mu\mu}) = \sum_i^n a_i \cdot e^{b_i\cdot m_{\mu\mu}} \tag{13}$$

DRAFT

In addition, another non Breit-Wigner like function, able to fit well the background spectrum in each category, is given by a Landau PDF modulated by polynomials, defined as follows:

$$\text{Landau} \times \text{Bernstein}(m_{\mu\mu}) = \text{Landau}(m_{\mu\mu}, m_{\text{Z}}, a) \times \text{Bern}_n(m_{\mu\mu}) \qquad (14)$$

The number of degrees of freedom for the polynomial terms used to modulate some of the physics-inspired functions, as well as the order of agnostic models like Bernstein polynomials, series of power-laws and exponentials, are selected by means of a likelihood ratio test as recommended by the CMS statistics committee [66]. This test works as follows: assuming that, within a given family of functions, n parameters are enough to describe the $m_{\mu\mu}$ spectrum in a chosen category, we first fit the $m_{\mu\mu}$ distribution in data with n and $n+1$ degrees of freedom. Then, we compute $2 \cdot (\text{Log}\mathcal{L}_{n+1} - \text{Log}\mathcal{L}_n)$ and, under the assumption that this quantity is distributed according to the χ^2 distribution with one degree of freedom, the corresponding p-value is used to accept or reject the null hypothesis. If the p-value is smaller than 5%, the null hypothesis is rejected and we apply the same test to the function with $n+1$ parameters. In contrast, if the p-value is larger than 5%, a good compatibility is observed between the two models, therefore the function with n degrees of freedom is selected.

In the cases of BWZ × Bernstein($m_{\mu\mu}$), FEWZ × Bernstein($m_{\mu\mu}$), and Landau × Bernstein($m_{\mu\mu}$), this test indicates that, depending on the category, either one or two or three polynomial degrees of freedom are enough to have an accurate description of the $m_{\mu\mu}$ spectrum. On the other hand, for what concerns sum of power-laws and exponentials, a series of order two ($n = 2$) is selected via the likelihood ratio test. Finally, when Bernstein polynomials are used to fit the $m_{\mu\mu}$ spectrum in each category, the likelihood ratio test suggests that from four to six degrees of freedom are necessary in order to have a good description of the observed spectra. Given that all the other families require about three shape degrees of freedom per category, plain Bernstein polynomials appear as an unnatural set of functions for modelling the total expected background, and therefore they are not considered in the studies presented in later Sections.

Table 27 reports for each category defined along ggH and VBF BDT outputs and for each family of functions the chosen number of shape degrees of freedom along with the χ^2/dof obtained from a background-only fit to the data. In each category and for each function the associated χ^2 is computed with the binning reported in Fig.. 45 and 46. Moreover, in order to correctly account for the low statistics regime of the high score VBF categories, the Baker-Cousins χ^2 defintion is adopted [67]. In each category, the χ^2/dof is about one for the chosen functions, indicating that they can well describe the dimuon mass spectrum in the range $110 < m_{\mu\mu} < 150\,\text{GeV}$. Moreover, Fig. 45 and 46 show the results of b-only fits performed independently in the ggH and VBF categories with the functions listed in Table 27. The lower pad in each plot reports the ratio between the background prediction obtained with the aforementioned functions and the one coming from a fit via the BWZRedux.

In each category and for each function selected by the F-test and reported in Table 27, it is important to check the behaviour of the post-fit shape uncertainty on the total background as a function of $m_{\mu\mu}$, as obtained from a background-only fit to the data. These comparisons are reported in Appendix C. As expected, the relative size of the post-fit shape uncertainties is proportional to the amount of data events in the category as well as to the number degrees of freedom of the chosen analytical function. Breit-Wigner and power-law based function not modulated by polynomials show a "pole" in the mass region between 120–130 GeV, therefore they are expected to lead to a better performance in terms of significance. On the other hand, functions like FEWZ × Bernstein, BWZ × Bernstein, and sum of exponentials are showing a

DRAFT

Table 27: Number of shape degrees of freedom selected, in each category and for each family of functions previously described, by means of a likelihood ratio test. The χ^2/dof obtained for each function from a background-only fit to the data in the range $110 < m_{\mu\mu} < 150\,\text{GeV}$ and computed following the recommendations of Ref. [67] is also reported.

Category	BWZ × Bernstein	BWZGamma	BWZRedux	FEWZ × Bernstein	Landau × Bernstein	S-Exponential	S-PowerLaw
ggH-cat0	3-dof, $\chi^2 = 1.07$	2-dof, $\chi^2 = 1.05$	3-dof, $\chi^2 = 1.09$	2-dof, $\chi^2 = 1.07$	3-dof, $\chi^2 = 1.07$	3-dof, $\chi^2 = 1.12$	3-dof, $\chi^2 = 1.06$
ggH-cat1	3-dof, $\chi^2 = 1.10$	2-dof, $\chi^2 = 1.05$	3-dof, $\chi^2 = 1.13$	2-dof, $\chi^2 = 1.09$	3-dof, $\chi^2 = 1.10$	3-dof, $\chi^2 = 1.09$	3-dof, $\chi^2 = 1.06$
ggH-cat2	3-dof, $\chi^2 = 1.18$	2-dof, $\chi^2 = 1.16$	3-dof, $\chi^2 = 1.19$	2-dof, $\chi^2 = 1.16$	3-dof, $\chi^2 = 1.19$	3-dof, $\chi^2 = 1.16$	3-dof, $\chi^2 = 1.16$
ggH-cat3	3-dof, $\chi^2 = 0.84$	2-dof, $\chi^2 = 0.84$	3-dof, $\chi^2 = 0.84$	2-dof, $\chi^2 = 0.84$	3-dof, $\chi^2 = 0.84$	3-dof, $\chi^2 = 0.84$	3-dof, $\chi^2 = 0.84$
ggH-cat4	3-dof, $\chi^2 = 0.95$	2-dof, $\chi^2 = 0.95$	3-dof, $\chi^2 = 0.95$	2-dof, $\chi^2 = 0.95$	3-dof, $\chi^2 = 0.97$	3-dof, $\chi^2 = 0.96$	3-dof, $\chi^2 = 0.94$
VBF-cat0	3-dof, $\chi^2 = 1.11$	2-dof, $\chi^2 = 1.10$	3-dof, $\chi^2 = 1.11$	2-dof, $\chi^2 = 1.11$	3-dof, $\chi^2 = 1.13$	3-dof, $\chi^2 = 1.10$	3-dof, $\chi^2 = 1.11$
VBF-cat1	3-dof, $\chi^2 = 0.85$	2-dof, $\chi^2 = 0.84$	3-dof, $\chi^2 = 0.85$	2-dof, $\chi^2 = 0.84$	3-dof, $\chi^2 = 0.87$	3-dof, $\chi^2 = 0.85$	3-dof, $\chi^2 = 0.85$
VBF-cat2	3-dof, $\chi^2 = 1.02$	2-dof, $\chi^2 = 1.01$	3-dof, $\chi^2 = 1.02$	2-dof, $\chi^2 = 1.01$	3-dof, $\chi^2 = 1.04$	3-dof, $\chi^2 = 1.02$	3-dof, $\chi^2 = 1.02$
VBF-cat3	2-dof, $\chi^2 = 1.03$	2-dof, $\chi^2 = 1.02$	3-dof, $\chi^2 = 1.03$	2-dof, $\chi^2 = 1.02$	2-dof, $\chi^2 = 1.03$	3-dof, $\chi^2 = 1.03$	3-dof, $\chi^2 = 1.03$

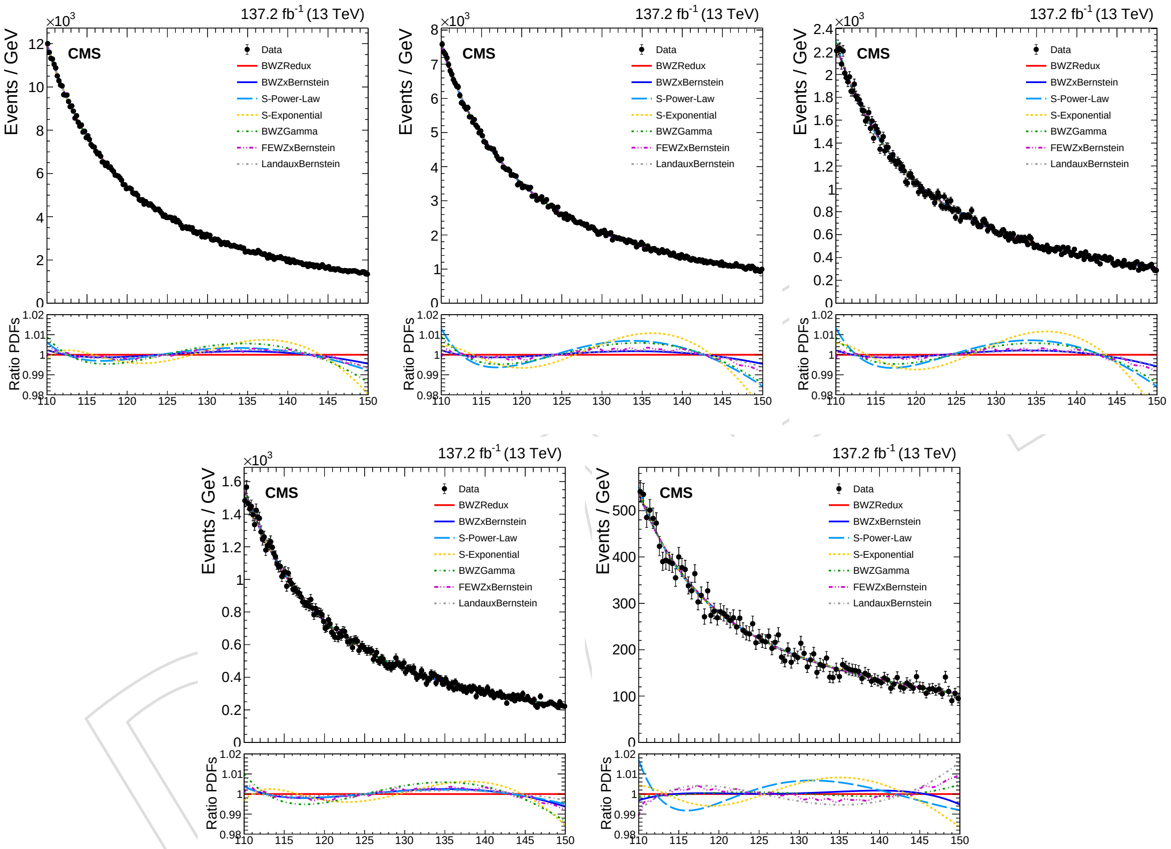

Figure 45: Dimuon invariant mass distributions in data for events belonging to the ggH BDT categories. The solid lines correspond to alternative background predictions obtained by fitting the mass spectra, under the background-only hypothesis, with the functional forms listed in Table 27. Top row, from left to right: ggH-cat0, ggH-cat1, and ggH-cat2. Bottom row, from left to right: ggH-cat3 and ggH-cat4. The lower pad in each plot reports the ratio between the background prediction obtained with each function and the one coming from a fit via the BWZRedux.

shape uncertainty less dependent on $m_{\mu\mu}$ depending on the number of polynomial degrees of freedom. However, it is important to remind that background shape uncertainties have a “important” effect only in categories with high data statistics, i.e. those for which the total background normalization can be precisely constrained by the $m_{\mu\mu}$ sidebands. In contrast, the performance of the high score VBF categories (VBF-cat1, VBF-cat2, and VBF-cat3) is mostly

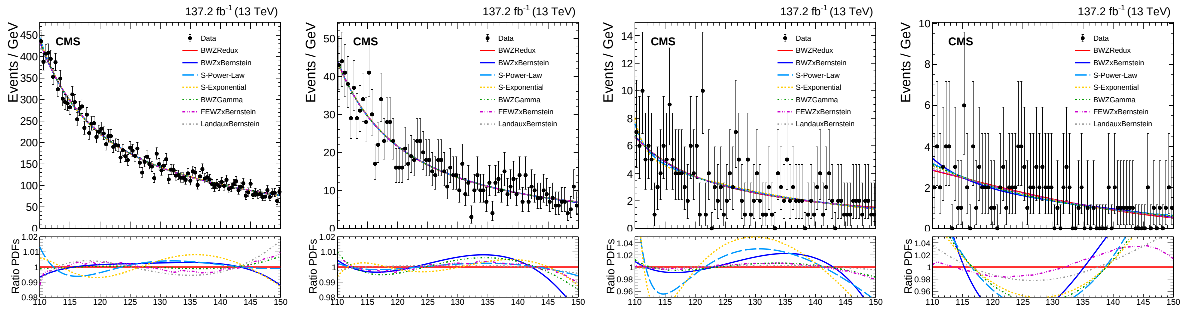

Figure 46: Dimuon invariant mass distributions in data for events belonging to the VBF BDT categories. The solid lines correspond to alternative background predictions obtained by fitting the mass spectra, under the background-only hypothesis, with the functional forms listed in Table 27. From left to right: VBF-cat0, VBF-cat1, VBF-cat2, and VBF-cat3. The lower pad in each plot reports the ratio between the background prediction obtained with each function and the one coming from a fit via the BWZRedux.

limited by the large uncertainty on the total background prediction, rather than to the choice of the background functional shape and the related shape uncertainty.

Finally, Table 28 reports the expected significances in each category for a SM Higgs boson with $m_{\mathrm{H}} = 125\,\mathrm{GeV}$, extracted from a s+b fit to Asimov toys in which the background predictions are taken from b-only fits to data with the functions reported in Table 27. Similarly, Table 29 reports the expected significances in each category extracted from s+b fit to Asimov toys in which the background predictions are taken from simulated events. The expected significance obtained from simulated events is used as figure of merit when choosing of the best performing functional form in each category. The prediction from simulation is found to be in good agreement with data as shown in Fig. 35 and 36. This reflects in very similar expected significances obtained in each category and for each functional form. Looking at the combined ggH and VBF significances, BWZRedux and S-Exponential are showing very similar performances, compatible within ±2%. The BWZGamma improves the performance of the BWZRedux in fitting the VBF categories. The FEWZ × Bernstein instead shows a small improvement of about 2% across the ggH categories. In both ggH and VBF categories the BWZ × Bernstein parametrization improves the performance obtained with the BWZRedux function by about 4–5%. These results, along with the outcome of the bias studies presented in Section 8.1.2, will be used to choose the background fit function in each category. In fact, in each category, the function providing the best expected significance without inducing a significant bias in the estimate of the signal strength is the one chosen for the final fit.

8.1 Uncertainty on the background model

As presented in Table 27, the functional forms used to model the dimuon mass distribution in background events belong to two main classes: physics-inspired models and agnostic ones. Within these families, the choice of particular analytical function to model the background in a given category may potentially inject a bias in the estimated signal strength, when a signal plus background fit is performed. Therefore, as done in earlier iteration of this analysis [13] as well as in other searches following a similar approach for the signal extraction [68], a detailed set of *bias studies* are performed.

DRAFT

Table 28: A posteriori expected significance in each category and for each family of functions obtained for a SM Higgs boson with $m_{\mathrm{H}} = 125\,\mathrm{GeV}$. Systematic uncertainties on the signal prediction have not be added into the datacards used to extract these significances, however their impact is expected to be quite small at the level of few %.

Category	BWZ × Bernstein	BWZGamma	BWZRedux	FEWZ × Bernstein	Landau × Bernstein	S-Exponential	S-PowerLaw
ggH-cat0	$\sigma = 0.48$	$\sigma = 0.47$	$\sigma = 0.45$	$\sigma = 0.49$	$\sigma = 0.47$	$\sigma = 0.47$	$\sigma = 0.48$
ggH-cat1	$\sigma = 0.88$	$\sigma = 0.85$	$\sigma = 0.83$	$\sigma = 0.87$	$\sigma = 0.85$	$\sigma = 0.84$	$\sigma = 0.86$
ggH-cat2	$\sigma = 0.73$	$\sigma = 0.71$	$\sigma = 0.70$	$\sigma = 0.72$	$\sigma = 0.71$	$\sigma = 0.70$	$\sigma = 0.71$
ggH-cat3	$\sigma = 0.85$	$\sigma = 0.85$	$\sigma = 0.85$	$\sigma = 0.86$	$\sigma = 0.85$	$\sigma = 0.84$	$\sigma = 0.85$
ggH-cat4	$\sigma = 0.84$	$\sigma = 0.81$	$\sigma = 0.81$	$\sigma = 0.82$	$\sigma = 0.81$	$\sigma = 0.80$	$\sigma = 0.80$
VBF-cat0	$\sigma = 0.41$	$\sigma = 0.39$	$\sigma = 0.39$	$\sigma = 0.39$	$\sigma = 0.39$	$\sigma = 0.38$	$\sigma = 0.38$
VBF-cat1	$\sigma = 0.62$	$\sigma = 0.60$	$\sigma = 0.60$	$\sigma = 0.61$	$\sigma = 0.60$	$\sigma = 0.60$	$\sigma = 0.60$
VBF-cat2	$\sigma = 0.74$	$\sigma = 0.75$	$\sigma = 0.75$	$\sigma = 0.75$	$\sigma = 0.75$	$\sigma = 0.72$	$\sigma = 0.78$
VBF-cat3	$\sigma = 1.08$	$\sigma = 1.11$	$\sigma = 0.98$	$\sigma = 1.00$	$\sigma = 1.00$	$\sigma = 1.11$	$\sigma = 1.12$
Combined ggH	$\sigma = 1.71$	$\sigma = 1.68$	$\sigma = 1.66$	$\sigma = 1.71$	$\sigma = 1.69$	$\sigma = 1.67$	$\sigma = 1.68$
Combined VBF	$\sigma = 1.51$	$\sigma = 1.52$	$\sigma = 1.43$	$\sigma = 1.44$	$\sigma = 1.45$	$\sigma = 1.50$	$\sigma = 1.51$

Table 29: A posteriori expected significance in each category and for each family of functions obtained for a SM Higgs boson with $m_{\mathrm{H}} = 125\,\mathrm{GeV}$. Systematic uncertainties on the signal prediction have not be added into the datacards used to extract these significances, however their impact is expected to be quite small at the level of few %.

Category	BWZ × Bernstein	BWZGamma	BWZRedux	FEWZ × Bernstein	Landau × Bernstein	S-Exponential	S-PowerLaw
ggH-cat0	$\sigma = 0.46$	$\sigma = 0.46$	$\sigma = 0.44$	$\sigma = 0.48$	$\sigma = 0.46$	$\sigma = 0.47$	$\sigma = 0.47$
ggH-cat1	$\sigma = 0.83$	$\sigma = 0.83$	$\sigma = 0.82$	$\sigma = 0.85$	$\sigma = 0.84$	$\sigma = 0.83$	$\sigma = 0.84$
ggH-cat2	$\sigma = 0.70$	$\sigma = 0.70$	$\sigma = 0.69$	$\sigma = 0.71$	$\sigma = 0.70$	$\sigma = 0.70$	$\sigma = 0.70$
ggH-cat3	$\sigma = 0.85$	$\sigma = 0.84$	$\sigma = 0.84$	$\sigma = 0.86$	$\sigma = 0.85$	$\sigma = 0.84$	$\sigma = 0.85$
ggH-cat4	$\sigma = 0.86$	$\sigma = 0.83$	$\sigma = 0.83$	$\sigma = 0.83$	$\sigma = 0.83$	$\sigma = 0.82$	$\sigma = 0.83$
VBF-cat0	$\sigma = 0.37$	$\sigma = 0.39$	$\sigma = 0.39$	$\sigma = 0.39$	$\sigma = 0.38$	$\sigma = 0.38$	$\sigma = 0.39$
VBF-cat1	$\sigma = 0.63$	$\sigma = 0.61$	$\sigma = 0.61$	$\sigma = 0.61$	$\sigma = 0.60$	$\sigma = 0.60$	$\sigma = 0.61$
VBF-cat2	$\sigma = 0.72$	$\sigma = 0.70$	$\sigma = 0.70$	$\sigma = 0.70$	$\sigma = 0.70$	$\sigma = 0.69$	$\sigma = 0.70$
VBF-cat3	$\sigma = 1.21$	$\sigma = 1.22$	$\sigma = 1.08$	$\sigma = 1.09$	$\sigma = 1.07$	$\sigma = 1.15$	$\sigma = 1.18$
Combined ggH	$\sigma = 1.69$	$\sigma = 1.68$	$\sigma = 1.66$	$\sigma = 1.70$	$\sigma = 1.68$	$\sigma = 1.67$	$\sigma = 1.68$
Combined VBF	$\sigma = 1.51$	$\sigma = 1.54$	$\sigma = 1.47$	$\sigma = 1.48$	$\sigma = 1.46$	$\sigma = 1.49$	$\sigma = 1.50$

8.1.1 Toy generation for bias studies

The first step in performing bias studies for the signal extraction consists in generating toy experiments for each category for each functional form selected as a possible *truth* function that describes the $m_{\mu\mu}$ spectrum in background events. Starting from the background shapes reported in Fig. 45 and 46, which have been obtained from a b-only fit to the data, and the expected signal shapes shown in Fig. 39 and 40, 2000 toys are generated for each background functional form in each category. The number of toys is chosen in order to have an uncertainty of about 2% on the estimated bias. Toys are generated by injecting an amount of signal event proportional to the expected rate in each category for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$. The same amount of toy data have been generated also for other Higgs boson masses of 120 and 130 GeV. Since we are only interested in estimating the bias on the signal strength induced by the assumption of a specific background shape, systematic uncertainties affecting the signal prediction, described in Section 7.2.2 and 7.2.3 are not considered in the bias studies.

8.1.2 Bias of the various background functions

The dimuon mass distributions obtained from the toy-experiments previously described are fitted with a signal plus background model, where the background function is one of those reported in Table 27. In each fit, the bias on the signal strength is defined as follows:

$$\mathrm{Bias}(\mu_{fit}, \hat{\mu}_{inj}, \sigma_{\mu}) = \frac{\mu_{fit} - \hat{\mu}_{inj}}{\sigma_{\mu}} \qquad (15)$$

where μ_{fit} is the value of the signal strength extracted from the fit, $\hat{\mu}_{inj}$ is the signal strength

DRAFT

injected is in each toy ($\hat{\mu}_{inj} = 1$)[4], and σ_μ is its post-fit uncertainty estimated as the crossing at $-2\Delta\text{Log}\mathcal{L} = 1$ of the likelihood scan as a function of μ. In absence of systematic effects, the bias is expected to be Gaussian distributed, centred at zero, and with a width of one. The mean value of the bias distribution is used to quote the bias associated to a given fit function. Good functional forms, which can be used in the final signal extraction, are those characterized by a small bias on the signal strength in each category ($<$ 20% as defined in Equation 15). The threshold of 20% is chosen because such bias guarantees that the systematic uncertainty on the quoted upper limit or on the expected significance is smaller than 2%.

Figure 47 and 48 show the measured bias on the signal strength for $m_\text{H} = 125\,\text{GeV}$ in ggH and VBF categories, respectively, obtained by fitting the toys generated via several truth function with BWZRedux, BWZGamma, FEWZ × Bernstein, S-Exponential, S-PowerLaw, and Landau × Bernstein. The number of degrees of freedom considered for each functional form and in each category are listed in Table 27. In the ggH categories, a large tension is observed between the series of exponentials and all the other truth models. This tension is substantially reduced in the VBF categories thanks to a better signal-to-background ratio, a smaller statistical constraint from events in the mass sidebands, and a larger contamination from combinatorial backgrounds that can be usually well described by a series of exponentials. BWZRedux, BWZGamma, BWZ × Bernstein, and FEWZ × Bernstein exhibit a bias in single categories within 20% against non exponential truth models. The bias w.r.t. the sum of exponentials is larger than 50% in ggH-cat0, ggH-cat1, and ggH-cat2, while it is larger than 20% in ggH-cat3 and ggH-cat4.

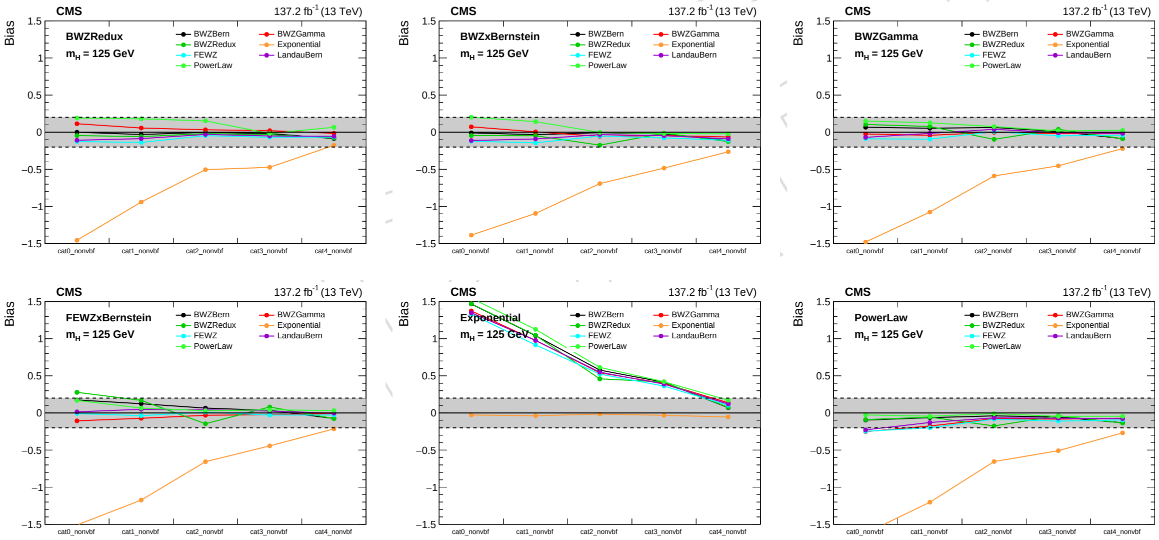

Figure 47: Mean value of the bias distribution in ggH BDT categories obtained by fitting toys, generated according to different background truth models and injecting a nominal signal at $m_\text{H} = 125\,\text{GeV}$, with the following functional forms: BWZRedux (top left), BWZxBernstein (top middle), BWZGamma (top right), FEWZ × Bernstein (bottom right), S-Exponential (bottom middle), and S-PowerLaw (bottom right).

An additional concern is the potential bias on the signal strength extracted from the simultaneous fit performed across multiple categories (*correlated bias*). In principle, since each category represents an independent entity from the background estimate point of view, we would need to generate toys according to all possible combinations of truth background functions. However, having nine event categories and at least six alternative truth models makes this test

---

[4] In each toy, the amount of signal events injected correspond to a Poissonian "smearing" around a mean value given by the expected signal rate for $\mu = 1$

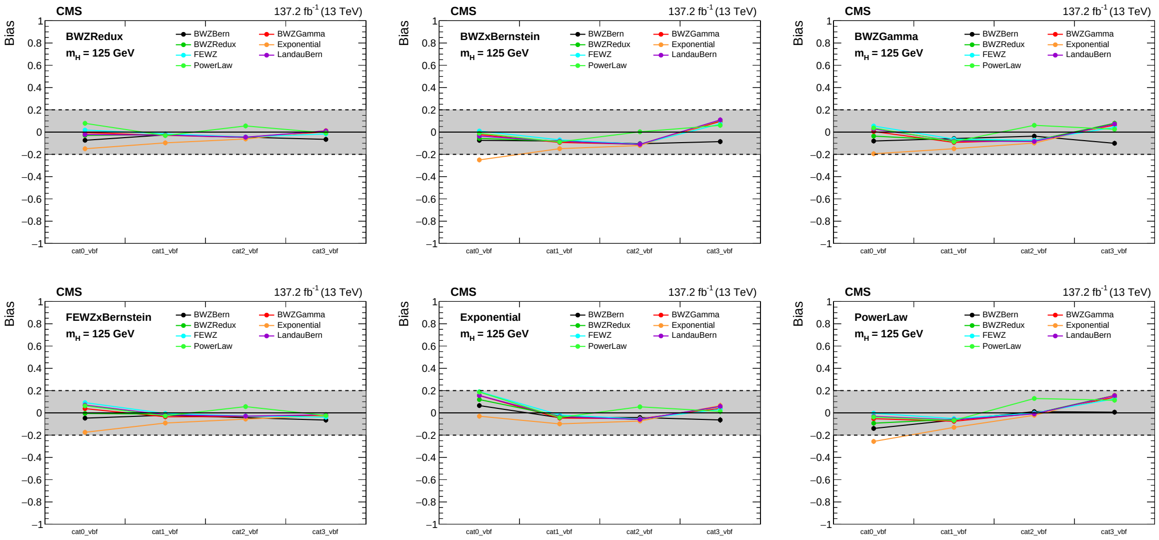

Figure 48: Mean value of the bias distribution in VBF BDT categories obtained by fitting toys, generated according to different truth background models and injecting a nominal signal at $m_{\mathrm{H}} = 125\,\mathrm{GeV}$, with the following functional forms:BWZRedux (top left), BWZxBernstein (top middle), BWZGamma (top right), FEWZ × Bernstein (bottom right), S-Exponential (bottom middle), and S-PowerLaw (bottom right).

computationally unfeasible. Therefore, we consider only the cases in which the type of function used to generate and fit the toys is the same across all categories. For example, when the BWZRedux is chosen as the truth model, pseudo-experiments are thrown using this background shape in all categories. On the other hand, when the BWZRedux is tested as fit function, pseudo-experiments generated according to every possible truth function are fitted with a signal-plus-background model in which the BWZRedux parametrization is used to describe the background in all categories. Figure 49 shows the estimated bias on the signal strength for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ obtained when the ggH categories are simultaneously fitted using either a BWZRedux (left), a BWZGamma (middle), or a FEWZ × Bernstein (right) function to model the background. Similarly, Fig. 50 reports the bias on the signal strength for a combined fit performed across VBF categories. As expected from the studies presented in Fig. 48, when all VBF categories are simultaneously fitted no significant bias is observed for any of the truth model. In contrast, when ggH categories are fitted, a large bias is observed against the series of exponentials. This was already expected from the measurements reported in Fig. 47, whereas the bias w.r.t all the other truth models can be controlled within 20%.

DRAFT

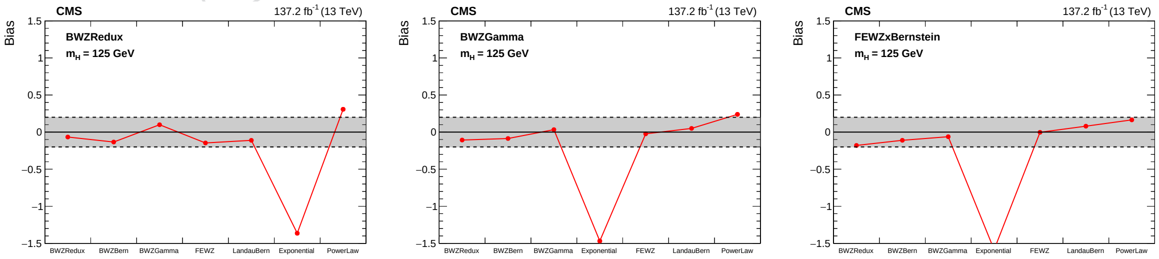

Figure 49: Mean value of the bias distribution on the signal strength obtained by fitting toys, generated across all the ggH categories according to alternative bakground truth models and injecting a nominal signal at $m_{\mathrm{H}} = 125\,\mathrm{GeV}$, with the following functional forms: BWZRedux (left), BWZGamma (middle), and FEWZ × Bernstein (right).

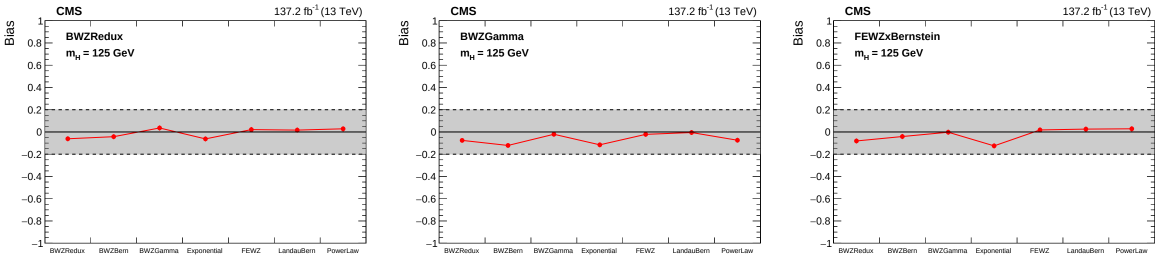

Figure 50: Mean value of the bias distribution on the signal strength obtained by fitting toys, generated across all the VBF categories according to alternative bakground truth models and injecting a nominal signal at $m_{\mathrm{H}} = 125\,\mathrm{GeV}$, with the following functional forms: BWZRedux (left), BWZGamma (middle), and FEWZ $\times$ Bernstein (right).

In conclusion, in ggH categories the bias on the signal strength cannot be controlled within an acceptable level by using single functions to fit for the background in each category. Therefore, if possible, an alternative bias-free strategy to estimate the total background is required that can control better the bias against the series of exponentials. In contrast, the bias in VBF categories can be well controlled by using different alternative fit functions. Based on the performance listed in Table 29, as well as the observed bias reported in Fig. 50 and Fig. 50, the following functions are chosen in each VBF category to estimate the total background: BWZRedux in VBF-cat0 and VBF-cat1, while BWZGamma in VBF-cat2 and VBF-cat3.

Finally, Appendix E and F show results from bias studies performed in both ggH and VBF categories for different mass points of $m_{\mathrm{H}} = 120\,\mathrm{GeV}$ and $m_{\mathrm{H}} = 130\,\mathrm{GeV}$, which correspond to the smallest and largest Higgs boson mass hypothesis tested by this search, respectively. Overall, similar results and conclusions to those presented for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ are found.

8.2 The core-PDF method

The simplest way to estimate the background via analytical fits to the observed $m_{\mu\mu}$ distributions relies on using independent functions for each category. However, as described in Section 8.1.2, this approach does not allow to obtain a robust control of the bias on the signal strength in the ggH categories. Therefore, we have developed an alternative strategy to estimate the background in these categories that tries to maximize the expected sensitivity while introducing a negligible bias on the measured signal yield. This method is referred to as *core-PDF*.

The main motivation for this method originates from the fact that the background composition across categories is quite stable, i.e. the background purity of dimuon decays of Z bosons (DY+jets or Z-EW) is fairly constant along the BDT outputs. Therefore, it is natural to expect that background shape parameters can be correlated across BDT categories. On the other hand, this correlation can only be partial because, as shown in Fig. 23, going from low to high score along the BDT outputs one moves from events with small p_{T} of the dimuon system to more boosted signatures, and this has a direct effect on the $m_{\mu\mu}$ shape of background events. In fact, the $m_{\mu\mu}$ distribution of Z/γ^{*}+jets events becomes less steep for events with larger $p_{\mathrm{T}}^{\mu\mu}$. The *core-PDF* method tries to resolve this difference in the following way:

- The background in the ggH categories is extracted using a model that contains a common function, called *core-function*, whose parameters are estimated from a simultaneous fit to the data across all the categories. Therefore, these parameters are fully correlated across categories.

DRAFT

- In order to give to the background function additional flexibility to better fit the data in each category, the $core\text{-}function$ is modulated (multiplied) in each category by a polynomial term called "shape modifier function (SMF)". We start with a first order polynomial but the ultimate order of the SMFs is selected by performing bias studies for the signal strength.

Figure 51 shows a comparison between the background shapes in ggH (left) and VBF (right) BDT categories as obtained from a background-only fit to the data via a BWZRedux (BWZGamma) function in the ggH (VBF) categories. This confirms that the $m_{\mu\mu}$ distribution in background events becomes progressively "softer" moving from low to high score, and that this effect is mainly correlated with an increase in the dimuon p_{T}. Moreover, Fig. 52 shows proxies for the SMFs in data for the ggH categories. These SMFs are defined as the bin-by-bin ratio between the $m_{\mu\mu}$ spectrum in the chosen category and inclusive obtained from by summing together all the events belonging to the ggH phase space. Results from a fit to the SMF using either a third or a second order polynomial are also reported.

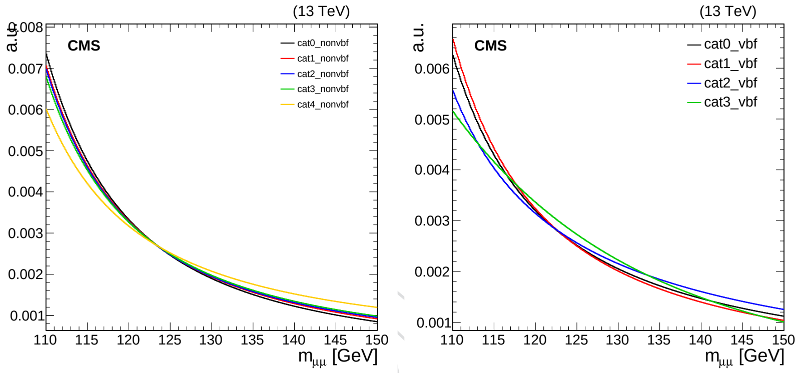

Figure 51: Comparison between the $m_{\mu\mu}$ shapes obtained from a background-only fit to the data using a BWZRedux function in ggH categories (left) and a BWZGamma PDF in the VBF ones (right). In both cases, the $m_{\mu\mu}$ distrubution becomes softer moving from low (cat0) to high (cat3 or cat4) BDT score.

In order to measure the bias of the $core\text{-}PDF$ model, which always performs a fit across multiple categories, one needs to test both the robustness of the method against alternative truth functions as well as the fact that polynomial SMFs are enough to model differences in the background shape across categories. Pseudo-experiments are generated following the same approach described in Section 8.1.2 in the context of testing the bias for the combined fit (*correlated bias*). Toys are generated independently in each category according to a chosen truth function. When we want to measure the bias in a single category, the following procedure is adopted:

- In the toy generation, the signal is only injected in that specific category while, in the other ones, background-only $m_{\mu\mu}$ distributions are produced.
- Toys are then fitted with a signal-plus-background model which uses a $core\text{-}PDF$ to describe the background and the presence of the signal is only allowed in the category where the signal injection was performed. In this way, the parameters of the $core\text{-}function$ are constrained by all the data included in the fit, while the signal strength is extracted only by looking at the category of interest.

On the other hand, when the bias on the overall signal strength is tested, the injection is performed in every category and therefore the presence of the signal in the fit is allowed everywhere.

DRAFT

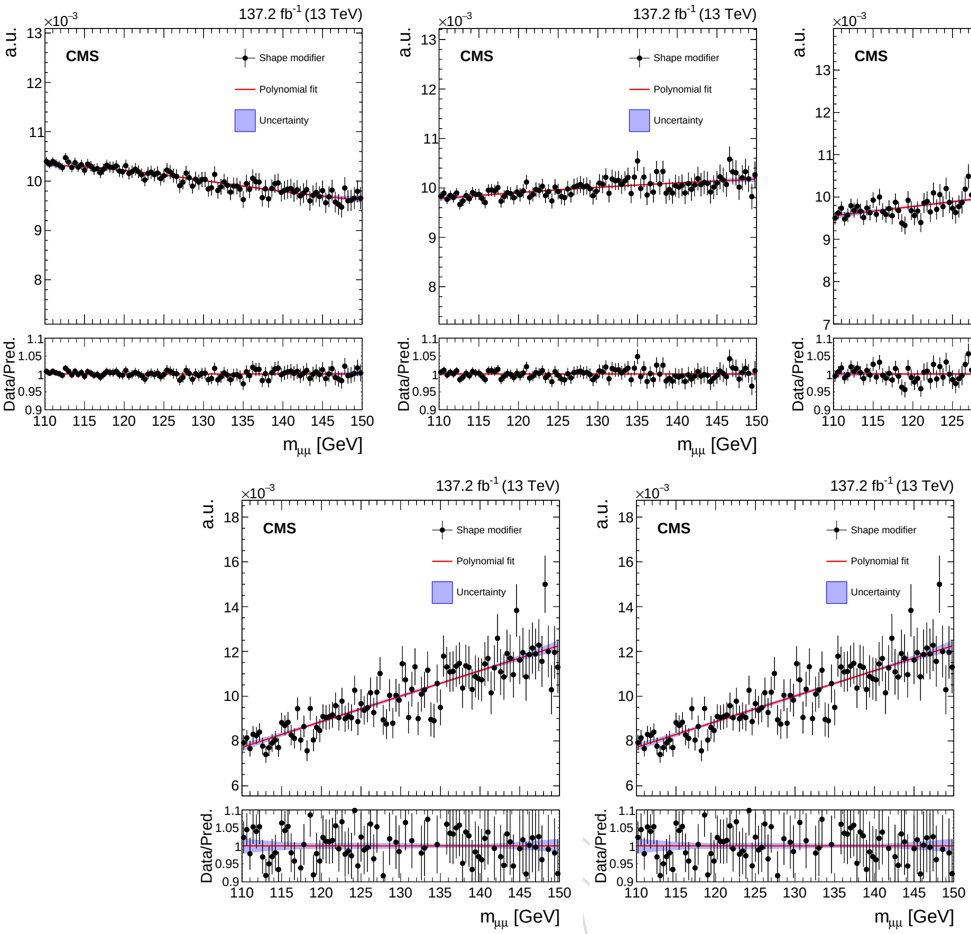

Figure 52: Shape modifiers in data used in the *core-PDF* for events belonging to the ggH categories. These shape modifiers are defined as the bin-by-bin ratio between the $m_{\mu\mu}$ spectrum in a given category and the inclusive one, obtained by merging the ggH categories into a single one. Top row: ggH-cat0 (left), ggH-cat1 (middle), and ggH-cat2 (right). Bottom row: ggH-cat3 (left), and ggH-cat4 (right). The solid red lines indicate the results of a fit to the shape modifiers via third-order (ggH-cat0 and ggH-cat1) or second-order polynomials, while the blue area corresponds to the 68% uncertainty band from the fit parameters.

DRAFT

In order to control the bias in the ggH categories within an acceptable threshold of 20%, the following setup is used:

- As described in Fig. 52, in both ggH-cat0 and ggH-cat1, SMFs are modelled with a third-order polynomial while in the other three categories second-order polynomials are considered.
- With the goal of substantially reducing the bias between Breit-Wigner like functions and sum of exponentials, an envelope of three PDFs with three degrees of freedom is used as *core-function*: BWZRedux, FEWZ × Bernstein, and S-Exponential. The discrete profiling method is described in more details in Ref. [69]. This method was designed in the context of the $H \rightarrow \gamma\gamma$ search in order to estimate the systematic uncertainty associated to the choice of a particular analytical function used to fit for the background. The method treats the choice of the background function as a discrete nuisance parameter in the likelihood fit to the data. This allows to dynamically select the best core-function that can be used to model the background at each point

of the likelihood scan as a function of the signal strength. Given that the three functions considered in the envelope have the same number of freely floating parameters (3-dof), no penalization term as a function of the degrees of freedom is used.

Figure 53 (left) shows the measured bias on the signal strength for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ in each of the ggH categories when the background is modelled using the *core-PDF* method. The bias is observed to be smaller than 20% in each category and for each truth function used to generate the toys. In addition, Fig. 53 (right) reports the bias measured in toys where the signal has been injected in all the ggH categories. Also in this case, the bias measured as a function of different truth functions can be controlled within 20%.

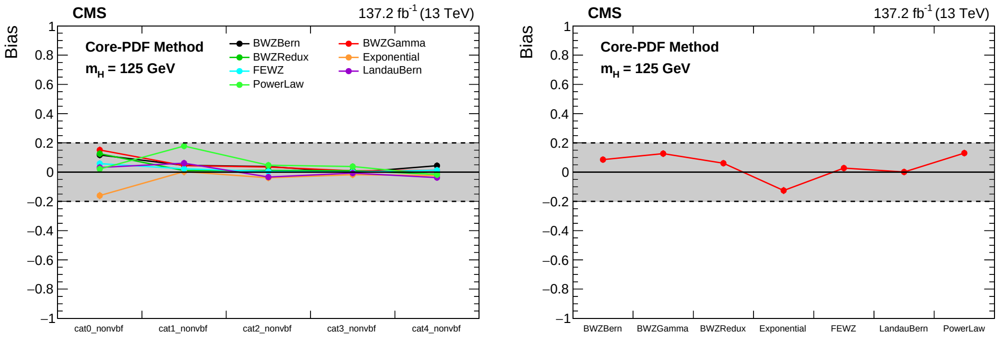

Figure 53: Mean bias on the signal strength obtained from the *core-PDF* method used to model the background in ggH categories. On the left, the measured bias in each of the ggH categories when a nominal signal at $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ is injected. On the right, the bias measured when the signal is injected in all the ggH categories simultaneously.

Given that a discrete likelihood profile is used to define the *core-PDF* component, the coverage of this method is checked on the same toys used for the bias studies. The coverage within the 68% CL ($\approx 1\sigma$) interval is defined as the ratio between the observed number of toys for which $-2 \times \Delta Log\mathcal{L}(\hat{\mu}_{inj}) < 1$ and the total number of toys generated, where a signal with $\hat{\mu}_{inj} = 1$ and $m_{\{H} = 125\,\mathrm{GeV}$ is injected. Similarly, the coverage for 16% CL ($\approx 0.2\sigma$), 38% CL ($\approx 0.5\sigma$), 86% CL ($\approx 1.5\sigma$), and 95% CL ($\approx 2\sigma$) intervals have also been tested. Figure 54 shows that the method designed for the background estimation has good coverage properties for the tested intervals. The *core-PDF* fit shows, as expected from a discrete likelihood profile, a small overcoverage (by few %) in categories and against some truth models.

Figure 55 shows the bias measured in toys where the signal has been injected in both ggH and VBF categories. Toys are fitted with a signal-plus-background model in which the presence of the signal is allowed in each categories, instead the background is estimated as follows:

- The *core-PDF* previously described is used to model the background in the ggH categories.
- In the VBF categories, the background is modelled via single and independent functions: BWZRedux in VBF-cat0 and VBF-cat1, while BWZGamma is used in VBF-cat2 and VBF-cat3.

The bias measured as a function of different truth functions can be controlled within 20%.

Finally, Appendix E and F show results from bias studies performed in ggH categories with the core-PDF method for different mass points of $m_{\mathrm{H}} = 120\,\mathrm{GeV}$ and $m_{\mathrm{H}} = 130\,\mathrm{GeV}$, which correspond to the smallest and largest Higgs boson mass hypothesis tested by this search, respectively. Overall, similar results and conclusions to those presented for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ are

DRAFT

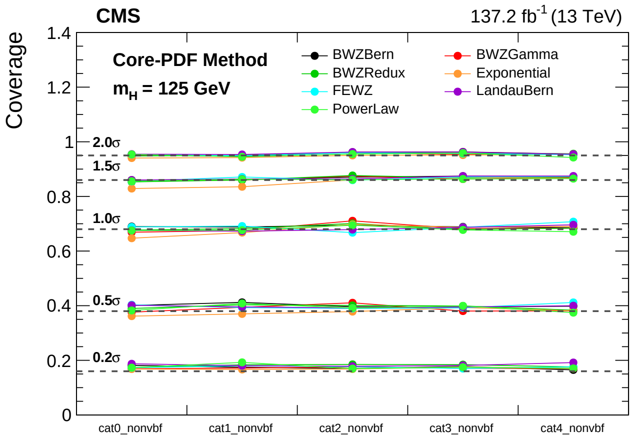

Figure 54: Coverage observed with the *core-PDF* method for 16%, 38%, 68%, 86%, and 95% CL intervals in toys where a signal at $m_{\{H} = 125\,\text{GeV}$ is injected with $\hat{\mu}_{inj} = 1$.

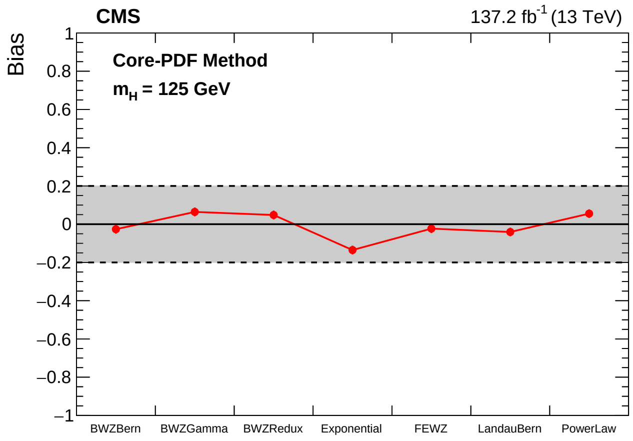

Figure 55: Mean bias on the signal strength obtained from the *core-PDF* method used to model the background in all categories in toys where the signal has been injected in all categories simultaneously.

Draft

found.

9 Results

In order to extract the final results, binned maximum-likelihood fits are performed to the observed $m_{\mu\mu}$ distributions of each category, in the range $110 < m_{\mu\mu} < 150\,\text{GeV}$, where the expected signal from the different production modes is scaled by common strength modifier (μ). Fits are performed fixing the expected Higgs boson mass to 125 GeV. Systematic uncertainties from theoretical and experimental sources, as described in Section 7, are included in the likelihood via Log-Normal nuisance parameters. Binned fits are used to speedup the computation, and the chosen bin size of 50 MeV is sufficiently small compared to the mass resolution that no information is lost. The data and the result from a signal-plus-background model performed simultaneously across categories are shown in Fig. 56 and 57 for ggH and VBF categories, respectively. The one (green) and two (yellow) standard deviation bands shown for the background component of the fit include the uncertainty in the fitted parameters. The background in the ggH categories is modelled via the *core-PDF* method detailed in Section 8.2, while in the VBF categories the functions listed in Section 8.1.2 are adopted. In addition, Fig. 58 shows the background component and the signal-plus-background fit to the data combining ggH and VBF categories, weighting them by the expected ratio between signal and signal-plus-background expectations in each category.

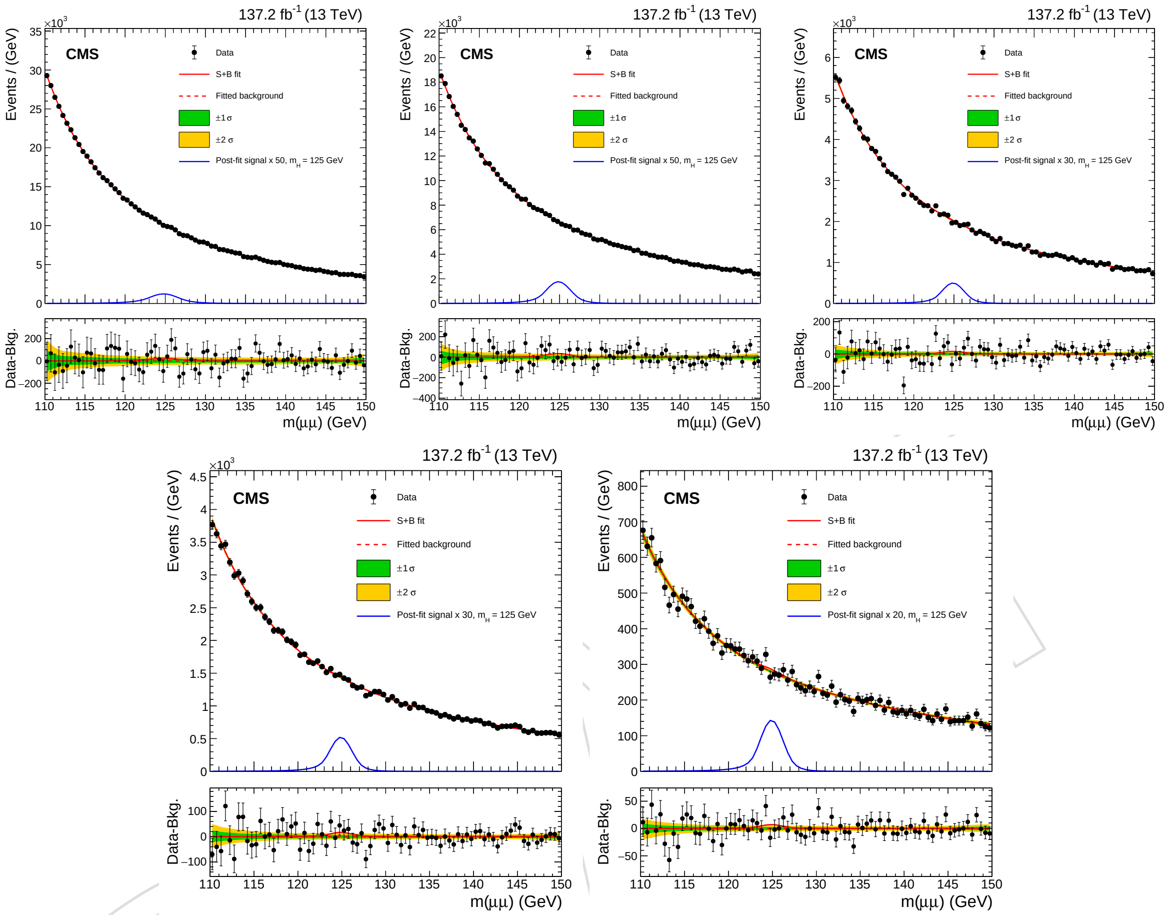

Figure 56: Comparison between observed data and the total background extracted from a signal-plus-background fit performed across all event categories. Top row: ggH-cat0 (left), ggH-cat1 (middle), and ggH-cat2 (right). Bottom row: ggH-cat3 (left), and ggH-cat4 (right). The one (green) and two (yellow) standard deviation bands include the uncertainties in the background component of the fit. The lower panel in each plot shows the residuals after the background subtraction.

Likelihood scans of the signal strength modifier are performed for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$, in which the other parameters of the signal and background models are allowed to vary. Systematic uncertainties are included in the form of nuisance parameters. Figure 59 (left) shows the expected likelihood scan obtained by fitting an Asimov dataset, defined by fixing the nuisances parameters to their maximum likelihood estimate obtained from a fit to the data in which $\mu = 0$ is assumed. The likelihood scan crossing at $2 \cdot \Delta \mathrm{Log}\mathcal{L} = 3.84$ corresponds to the expected asymptotic upper limit. In contrast, Fig. 59 (right) reports the expected likelihood scan obtained by fitting an alternative Asimov dataset in which nuisance parameters are still estimated from a background-only fit to the data but a signal is also injected, with a rate corresponding to the one predicted by the SM. The crossing point of this likelihood scan at $\mu = 0$ gives the expected significance of the signal-plus-background hypothesis w.r.t. the background-only one. In Appendix G, the same expected likelihood scans, performed either in presence or in absence of a

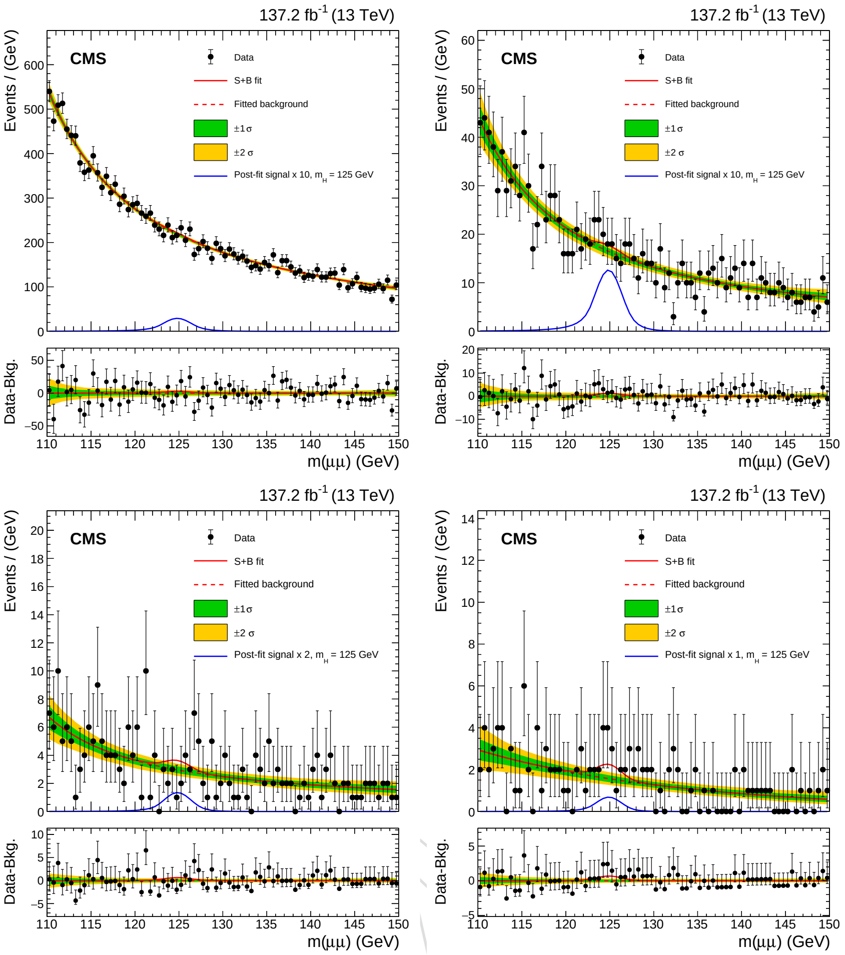

Figure 57: Comparison between observed data and the total background extracted from a signal-plus-background fit performed across all event categories. Top row: VBF-cat0 (left), and VBF-cat1 (right). Bottom row: VBF-cat2 (left), and VBF-cat3 (right). The one (green) and two (yellow) standard deviation bands include the uncertainties in the background component of the fit. The lower panel in each plot shows the residuals after the background subtraction.

SM Higgs boson signal in the toy, are presented for each of the nine event categories defined along the ggH and VBF BDT outputs. In addition, Figure 60 reports the observed likelihood scans obtained by fitting ggH (left), VBF (middle), and ggH+VBF categories simultaneously. In Appendix G, observed scans are reported in each individual categories.

The best fit signal strength for the Higgs boson with mass of 125 GeV, and the corresponding 68% CL interval, is $\hat{\mu} = 0.95\,^{+0.48}_{-0.46}$. The uncertainty in the measured signal rate is dominated by the limited number of events in data. Figure 61 (left) reports a summary of the best fit values for the signal strength and the corresponding 68% CL intervals obtained from a profile likelihood scan in each event category. The best fit signal strengths in each category are consistent with the combined fit result as well as the SM expectation.

The systematic uncertainties affecting the signal prediction in each category have a small impact on the ultimate analysis sensitivity, causing a loss of about 3% in both the expected significance and expected upper limit. To assess the importance of each source of systematic uncertainty (constrained nuisance parameter) on the signal strength, extracted from the combined across ggH and VBF BDT categories, a signal-plus-background fit is performed to estimate the best fit value of each nuisance parameter, along with their related post-fit uncertainty. Data

DRAFT

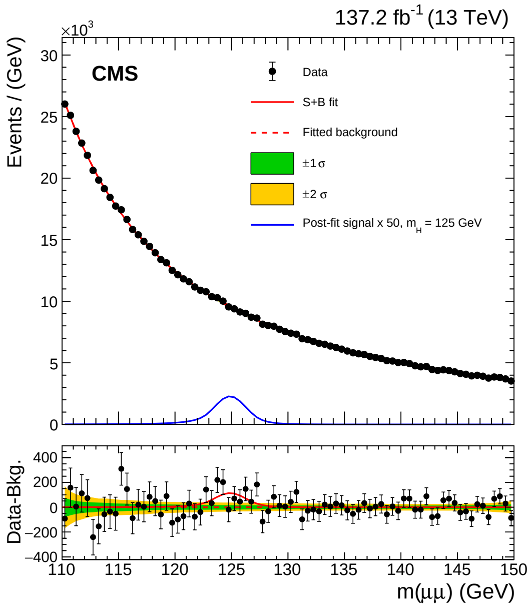

Figure 58: Comparison between observed data and the total background from a signal-plus-background fit performed across all event categories. Events are weighted according to the expected ratio of signal to signal-plus-background in the category to which they belong. The solid blue line represents the expectation from the SM Higgs boson multiplied by a factor of 10. The one (green) and two (yellow) standard deviation bands include the uncertainties in the background component of the fit. The lower panel in each plot shows the residuals after the background subtraction.

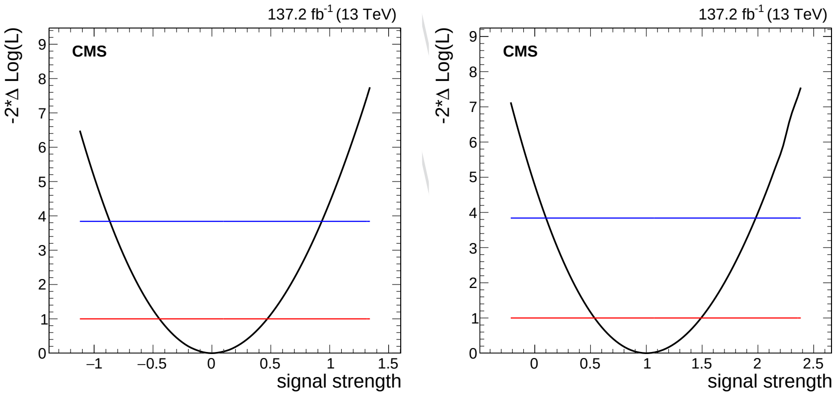

Figure 59: Expected profile likelihood ratios as a function of the signal strength modifier (μ) for the combined signal-plus-background fit performed across all event categories. On the left, likelihood scan obtained from an Asimov dataset defined by fixing nuisances parameters to the values extracted from a background-only fit to the data. On the right, the likelihood profile vs μ is obtained by fitting an Asimov dataset in which the expected Higgs boson signal, assuming SM production cross sections, has been injected in the toy.

are then re-fitted shifting independently each nuisance parameter by ±1-σ (post-fit) with respect to the best fit value. The impact of each source of uncertainty on the signal strength is defined as the difference between the values of μ obtained after moving the nuisance by ±1-σ and the best fit result. Figure 62 (left) shows the result of the impact study in the context of the combined fit on the observed data, where nuisances are ranked according to their impact on the measured signal strength. None of these nuisance parameters are significantly pulled

DRAFT

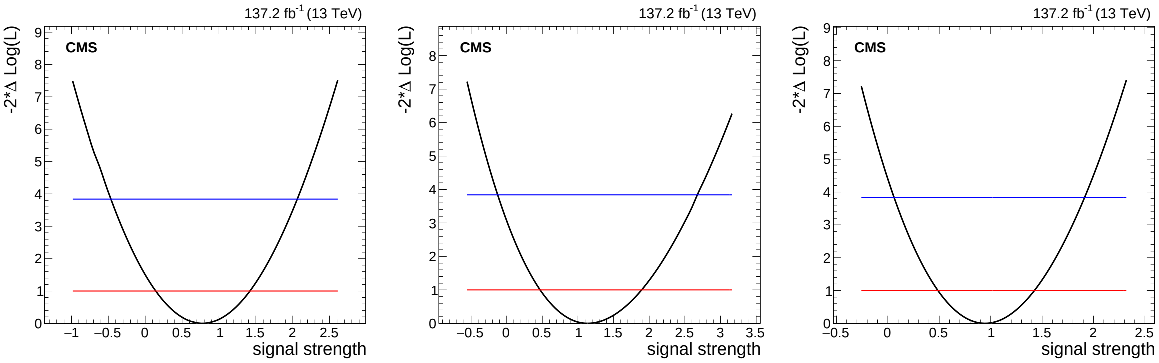

Figure 60: Observed profile likelihood ratios as a function of the signal strength modifier (μ) for the combined signal-plus-background fit performed across ggH (left), VBF (middle), and all (right) event categories.

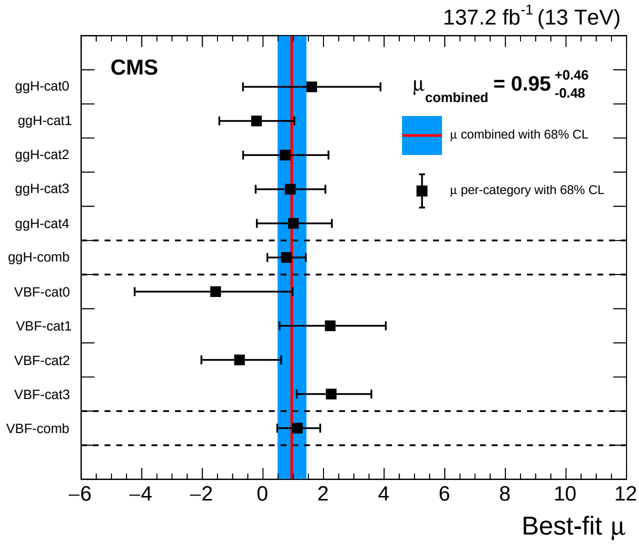

Figure 61: Signal strength modifiers measured for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ in each event category (black points) are compared to the result of the combined fit (solid red line) and the SM expectation (dashed grey line).

Draft

or constrained by the signal-plus-background fit. The most important systematics are represented by the uncertainty in the dimuon mass resolution (10% uncertainty correlated across categories as described in Section 7.2.1), the uncertainties in the ggH inclusive cross section, the uncertainty related to the parton shower parameters, and those affecting the ggH and VBF acceptance from the STXS stage 1.2 prescription. Finally, the uncertainties in measurement of the integrated luminosity and in the prediction of $\mathcal{B}(\mathrm{H} \rightarrow \mu\mu)$ are also playing a relevant role. On the other hand, Fig. 62 (right) reports the result of the impact study in the combined fit to the observed data when also freely floating background parameters (normalization and shape) are considered. The parameters with the largest impact on the performance of the final fit are the background normalization in each category, as well as the BWZGamma shape parameters for VBF cat-3 and the polynomial transfer factors in ggH low score categories.

Figure 63 shows the results of a goodness-of-fit test for the combined fit performed across ggH (left) and VBF (right) categories, in which the saturated likelihood ratio is used as test-statistics. The expected distribution for the test-statistics is obtained by performing signal-plus-background fits, with floating μ, on toys thrown from the best-fit estimate of the nuisance

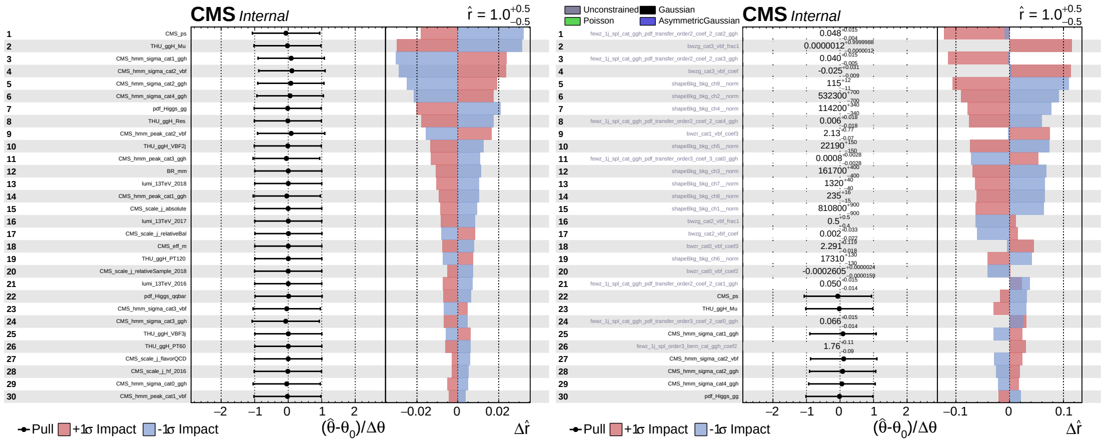

Figure 62: Result of the impact study on μ for the combined fit performed across ggH and VBF categories on the observed data.

parameters (a-posteriori toys). The observed p-value, i.e. probability of having a goodness-of-fit value larger than the observed one, is 0.23 and 0.08 for the ggH and VBF combined fits, respectively. This indicates a good compatibility between the observed data and the signal-plus-background model.

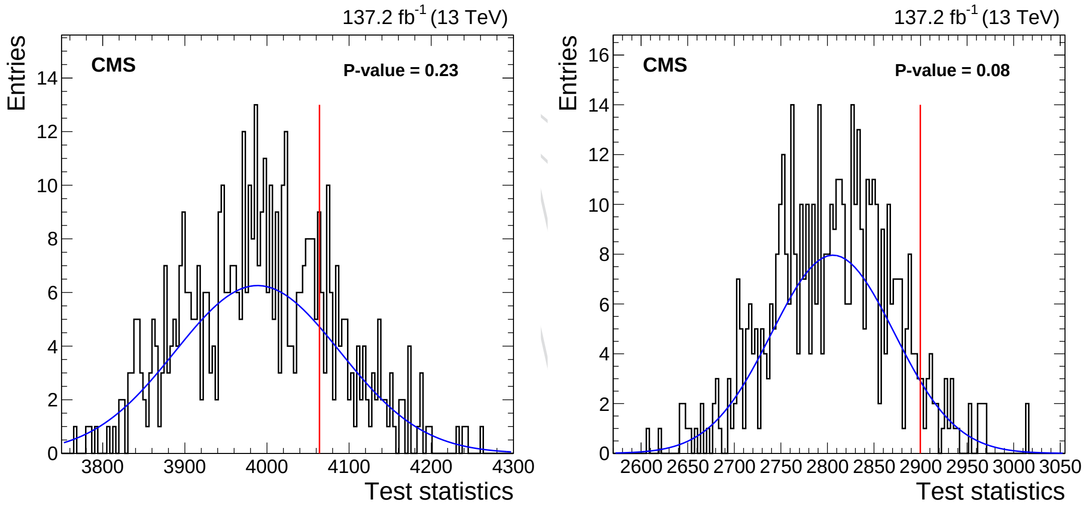

Figure 63: Results from a goodness-of-fit test for the combined fit performed across ggH (left) and VBF (right) categories. The observed value for the test-statistics is indicated by the blue arrow, while the histogram (solid black) indicates the expected distribution obtained in signal-plus-background toys. The observed p-value, i.e. probability of having a goodness-of-fit value larger than the observed one, is 0.23 and 0.08 for the ggH and VBF combined fits, respectively.

Table 30 lists both the expected upper limits on the signal strength modifier, computed at the 95% CL with the Asymptotic CL_s method [70–72], and the significance of the incompatibility between the expected signal-plus-background hypothesis and the background-only hypothesis for each BDT category, as well as for their combination. The *core-PDF* allows to control the bias on the signal strength within 20% in each category at the price of loosing about 6% sensitivity in terms of expected significance and upper limit. The expected upper limit in the absence of a SM decay in this channel is $\mu < 0.93 \times SM$, which corresponds to an expected significance of the Higgs boson decaying into two muons of 2.18 standard deviations. Moreover, in Table 35 of Appendix D expected asymptotic upper limits on the signal strength are reported independently for each era and in each category. They have been computed with simplified datacards neglecting systematic uncertainties in the signal prediction and modelling the background via a BWZRedux or a BWZ × Bernstein function depending on the analysis categories.

Table 30: Summary of the expected and observed upper limits on μ, computed at the 95% CL for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$, as well as the expected and observed signal significance for each BDT category and for their combination.

Category	Expected UL	Observed UL	Expected Signif.	Observed Signif.
ggH-cat0	< 4.641	< 5.750	0.440σ	0.707σ
ggH-cat1	< 2.477	< 2.340	0.806σ	—
ggH-cat2	< 2.773	< 3.383	0.724σ	0.532σ
ggH-cat3	< 2.305	< 2.997	0.877σ	0.787σ
ggH-cat4	< 2.477	< 3.345	0.829σ	0.827σ
VBF-cat0	< 5.359	< 4.305	0.380σ	—
VBF-cat1	< 3.422	< 5.404	0.606σ	1.329σ
VBF-cat2	< 2.789	< 2.399	0.739σ	—
VBF-cat3	< 2.195	< 4.512	1.120σ	2.216σ
Combination ggH	< 1.266	< 1.885	1.571σ	1.226σ
Combination VBF	< 1.418	< 2.422	1.519σ	1.751σ
Combination ggH and VBF	< 0.926	< 1.756	2.183σ	2.116σ

The 95% CL upper limits on the Higgs boson production cross section times $\mathcal{B}(\mathrm{H} \to \mu\mu)$ are computed in the range $120 < m_{\mathrm{H}} < 130\,\mathrm{GeV}$ using an asymptotic approximation [72] of the $\mathrm{CL_s}$ method [71], and are shown in Figure 64 (left). The observed and expected upper limits for a 125 GeV $\mathrm{H} \to \mu\mu$ signal, assuming SM production rates, are found to be 1.76 and 0.93 times the SM expectation, respectively. Figure 64 (right) shows the observed and expected local p-values computed in the mass range $120 < m_{\mathrm{H}} < 130\,\mathrm{GeV}$ for a 125 GeV Higgs boson signal. The expected significance is found to be 2.18 standard deviations, while the observed significance in data is 2.12 standard deviations.

9.1 Effect of systeamtic uncertainties on the background model

As presented in Section 8.1.2 and 8.2, no significant bias (< 20%) on the signal strength from the chosen parametric background model in each category is observed, therefore no extra nuisance parameters are introduced in the final fit to account the residual per-category bias. However, in order to verify that this missing systematic uncertainty has a small impact (< 2%) on the overall significance, spurious signals are injected in each category proportionally to the maximum observed bias. Table 31 reports, for each category, the chosen function used in the signal extraction, the observed post-fit uncertainty on the signal strength ($\Delta\mu$), and the maximum bias w.r.t. the chosen functional form.

The spurious signal considered in each category to account for the measured bias on the signal strength is added as a new background process and its $m_{\mu\mu}$ shape is chosen be identical to the one expected for the ggH (VBF) process in the ggH (VBF) categories. The total pre-fit normalization for this spurious process is set to be zero in each category. The normalization is allowed to vary in the fit to the data within a Gaussian prior with width (ΔB) defined, for each category, as:

$$\Delta\mathrm{B} = \hat{b} \times \Delta\mu \times \mathrm{N_{sig}} \tag{16}$$

where $\hat{b}$ is the measured bias on μ, $\Delta\mu$ is the post-fit uncertainty on the signal strength, and $\mathrm{N_{sig}}$ is the expected number of signal events in the chosen category. Table 32 reports the difference in the expected significance for Higgs boson decay to muon pairs with $m_{\mathrm{H}} = 125\,\mathrm{GeV}$

DRAFT

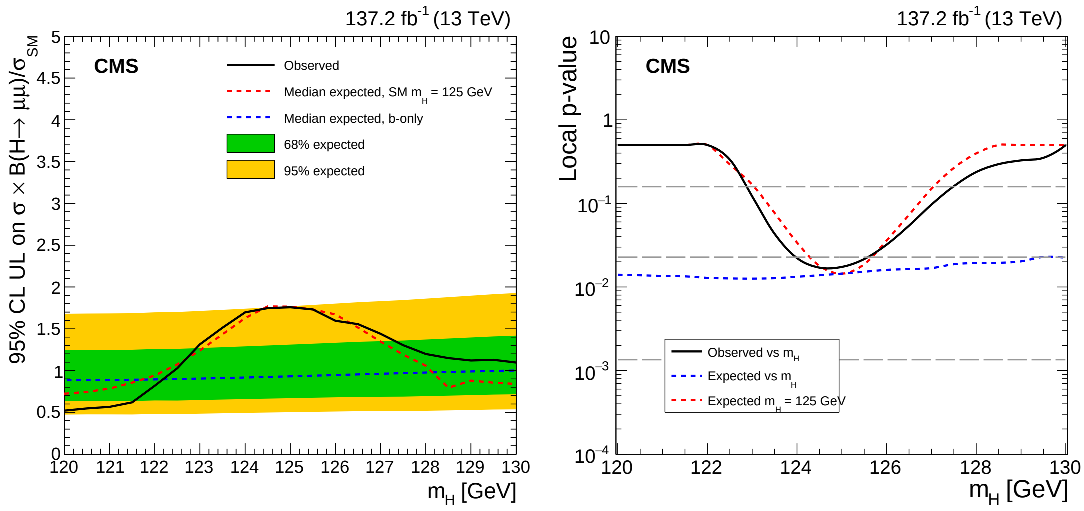

Figure 64: On the left, the observed 95% CL upper limit on the on the Higgs boson production cross section times $\mathcal{B}(\mathrm{H} \to \mu\mu)$, relative to the SM expectation, in the region around the Higgs boson mass along with the expected exclusion limit obtained in the background-only hypothesis (dashed blue line) and in the signal-plus-background one (dashed red line) for the SM Higgs boson with $m_{\mathrm{H}} = 125\,\mathrm{GeV}$. On the right, the observed and expected local p-values as a function of m_{H} in the range $120 < m_{\mathrm{H}} < 130\,\mathrm{GeV}$.

Table 31: In each category, the chosen function used in the signal extraction, the observed post-fit uncertainty on the signal strength ($\Delta\mu$), and the maximum bias for the selected function form as measured in the studies presented in Section 8.1.2 and 8.2.

Category	Functional form	Post-fit $\Delta\mu$	Maximum bias (%)
ggH-cat0	Core–PDF	2.09	16.1%
ggH-cat1	Core–PDF	1.18	17.4%
ggH-cat2	Core–PDF	1.47	4.7%
ggH-cat3	Core–PDF	1.18	3.8%
ggH-cat4	Core–PDF	1.25	3.7%
VBF-cat0	BWZRedux	2.56	15%
VBF-cat1	BWZRedux	1.71	9.7%
VBF-cat2	BWZGamma	1.32	9.9%
VBF-cat3	BWZGamma	1.24	7.9%

between the “baseline” result already presented in Table 30 and the one obtained considering the uncertainty in the background model (implemented via spurious signals). The difference in the combined expected significance is found to be 0.6% and therefore neglected in the final result.

DRAFT

Table 32: Difference in the expected significance to $\mathrm{H} \rightarrow \mu\mu$ decays for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ in each category between the “baseline” result, which ignores systematic uncertainties in the background model, and the one obtained adding spurious signals with rate proportional to the measured bias on μ from the studies presented in Section 8.1.2 and 8.2.

Category	Signif. w/o bias	Signif. with bias	Difference (%)
ggH-cat0	0.440σ	0.435σ	1.15%
ggH-cat1	0.806σ	0.794σ	1.51%
ggH-cat2	0.724σ	0.723σ	0.14%
ggH-cat3	0.877σ	0.876σ	0.11%
ggH-cat4	0.829σ	0.828σ	0.13%
VBF-cat0	0.380σ	0.378σ	0.53%
VBF-cat1	0.606σ	0.604σ	0.33%
VBF-cat2	0.739σ	0.736σ	0.41%
VBF-cat3	1.120σ	1.110σ	0.09%
Combination ggH	1.571σ	1.565σ	0.38%
Combination VBF	1.519σ	1.509σ	0.66%
Combination ggH and VBF	2.183σ	2.170σ	0.59%

DRAFT

A Comparison between data and simulation in each era

In this Section, the accuracy of the MC prediction in each era is tested by means of comparing it with the observation in data.

A.1 Comparisons for events belonging to the ggH BDT

From Fig. 65 to 83 comparisons are reported for events belonging to the ggH phase space. The dimuon p_T distribution for simulated Drell-Yan events has been re-weighted to match the observed one in data using events with $m_{\mu\mu}$ between 70 and 110 GeV, as described in Sec. 4.4. Overall, the agreement between data and simulation is good apart from local discrepancies in known distributions.

Looking at the leading and subleading jet η distributions, shown in Fig. 75 and 77, good agreement between data and simulation is observed in the central region of the detector $|\eta| < 2.7$ across the three years. In the 2016 era, a spike in data is observed in the transition region between ECAL/HE and HF detectors, $|\eta| \approx 3.0$, while an higher prediction from MC by about 15-20% is present in the HF region, $|\eta_j| > 3.1$. In contrast, in the 2017 and 2018 eras, the observed trends in the HF region are pretty different from 2016: higher occupancy in data in the region $3.1 < |\eta_j| < 3.7$, and a higher prediction from simulation for jets with $|\eta_j| > 3.8$. In the forward region of the ECAL detector, $2.6 < |\eta| < 3.0$, in the 2017 era an excess in data is observed with respect to the expectation from MC (due to large detector noise), while in 2018 a higher prediction from MC is observed (due to tighter online/offline noise thresholds used during the data-taking). Finally, in both 2017 and 2018 eras, we observe a discrepancy between data and MC in the tail of the number of jets distribution ($\mathrm{N_{jets}} \geq 4$).

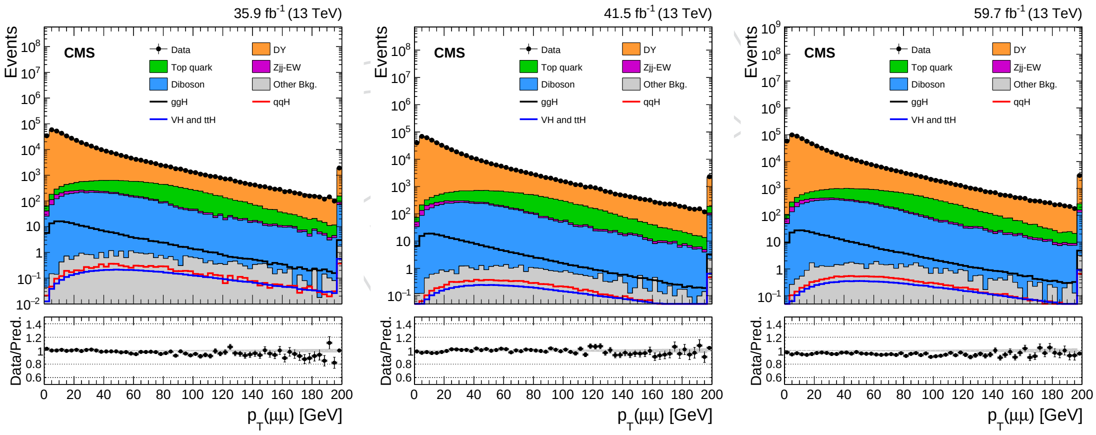

Figure 65: Dimuon p_T distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

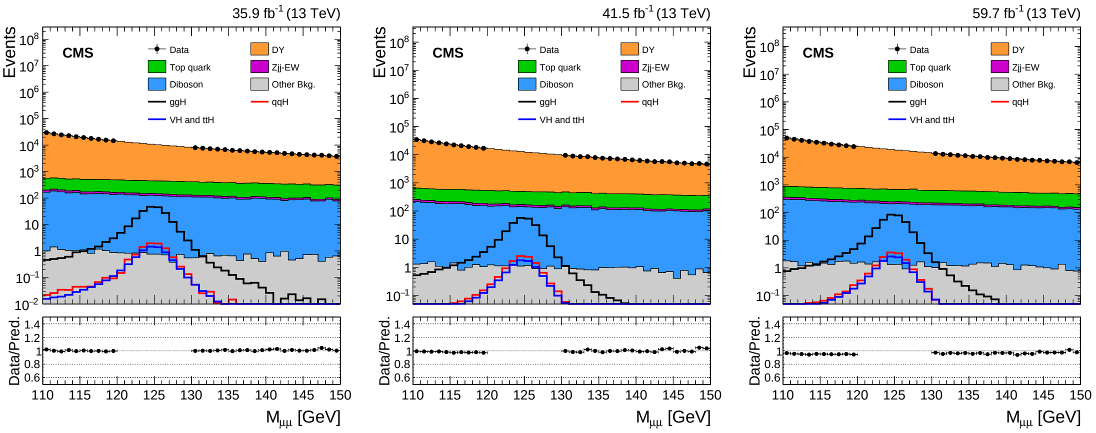

Figure 66: Dimuon mass distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

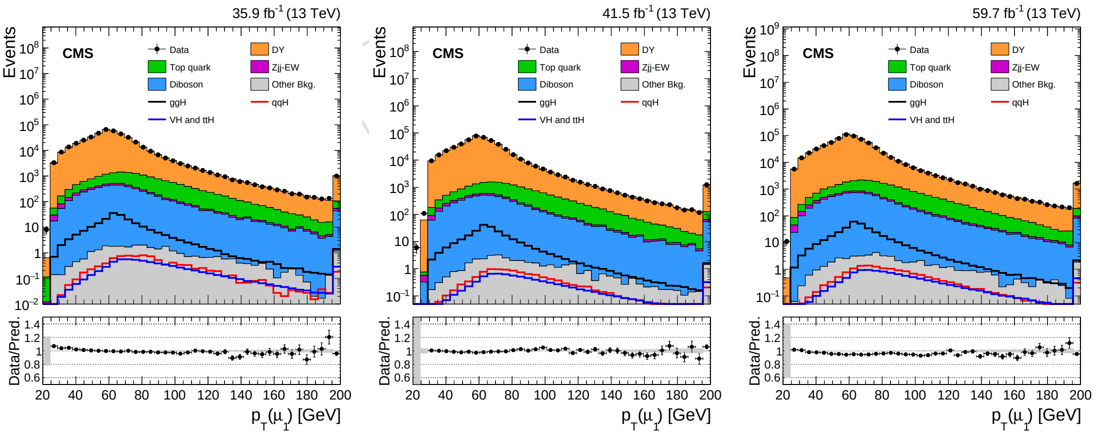

Figure 67: Leading muon p_{T} distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

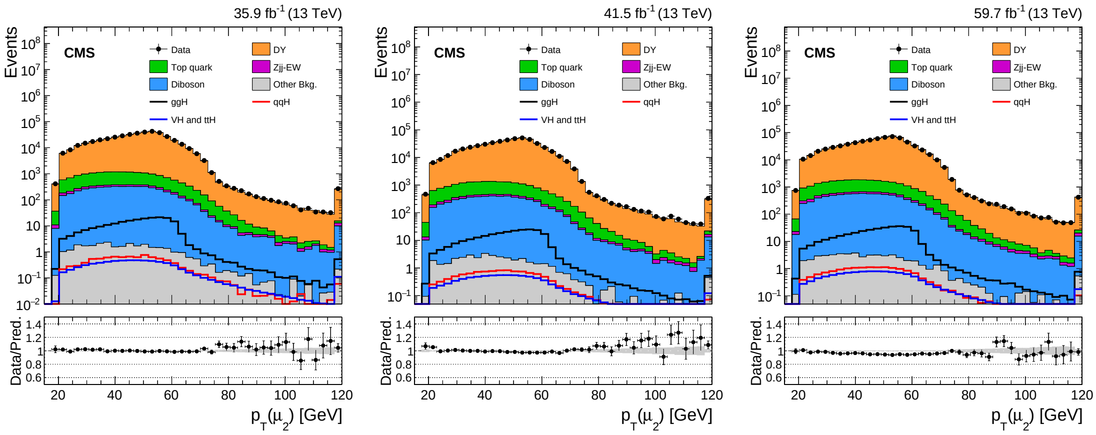

Figure 68: Subleading muon p_T distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

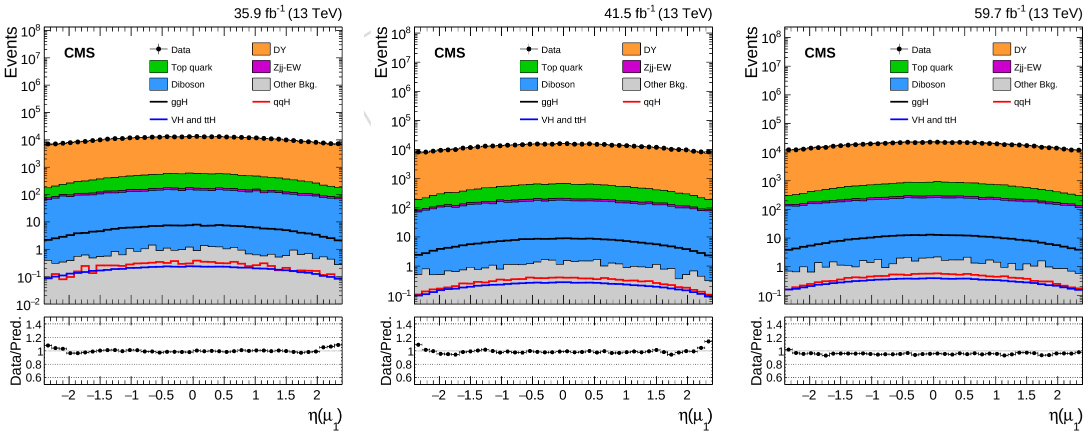

Figure 69: Subleading muon p_T distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

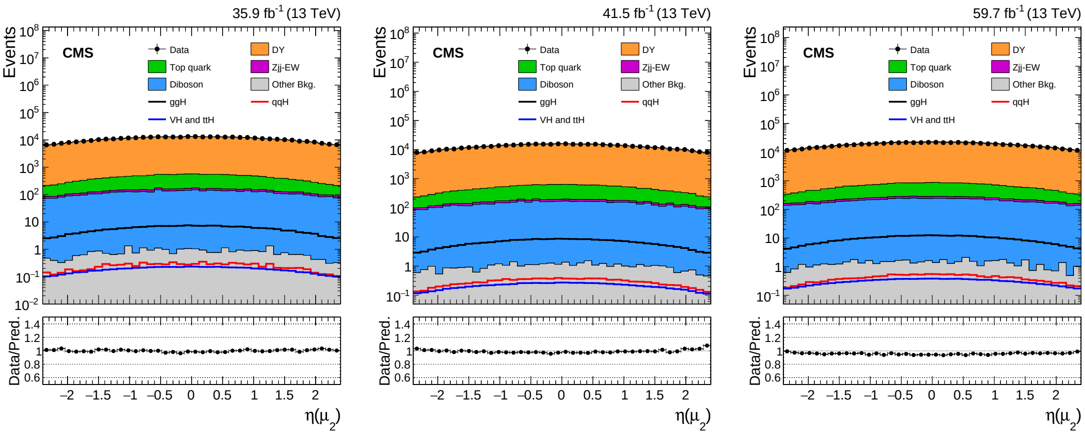

Figure 70: Subleading muon p_T distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

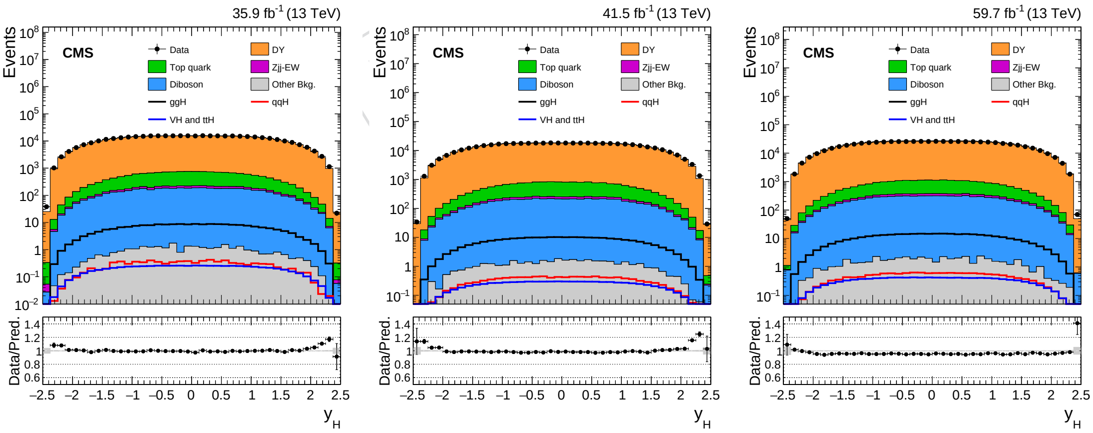

Figure 71: Dimuon rapidity distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

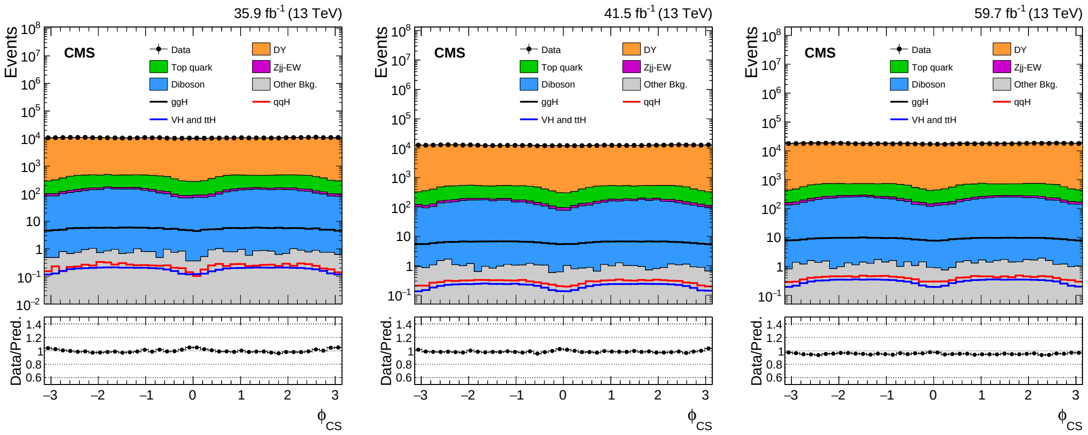

Figure 72: Dimuon ϕ Collins-Soper distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

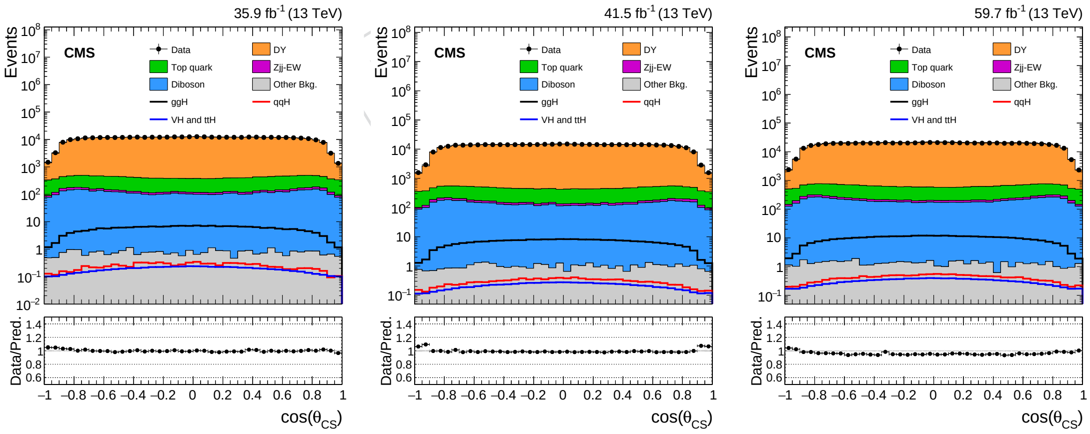

Figure 73: Dimuon $\cos\theta$ Collins-Soper distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

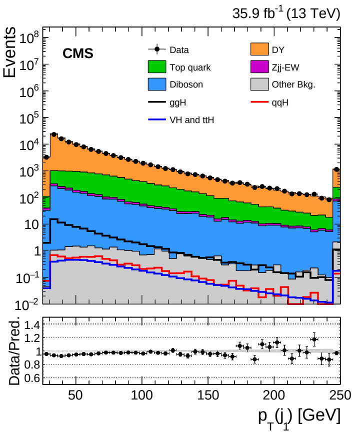

Figure 74: Leading jet p_T distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

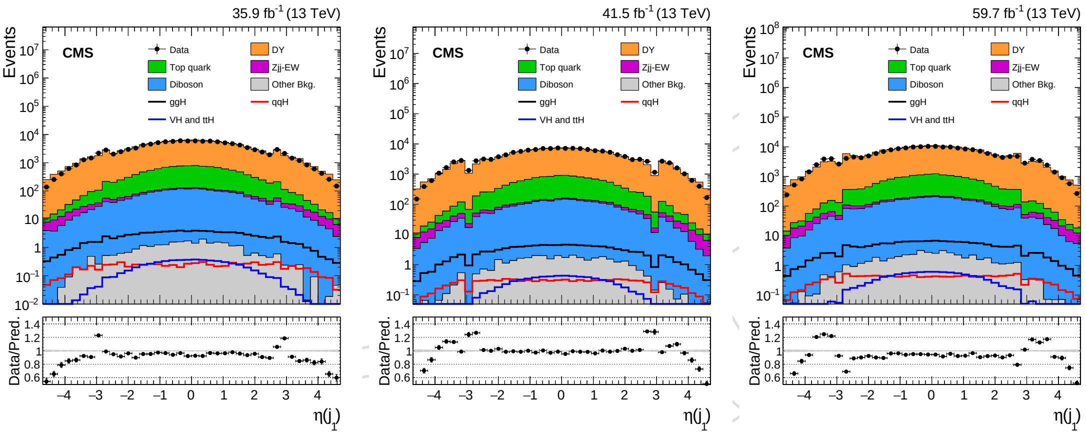

Figure 75: Leading jet η distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

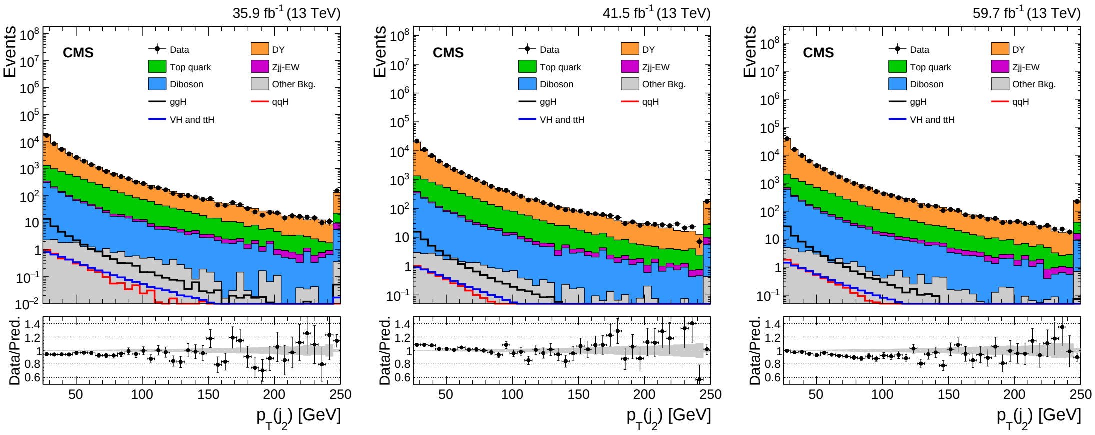

Figure 76: Subleading jet p_T distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

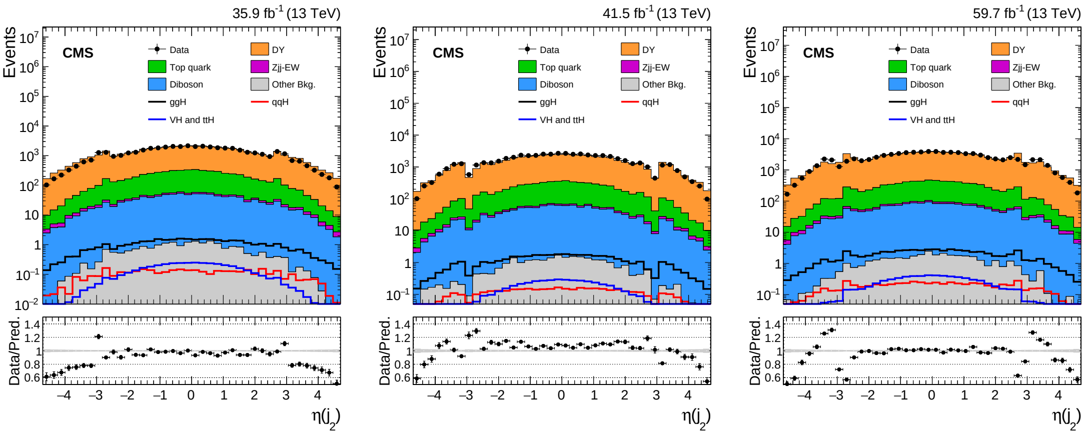

Figure 77: Subleading jet η distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

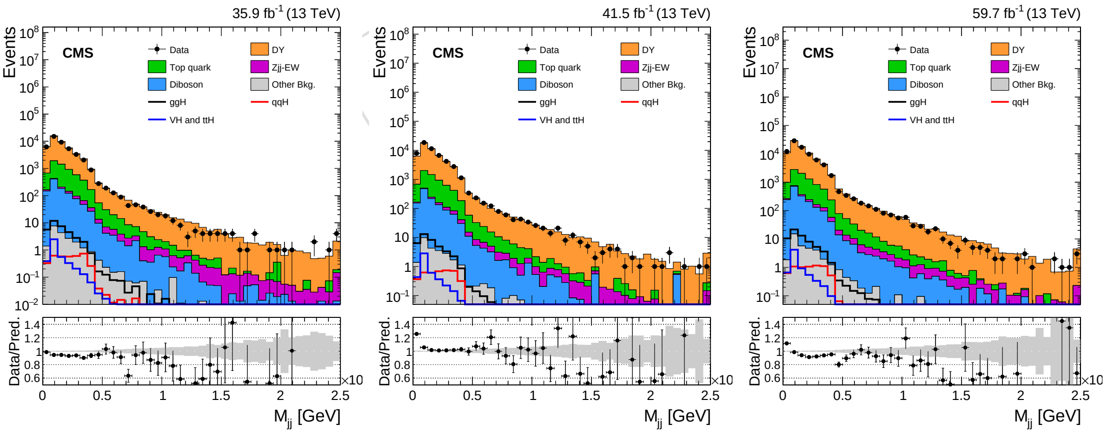

Figure 78: Dijet invariant mass (m_{jj}) distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

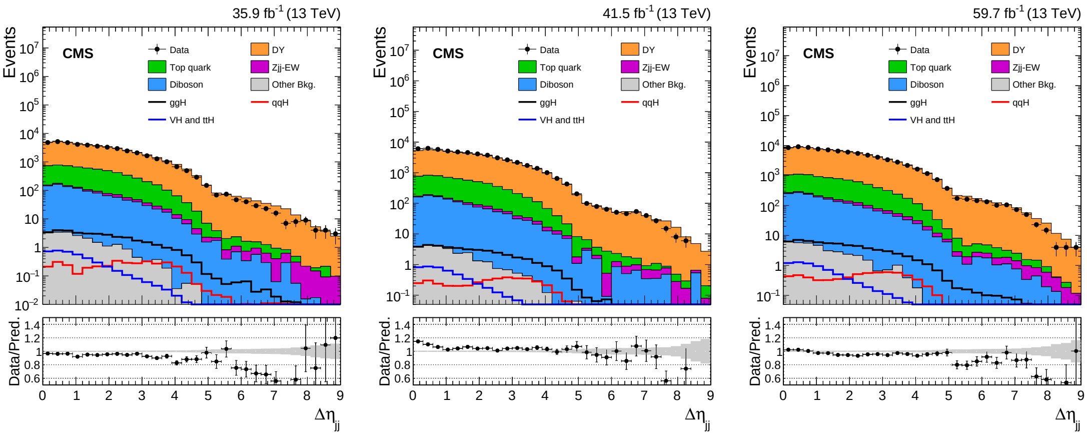

Figure 79: Dijet η separation ($\Delta\eta_{\mathrm{jj}}$) for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

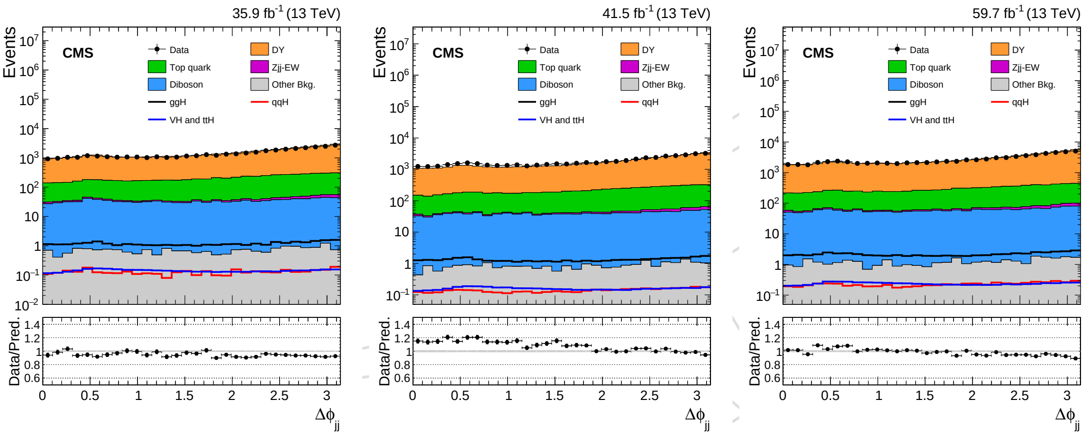

Figure 80: Dijet azimuthal angle separation ($\Delta\phi_{\mathrm{jj}}$) for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

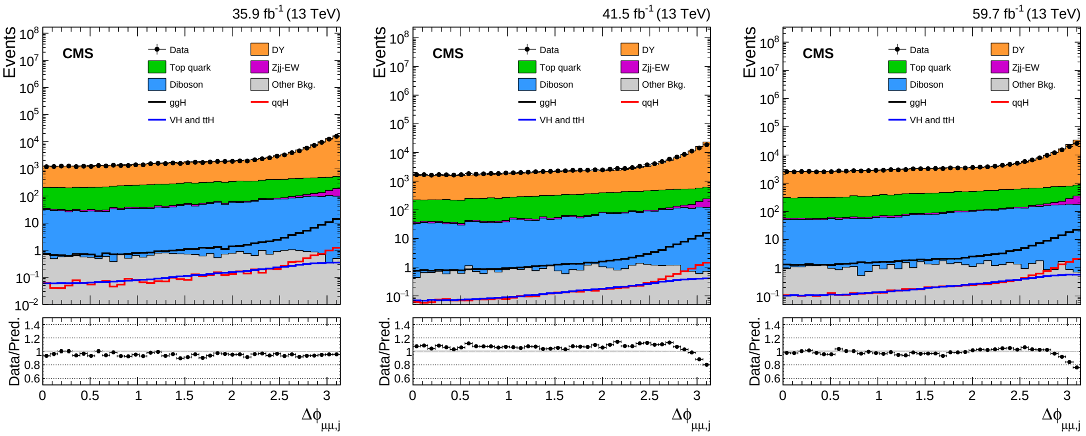

Figure 81: $\Delta\phi(\mu\mu, j)$ distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

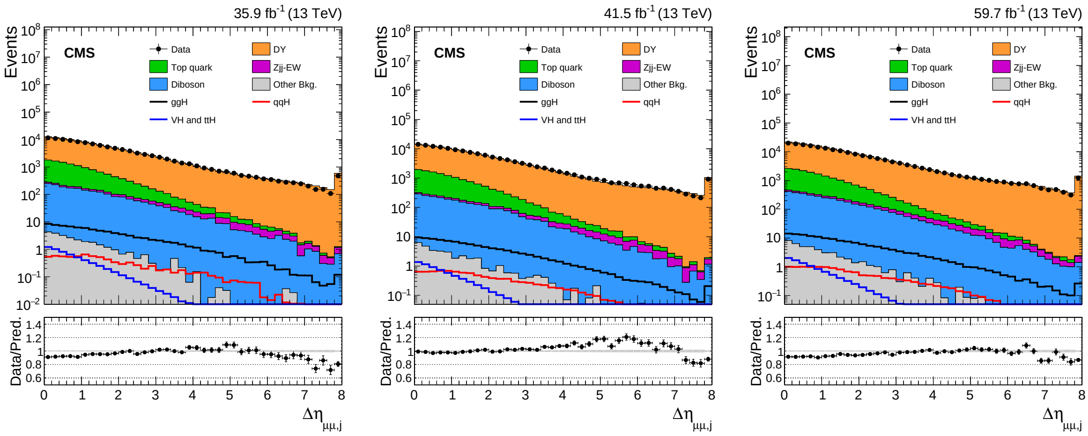

Figure 82: $\Delta\eta(\mu\mu, j)$ distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

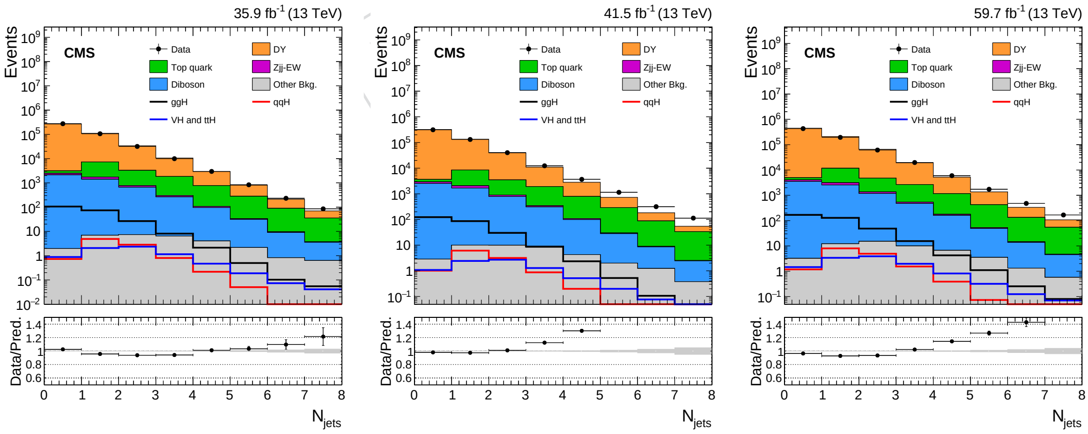

Figure 83: N_{jets} distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region of the ggH phase space. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

A.2 Comparisons for events belonging to the VBF BDT

From Fig. 84 to 104 comparisons are shown for events with two or more jets in the final state with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$ and $\Delta\eta_{\mathrm{jj}} > 2.5$ and leading jet p_{T} larger than 35 GeV. The dimuon p_{T} distribution for simulated Drell-Yan events has been re-weighted to match the observed one in data using events with $m_{\mu\mu}$ between 70 and 110 GeV, as described in Sec. 4.4. Overall, the agreement between data and simulation is good apart from local discrepancies in known distributions. The trends observed in the leading and subleading jet η distributions are similar to those observed and described in Sec. A.1. A residual bias in the 2017 jet energy corrections of forward jets with p_{T} smaller than 60 GeV causes a difference in absolute normalization between data and simulation of about 10%, as proved by Fig. 95 and Fig. 98. Moreover, clear trends are observed in the $\mathrm{R}_{p_{\mathrm{T}}}$ distributions in 2017 and 2018 eras, which point to non perfect residual jet-energy-calibrations (JECs). This hypothesis is also supported by the worse agreement between data and MC in the $\mathrm{N}_{\mathrm{jets}}$ distribution observed in 2017 and 2018 with respect to the 2016 data taking (see Fig.103). Finally, a discrepancy in the tail of the soft-jet distribution is observed in both 2017 and 2018 eras, which is most-likely due to the PYTHIA tune (TuneCP5) used in 2017 and 2018 MC productions.

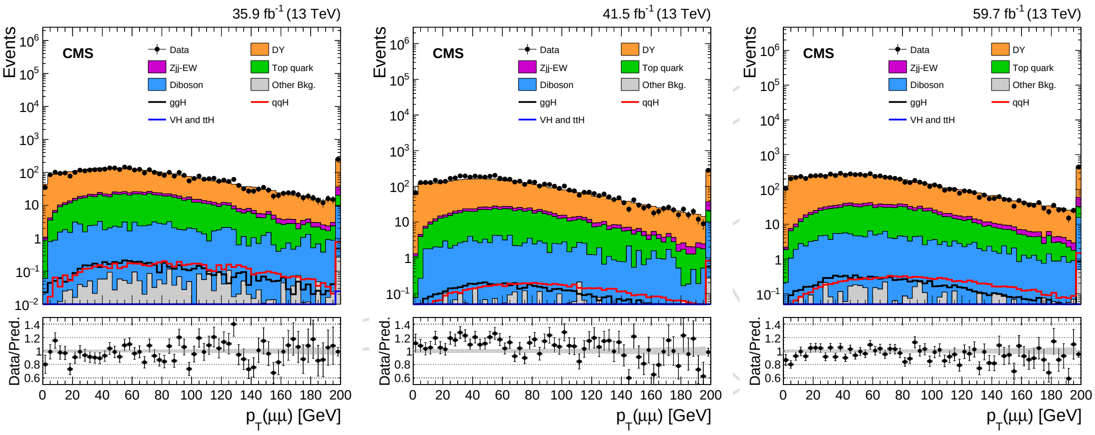

Figure 84: Dimuon p_{T} distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

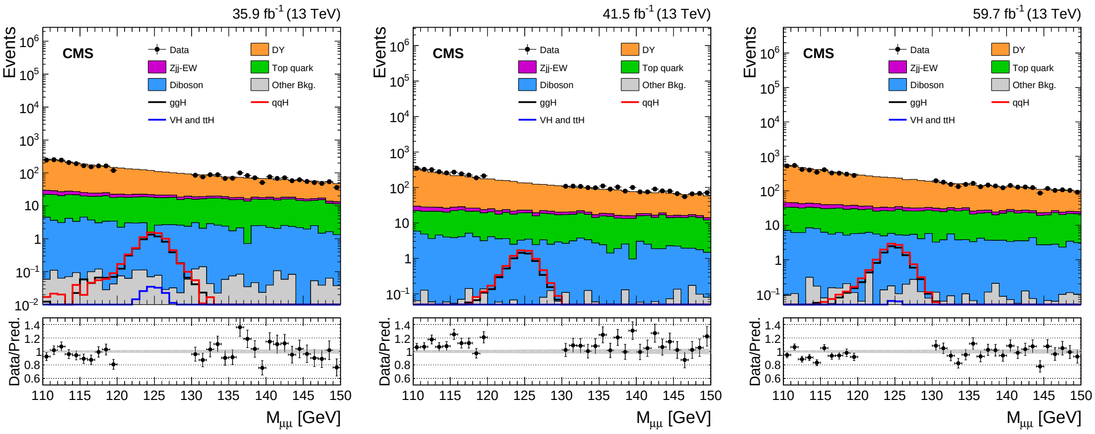

Figure 85: Dimuon mass distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

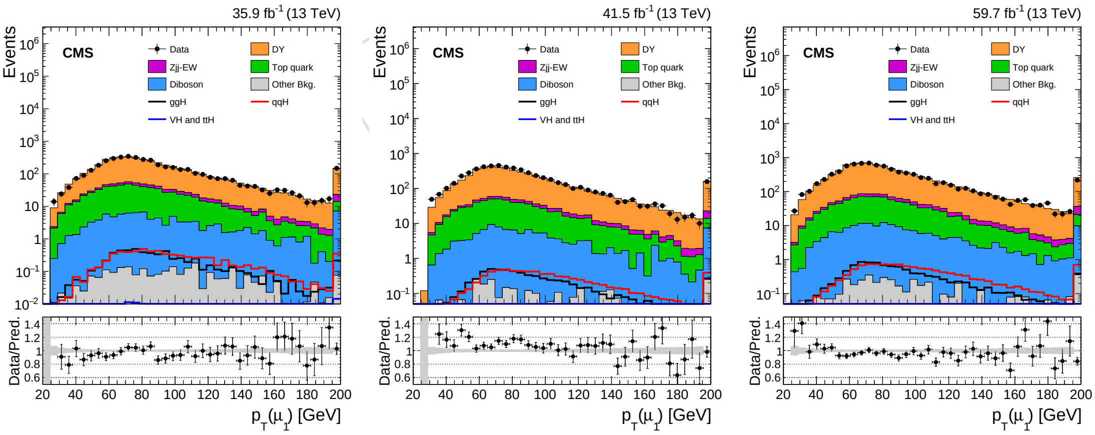

Figure 86: Leading muon p_{T} distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

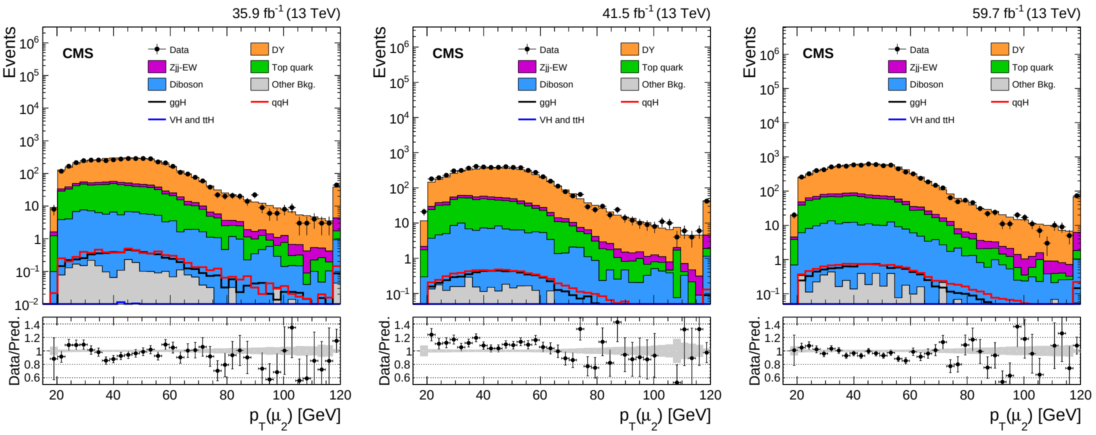

Figure 87: Subleading muon p_T distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{jj} > 400\,\text{GeV}$, $\Delta\eta_{jj} > 2.5$, and $p_T^{j1} > 35\,\text{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

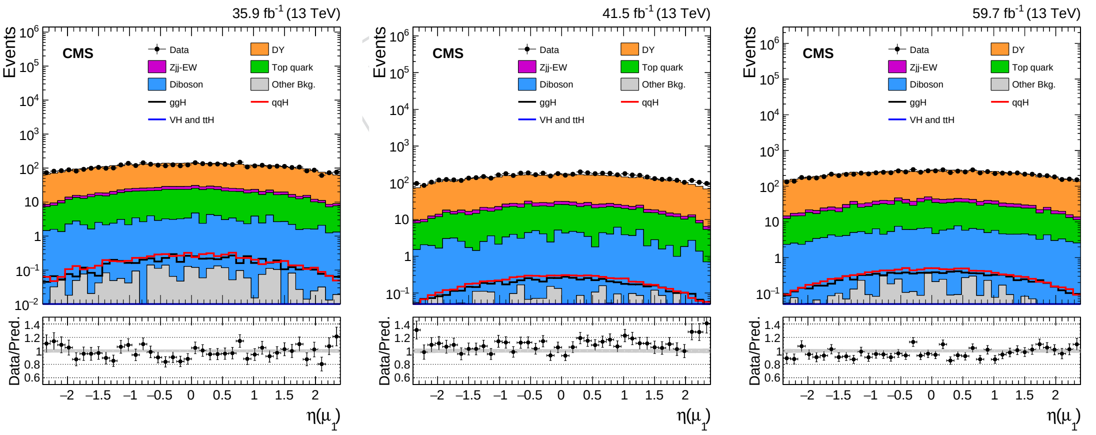

Figure 88: Subleading muon p_T distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{jj} > 400\,\text{GeV}$, $\Delta\eta_{jj} > 2.5$, and $p_T^{j1} > 35\,\text{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

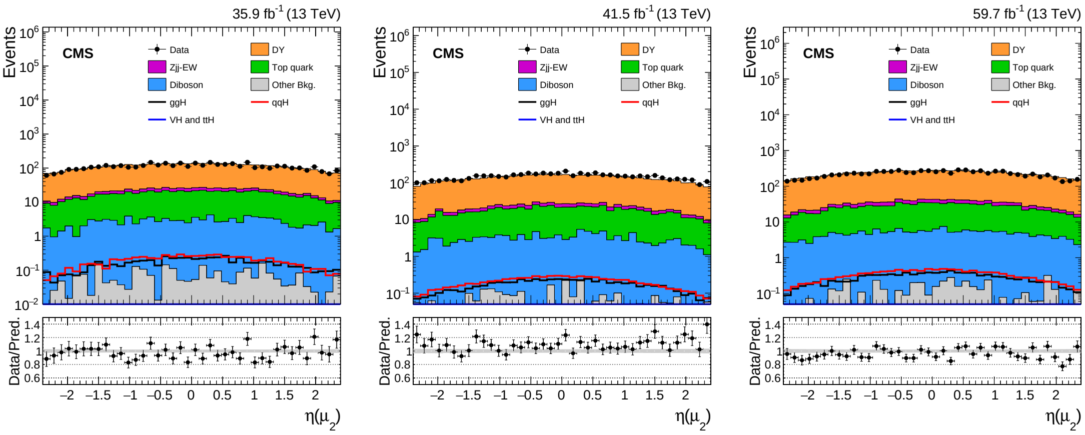

Figure 89: Subleading muon p_{T} distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

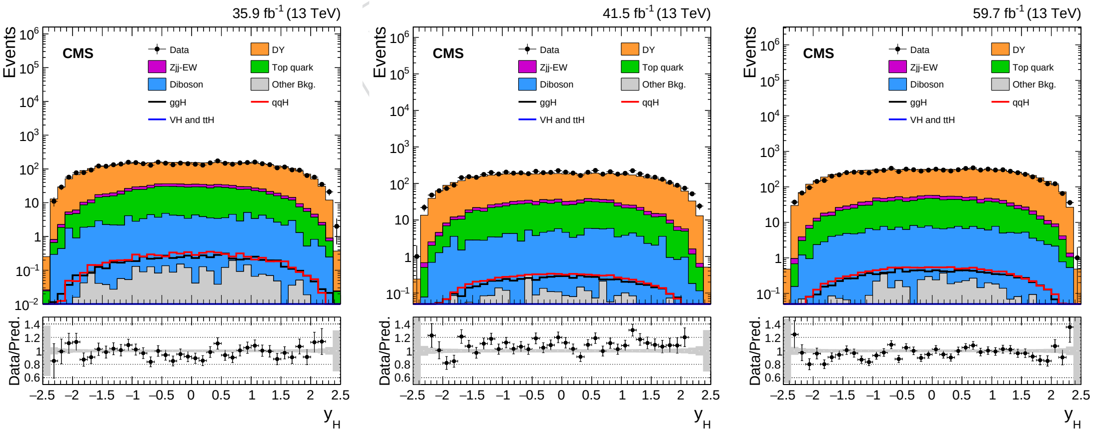

Figure 90: Dimuon rapidity distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $N_{\mathrm{jets}} \geq 2$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

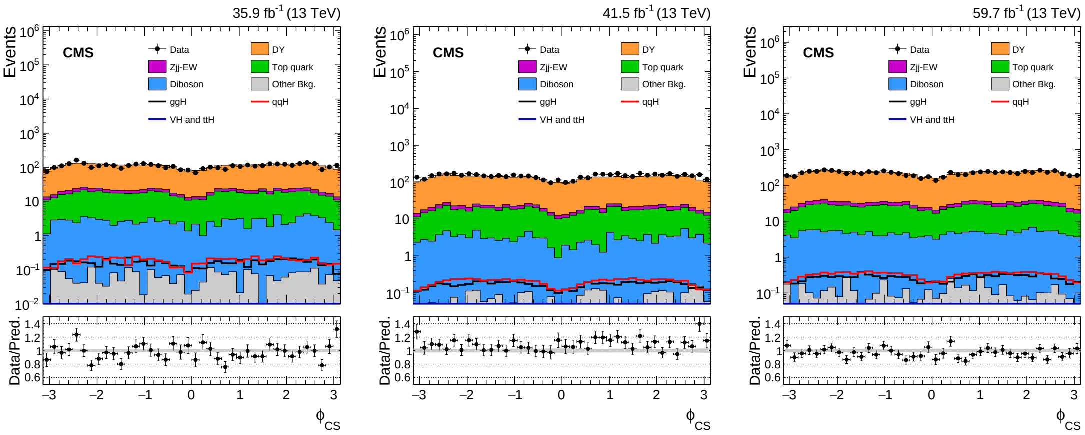

Figure 91: Dimuon ϕ Collins-Soper distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $N_{jets} \geq 2$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

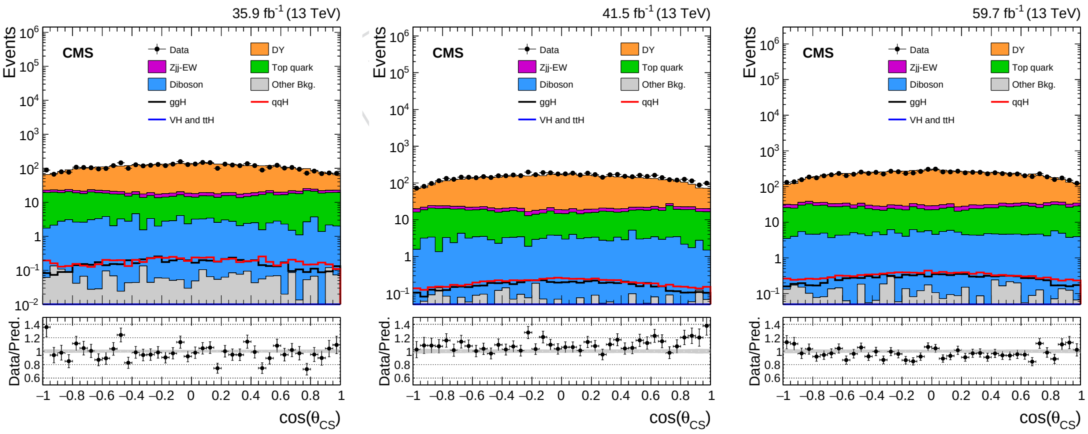

Figure 92: Dimuon $\cos\theta$ Collins-Soper distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $N_{jets} \geq 2$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

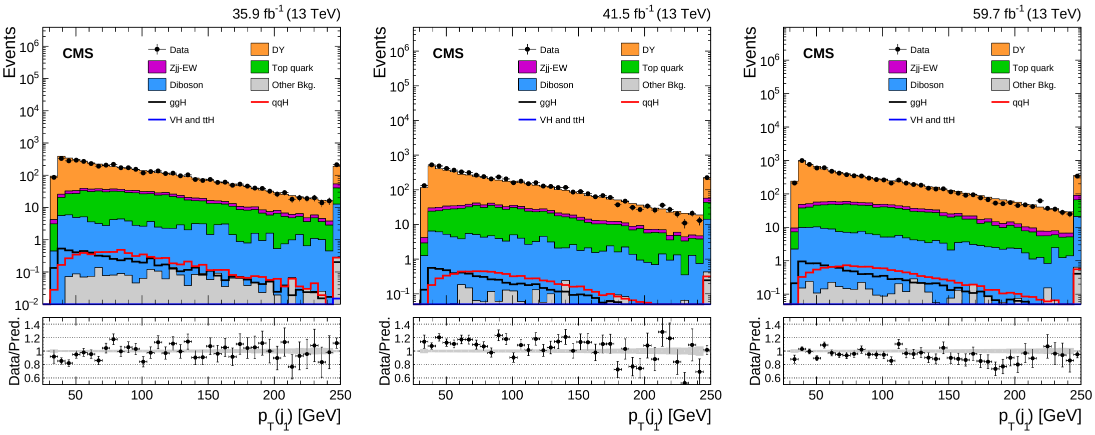

Figure 93: Leading jet p_{T} distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

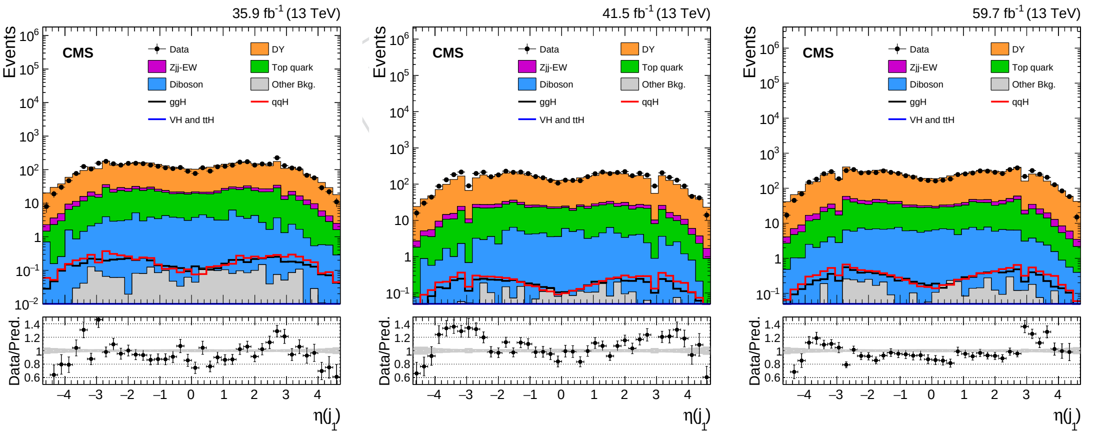

Figure 94: Subleading jet η distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

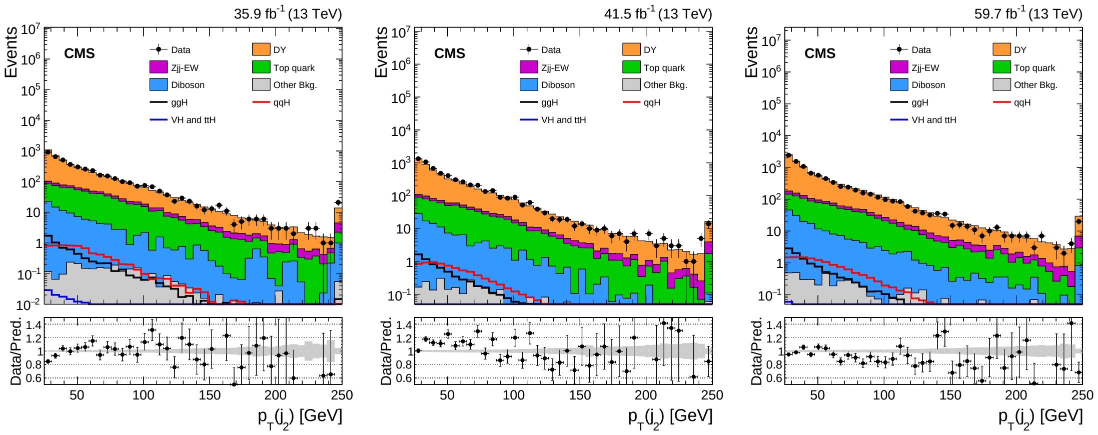

Figure 95: Leading jet p_T distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{jj} > 400\,\text{GeV}$, $\Delta\eta_{jj} > 2.5$, and $p_T^{j1} > 35\,\text{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

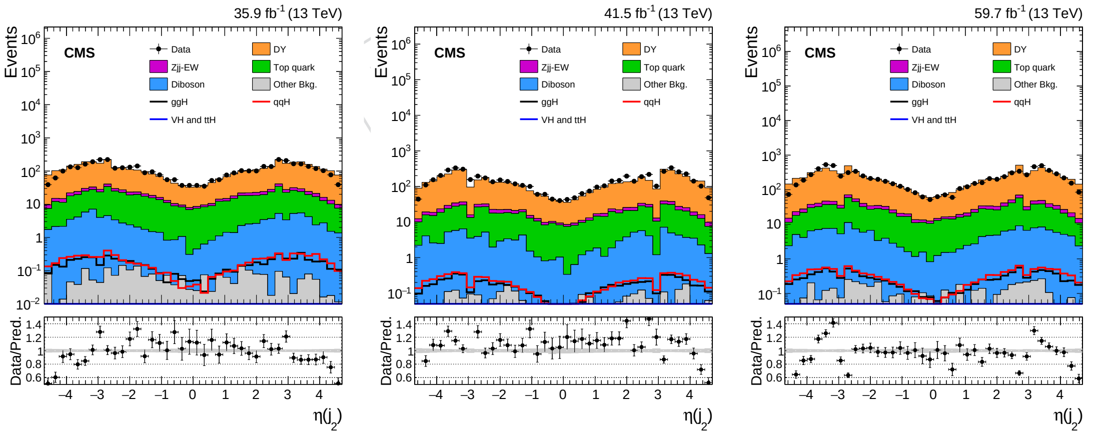

Figure 96: Subleading jet η distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{jj} > 400\,\text{GeV}$, $\Delta\eta_{jj} > 2.5$, and $p_T^{j1} > 35\,\text{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

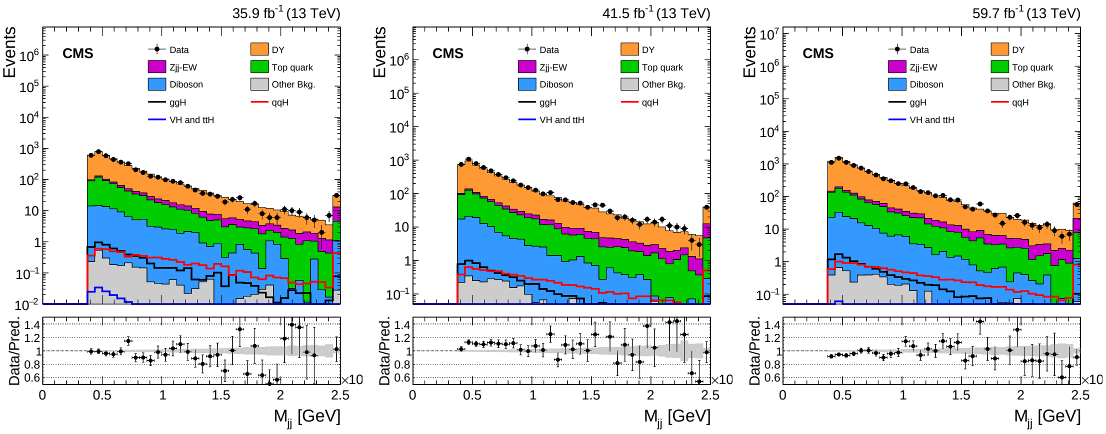

Figure 97: Dijet invariant mass (m_{jj}) distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

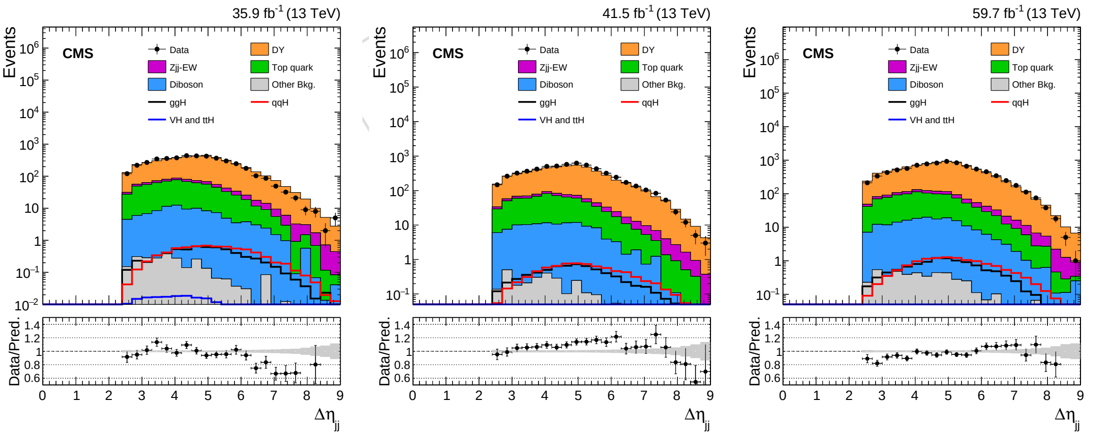

Figure 98: Dijet η separation ($\Delta\eta_{\mathrm{jj}}$) for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

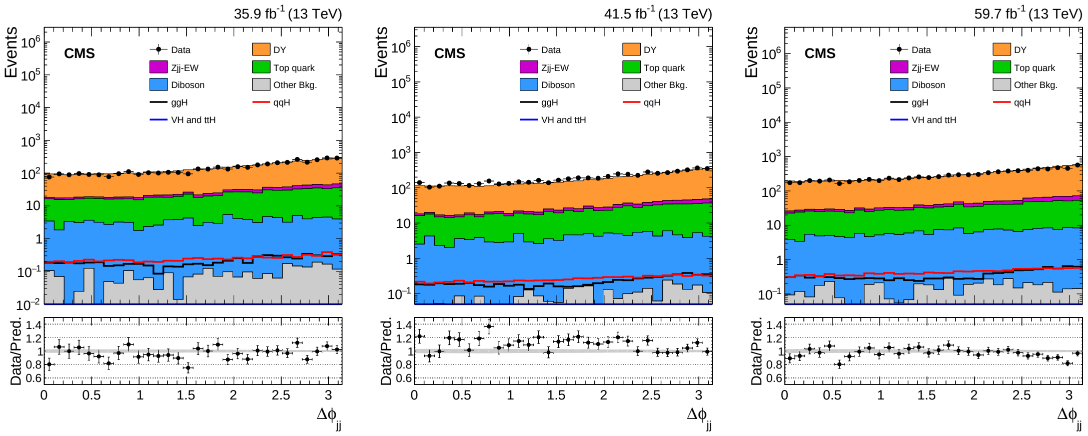

Figure 99: Dijet azimuthal angle separation ($\Delta\phi_{\mathrm{jj}}$) for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

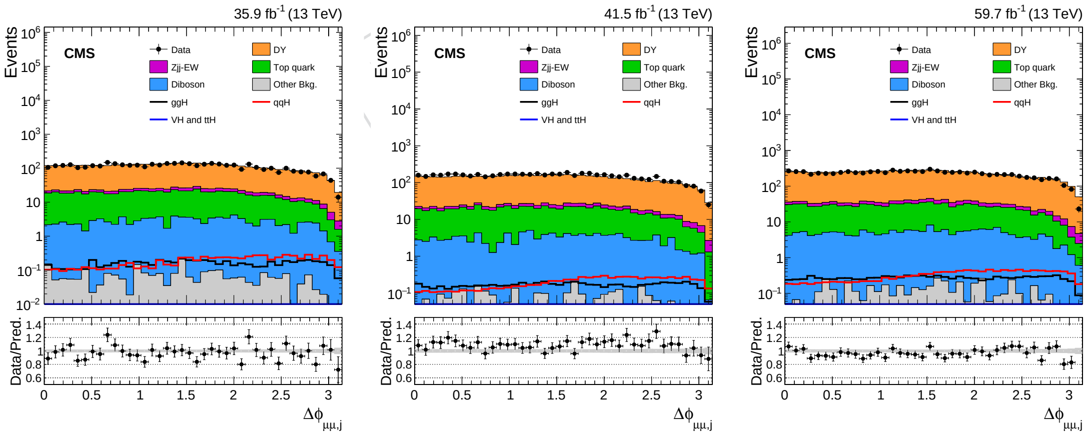

Figure 100: $\Delta\phi(\mu\mu, j)$ for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

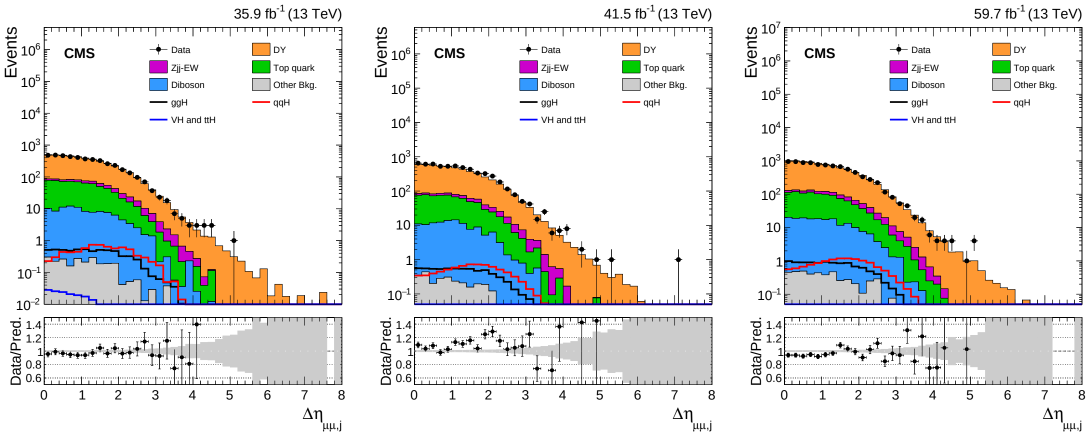

Figure 101: $\Delta\eta(\mu\mu, j)$ for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

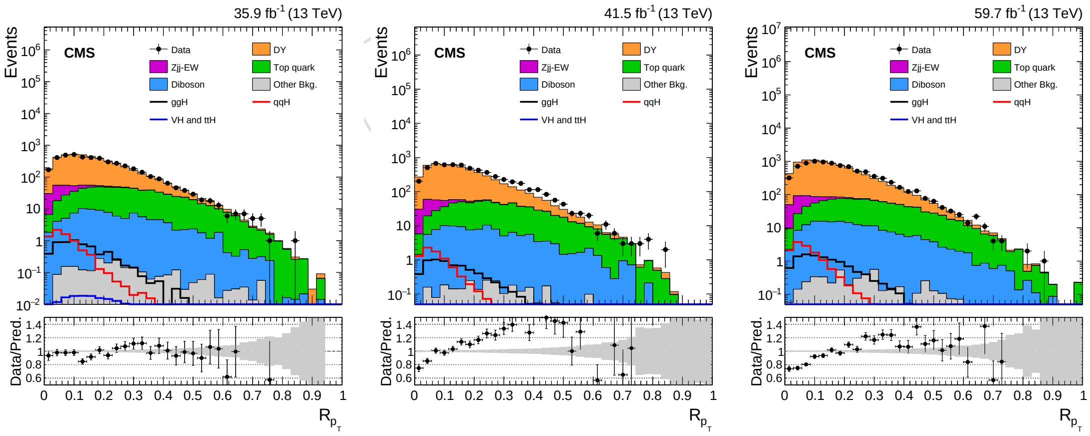

Figure 102: $R_{p_{\mathrm{T}}}$ for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

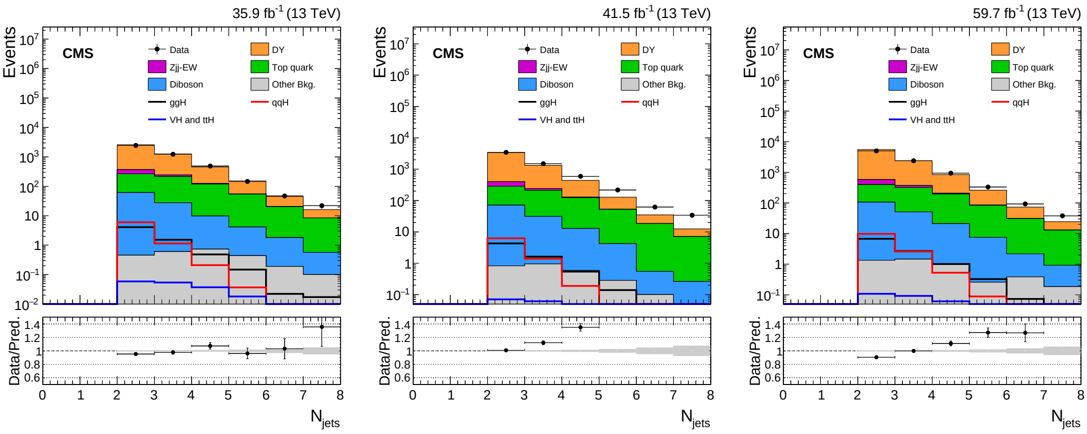

Figure 103: N_{jets} distributions for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

DRAFT

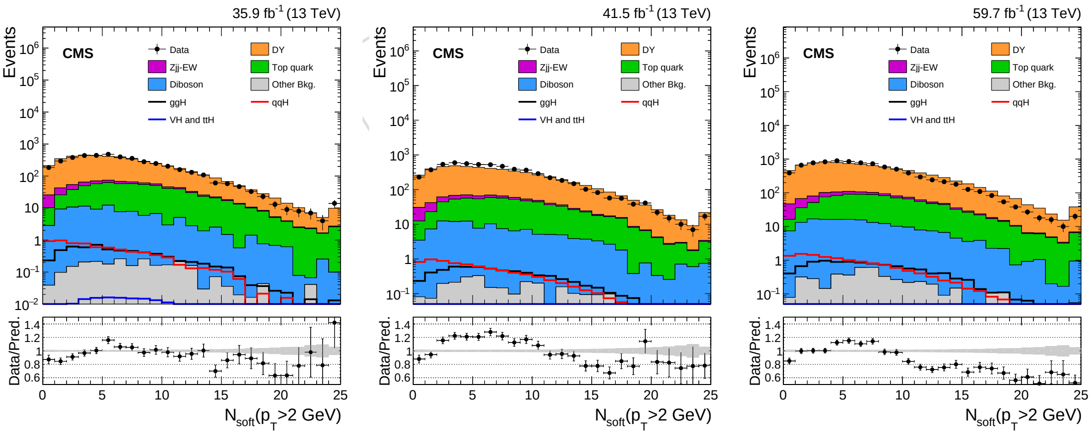

Figure 104: Number of soft-jets with p_{T} larger than 2 GeV for data (black points) and simulation (solid histograms) obtained from events in the signal fit region with $m_{\mathrm{jj}} > 400\,\mathrm{GeV}$, $\Delta\eta_{\mathrm{jj}} > 2.5$, and $p_T^{j1} > 35\,\mathrm{GeV}$. Comparisons are shown independently for each data taking period (era): 2016 (left), 2017 (middle), and 2018 (right).

B Jet energy scale uncertainties: full splitting

Table 33 and Table 34 list, in each BDT category (columns), the estimated values for each jet-energy-scale and jet-energy-resolution uncertainty source (rows) for the ggH and VBF processes, respectively. Uncertainties in the jet energy scale cause significant migration of events between the ggH and the VBF categories, therefore the effect of these nuisances is found to be largely anti-correlated across N_{jet} bins. The sources showing the largest impact on the predicted signal yields are: flavorQCD, relativeBal, relativeSample, pileupDataMC, and relativeFSR. According to Table 33, the total uncertainty due to jet-energy-scale affecting the ggH process ranges between 0.3–2.5% as a function of the ggH BDT-score, increasing up to 12–16% in the VBF categories. In contrast, the total uncertainty on the predict VBF signal yields due to jet-energy-scale is found to be larger than ggH in the ggH categories, ranging from 1.5 to 7%, and significantly smaller in the VBF ones ranging between 3–5%. On the other hand, the uncertainty due to jet energy resolution has a slightly smaller impact on the predicted signal yields. In the ggH categories, the total jet-energy-resolution uncertainty affecting the ggH (VBF) pre diction ranges between 0.1–1% (0.1–2.5%) as a function of the BDT score. In contrast, the effect on the ggH (VBF) prediction in VBF categories moves from 10–5% (4–6%) as a function of the VBF BDT score.

Table 33: Uncertainties for the ggH process in both the ggH and VBF BDT categories obtained by propagating the effect of each source of uncertainty related to jet energy scale and jet energy resolution to the expected signal yields. Uncertainties smaller than 0.1% are not reported.

Source of Uncertainty	ggH Categories					VBF Categories			
Jet energy scale	cat0	cat1	cat2	cat3	cat4	cat0	cat1	cat2	cat3
flavorQCD	0.994/1.006	—	1.008/0.992	1.011/0.989	1.013/0.989	0.957/1.043	0.965/1.024	0.948/1.019	0.994/1.043
pileupDataMC	0.999/1.001	1.001/0.999	1.006/0.994	1.005/0.994	1.001/0.999	0.948/1.061	0.945/1.044	0.931/1.044	0.989/1.057
relativeBal	—	0.999/1.001	0.998/1.003	0.996/1.004	0.998/1.002	1.044/0.960	1.029/0.957	1.041/0.965	1.062/0.975
relativeSample	—	0.999/1.001	0.995/1.005	0.995/1.005	0.998/1.003	1.063/0.944	1.032/0.954	1.084/0.907	1.014/0.994
relativeFSR	1.001/0.999	—	0.998/1.002	0.999/1.001	0.999/1.003	1.013/0.989	1.008/0.984	1.006/0.974	1.032/0.990
absStat	0.999/1.001	—	1.001/0.999	1.001/0.999	1.002/0.999	0.994/1.005	0.996/1.004	0.987/1.001	0.998/1.002
absScale	—	—	1.001/0.999	—/0.999	1.001/0.999	0.997/1.004	0.997/1.001	0.991/1.002	1.009/0.996
absMPFBias	0.999/1.001	—	1.002/0.999	1.002/0.998	1.003/0.998	0.993/1.009	0.993/1.001	0.984/1.006	1.002/1.016
singlePionECAL	1.001/0.999	—	0.998/1.002	0.998/1.002	0.998/1.003	1.008/0.992	1.003/0.991	1.005/0.990	1.011/1.002
singlePionHCAL	1.001/0.999	—	0.999/1.001	0.999/1.001	0.999/1.002	1.006/0.996	1.003/0.996	1.002/0.993	1.009/—
fragmentation	0.999/1.001	—	1.001/0.999	1.002/0.998	1.002/0.998	0.994/1.007	0.994/1.003	0.990/1.005	1.002/1.006
timePtEta	1.001/0.997	—/0.999	0.999/1.002	0.997/1.004	0.998/1.003	1.001/0.998	1.001/0.997	0.998/0.982	1.001/1.024
relativeJEREC1	—	—	—	1.001/0.999	1.001/0.999	0.999/1.003	0.998/—	0.999/1.005	1.002/0.997
relativeJEREC2	—	—	1.001/0.999	1.001/0.999	1.001/—	0.988/1.013	0.992/1.004	0.992/1.001	0.995/1.019
relativeJEREHF	—/0.999	—	1.001/0.999	—	0.999/1.002	0.988/1.011	0.991/1.011	0.980/1.002	0.989/1.037
relativePtBB	—	—	—	—	—/1.001	—	—	0.999/1.001	1.001/0.999
relativePtEC1	—	—	—	—	—	—/1.001	—/1.001	—/1.002	1.001/0.999
relativePtEC2	—	—	—	1.001/0.999	1.001/0.999	0.994/1.006	0.994/1.003	0.990/1.004	0.992/1.001
relativePtHF	—	—	—	—	—/1.001	0.990/1.009	0.990/1.006	0.989/1.004	0.998/1.030
relativeStatFSR	—	—	—	—	—	0.996/1.005	0.996/1.003	0.989/1.002	1.002/1.011
relativeStatEC	—	—	—/0.999	1.001/0.999	1.001/0.999	0.996/1.005	0.994/1.001	1.001/1.002	0.997/1.016
relativeStatHF	—	—	—	—	—/1.001	0.992/1.008	0.991/1.005	0.975/1.004	0.986/1.027
pileupPtRef	0.999/1.001	—	1.001/0.999	1.002/0.999	1.002/0.998	0.994/1.007	0.994/1.005	0.988/1.004	1.003/1.013
pileupPtBB	—/1.001	—	—	1.001/0.999	1.001/—	—	—/0.998	0.999/1.002	1.001/0.999
pileupPtEC1	0.999/1.001	—	1.001/0.999	1.002/0.998	1.003/0.997	0.999/1.005	0.995/1.004	0.995/1.003	0.991/1.002
pileupPtEC2	—	—	—	1.001/0.999	—	0.994/1.007	0.993/1.003	0.999/1.007	0.991/1.002
pileupPtHF	1.001/0.999	1.001/0.999	1.001/0.999	0.999/—	0.999/1.003	0.986/1.016	0.986/1.010	0.981/1.008	1.001/1.025
Jet energy resolution	cat0	cat1	cat2	cat3	cat4	cat0	cat1	cat2	cat3
jerBarrel	—/1.005	—	—/0.996	—/0.994	—/0.994	—/1.007	—/0.998	—/0.986	—/1.012
jerEC1	—/1.001	—	—/0.999	—/0.997	—/0.997	—/1.010	—/1.006	—/0.986	—/1.013
jerEC2low	—	—/0.999	—/0.997	—/0.997	—/0.995	—/1.068	—/1.038	—/1.009	—/1.007
jerEC2high	—	—	—/0.999	—	—/0.999	—/1.002	—/1.002	—/0.983	—/0.977
jerHFlow	—/0.998	—/0.999	—	—	—/1.004	—/1.026	—/1.025	—/1.014	—/0.998
jerHFhigh	—	—/0.999	—	—	—	—/1.001	—/0.998	—/0.997	—/0.996

DRAFT

Table 34: Uncertainties for the VBF process in both the ggH and VBF BDT categories obtained by propagating the effect of each source of uncertainty related to jet energy scale and jet energy resolution to the expected signal yields. Uncertainties smaller than 0.1% are not reported.

Source of Uncertainty	ggH Categories					VBF Categories			
Jet energy scale	cat0	cat1	cat2	cat3	cat4	cat0	cat1	cat2	cat3
flavorQCD	0.999/1.001	1.008/0.995	1.013/0.987	1.011/0.977	1.028/0.976	0.979/1.022	0.991/1.013	0.987/1.010	0.988/1.008
pileupDataMC	1.003/0.995	1.008/0.995	1.015/0.990	1.013/0.980	1.024/0.976	0.980/1.019	0.987/1.016	0.985/1.017	0.982/1.014
relativeBal	0.996/1.005	1.012/0.990	0.987/1.010	0.983/1.002	0.982/1.023	1.020/0.986	1.011/0.988	1.010/0.991	1.014/0.980
relativeSample	0.992/1.009	0.991/1.015	0.981/1.012	0.984/1.012	0.978/1.027	1.025/0.978	1.013/0.988	1.021/0.981	1.013/0.969
relativeFSR	0.999/—	1.002/0.999	0.992/1.003	0.994/1.001	0.995/1.006	1.006/0.995	1.004/—	1.001/0.996	1.008/0.994
absStat	0.999/—	1.002/—	1.002/0.997	1.001/0.998	1.003/0.997	0.997/1.002	0.998/1.003	—/1.001	0.998/1.001
absScale	0.999/1.001	1.001/—	—/0.997	—/0.999	1.003/0.998	0.998/1.002	—/1.002	0.999/—	0.998/—
absMPFBias	0.999/—	1.002/0.999	1.003/0.995	1.001/0.996	1.005/0.996	0.995/1.005	0.999/1.003	0.999/1.001	0.996/1.001
singlePionECAL	0.999/—	0.999/1.003	0.998/1.003	0.997/1.001	0.996/1.005	1.004/0.996	1.002/0.999	1.002/0.997	1.002/0.999
singlePionHCAL	—/0.999	0.999/1.002	0.998/1.002	0.998/1.001	0.997/1.003	1.002/0.998	1.002/0.999	1.001/0.998	1.001/—
fragmentation	0.999/—	1.002/0.999	1.003/0.997	1.001/0.998	1.004/0.997	0.997/1.003	0.998/1.002	0.999/1.001	0.999/1.001
timePtEta	cat0	1.002/0.997	—	0.995/—	0.998/1.005	1.001/—	1.002/0.999	1.003/0.998	0.996/0.998
relativeJEREC1	0.999/—	—	1.002/0.999	—/0.996	1.002/—	0.999/1.003	—/1.001	—/0.999	0.999/—
relativeJEREC2	—	1.003/0.999	—/0.995	1.002/0.998	1.007/0.995	0.995/1.006	0.999/1.001	0.996/1.006	0.996/0.999
relativeJEREHF	1.002/0.998	1.002/0.998	1.002/0.996	1.002/0.998	1.004/0.996	0.997/1.004	0.997/1.004	0.997/1.004	0.996/1.001
relativePtBB	—/0.999	—/1.001	1.001/0.999	—	—	—	—/1.001	—	—
relativePtEC1	—	1.001/—	—	—	1.001/—	—	—	—/1.001	0.999/—
relativePtEC2	—	1.001/—	1.002/0.996	—/0.998	1.004/0.998	0.995/1.006	0.998/1.001	0.998/1.004	0.998/1.001
relativePtHF	1.002/0.998	1.002/0.999	1.002/0.997	1.003/0.997	1.002/0.999	0.997/1.004	0.998/1.001	—/1.002	0.995/1.005
relativeStatFSR	0.999/—	1.002/0.999	0.992/1.003	—/0.999	1.002/0.999	0.999/1.002	—/1.001	1.001/0.996	0.999/—
relativeStatEC	0.999/1.001	1.001/—	1.001/0.998	1.001/0.998	1.003/0.998	0.997/1.003	0.999/1.001	0.999/1.002	—
relativeStatHF	1.001/—	1.002/0.998	1.003/0.998	—/0.999	1.003/0.997	0.998/1.002	0.996/1.002	0.999/1.004	—
pileupPtRef	0.999/—	1.002/—	1.003/0.998	1.002/0.998	1.003/0.997	0.997/1.001	0.999/1.002	0.999/1.003	0.997/1.002
pileupPtBB	0.999/—	0.999/1.001	1.002/—	—/0.999	—	—	—	—	—
pileupPtEC1	0.999/—	1.001/—	1.002/0.998	1.001/0.998	1.004/0.995	0.998/1.003	0.999/1.001	0.999/—	0.998/1.001
pileupPtEC2	0.999/—	1.002/0.999	1.002/0.996	—/0.998	1.004/0.997	0.996/1.003	—/1.001	0.999/1.003	0.997/1.001
pileupPtHF	1.001/0.999	1.002/0.998	1.005/0.997	—/0.996	1.003/0.996	0.996/1.005	0.996/1.003	0.999/1.003	0.998/1.002
Jet energy resolution	cat0	cat1	cat2	cat3	cat4	cat0	cat1	cat2	cat3
jerBarrel	—/1.001	—	—/1.006	—/0.994	—/0.995	—/1.012	—/0.999	—/0.992	—/0.993
jerEC1	—	—	—/0.997	—/0.997	—/0.998	—/1.010	—/1.003	—/0.993	—/0.991
jerEC2low	—/1.001	—/1.002	—/1.002	—/0.990	—/0.996	—/1.012	—/1.002	—/0.998	—/0.989
jerEC2high	—/1.001	—/0.999	—/0.999	—/0.999	—	—/1.015	—/1.003	—/0.998	—/0.973
jerHFlow	—/0.999	—	—/1.004	—/0.992	—/0.994	—/1.010	—/0.999	—/1.006	—/0.997
jerHFhigh	—	—	—/0.999	—/1.001	—/0.999	—/1.004	—/1.002	—/1.002	—/0.988

DRAFT

C Post-fit shape uncertainties vs background models

Figures 105–109 show the evolution of the the post-fit shape uncertainty on the total background in each ggH category as a function of $m_{\mu\mu}$, as obtained from a background-only fit to the data performed using the analytical functions reported in Table 27. Similarly, Figs 110–113 report the same trends for the VBF BDT categories. As expected, the relative size of the post-fit shape uncertainties is proportional to the amount of data events in the category as well as to the number degrees of freedom of the chosen analytical function. Breit-Wigner and power-law based function not modulated by polynomials show a “pole” in the mass region between 120–130 GeV, therefore they are expected to lead to a better performance in terms of significance. On the other hand, functions like FEWZ × Bernstein, BWZ × Bernstein, and sum of exponentials are showing a shape uncertainty less dependent on $m_{\mu\mu}$ depending on the number of polynomial degrees of freedom.

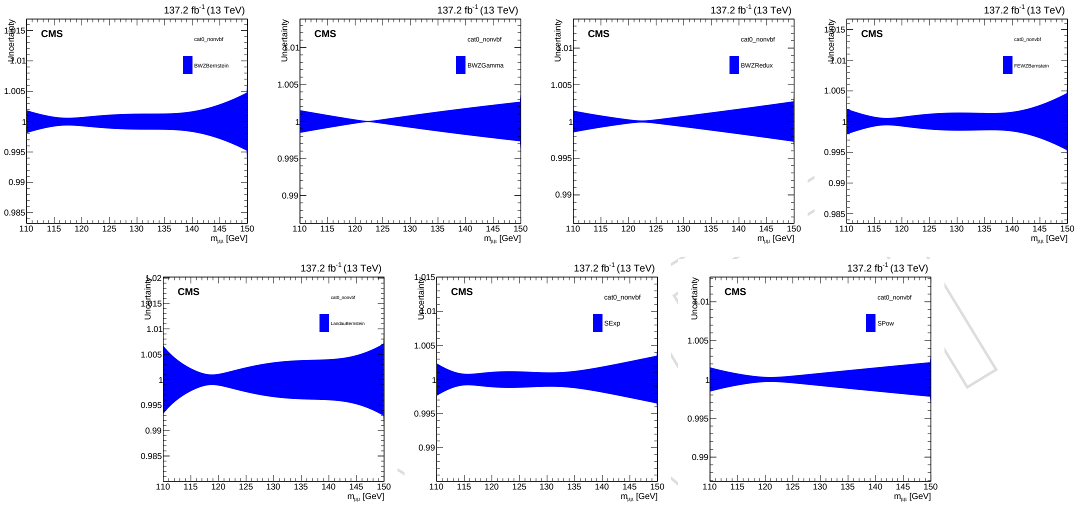

Figure 105: Post-fit shape uncertainty on the total background estimate obtained from a background-only fit to the data in ggH-cat0 for the following analytical functions (from left to right): BWZ×Bernstein, BWZGamma, BWZRedux, FEWZ×Bernstein, Landau×Bernstein, sum-of-exponentials, and sum-of-power-laws.

DRAFT

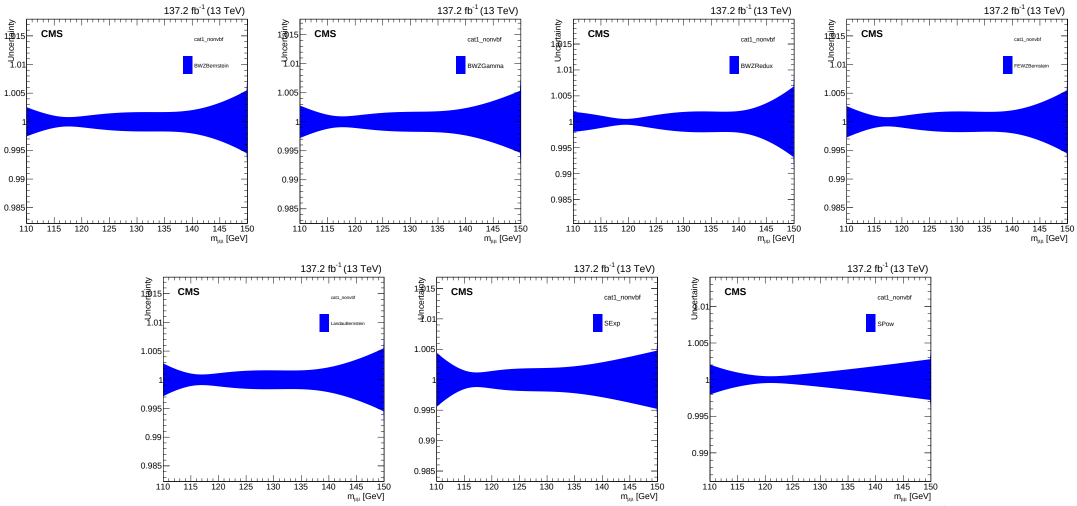

Figure 106: Post-fit shape uncertainty on the total background estimate obtained from a background-only fit to the data in ggH-cat1 for the following analytical functions (from left to right): BWZ×Bernstein, BWZGamma, BWZRedux, FEWZ×Bernstein, Landau×Bernstein, sum-of-exponentials, and sum-of-power-laws.

DRAFT

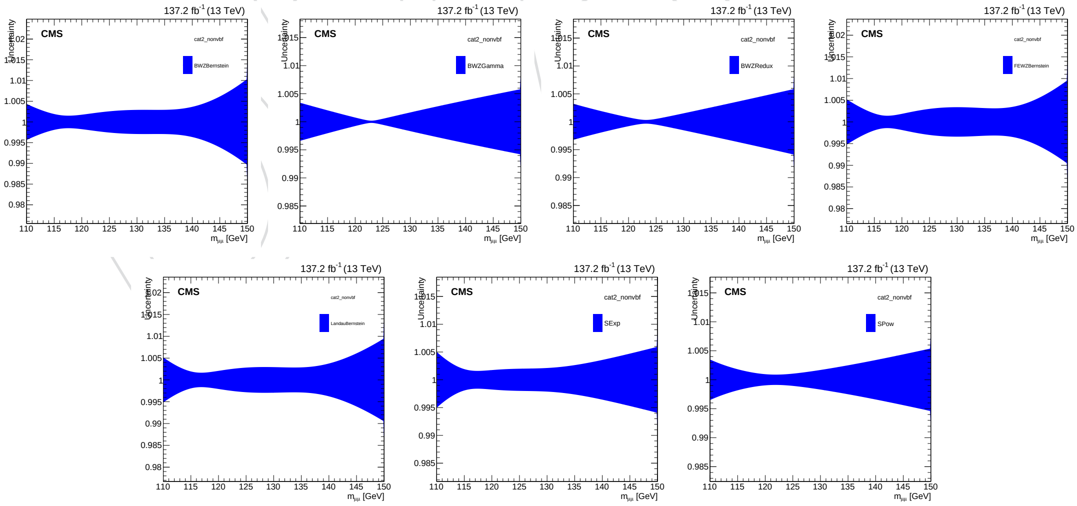

Figure 107: Post-fit shape uncertainty on the total background estimate obtained from a background-only fit to the data in ggH-cat2 for the following analytical functions (from left to right): BWZ×Bernstein, BWZGamma, BWZRedux, FEWZ×Bernstein, Landau×Bernstein, sum-of-exponentials, and sum-of-power-laws.

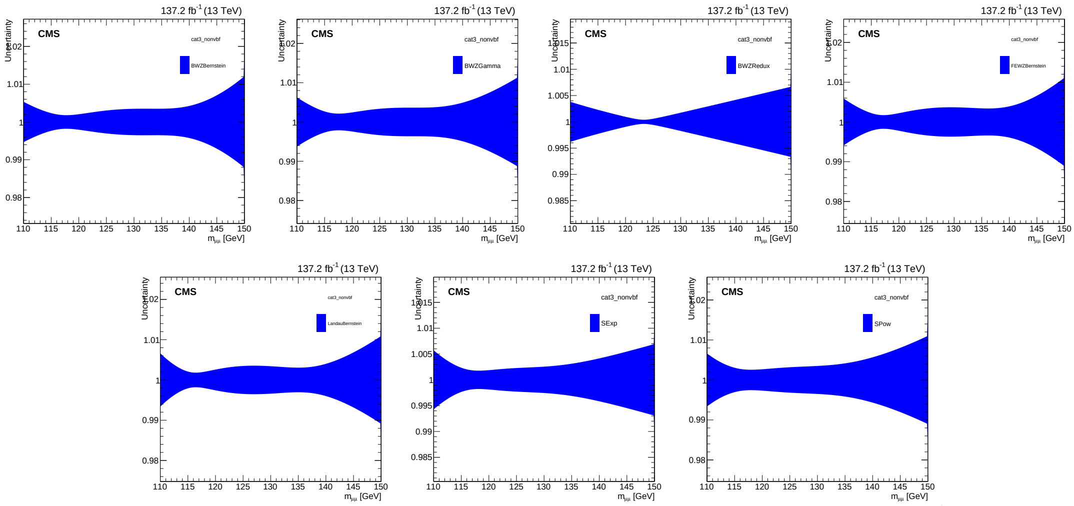

Figure 108: Post-fit shape uncertainty on the total background estimate obtained from a background-only fit to the data in ggH-cat3 for the following analytical functions (from left to right): BWZ×Bernstein, BWZGamma, BWZRedux, FEWZ×Bernstein, Landau×Bernstein, sum-of-exponentials, and sum-of-power-laws.

DRAFT

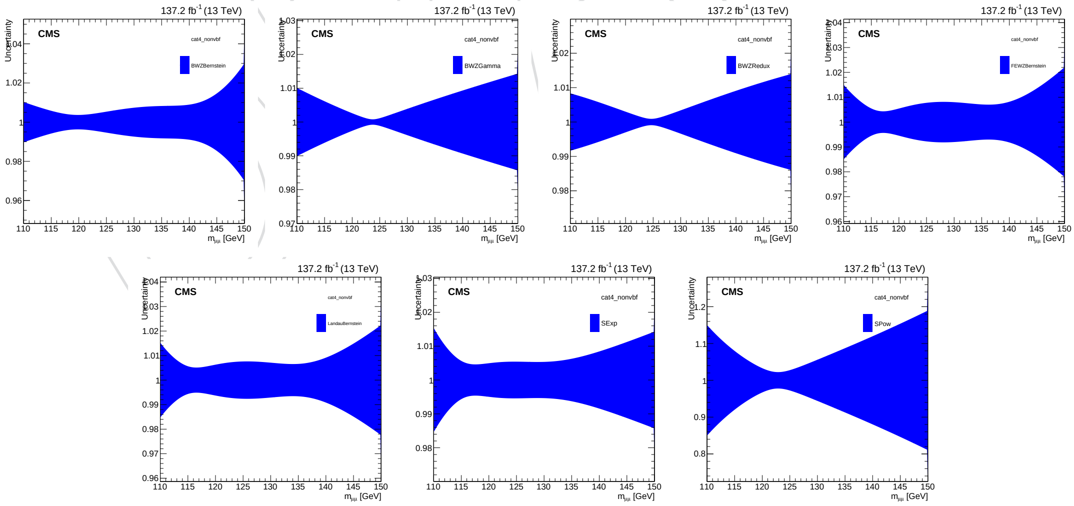

Figure 109: Post-fit shape uncertainty on the total background estimate obtained from a background-only fit to the data in ggH-cat4 for the following analytical functions (from left to right): BWZ×Bernstein, BWZGamma, BWZRedux, FEWZ×Bernstein, Landau×Bernstein, sum-of-exponentials, and sum-of-power-laws.

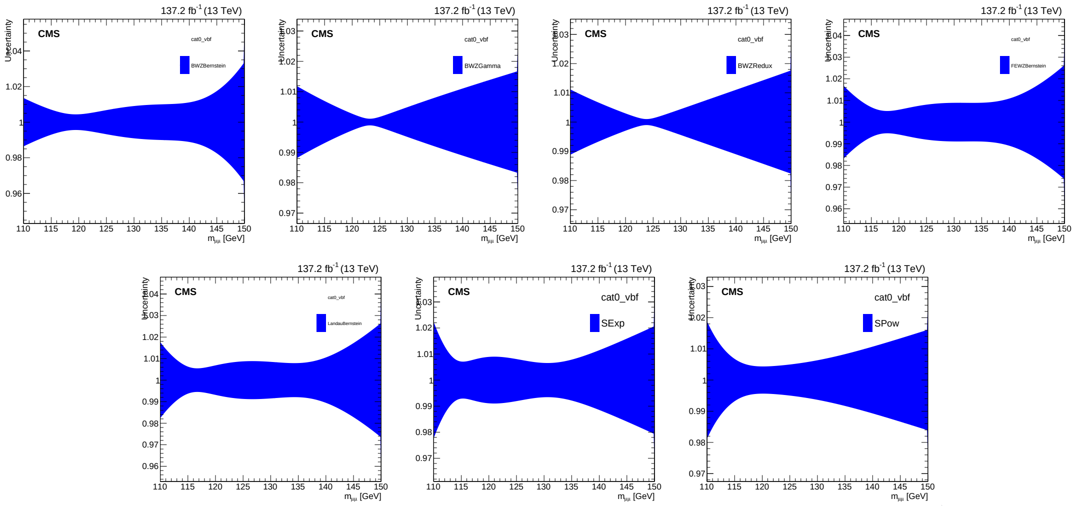

Figure 110: Post-fit shape uncertainty on the total background estimate obtained from a background-only fit to the data in ggH-cat0 for the following analytical functions (from left to right): BWZ×Bernstein, BWZGamma, BWZRedux, FEWZ×Bernstein, Landau×Bernstein, sum-of-exponentials, and sum-of-power-laws.

DRAFT

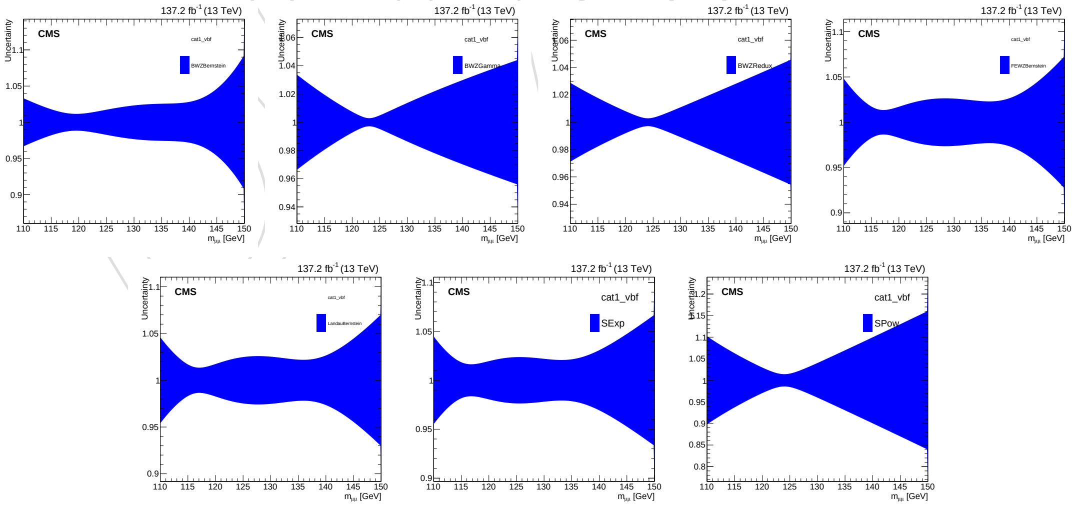

Figure 111: Post-fit shape uncertainty on the total background estimate obtained from a background-only fit to the data in ggH-cat1 for the following analytical functions (from left to right): BWZ×Bernstein, BWZGamma, BWZRedux, FEWZ×Bernstein, Landau×Bernstein, sum-of-exponentials, and sum-of-power-laws.

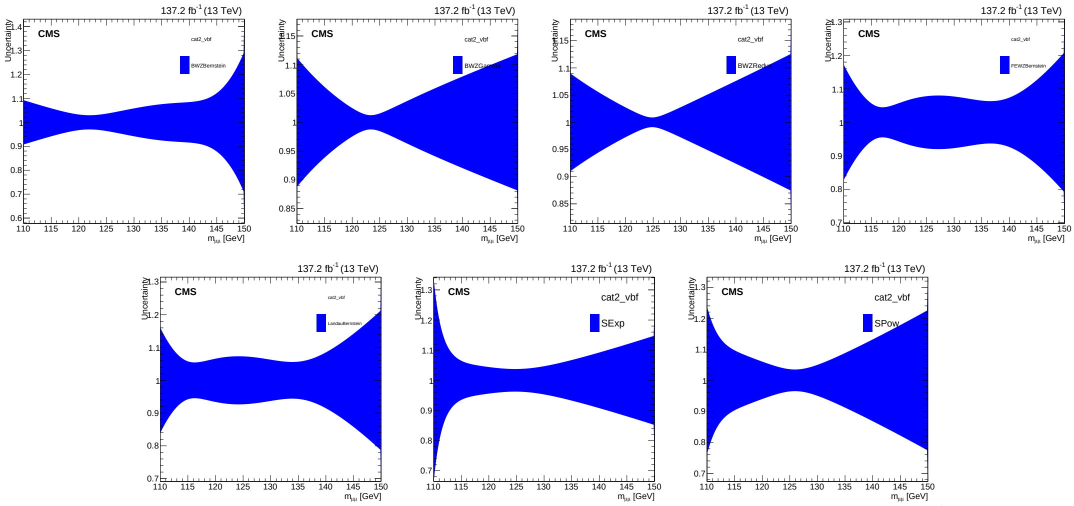

Figure 112: Post-fit shape uncertainty on the total background estimate obtained from a background-only fit to the data in ggH-cat2 for the following analytical functions (from left to right): BWZ×Bernstein, BWZGamma, BWZRedux, FEWZ×Bernstein, Landau×Bernstein, sum-of-exponentials, and sum-of-power-laws.

DRAFT

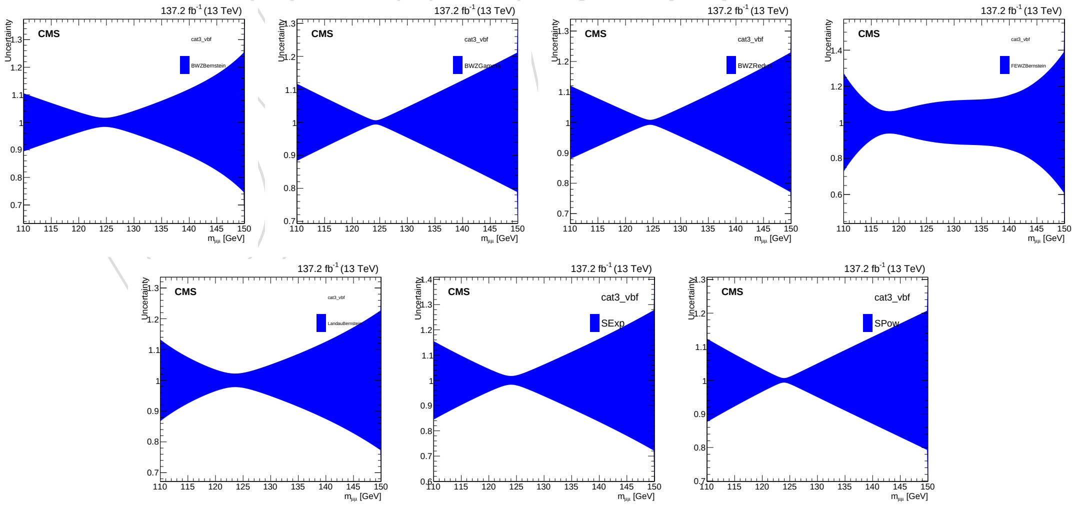

Figure 113: Post-fit shape uncertainty on the total background estimate obtained from a background-only fit to the data in ggH-cat3 for the following analytical functions (from left to right): BWZ×Bernstein, BWZGamma, BWZRedux, FEWZ×Bernstein, Landau×Bernstein, sum-of-exponentials, and sum-of-power-laws.

D Analysis performance in each era

In this Section, the 95% CL expected upper limits on the Higgs boson production cross section times $\mathcal{B}(\mathrm{H} \to \mu\mu)$ and the expected significance for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ are used as figure of merit to assess the evolution of the performance of each category across eras. Table 35 lists these expected upper limits and significances for each era and each category. Results in single categories don't exactly scale with luminosity because selection efficiency and detector performance were not stable across eras, as well as because BDTs have been trained separately for each year but the final categories have been optimized looking at the total combined significance. The combined limit from ggH categories scales according to the difference in the integrated luminosity of each data taking period ($35.9\,\mathrm{fb}^{-1}$ for 2016, $41.5\,\mathrm{fb}^{-1}$ for 2017, and $59.7\,\mathrm{fb}^{-1}$ for 2018). In contrast, in VBF categories, the performance in 2016 outperforms those based on 2017 and 2018 eras.

Table 35: Summary of expected 95% CL upper limits on μ and expected significances for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ in single BDT categories as well as for their combination. Upper bounds are obtained by modelling the background in ggH category via a BWZRedux function, while the BWZGamma is chosen for the VBF categories. Systematic uncertainties affecting the signal prediction in each category are neglected.

Category	UL 2016	UL 2017	UL 2018	σ_{exp} 2016	σ_{exp} 2017	σ_{exp} 2018
ggH-cat0	< 8.16	< 7.28	< 5.90	0.21	0.24	0.29
ggH-cat1	< 4.61	< 4.22	< 3.41	0.41	0.44	0.55
ggH-cat2	< 5.57	< 5.10	< 4.16	0.35	0.38	0.46
ggH-cat3	< 4.86	< 4.22	< 3.45	0.40	0.46	0.56
ggH-cat4	< 5.10	< 4.47	< 3.64	0.39	0.44	0.54
VBF-cat0	< 9.49	< 9.96	< 7.23	0.20	0.19	0.27
VBF-cat1	< 6.05	< 5.62	< 4.39	0.34	0.38	0.48
VBF-cat2	< 4.74	< 4.73	< 4.51	0.47	0.49	0.48
VBF-cat3	< 5.23	< 3.83	< 3.68	0.45	0.64	0.66
Comb. inclusive	< 2.36	< 2.12	< 1.72	0.80	0.89	1.10
Comb. VBF	< 2.73	< 2.35	< 2.18	0.78	0.86	0.95
Combined total	< 1.76	< 1.54	< 1.33	1.11	1.25	1.46

DRAFT

E Bias studies for $m_{\mathrm{H}} = 120\,\mathrm{GeV}$

Figure 114 (left) shows the measured bias on the signal strength for $m_{\mathrm{H}} = 120\,\mathrm{GeV}$ in each of the ggH categories when the background is modelled using the *core-PDF* method. The bias is observed to be smaller than 20% in each category and for each truth function used to generate the toys. In addition, Fig. 114 (right) reports the bias measured in toys where the signal has been injected in all the ggH categories. Also in this case, the bias measured as a function of different truth functions can be controlled within 20%. Bias studies have been performed using the same strategy/method detailed in Section 8.2, where the only difference is the tested Higgs boson mass hypothesis. Even though the bias is still within the upper threshold of 20%, the maximum bias in each category is larger for $m_{\mathrm{H}} = 120\,\mathrm{GeV}$ than $m_{\mathrm{H}} = 125\,\mathrm{GeV}$. This is expected because of two reasons:

- At $m_{\mathrm{H}} = 120\,\mathrm{GeV}$ the resonance is not placed anymore in the center of the fit range (110–150 GeV).
- The background shape around 120 GeV is steeper and the corresponding background prediction from the tested functional forms appear to have a larger spread in that region.

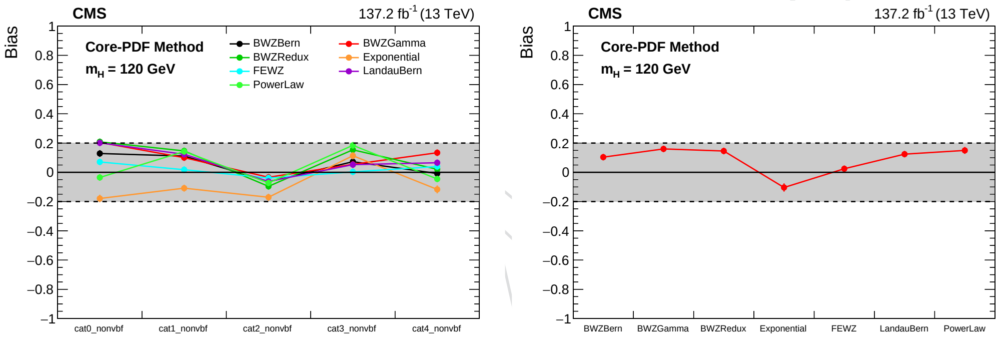

Figure 114: Mean bias on the signal strength obtained from the *core-PDF* method used to model the background in ggH categories. On the left, the measured bias in each of the ggH categories when a nominal signal at $m_{\mathrm{H}} = 120\,\mathrm{GeV}$ is injected. On the right, the bias measured when the signal is injected in all the ggH categories simultaneously.

The coverage within the 68% CL ($\approx 1\sigma$) interval is defined as the ratio between the observed number of toys for which $-2 \times \Delta Log\mathcal{L}(\hat{\mu}_{inj}) < 1$ and the total number of toys generated, where a signal with $\hat{\mu}_{inj} = 1$ and $m_{\{H} = 120\,\mathrm{GeV}$ is injected. Similarly, the coverage for 16% CL ($\approx 0.2\sigma$), 38% CL ($\approx 0.5\sigma$), 86% CL ($\approx 1.5\sigma$), and 95% CL ($\approx 2\sigma$) intervals have also been tested. Figure 115 shows that the method designed for the background estimation has good coverage properties for the tested intervals. The *core-PDF* fit shows, as expected from a discrete likelihood profile, a small overcoverage (by few %) in categories and against some truth models.

Figure 116 shows the measured bias on the signal strength for $m_{\mathrm{H}} = 120\,\mathrm{GeV}$ in the VBF categories obtained by fitting the toys generated via several truth function with BWZRedux, BWZGamma, and FEWZ × Bernstein. The number of degrees of freedom considered for each functional form and in each category are listed in Table 27, optimized for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$. The BWZRedux, BWZGamma, and FEWZ × Bernstein exhibit a bias within 20% against all possible truth models in VBF cat-1, VBF-cat2, and VBF-cat3. Differently from what was already observed for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ in Section 8.1.2, the maximum bias measured in VBF cat-0 (against sum of exponential) is around 30% for BWZRedux, 35% for BWZGamma, and 25% for

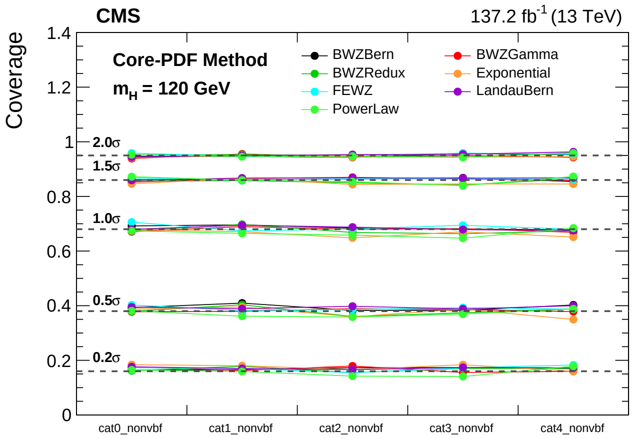

Figure 115: Coverage observed with the *core-PDF* method for 16%, 38%, 68%, 86%, and 95% CL intervals in toys where a signal at $m_{\{H} = 120\,\text{GeV}$ is injected with $\hat{\mu}_{inj} = 1$.

1652 FEWZ × Bernstein.

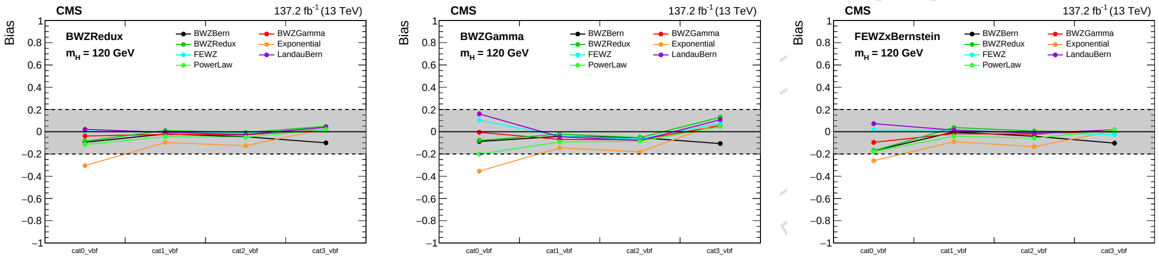

Figure 116: Mean value of the bias distribution in VBF BDT categories obtained by fitting toys, generated according to different truth background models and injecting a nominal signal at $m_{\text{H}} = 120\,\text{GeV}$, with the following functional forms: BWZRedux (left), BWZGamma (middle), and FEWZ × Bernstein (right).

1653 Figure 117 shows the estimated bias on the signal strength for $m_{\text{H}} = 120\,\text{GeV}$ obtained when 1654 the VBF categories are simultaneously fitted using either a BWZRedux (left), a BWZGamma 1655 (middle), or a FEWZ × Bernstein (right) function to model the background. When all VBF 1656 categories are simultaneously fitted no significant bias is observed for any of the truth model 1657 despite the tension observed in VBF cat-0 as reported in the studies presented in Fig. 116.

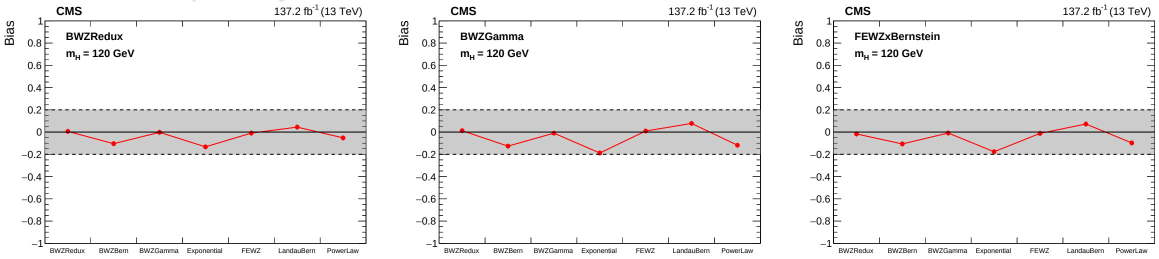

Figure 117: Mean value of the bias distribution on the signal strength obtained by fitting toys, generated across all the VBF categories according to alternative bakground truth models and injecting a nominal signal at $m_{\text{H}} = 120\,\text{GeV}$, with the following functional forms: BWZRedux (left), BWZGamma (middle), and FEWZ × Bernstein (right).

F Bias studies for $m_{\mathrm{H}} = 130\,\mathrm{GeV}$

Figure 118 (left) shows the measured bias on the signal strength for $m_{\mathrm{H}} = 130\,\mathrm{GeV}$ in each of the ggH categories when the background is modelled using the *core-PDF* method. The bias is observed to be smaller than 20% in each category and for each truth function used to generate the toys. In addition, Fig. 118 (right) reports the bias measured in toys where the signal has been injected in all the ggH categories. Also in this case, the bias measured as a function of different truth functions can be controlled within 20%. Bias studies have been performed using the same strategy/method detailed in Section 8.2, where the only difference is the tested Higgs boson mass hypothesis.

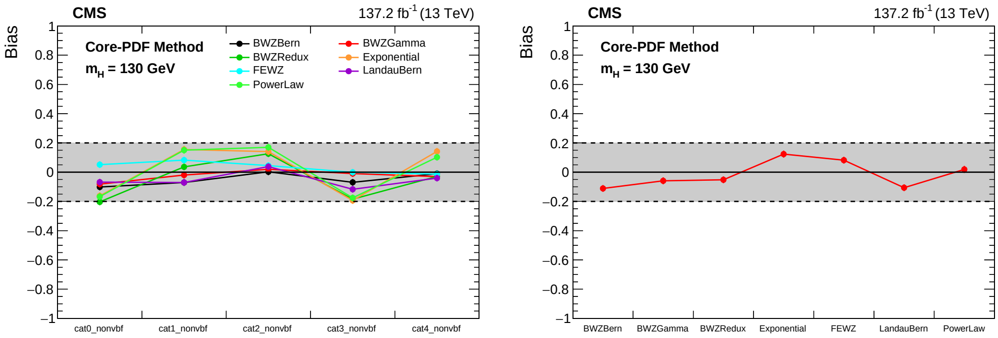

Figure 118: Mean bias on the signal strength obtained from the *core-PDF* method used to model the background in ggH categories. On the left, the measured bias in each of the ggH categories when a nominal signal at $m_{\mathrm{H}} = 130\,\mathrm{GeV}$ is injected. On the right, the bias measured when the signal is injected in all the ggH categories simultaneously.

The coverage within the 68% CL ($\approx 1\sigma$) interval is defined as the ratio between the observed number of toys for which $-2 \times \Delta Log\mathcal{L}(\hat{\mu}_{inj}) < 1$ and the total number of toys generated, where a signal with $\hat{\mu}_{inj} = 1$ and $m_{\{H} = 130\,\mathrm{GeV}$ is injected. Similarly, the coverage for 16% CL ($\approx 0.2\sigma$), 38% CL ($\approx 0.5\sigma$), 86% CL ($\approx 1.5\sigma$), and 95% CL ($\approx 2\sigma$) intervals have also been tested. Figure 119 shows that the method designed for the background estimation has good coverage properties for the tested intervals. The *core-PDF* fit shows, as expected from a discrete likelihood profile, a small overcoverage (by few %) in categories and against some truth models.

Figure 120 shows the measured bias on the signal strength for $m_{\mathrm{H}} = 130\,\mathrm{GeV}$ in the VBF categories obtained by fitting the toys generated via several truth function with BWZRedux, BWZGamma, FEWZ × Bernstein. The number of degrees of freedom considered for each functional form and in each category are listed in Table 27, optimized for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$. The BWZRedux, BWZGamma, and FEWZ × Bernstein exhibit a bias in single categories within 20% against all possible truth models, as already observed for $m_{\mathrm{H}} = 125\,\mathrm{GeV}$ in Section 8.1.2.

Figure 121 shows the estimated bias on the signal strength for $m_{\mathrm{H}} = 130\,\mathrm{GeV}$ obtained when the VBF categories are simultaneously fitted using either a BWZRedux (left), a BWZGamma (middle), or a FEWZ × Bernstein (right) function to model the background. As expected from the studies presented in Fig. 120, when all VBF categories are simultaneously fitted no significant bias is observed for any of the truth model.

DRAFT

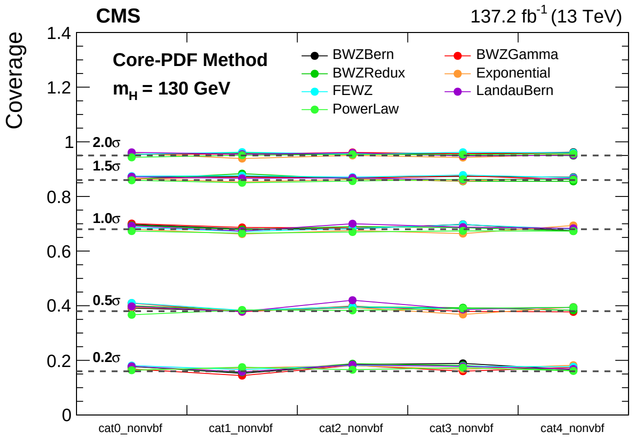

Figure 119: Coverage observed with the *core-PDF* method for 16%, 38%, 68%, 86%, and 95% CL intervals in toys where a signal at $m_{\{H} = 130\,\text{GeV}$ is injected with $\hat{\mu}_{inj} = 1$.

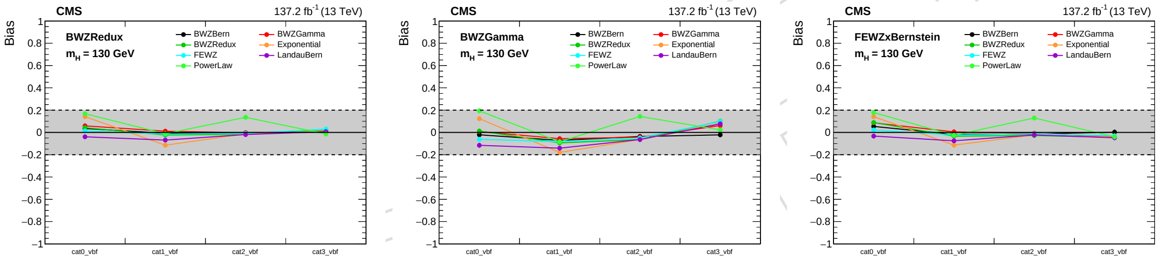

Figure 120: Mean value of the bias distribution in VBF BDT categories obtained by fitting toys, generated according to different truth background models and injecting a nominal signal at $m_{\mathrm{H}} = 130\,\text{GeV}$, with the following functional forms:BWZRedux (left), BWZGamma (middle), and FEWZ $\times$ Bernstein (right).

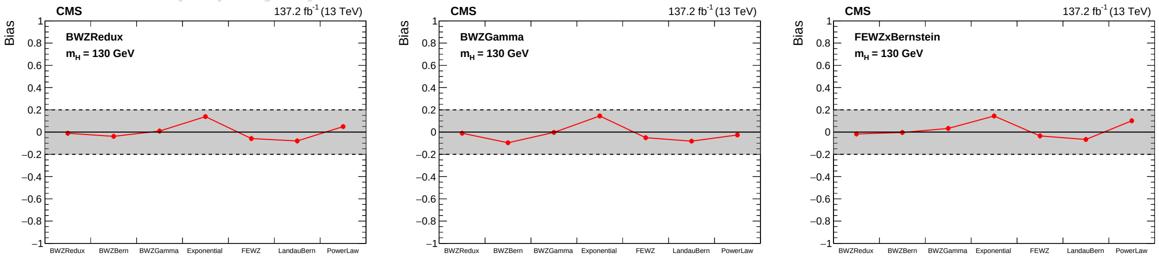

Figure 121: Mean value of the bias distribution on the signal strength obtained by fitting toys, generated across all the VBF categories according to alternative bakground truth models and injecting a nominal signal at $m_{\mathrm{H}} = 130\,\text{GeV}$, with the following functional forms: BWZRedux (left), BWZGamma (middle), and FEWZ $\times$ Bernstein (right).

G Likelihood scans in each category

In this Section, likelihood scans of the signal strength modifier in each category of the analysis are reported, with other parameters of the signal and background models allowed to vary. Systematic uncertainties are included in the form of nuisance parameters. Figure 122 shows the expected likelihood scans obtained by fitting Asimov datasets, defined by fixing the nuisances parameters to their maximum likelihood estimate obtained from a fit to the data in which $\mu = 0$ is assumed, whose crossing at $2 \cdot \Delta\mathrm{Log}\mathcal{L} = 3.84$ roughly corresponds to the expected asymptotic upper limit. In contrast, Fig. 123 reports the expected likelihood scans obtained by fitting alternative Asimov datasets in which nuisance parameters are still estimated from a background-only fit to the data but a signal is also injected, with a rate corresponding to the one predicted by the SM. The crossing point of this likelihood scan at $\mu = 0$ gives the expected significance of the signal-plus-background hypothesis w.r.t. the background-only hypothesis. Figure 124 shows the observed likelihood scans obtained by fitting each category independently.

DRAFT

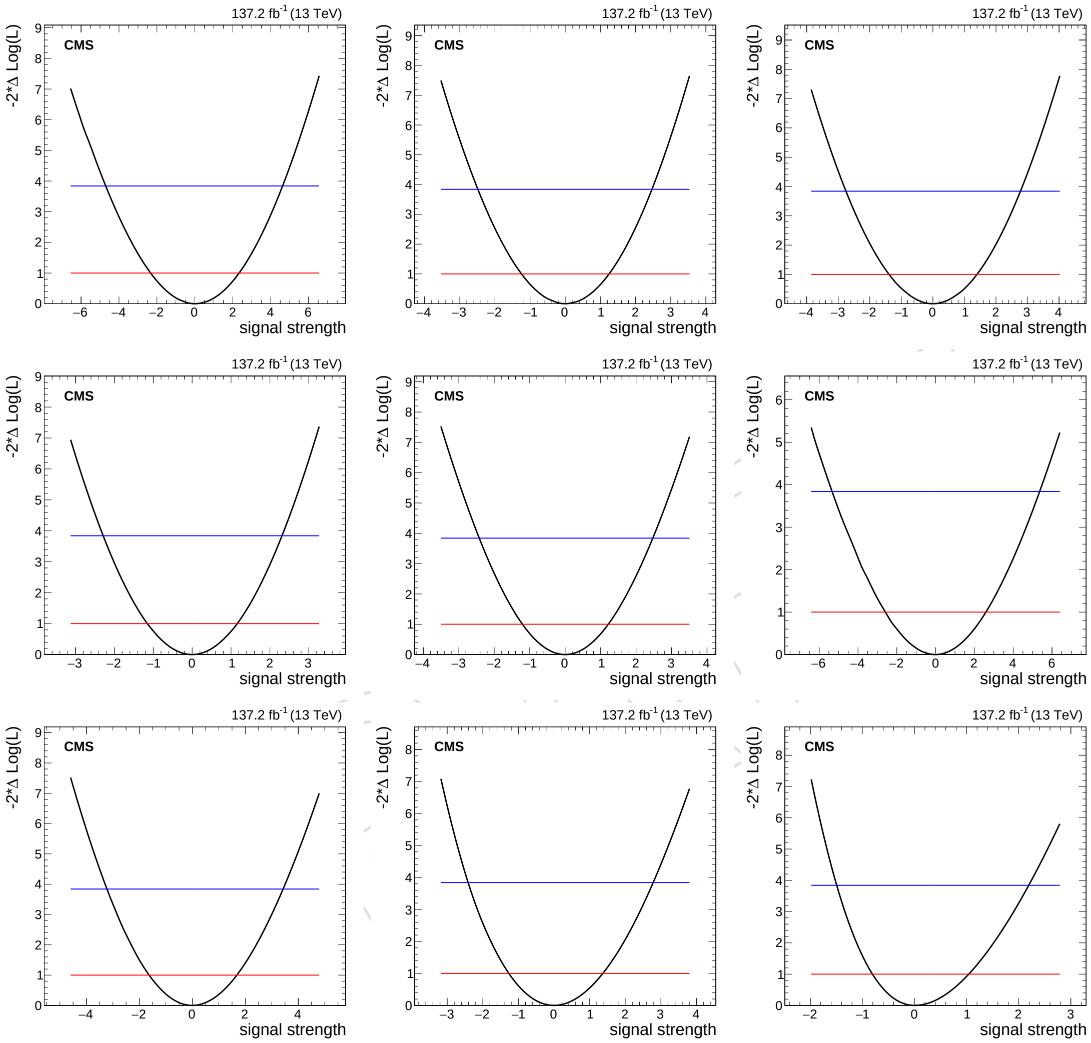

Figure 122: Expected profile likelihood ratios as a function of the signal strength modifier (μ) for signal-plus-background fits performed in individual categories. Likelihood scan obtained from an Asimov dataset defined by fixing nuisances parameters to the values extracted from a background-only fit to the data. First row: ggH-cat0 (left), ggH-cat1 (middle), and ggH-cat2 (right). Second row: ggH-cat3 (left), ggH-cat4 (middle), and VBF-cat0 (right). Third row: VBF-cat1 (left), VBF-cat2 (middle), and VBF-cat3 (right).

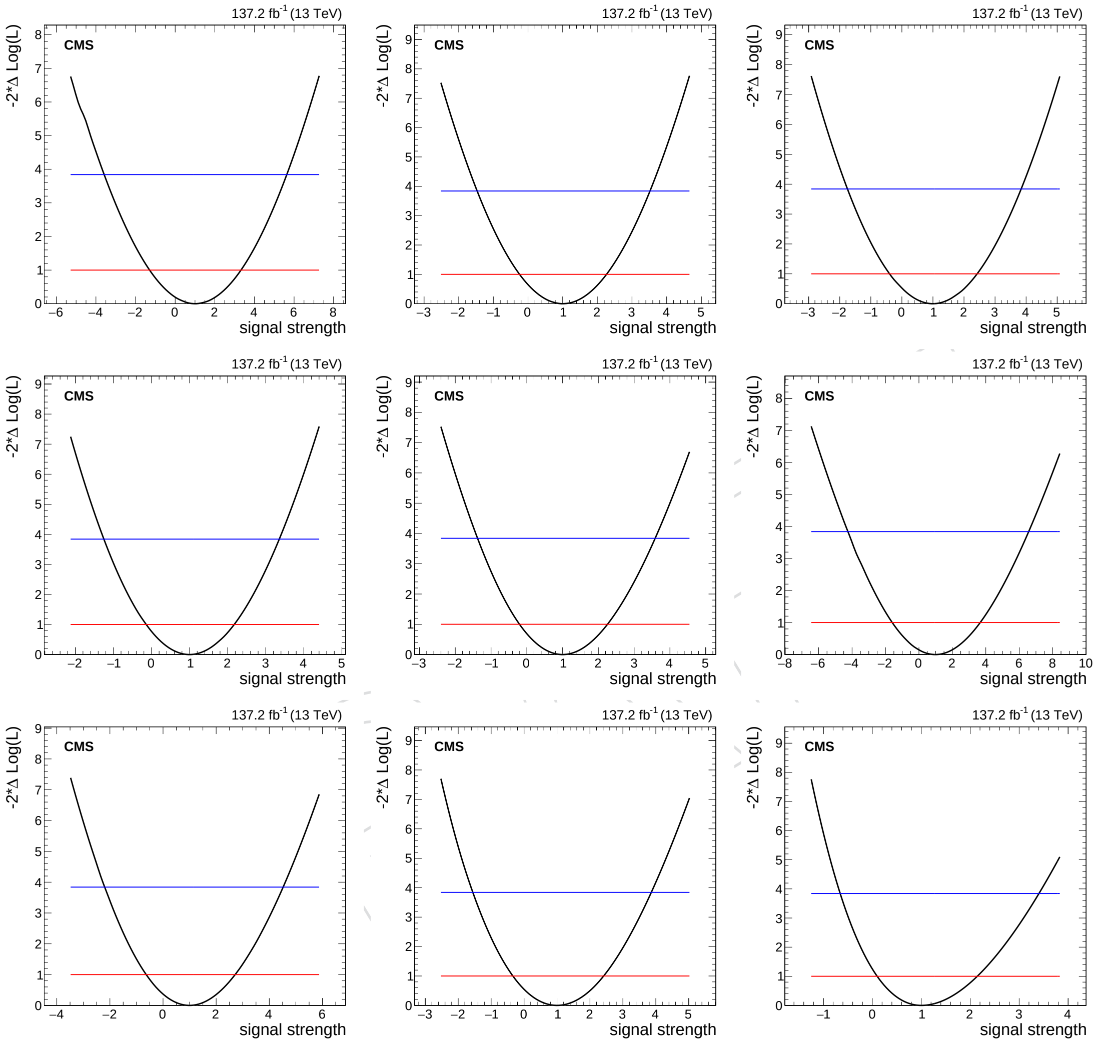

Figure 123: Expected profile likelihood ratios as a function of the signal strength modifier (μ) for signal-plus-background fits performed in individual categories. Likelihood scan obtained by fitting an Asimov dataset in which the expected Higgs boson signal, assuming SM production cross sections, has been injected. First row: ggH-cat0 (left), ggH-cat1 (middle), and ggH-cat2 (right). Second row: ggH-cat3 (left), ggH-cat4 (middle), and VBF-cat0 (right). Third row: VBF-cat1 (left), VBF-cat2 (middle), and VBF-cat3 (right).

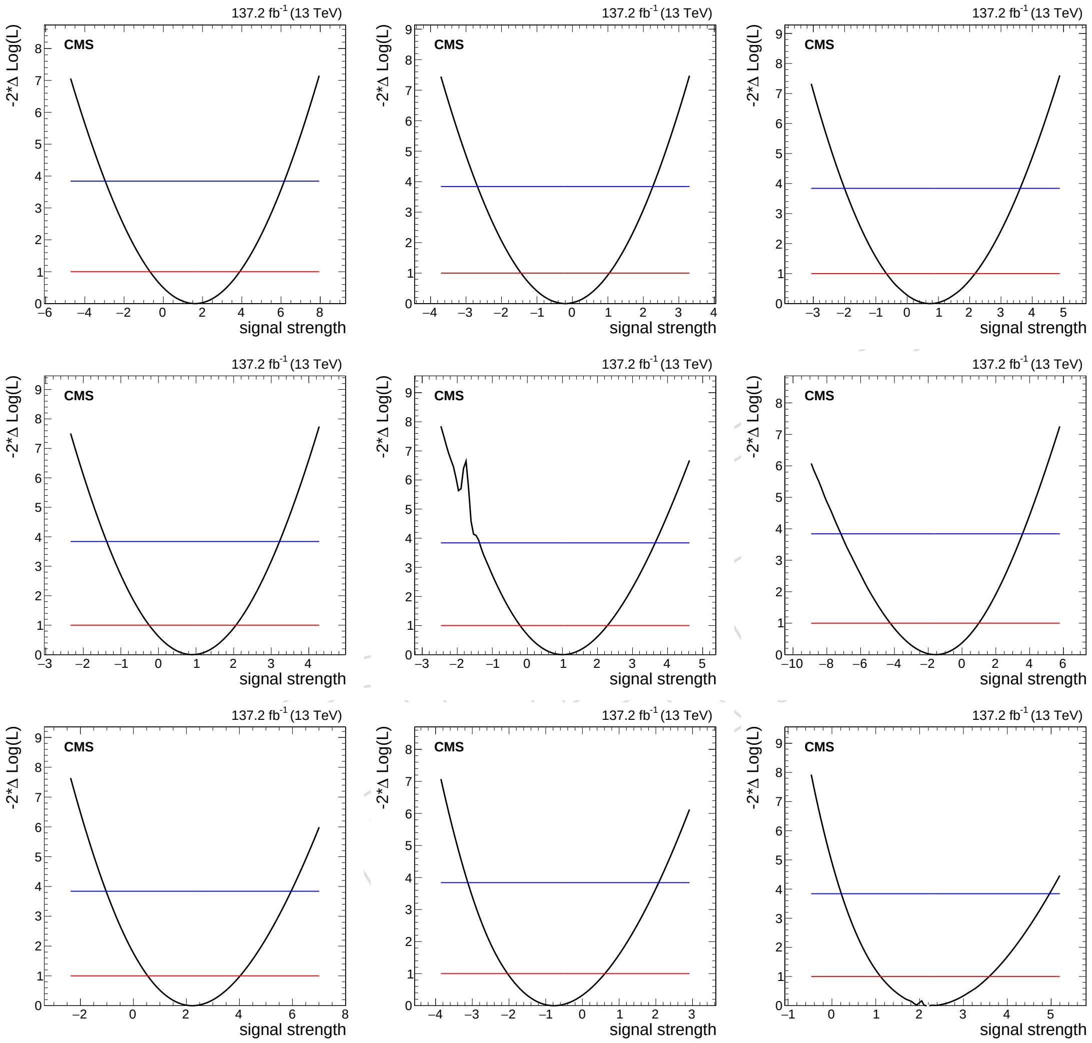

Figure 124: Observed profile likelihood ratios as a function of the signal strength modifier (μ) for signal-plus-background fits performed in individual categories. First row: ggH-cat0 (left), ggH-cat1 (middle), and ggH-cat2 (right). Second row: ggH-cat3 (left), ggH-cat4 (middle), and VBF-cat0 (right). Third row: VBF-cat1 (left), VBF-cat2 (middle), and VBF-cat3 (right).

References

[1] ATLAS Collaboration, “Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC”, *Phys. Lett.* **B716** (2012) 1–29, `doi:10.1016/j.physletb.2012.08.020`, `arXiv:1207.7214`.

[2] CMS Collaboration, “Observation of a new boson at a mass of 125 GeV with the CMS experiment at the LHC”, *Phys. Lett.* **B716** (2012) 30–61, `doi:10.1016/j.physletb.2012.08.021`, `arXiv:1207.7235`.

[3] CMS Collaboration, “Observation of the Higgs boson decay to a pair of τ leptons with the CMS detector”, *Phys. Lett.* **B779** (2018) 283–316, `doi:10.1016/j.physletb.2018.02.004`, `arXiv:1708.00373`.

[4] CMS Collaboration, “Observation of $\mathrm{t\bar{t}H}$ production”, *Phys. Rev. Lett.* **120** (2018), no. 23, 231801, `doi:10.1103/PhysRevLett.120.231801`, `arXiv:1804.02610`.

[5] CMS Collaboration, “Observation of Higgs boson decay to bottom quarks”, *Phys. Rev. Lett.* **121** (2018), no. 12, 121801, `doi:10.1103/PhysRevLett.121.121801`, `arXiv:1808.08242`.

[6] CMS Collaboration, “Combined measurements of Higgs boson couplings in proton-proton collisions at $\sqrt{s} = 13$ TeV”, *Eur. Phys. J.* **C79** (2019), no. 5, 421, `doi:10.1140/epjc/s10052-019-6909-y`, `arXiv:1809.10733`.

[7] ATLAS Collaboration, “Cross-section measurements of the Higgs boson decaying into a pair of τ-leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Phys. Rev.* **D99** (2019) 072001, `doi:10.1103/PhysRevD.99.072001`, `arXiv:1811.08856`.

[8] ATLAS Collaboration, “Observation of Higgs boson production in association with a top quark pair at the LHC with the ATLAS detector”, *Phys. Lett.* **B784** (2018) 173–191, `doi:10.1016/j.physletb.2018.07.035`, `arXiv:1806.00425`.

[9] ATLAS Collaboration, “Observation of $H \to b\bar{b}$ decays and VH production with the ATLAS detector”, *Phys. Lett.* **B786** (2018) 59–86, `doi:10.1016/j.physletb.2018.09.013`, `arXiv:1808.08238`.

[10] ATLAS Collaboration, “Measurement of the Higgs boson mass in the $H \to ZZ^* \to 4\ell$ and $H \to \gamma\gamma$ channels with $\sqrt{s} = 13$ TeV pp collisions using the ATLAS detector”, *Phys. Lett.* **B784** (2018) 345–366, `doi:10.1016/j.physletb.2018.07.050`, `arXiv:1806.00242`.

[11] ATLAS Collaboration, “Measurements of gluon-gluon fusion and vector-boson fusion Higgs boson production cross-sections in the $H \to WW^* \to e\nu\mu\nu$ decay channel in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector”, *Phys. Lett.* **B789** (2019) 508–529, `doi:10.1016/j.physletb.2018.11.064`, `arXiv:1808.09054`.

[12] LHC Higgs Cross Section Working Group Collaboration, “Handbook of LHC Higgs Cross Sections: 4. Deciphering the Nature of the Higgs Sector”, `doi:10.23731/CYRM-2017-002`, `arXiv:1610.07922`.

[13] CMS Collaboration, “Search for the Higgs boson decaying to two muons in proton-proton collisions at $\sqrt{s} = 13$ TeV”, *Phys. Rev. Lett.* **122** (2019), no. 2, 021801, `doi:10.1103/PhysRevLett.122.021801`, `arXiv:1807.06325`.

DRAFT

[14] ATLAS Collaboration Collaboration, "A search for the rare decay of the Standard Model Higgs boson to dimuons in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS Detector", Technical Report ATLAS-CONF-2018-026, CERN, Geneva, Jul, 2018.

[15] ATLAS Collaboration, "Search for the dimuon decay of the Higgs boson in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector", *Phys. Rev. Lett.* **119** (2017), no. 5, 051802, doi:10.1103/PhysRevLett.119.051802, arXiv:1705.04582.

[16] J. Alwall et al., "The automated computation of tree-level and next-to-leading order differential cross sections, and their matching to parton shower simulations", *JHEP* **07** (2014) 079, doi:10.1007/JHEP07(2014)079, arXiv:1405.0301.

[17] P. Nason, "A New method for combining NLO QCD with shower Monte Carlo algorithms", *JHEP* **11** (2004) 040, doi:10.1088/1126-6708/2004/11/040, arXiv:hep-ph/0409146.

[18] T. Sjostrand, S. Mrenna, and P. Z. Skands, "A Brief Introduction to PYTHIA 8.1", *Comput. Phys. Commun.* **178** (2008) 852–867, doi:10.1016/j.cpc.2008.01.036, arXiv:0710.3820.

[19] CMS Collaboration, "Event generator tunes obtained from underlying event and multiparton scattering measurements", *Eur. Phys. J.* **C76** (2016), no. 3, 155, doi:10.1140/epjc/s10052-016-3988-x, arXiv:1512.00815.

[20] CMS Collaboration, "Extraction and validation of a new set of CMS PYTHIA8 tunes from underlying-event measurements", arXiv:1903.12179.

[21] J. Alwall et al., "Comparative study of various algorithms for the merging of parton showers and matrix elements in hadronic collisions", *Eur. Phys. J.* **C53** (2008) 473–500, doi:10.1140/epjc/s10052-007-0490-5, arXiv:0706.2569.

[22] R. Frederix and S. Frixione, "Merging meets matching in MC@NLO", *JHEP* **12** (2012) 061, doi:10.1007/JHEP12(2012)061, arXiv:1209.6215.

[23] NNPDF Collaboration, "Parton distributions for the LHC Run II", *JHEP* **04** (2015) 040, doi:10.1007/JHEP04(2015)040, arXiv:1410.8849.

[24] NNPDF Collaboration, "Parton distributions from high-precision collider data", *Eur. Phys. J.* **C77** (2017), no. 10, 663, doi:10.1140/epjc/s10052-017-5199-5, arXiv:1706.00428.

[25] S. Agostinelli et al., "Geant4 : a simulation toolkit", *Nuclear Instruments and Methods in Physics Research Section A: Accelerators, Spectrometers, Detectors and Associated Equipment* **506** (2003), no. 3, 250 – 303, doi:https://doi.org/10.1016/S0168-9002(03)01368-8.

[26] K. Hamilton, P. Nason, E. Re, and G. Zanderighi, "NNLOPS simulation of Higgs boson production", *JHEP* **10** (2013) 222, doi:10.1007/JHEP10(2013)222, arXiv:1309.0017.

[27] K. Hamilton, P. Nason, and G. Zanderighi, "Finite quark-mass effects in the NNLOPS POWHEG+MiNLO Higgs generator", *JHEP* **05** (2015) 140, doi:10.1007/JHEP05(2015)140, arXiv:1501.04637.

DRAFT

[28] J. Bellm et al., "Herwig 7.0/Herwig++ 3.0 release note", *Eur. Phys. J.* **C76** (2016), no. 4, 196, `doi:10.1140/epjc/s10052-016-4018-8`, `arXiv:1512.01178`.

[29] B. Jäger et al., "Parton-shower effects in Higgs production via Vector-Boson Fusion", `arXiv:2003.12435`.

[30] CMS Collaboration Collaboration, "Template-based search for Higgs boson decay to muons targeting the VBF production mode with full Run-2 data", Technical Report AN-19-205, Geneva, 2019.

[31] CMS Collaboration, "Electroweak production of two jets in association with a Z boson in proton–proton collisions at $\sqrt{s} = 13$ TeV", *Eur. Phys. J.* **C78** (2018), no. 7, 589, `doi:10.1140/epjc/s10052-018-6049-9`, `arXiv:1712.09814`.

[32] Y. Li and F. Petriello, "Combining QCD and electroweak corrections to dilepton production in FEWZ", *Phys. Rev.* **D86** (2012) 094034, `doi:10.1103/PhysRevD.86.094034`, `arXiv:1208.5967`.

[33] D. Bourilkov, "Photon-induced Background for Dilepton Searches and Measurements in pp Collisions at 13 TeV", `arXiv:1606.00523`.

[34] D. Bourilkov, "Exploring the LHC Landscape with Dileptons", `arXiv:1609.08994`.

[35] M. Czakon, P. Fiedler, and A. Mitov, "Total Top-Quark Pair-Production Cross Section at Hadron Colliders Through $O(\frac{4}{S})$", *Phys. Rev. Lett.* **110** (2013) 252004, `doi:10.1103/PhysRevLett.110.252004`, `arXiv:1303.6254`.

[36] T. Gehrmann et al., "W^+W^- Production at Hadron Colliders in Next to Next to Leading Order QCD", *Phys. Rev. Lett.* **113** (2014), no. 21, 212001, `doi:10.1103/PhysRevLett.113.212001`, `arXiv:1408.5243`.

[37] M. Grazzini, S. Kallweit, D. Rathlev, and M. Wiesemann, "$W^{\pm}Z$ production at the LHC: fiducial cross sections and distributions in NNLO QCD", *JHEP* **05** (2017) 139, `doi:10.1007/JHEP05(2017)139`, `arXiv:1703.09065`.

[38] M. Grazzini, S. Kallweit, and D. Rathlev, "ZZ production at the LHC: fiducial cross sections and distributions in NNLO QCD", *Phys. Lett. B* **750** (2015) 407–410, `doi:10.1016/j.physletb.2015.09.055`, `arXiv:1507.06257`.

[39] CMS Collaboration, "Measurements of properties of the Higgs boson decaying into the four-lepton final state in pp collisions at $\sqrt{s} = 13$ tev", *JHEP* **11** (2017) 047, `doi:10.1007/JHEP11(2017)047`, `arXiv:1706.09936`.

[40] CMS Muon POG, "Baseline muon selections for Run-II – Medium ID". `https://twiki.cern.ch/twiki/bin/viewauth/CMS/SWGuideMuonIdRun2#Medium_Muon`.

[41] CMS Muon POG, "Baseline muon selections for Run-II – Particle flow isolation". `https://twiki.cern.ch/twiki/bin/viewauth/CMS/SWGuideMuonIdRun2#Particle_Flow_isolation`.

[42] CMS Collaboration, "Particle-flow reconstruction and global event description with the CMS detector", *JINST* **12** (2017), no. 10, P10003, `doi:10.1088/1748-0221/12/10/P10003`, `arXiv:1706.04965`.

DRAFT

[43] CMS Collaboration, "Search for a new scalar resonance decaying to a pair of Z bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV", *JHEP* **06** (2018) 127, `doi:10.1007/JHEP06(2018)127,10.1007/JHEP03(2019)128`, `arXiv:1804.01939`. [Erratum: JHEP03,128(2019)].

[44] O. Rieger, T. Kramer, and P. Schleper, "Report on Muon ID/Isolation and FSR Recovery". `https://indico.cern.ch/event/788758`.

[45] M. Cacciari, G. P. Salam, and G. Soyez, "The anti-k_t jet clustering algorithm", *JHEP* **04** (2008) 063, `doi:10.1088/1126-6708/2008/04/063`, `arXiv:0802.1189`.

[46] M. Cacciari and G. P. Salam, "Pileup subtraction using jet areas", *Phys. Lett. B* **659** (2008) 119, `doi:10.1016/j.physletb.2007.09.077`, `arXiv:0707.1378`.

[47] CMS Collaboration, "Jet energy scale and resolution in the CMS experiment in pp collisions at 8 TeV", *JINST* **12** (2017) P02014, `doi:10.1088/1748-0221/12/02/P02014`, `arXiv:1607.03663`.

[48] CMS JetMET POG, "Recommended Jet Energy Corrections and Uncertainties For Data and MC". `https://twiki.cern.ch/twiki/bin/view/CMS/JECDataMC`.

[49] Raffaele Gerosa, ""PUjetID in VBF H → $\mu\mu$ analysis"". `https://indico.cern.ch/event/860457/contributions/3623773/attachments/1939388/3215059/Raffaele_05_11_2019.pdf`.

[50] CMS JetMET POG, "Recommended MET filters". `https://twiki.cern.ch/twiki/bin/view/CMS/MissingETOptionalFiltersRun2`.

[51] CMS EGM POG, "Multivariate Electron Identification for Run2". `https://twiki.cern.ch/twiki/bin/view/CMS/MultivariateElectronIdentificationRun2`.

[52] CMS BTV POG, "Usage of b/c Tag Objects for 13 TeV Data in 2016 data and simulation". `https://twiki.cern.ch/twiki/bin/viewauth/CMS/BtagRecommendation2016Legacy`.

[53] CMS BTV POG, "Heavy flavour tagging for 13 TeV data in 2017 data and simulation". `https://twiki.cern.ch/twiki/bin/viewauth/CMS/BtagRecommendation94X`.

[54] CMS BTV POG, "Heavy flavour tagging for 13 TeV data in 2018 data and simulation". `https://twiki.cern.ch/twiki/bin/viewauth/CMS/BtagRecommendation102X`.

[55] CMS Lumi POG, "Utilities for Accessing Pileup Information for Data". `https://twiki.cern.ch/twiki/bin/view/CMS/PileupJSONFileforData`.

[56] CMS BTV POG, "Methods to apply b-tagging efficiency scale factors". `https://twiki.cern.ch/twiki/bin/view/CMS/BTagSFMethods`.

[57] CMS Muon POG, "Rochester Correction". `https://twiki.cern.ch/twiki/bin/viewauth/CMS/RochcorMuon`.

[58] CMS Collaboration Collaboration, "H → $\mu\mu$ with the Run 2 datasets", Technical Report AN-19-124, Geneva, 2019.

DRAFT

[59] J. C. Collins and D. E. Soper, "Angular distribution of dileptons in high-energy hadron collisions", *Phys. Rev. D* **16** (Oct, 1977) 2219–2225, `doi:10.1103/PhysRevD.16.2219`.

[60] J. R. Andersen, K. Arnold, and D. Zeppenfeld, "Azimuthal angle correlations for higgs boson plus multi-jet events", *Journal of High Energy Physics* **2010** (Jun, 2010) 91, `doi:10.1007/JHEP06(2010)091`.

[61] D. Rainwater, R. Szalapski, and D. Zeppenfeld, "Probing color-singlet exchange in $z + 2$-jet events at the cern lhc", *Phys. Rev. D* **54** (Dec, 1996) 6680–6689, `doi:10.1103/PhysRevD.54.6680`.

[62] LHC Higgs cross section working group I, "Uncertainties for ggH process in different kinematic regions". `https://twiki.cern.ch/twiki/bin/view/LHCPhysics/LHCHXSWGGGF_RUN2`.

[63] LHC Higgs cross section working group II, "Uncertainties for VBF-H process in different kinematic regions". `https://gitlab.cern.ch/LHCHIGGSXS/LHCHXSWG2/STXS/VBF-Uncertainties`.

[64] CMS JetMET POG, "Sources of uncertainty affecting jet energy scale in CMS". `https://twiki.cern.ch/twiki/bin/view/CMS/JECUncertaintySources`.

[65] CMS JetMET POG, "Reduced set of jet energy uncertainties for Run2 searches". `https://twiki.cern.ch/twiki/bin/viewauth/CMS/JECUncertaintySources#Run_2_reduced_set_of_uncertainty`.

[66] C. S. Committee, "Report on parametric density estimator". `https://twiki.cern.ch/twiki/bin/view/CMS/DensityEstimation`.

[67] S. Baker and R. D. Cousins, "Clarification of the Use of Chi Square and Likelihood Functions in Fits to Histograms", *Nucl. Instrum. Meth.* **221** (1984) 437–442, `doi:10.1016/0167-5087(84)90016-4`.

[68] CMS Collaboration, "Measurements of Higgs boson properties in the diphoton decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV", *JHEP* **11** (2018) 185, `doi:10.1007/JHEP11(2018)185`, `arXiv:1804.02716`.

[69] P. D. Dauncey, M. Kenzie, N. Wardle, and G. J. Davies, "Handling uncertainties in background shapes", *JINST* **10** (2015), no. 04, P04015, `doi:10.1088/1748-0221/10/04/P04015`, `arXiv:1408.6865`.

[70] T. Junk, "Confidence level computation for combining searches with small statistics", *Nucl. Instrum. Meth.* **A434** (1999) 435–443, `doi:10.1016/S0168-9002(99)00498-2`, `arXiv:hep-ex/9902006`.

[71] A. L. Read, "Presentation of search results: theCLstechnique", *Journal of Physics G: Nuclear and Particle Physics* **28** (sep, 2002) 2693–2704, `doi:10.1088/0954-3899/28/10/313`.

[72] G. Cowan, K. Cranmer, E. Gross, and O. Vitells, "Asymptotic formulae for likelihood-based tests of new physics", *Eur. Phys. J.* **C71** (2011) 1554, `doi:10.1140/epjc/s10052-011-1554-0`, `10.1140/epjc/s10052-013-2501-z`, `arXiv:1007.1727`. [Erratum: Eur. Phys. J.C73,2501(2013)].

DRAFT